

The Igusa Square Root

The Borchers determinant of $\phi_{o,i}$, its denominator algebra, and the $K_3 \times E$ realization problem

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Abstract

For $X = K_3 \times E$ the theorem-level physical input used here is the protected index. Right-moving localization in the K_3 Ramond sector gives

$$Z_{K_3} = 2\phi_{o,i}, \quad \phi_{o,i} = \sum f(n, l) q^n r^l.$$

Replacing the protected K_3 index by its Borchers half produces only a virtual K_o -determinant package. It does not produce a microscopic Hilbert space. Unitarity is used to localize the right-moving K_3 supertrace before the index is taken; the index itself is a charge-graded virtual object that no longer requires Hilbert positivity. For virtual super vector spaces $\mathbb{U}_{n,l}$ with $\text{sdim } \mathbb{U}_{n,l} = f(n, l)$, set $\mathcal{V}_{(n,l,m)} = \mathbb{U}_{nm,l}$. In the Borchers cusp chamber the determinant is

$$\mathcal{D}_X(Z) = 64 q^{1/2} r^{1/2} s^{1/2} \prod_{(n,l,m) \in \Gamma_{\text{eff}}} (1 - q^n r^l s^m)^{f(nm,l)} = \Delta_5(Z).$$

The constant 64 is the theta-constant normalization $[q^{1/2} r^{1/2} s^{1/2}] \Delta_5 = 64$; hence the monic Borchers product is $D_5 = 64^{-1} \Delta_5$.

The same coefficients are the signed root supermultiplicities in the Gritsenko–Nikulin Borchers–Kac–Moody denominator algebra. Under

$$\alpha(n, l, m) = 2nf_2 - lf_3 + 2mf_{-2}$$

one has

$$\text{den}(\mathfrak{g}_{\Delta_5}) = 64^{-1} \Delta_5(2Z), \quad \text{smult } \mathfrak{g}_{\Delta_5, \alpha(n,l,m)} = f(nm, l)$$

on the active support. Thus the denominator algebra supplies the algebraic square root target and its target-internal Borchers bracket. It does not supply a compact BPS Hilbert space, compact Hall correspondences, an orientation, or a BPS operator product/local collision map.

The square root in the title is meant in Dirac's sense. The scalar OP/DT branch $Z_{\text{OP/DT}}^X = -4096 \Delta_5^{-2}$ is orientation-forgetful: the sign character is squared away. The recognition target is a first-order protected operator/algebra $\mathfrak{D}_X$ with protected Pfaffian $\text{Pf}_{\text{prot}}(\mathfrak{D}_X) = \Delta_5$ and scalar trace $\text{Tr}_{\text{prot}}((-1)^F \mathfrak{D}_X^{-2}) = -4096 \Delta_5^{-2}$; the orientation/Pfaffian datum is the algebraic input that distinguishes Δ_5 from $-\Delta_5$ and that the squared scalar cannot recover. Constructing $\mathfrak{D}_X$ is the open recognition problem; this paper formulates the target and the orientation-character constraint $\varepsilon_o = \nu_{\Delta_5}$ on $W^{(2)}(\Lambda_{II}^{2,1})$.

The scalar Behrend-weighted branch is imported from Oberdieck–Pixton, Oberdieck–Pandharipande, and Oberdieck. In OP’s normalization $\chi_{\text{io}}^{\text{OP}} = D_5^2$, so

$$\chi_{\text{io}}^{\text{OP}} = (64^{-1} \Delta_5)^2 = 4096^{-1} \Delta_5^2, \quad Z_{\text{DT}}^X = Z_{\text{OP}}^X = -\left(\chi_{\text{io}}^{\text{OP}}\right)^{-1} = -4096 \Delta_5^{-2}.$$

The factor $4096 = 64^2$ is a normalization conversion; the minus sign is the OP scalar convention. Neither fixes the automorphic reflection character or the compact orientation monodromy.

A compact realization would have to begin with an additive physical charge lattice Γ_X^{phys} , the Gram map $\Pi_X : \Gamma_X^{\text{phys}} \rightarrow \Gamma_{\text{gram}}$, and a holomorphic E_3 -factorization algebra on $K_3 \times E$, with a specified formality point, S^3 -framing, quantum master equation, anomaly control, descent, and finite-first HN completion. It would then specialize along the K_3 direction to a chiral object on $\text{Ran}(E)$, solve the $\text{Ran}(E)$ -support-locality obstruction in the K_3 -integrated chiral shadow by mixed local/wrapped correspondences, and identify the Gram-descended protected primitive bracket with $\mathfrak{g}_{\Delta_5}$. This paper proves the virtual determinant, records the imported scalar and denominator theorems, and formulates that compact Hall/factorization realization as an open recognition problem. The final part records the eight Clery–Gritsenko diagonal-divisor rows and the certificate required before any boundary row becomes a physical host.

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I INTRODUCTION AND CLAIM STRENGTH

The local object comes first. A compact BPS theory on $X = K_3 \times E$, if it exists in the form needed here, should produce protected local observables before any trace is taken. After twisting and passing to the protected index, the positive Hilbert space has collapsed to a charge-graded virtual object, but locality has not disappeared: disjoint insertions must still factorize and collisions must remember extensions.

Thus the natural order is

$$\begin{aligned} & \text{protected locality} \longrightarrow \text{compact/hybrid factorization} \longrightarrow \text{primitive recognition} \\ & \longrightarrow \text{Borcherds denominator} \longrightarrow \text{OP scalar square} \longrightarrow \text{eight-row host certificates.} \end{aligned}$$

The paper follows this order at the level of claims. Some arrows are proved in this manuscript, some are imported theorems, and some are open realization requirements.

1.0.0.1 Protected locality. The protected index is theorem-level. Right-moving localization of the unitary K_3 sigma model gives

$$Z_{K_3} = 2\phi_{0,1}, \quad \phi_{0,1} = \sum f(n, l) q^n r^l.$$

The half-index used by the Borcherds lift is represented only in $K_0(\text{sVect}_{\text{fd}})$. Its coefficients are signed superdimensions, not dimensions of a constructed compact D-brane Hilbert space.

1.0.0.2 Compact and hybrid factorization. A compact realization would first supply the additive physical charge lattice Γ_X^{phys} , the non-additive Gram map $\Pi_X : \Gamma_X^{\text{phys}} \rightarrow \Gamma_{\text{gram}}$, and a charge-completed holomorphic E_3 -factorization algebra

$$\mathcal{A}_X^{E_3}, \quad X = K_3 \times E,$$

then a chain-level K_3 -to- E specialization

$$\mathcal{F}_X \simeq \text{Sp}_{K_3, E}^{\text{ch}}(\mathcal{A}_X^{E_3}) \in \text{Fact}(\text{Ran}(E)).$$

This is not reconstructed from the scalar partition function or from the Borcherds product. There is also a geometric obstruction: a one-dimensional subscheme with positive elliptic degree projects onto all of E . Hence a naive support-local object on $\text{Ran}(E)$ sees only projection-finite sectors. The full Igusa s -direction requires hybrid local/global factorization for wrapped E -sectors. A separate global Hecke/Fock mode is only a determinant-level repair unless it is coupled to the projection-finite sector by mixed Hall actions and then tested at the primitive bracket level. This is open.

1.0.0.3 Primitive recognition. The microscopic algebra sought from such a compact realization is first the upstairs protected primitive bracket

$$[\ , \] : \text{Prim}_c^{\text{prot}}(\mathcal{F}_X) \otimes \text{Prim}_{c'}^{\text{prot}}(\mathcal{F}_X) \longrightarrow \text{Prim}_{c+c'}^{\text{prot}}(\mathcal{F}_X), \quad c, c' \in \Gamma_X^{\text{phys}}.$$

The recognition certificate proved later is conditional: if the compact Igusa datum $\mathfrak{R}_X$ exists, including the oriented Hall object, protected integration, Gram descent $\Pi_{X,*}$, chiral bar-cobar comparison, presentation-level primitive Lie comparison, no-extra-relations theorem, completed PBW compatibility, and radical quotient, then

$$H^0 \text{Prim}_{\text{prot}}(\Pi_{X,*}\mathcal{F}_X) \cong \mathfrak{g}_{\Delta_5}$$

and the chiral cobar object is the formal current envelope $\text{Den}_{\Delta_5, E}$. This is a recognition criterion, not an existence theorem.

1.0.0.4 Borchers denominator. The target algebra is imported from the Borchers–Gritsenko–Nikulin construction. With

$$\alpha(n, l, m) = 2nf_2 - lf_3 + 2mf_{-2},$$

the visible active coefficients satisfy

$$\text{smult } \mathfrak{g}_{\Delta_5, \alpha(n, l, m)} = f(nm, l), \quad \text{den}(\mathfrak{g}_{\Delta_5}) = 64^{-1} \Delta_5(2Z).$$

This determines signed root supermultiplicities and the denominator identity. It does not determine compact Hall constants.

1.0.0.5 OP scalar square. The reduced scalar branch is imported from Oberdieck–Pixton, Oberdieck–Pandharipande, and Oberdieck:

$$Z_{\text{DT}}^X(T, Q, P) = Z_{\text{OP}}^X(P, Q, T) = -4096 \Delta_5(Z)^{-2}.$$

The constants are fixed once. The theta-constant form satisfies

$$[q^{1/2} r^{1/2} s^{1/2}] \Delta_5 = 64, \quad D_5 = 64^{-1} \Delta_5.$$

OP's monic product is $\chi_{\text{io}}^{\text{OP}} = D_5^2$, hence

$$\chi_{\text{io}}^{\text{OP}} = 4096^{-1} \Delta_5^2, \quad Z_{\text{OP}}^X = -(\chi_{\text{io}}^{\text{OP}})^{-1} = -4096 \Delta_5^{-2}.$$

The number 64 is a theta-leading coefficient. The number -4096 is the OP scalar normalization. Neither proves the compact orientation character. The comparison

$$\epsilon_o = \nu_{\Delta_5}$$

between geometric orientation monodromy and the automorphic reflection character remains part of the open compact realization datum.

1.0.0.6 *Eight-row host certificates.* The diagonal-divisor boundary problem concerns the eight Clery–Gritsenko forms

$$\Delta_5, \Delta_2, \Delta_1, \Delta_{1/2}, \nabla_3, \nabla_2, \nabla_{3/2}, Q_1.$$

Rows 1–4 have Gritsenko–Nikulin denominator theorems. Rows 5–8 have Clery–Gritsenko products and Govindarajan CHL/WKB denominator theorems; $F_7 = \nabla_{3/2}$ is on the required μ_4 -multiplier cover. A physical host for any boundary row requires a certificate fixing the modular group or cover, character, divisor, cusp Jacobi inputs, product lattice, Weyl vector, real roots, reflection character, physical charge lattice, protected Gram-index, orientation, reduced scalar theorem if one is claimed, and primitive algebra comparison. Without that certificate the row is only an automorphic product plus any separately proved denominator theorem.

Structural level	Claim strength	Content
Virtual determinant	proved here from K_o data plus imported Borcherds product	$\mathcal{D}_X = \Delta_5$. No canonical half-Hilbert space or operator product is constructed.
Scalar Behrend square	imported theorem	Oberdieck–Pixton, Oberdieck–Pandharipande, and Oberdieck give $Z_{\text{DT}}^X = Z_{\text{OP}}^X = -4096\Delta_5^{-2}$ in the reduced quotient-by- E , Behrend-weighted setting.
BKM denominator	imported theorem	Gritsenko–Nikulin give $\mathfrak{g}_{\Delta_5}$ and $\text{den}(\mathfrak{g}_{\Delta_5}) = 64^{-1}\Delta_5(2Z)$.
Compact Hall/factorization realization	conditional/open	If $\mathfrak{R}_X$ exists, the primitive and chiral comparisons follow by the recognition criterion. Existence, $\text{Ran}(E)$ hybrid locality, protected integration, and orientation monodromy are open.
Eight-row host certificates	framework; mostly open	The catalogue and denominator rows are imported where cited. Physical row maps and scalar squares require row-by-row certificates.

All super vector spaces in the representation-theoretic constructions are over $\mathbb{C}$. Real vector spaces occur only after tensoring lattices with $\mathbb{R}$ to discuss cones, chambers, and period domains. Products may be written over Γ_{eff} ; the active factors are exactly the triples in Γ_{act} , where the exponent $f(nm, l)$ is non-zero.

Part I

The $N = 1$ Igusa square root

2 THE PHYSICAL QUESTION

A charged BPS state is not, in the first instance, a local field. It lies in a superselection sector. Its charge is measured at infinity, and its existence depends on a stability chamber. Thus the ordinary local operator algebra of an untwisted four-dimensional theory is not the primary object of this paper.

At the level of the protected index the situation is different. After taking the cohomology of the chosen supercharge, long multiplets cancel. The object which remains is not a positive Hilbert space but a charge-graded virtual object whose Euler characteristic is stable under deformation. The index remembers the cancellation law of a unitary theory, not its positive inner product.

For $X = K_3 \times E$ the expected local object in a compact realization is a supplied holomorphic E_3 -factorization algebra

$$\mathcal{A}_X^{E_3}, \quad X = K_3 \times E.$$

The auxiliary notation $\Phi_3^{\text{FA}}(X)$, when used, is only a placeholder for this source datum; no such functor is constructed in this paper. No external Φ_3^{FA} formalism is used here as an existence theorem unless it supplies the chain-level local observables, factorization maps, descent, anomaly/QME control, and compact charge completion required below. In a compact realization $\mathcal{A}_X^{E_3}$ is the object whose disjoint holomorphic insertions factorize. The elliptic curve E enters only after a supplied chain-level K_3 -specialization whose output is denoted

$$\mathcal{F}_X \in \text{Fact}(\text{Ran}(E)).$$

This is the first compact-realization obstruction. The projection-finite sector, where $p_E(\text{Supp } \mathcal{A})$ is a finite subset of E , is naturally local on $\text{Ran}(E)$. A one-dimensional sector of positive elliptic degree is not. If $C \subset K_3 \times E$ has positive elliptic degree, then $(p_E)_*[C]$ is a positive multiple of $[E]$; hence $p_E(C)$ is a one-dimensional closed subset of the irreducible curve E , and so $p_E(C) = E$. Such a class is wrapped or global over E , not localized at finitely many points of E .

This is the $\text{Ran}(E)$ -support-locality obstruction in the K_3 -integrated chiral shadow. Naive support-local factorization on $\text{Ran}(E)$ sees projection-finite sectors, but by itself it misses the positive s -direction of the Igusa product, the direction which records elliptic degree after the OP/Borcherds variable conversion. A compact realization must therefore choose at least one repair of the following kind: projection-finite local factorization together with separate global Fock/Hecke modes for the wrapped elliptic-degree sector, or a hybrid local/global wrapped factorization theory. This is a realization obstruction. It is not a defect in the Borcherds determinant or in the Gritsenko–Nikulin denominator identity.

Thus, in a realization, the curve-level rule is not imposed externally. It is the one-dimensional chiral shadow of the holomorphic E_3 -observable: well-separated protected insertions on E multiply, while collisions remember extension and bound-state data. In primitive form this is the expected bracket

$$[\ , \] : \text{Prim}_c^{\text{prot}}(\mathcal{F}_X) \otimes \text{Prim}_{c'}^{\text{prot}}(\mathcal{F}_X) \longrightarrow \text{Prim}_{c+c'}^{\text{prot}}(\mathcal{F}_X), \quad c, c' \in \Gamma_X^{\text{phys}}.$$

The Igusa square-root problem is to construct or recognize this primitive protected algebra and its Gram descent. The determinant below gives the decategorified signed Gram-index shadow that such a realization would have:

$$\text{sdim Prim}_\gamma^{\text{prot}}(\Pi_{X,*}\mathcal{F}_X) = f(nm, l), \quad \gamma = (n, l, m).$$

The Borcherds–Kac–Moody denominator algebra gives the algebraic candidate:

$$H^0 \text{Prim}_{\text{prot}}(\Pi_{X,*}\mathcal{F}_X) \stackrel{?}{\cong} \mathfrak{g}_{\Delta_\gamma}.$$

The theorem-level content of the paper is the determinant and denominator comparison. The displayed isomorphism is the compact E_3 /Hall/chiral realization problem.

The square root is therefore not an analytic ambiguity in Δ_3^2 . It is the normalized automorphic determinant of the virtual protected one-particle index. The scalar square forgets the automorphic reflection character; a compact Hall orientation realizing that character is part of the open datum.

Schema of the local protected observable algebra

The candidate local source object is the following proposal-only schema, marked here as an external assumption. All ingredients are taken from existing literature; no part of the construction is the deliverable of this paper.

Definition 2.1 (Local protected observable algebra; proposal-only schema). For an open $U \subset X = K_3 \times E$, let $\mathfrak{M}_c(U)$ denote the derived moduli stack of compactly supported B-branes on U of physical Mukai charge $c \in \Gamma_X^{\text{phys}} = \Lambda_S \oplus \Lambda_S$, in the sense of Pantev–Toen–Vaquié–Vezzosi [?PTVV2013], so that $\mathfrak{M}_c(U)$ carries a (-1) -shifted symplectic structure; let Φ_c denote the perverse sheaf of vanishing cycles of any local d-critical chart, in the sense of Brav–Bussi–Dupont–Joyce–Szendroi [?BBDJS2015] (BPS sheaf, alias DT sheaf), well defined globally as a pre-D-critical structure on $\mathfrak{M}_c(U)$; let o_c denote the orientation line, a square root of the determinant line of the cotangent complex of $\mathfrak{M}_c(U)$, supplied as part of the Hall datum. The *local protected observable algebra of X on U* is the proposal

$$\mathcal{A}_X(U) := \bigoplus_{c \in \Gamma_X^{\text{phys}}} H_*^{\text{BM}}(\mathfrak{M}_c(U), \Phi_c \otimes o_c),$$

graded by physical charge. For disjoint $U_1, \dots, U_k \subset X$, the proposed factorization map is

$$\mathcal{A}_X(U_1) \otimes \dots \otimes \mathcal{A}_X(U_k) \longrightarrow \mathcal{A}_X(U_1 \sqcup \dots \sqcup U_k),$$

defined chain-level by Massey–Saito Thom–Sebastiani for vanishing cycles

$$\Phi_{F_1 \oplus \dots \oplus F_k} \simeq \Phi_{F_1} \boxtimes \dots \boxtimes \Phi_{F_k}$$

on disjoint d-critical charts, together with the Ext-vanishing $\text{Ext}^*(\mathcal{F}, \mathcal{G}) = 0$ for branes $\mathcal{F}, \mathcal{G}$ with disjoint supports. This is the proposed compact E_3 source $\mathcal{A}_X^{E_3}$ of the present paper, in the sense that its disjoint-supports factorization on opens of $X = K_3 \times E$ is the holomorphic E_3 -factorization datum.

Remark 2.2 (Status of Definition 2.1). The components are external assumptions: PTVV (-1) -shifted symplectic structure on derived moduli of B-branes [?PTVV2013]; BBDJS vanishing-cycle perverse sheaf and its Thom–Sebastiani isomorphism (extending Massey 2001 and Saito 1990 to the d-critical setting) [?BBDJS2015]; orientation lines as in the Joyce–Upmeyer orientation theory [?JoyceUpmeyer] or in Kontsevich–Soibelman [?KSCoHA]. The factorization map relies on disjoint-supports Ext-vanishing for compactly supported branes (a classical statement; absent for non-compact supports, where one recovers chiral algebra structure rather than commutative factorization). Compactness on $K_3 \times E$ is itself an open problem because E is compact but the relevant moduli of branes are typically non-quasicompact in Λ_S -charge; the sectorial truncation of Proposition 10.31 is the proposed mechanism for cutting it down. None of this construction is supplied in the present paper; the schema is named here so that the (E_3) -slot of Definition 10.32 has an explicit candidate to point at.

3 THE ONE-PARTICLE INDEX AND ITS DETERMINANT

For the determinant calculation set

$$Z = \begin{pmatrix} z_1 & z_2 \\ z_2 & z_3 \end{pmatrix}, \quad q = e^{2\pi i z_1}, \quad r = e^{2\pi i z_2}, \quad s = e^{2\pi i z_3}.$$

The Gram-index lattice is

$$\Gamma_{\text{ind}} = \mathbb{Z}^3, \quad \gamma = (n, l, m), \quad x^\gamma = q^n r^l s^m.$$

The standard cusp polarization is the ordered limit

$$0 < |s| \ll |q| \ll |r|^{-1} \ll 1.$$

It determines the semigroup

$$\Gamma_{\text{eff}} = \{(n, l, m) \mid m > 0, n \geq 0, l \in \mathbb{Z}\} \cup \{(n, l, 0) \mid n > 0, l \in \mathbb{Z}\} \cup \{(0, l, 0) \mid l < 0\}.$$

Indeed, $m > 0$ is positive because s is the smallest parameter; if $m = 0$, then $n > 0$ is positive because q is next; if $m = n = 0$, positivity is $l < 0$ because $|r|^{-1} \ll 1$. This is a chamber convention for the completed product, not a microscopic charge positivity statement.

Definition 3.1 (Ramond-sector K_3 index and its half). Let $\mathcal{H}_{\text{RR}}(S)$ be the Ramond–Ramond Hilbert space of a unitary $\mathcal{N} = (4, 4)$ sigma model on a K_3 surface S . Its central charges are

$$c = \bar{c} = 6.$$

This Hilbert space is a supplied worldsheet input for the protected K_3 index. It is not a construction of the compact $K_3 \times E$ D-brane Hilbert space. Choose a right-moving Ramond supercharge Q_R with

$$Q_R^2 = 0, \quad \{Q_R, Q_R^\dagger\} = 2\left(\bar{L}_0 - \frac{\bar{c}}{24}\right).$$

The K_3 elliptic genus is the protected supertrace

$$Z_{K_3}(\tau, z) = \text{Tr}_{\mathcal{H}_{\text{RR}}(S)}(-1)^{F_L + F_R} q^{L_0 - c/24} \bar{q}^{\bar{L}_0 - \bar{c}/24} r^{J_0}.$$

The BPS space in degree (n, l) is

$$\mathcal{H}_{n,l}^{\text{BPS}}(S) := H^\bullet(\{\psi \mid (L_0 - c/24)\psi = n\psi, J_0\psi = l\psi\}, Q_R).$$

LEMMA 3.2 (*Right-moving localization*). Then

$$Z_{K_3}(\tau, z) = \sum_{n \geq 0, l \in \mathbb{Z}} \text{sdim } \mathcal{H}_{n,l}^{\text{BPS}}(S) q^n r^l.$$

Proof. If $\bar{L}_0 - \bar{c}/24 > 0$, the displayed anticommutator pairs states with opposite $(-1)^{F_R}$ parity. Their supertrace cancels. The unpaired states are the Q_R -harmonic representatives, hence the graded superdimension of Q_R -cohomology gives the elliptic genus. $\square$

Remark 3.3 (Where unitarity enters). Unitarity is used here before the index is taken: it gives the adjoint relation which pairs positive right-moving energy states and makes the supertrace localize. After passing to the protected index, the object is virtual. A coefficient $f(n, l)$ is a superdimension, not a positive dimension. Negative coefficients record an odd excess in the protected supertrace, or in the chosen K_o -representative. They do not determine the two parity dimensions without an additional Hilbert-space or denominator-algebra parity datum.

Definition 3.4 (Grothendieck half-index representative). Eichler–Zagier normalization [EZ: 1]:

$$Z_{K_3}(\tau, z) = 2\phi_{o,1}(\tau, z),$$

$$\phi_{o,1}(\tau, z) = \sum_{n \geq o, l \in \mathbb{Z}} f(n, l) q^n r^l.$$

Thus

$$\text{sdim } \mathcal{H}_{n,l}^{\text{BPS}}(S) = 2f(n, l).$$

The arbitrary Grothendieck representative

$$\mathbb{U}_{n,l} \in K_o(\text{sVect}_{\text{fd}}), \quad \text{sdim } \mathbb{U}_{n,l} = f(n, l),$$

is the normalized half-index class used by the Borchers lift. It is not asserted to be a canonical half of $\mathcal{H}_{n,l}^{\text{BPS}}(S)$ as a Hilbert space. As a K_o -model one may take $\mathbb{U}_{n,l}$ to be $\mathbb{C}_{\bar{o}}^{f(n,l)}$ when $f(n, l) \geq o$, and $\mathbb{C}_{\bar{1}}^{-f(n,l)}$ when $f(n, l) < o$. No canonical choice of basis, parity lift, or operator product is implied.

$$\phi_{o,1} = r^{-1} + 1o + r + O(q), \quad \frac{f(o, o)}{2} = 5.$$

LEMMA 3.5 (*Fock trace and superdeterminant*). Let

$$V = \bigoplus_{\lambda \in P} V_{\lambda}$$

be a P -graded super vector space with finite-dimensional homogeneous pieces and let $a_{\lambda} = \text{sdim } V_{\lambda}$. Put

$$\mathcal{F}(V) = \text{Sym}(V_{\bar{o}}) \otimes \bigwedge (V_{\bar{1}}).$$

As a formal character,

$$\text{sch } \mathcal{F}(V) := \sum_{\mu \in P} \text{sdim } \mathcal{F}(V)_{\mu} x^{\mu} = \prod_{\lambda \in P} (1 - x^{\lambda})^{-a_{\lambda}}.$$

The inverse character is the superdeterminant

$$\text{sdet}_V(1 - x) = \prod_{\lambda \in P} (1 - x^{\lambda})^{a_{\lambda}}.$$

Proof.

$$V_{\lambda} = \mathbb{C}_{\bar{o}} : \quad \text{sch } \text{Sym}(V_{\lambda}) = (1 - x^{\lambda})^{-1}.$$

$$V_{\lambda} = \mathbb{C}_{\bar{1}} : \quad \text{sch } \bigwedge (V_{\lambda}) = 1 - x^{\lambda}.$$

Tensor product over a homogeneous basis proves the product. □

When $P = \Gamma_{\text{eff}}$, all products below are taken in the cusp-completed semigroup algebra determined by $0 < |s| \ll |q| \ll |r|^{-1} \ll 1$. Every coefficient receives only finitely many contributions in this completion.

$$x^{(n,l,m)} = q^n r^l s^m.$$

We now form the virtual Borchers determinant package. The third index m is the second-quantized/Borchers degree. The positive m sector carries the Hecke–DMVV exponents $f(nm, l)$. The $m = 0$ sector is a cusp/Weyl boundary package carrying the repeated boundary indices $f(0, l)$. Neither sector is a microscopic one-particle Hilbert space.

$$\mathcal{V}^{\text{bulk}} := \bigoplus_{\substack{(n,l,m) \in \Gamma_{\text{eff}} \\ m > 0}} \mathbb{U}_{nm,l}, \quad \mathcal{V}^{\text{cusp}} := \bigoplus_{(n,l,0) \in \Gamma_{\text{eff}}} \mathbb{U}_{0,l}.$$

$$\mathcal{V} = \mathcal{V}^{\text{bulk}} \oplus \mathcal{V}^{\text{cusp}}, \quad \mathcal{V}_{(n,l,m)} = \mathcal{V}_{n,l,m} := \mathbb{U}_{nm,l}, \quad \text{sdim } \mathcal{V}_{n,l,m} = f(nm, l).$$

This is a definition in K_0 , not a microscopic state space. The $m > 0$ sector is the K_0 -level virtual half of the DMVV product [DMVV], obtained by replacing $Z_{K_3} = 2\phi_{0,1}$ by $\phi_{0,1}$. The normalized Fock character in the Borchers chamber is

$$\text{sch } \mathcal{F}(\mathcal{V}) = \prod_{\gamma \in \Gamma_{\text{eff}}} (1 - x^\gamma)^{-\text{sdim } \mathcal{V}_\gamma} = \prod_{(n,l,m) \in \Gamma_{\text{eff}}} (1 - q^n r^l s^m)^{-f(nm,l)}.$$

$$\text{sdet}_\mathcal{V}(1 - x) = \prod_{\gamma \in \Gamma_{\text{eff}}} (1 - x^\gamma)^{\text{sdim } \mathcal{V}_\gamma} = \prod_{(n,l,m) \in \Gamma_{\text{eff}}} (1 - q^n r^l s^m)^{f(nm,l)}.$$

The sector $m > 0$ is the normalized half of the second-quantized DMVV index. The terms with $m = 0$ are cusp and Weyl-chamber boundary factors of the Borchers product; the determinant reads only their signed indices $f(0, l)$. The normalized cusp-form section below proves the leading coefficient which fixes the Weyl monomial

$$v_0(Z) = 64q^{1/2}r^{1/2}s^{1/2}.$$

$$v_0 \notin \bigoplus_{\gamma \in \Gamma_{\text{eff}}} \mathcal{V}_\gamma.$$

Definition 3.6 (Normalized K_0 -determinant).

$$\mathcal{D}_X(Z) = v_0(Z) \text{sdet}_\mathcal{V}(1 - x) = 64q^{1/2}r^{1/2}s^{1/2} \prod_{(n,l,m) \in \Gamma_{\text{eff}}} (1 - q^n r^l s^m)^{f(nm,l)}.$$

$$\mathcal{Z}_{\text{Borch}}^{\text{vir}}(Z) = \mathcal{D}_X(Z)^{-1} = \frac{1}{64}q^{-1/2}r^{-1/2}s^{-1/2} \prod_{(n,l,m) \in \Gamma_{\text{eff}}} (1 - q^n r^l s^m)^{-f(nm,l)}.$$

The object constructed here is the virtual K_0 -determinant; compact BPS meaning is part of the separate realization problem.

PROPOSITION 3.7 (*Determinant normalization*).

$$[q^{1/2}r^{1/2}s^{1/2}] \mathcal{D}_X = 64,$$

$$\mathcal{Z}_{\text{Borch}}^{\text{vir}} = \frac{1}{64}q^{-1/2}r^{-1/2}s^{-1/2} \text{sch } \mathcal{F}(\mathcal{V}),$$

$$\mathcal{D}_X = 64q^{1/2}r^{1/2}s^{1/2} \text{sdet}_\mathcal{V}(1 - x).$$

Proof.

$$\begin{aligned} \text{sch } \mathcal{F}(\mathcal{V}) &= \prod_{(n,l,m) \in \Gamma_{\text{eff}}} (1 - q^n r^l s^m)^{-f(nm,l)}, & \text{sdet}_{\mathcal{V}}(1-x) &= \prod_{(n,l,m) \in \Gamma_{\text{eff}}} (1 - q^n r^l s^m)^{f(nm,l)}. \\ v_o(Z) \text{sdet}_{\mathcal{V}}(1-x) &= \mathcal{D}_X, & v_o(Z)^{-1} \text{sch } \mathcal{F}(\mathcal{V}) &= \mathcal{Z}_{\text{Borch}}^{\text{vir}}. \end{aligned}$$

□

PROPOSITION 3.8 (*Virtuality of the square root*). The displayed construction produces a class

$$[\mathcal{V}] \in K_o(\Gamma_{\text{eff}}\text{-graded finite-dimensional super vector spaces})$$

together with its Fock determinant. By itself this determinant does not construct a Hilbert space, a vertex algebra, a holomorphic E_3 -factorization algebra, or a chiral factorization algebra whose character is $\mathcal{Z}_{\text{Borch}}^{\text{vir}}$.

All denominator statements use only the equalities

$$\text{sdim } \mathcal{V}_{(n,l,m)} = f(nm, l), \quad \mathcal{D}_X = v_o \text{sdet}_{\mathcal{V}}(1-x).$$

The denominator algebra and formal current envelope are constructed later from the Borchers root datum, not from a canonical half of the K_3 Hilbert space.

Proof. By definition, $\mathcal{V}_{(n,l,m)} = \mathbb{U}_{nm,l}$ is specified only through its class in $K_o(\text{sVect}_{\text{fd}})$ and through the integer $\text{sdim } \mathbb{U}_{nm,l} = f(nm, l)$. The Fock trace and superdeterminant lemma depends only on these signed dimensions. No operation in the construction supplies composition maps, operator products, locality, or a Hilbert completion. □

PROPOSITION 3.9 (*Dirac/Pfaffian recognition target*). The class $[\mathcal{V}] \in K_o$ and its Fock determinant $\mathcal{D}_X = v_o \text{sdet}_{\mathcal{V}}(1-x) = \Delta_5$ are cosmetic in the following sense: they record signed Gram-degree multiplicities and a formal infinite-product, but they do not produce an operator with a spectrum. The structural upgrade demanded by the protected-locality side of the constellation is an *open recognition target*, not a theorem of this paper:

construct on the compact $K_3 \times E$ factorization datum of Definition 10.32 a first-order protected operator/algebra $\mathbb{D}_X$ acting on a chiral protected module, whose charge-graded Pfaffian satisfies

$$\text{Pf}_{\text{prot}}(\mathbb{D}_X) = \mathcal{D}_X = \Delta_5,$$

and whose scalar trace, in the OP normalization of Definition 6.3 below, satisfies

$$\text{Tr}_{\text{prot}}((-1)^F (\mathbb{D}_X)^{-2}) = -4096 \Delta_5^{-2}.$$

This is the Dirac/Pfaffian recognition target. Its existence is open; it is the structural witness that would convert the cosmetic K_o identity of Proposition 3.8 into a first-order protected statement. The target carries the sign-square ambiguity flagged in Conjecture 5.3 and the orientation monodromy of Open Problem 5.7; choosing a square root of $(\mathbb{D}_X)^{-2}$ requires the orientation datum of that open problem and is not fixed by the OP scalar.

THEOREM 3.10 (*Dirac/Pfaffian recognition under $K_3 \times E$ orientation existence*). Suppose the following two data items are granted on the reduced compact critical Hall/CoHA sector of $X = K_3 \times E$ of Definition 5.4.

(O1) *Strong multiplicative orientation datum*. There exists a Joyce–Upmeier-type orientation line [?JoyceUpmeier]

$$o_C \in \text{Pic}(\mathfrak{M}_X^{or}), \quad o_C^{\otimes 2} \simeq \det \text{RHom}(C, C),$$

in the sense of Definition 4.3, satisfying

$$o_{C \oplus C'} \simeq o_C \otimes o_{C'} \quad \text{compatibly with the extension correspondences,}$$

descending under the reduced E -quotient and lifting equivariantly along the Weyl action transported by Gram-projection $\Pi : \Gamma_X^{\text{phys}} \rightarrow \Gamma_{\text{ind}}$.

(O1)⁺ *Weyl-equivariant cocycle extension*. The Weyl-equivariant lift of (O1) is an honest 1-cocycle on $W^{(2)}(\Lambda_{II}^{2,1})$ with values in $\{\pm 1\}$, descending to a sign character $\varepsilon_o : W^{(2)}(\Lambda_{II}^{2,1}) \rightarrow \{\pm 1\}$ (the orientation monodromy character of Open Problem 5.7). Equivalently, the restriction of the Joyce–Upmeier multiplicative orientation cocycle [?JoyceUpmeier, ?JoyceTanakaUpmeier] to wall-reflection generators is a homomorphism on all of $W^{(2)}(\Lambda_{II}^{2,1})$, not merely on its Coxeter generators.

(O2) *Pfaffian local model on type-II walls*. The geometric reflection sign

$$\varepsilon_o : W^{(2)}(\Lambda_{II}^{2,1}) \longrightarrow \{\pm 1\}$$

of Open Problem 5.7 is computed, on a simple type-II reflection s_δ , as the parity of the count of oriented real normal modes of the orientation line at the wall $\mathbb{H}_\delta$; equivalently, on a single oriented normal mode s_δ acts as (-1) on the Pfaffian section and squares to the identity on the determinant section.

Then:

- (i) (*Pfaffian recognition*.) The recognition target of Proposition 3.9 is realized. There exists a first-order protected operator/algebra $\mathfrak{D}_X$ on the chiral protected module $\mathcal{A}_X$ over Γ_{eff} whose graded Pfaffian section coincides with the virtual determinant:

$$\text{Pf}_{\text{prot}}(\mathfrak{D}_X) = \mathcal{D}_X = \Delta_5,$$

and whose scalar trace, in the OP normalization of Definition 6.3, satisfies

$$\text{Tr}_{\text{prot}}((-1)^F \mathfrak{D}_X^{-2}) = -4096 \Delta_5^{-2}.$$

- (ii) (*Three-sign identification*.) For the three type-II simple reflections δ_i of the Weyl chamber $\mathcal{P}_{II}$ of Section The Weyl chamber and the Weyl vector (notation $\delta_1 = 2f_2 - f_3$, $\delta_2 = 2f_{-2} - f_3$, $\delta_3 = f_3$),

$$\varepsilon_o(s_{\delta_i}) = -1 = \nu_{\Delta_5}(s_{\delta_i}), \quad i = 1, 2, 3,$$

and on the three $\text{Aut}(\mathcal{P}_{II}) \simeq S_3$ chamber-permuting reflections $s_{f_{-2}-f_2}$, $s_{f_2-f_3}$, $s_{f_2+f_3}$,

$$\varepsilon_o(s) = +1 = \nu_{\Delta_5}(s).$$

In particular,

$$\varepsilon_o|_{W^{(2)}(\Lambda_{II}^{2,1})} = \nu_{\Delta_5}|_{W^{(2)}(\Lambda_{II}^{2,1})}.$$

- (iii) (*Categorical upgrade.*) The virtual K_0 -determinant package $[\mathcal{V}], \mathcal{D}_X$ of Proposition 3.8, formerly cosmetic in the sense of Proposition 3.9, is upgraded to a chiral protected first-order theory whose decategorification is the Borchers product

$$\mathrm{Pf}_{\mathrm{prot}}(\mathfrak{D}_X)|_{\mathrm{c}_{\infty}} = 64 q^{1/2} r^{1/2} s^{1/2} \prod_{(n,l,m) \in \Gamma_{\mathrm{eff}}} (1 - q^n r^l s^m)^{f(nm,l)} = \Delta_s,$$

and whose orientation monodromy character is the Maass character ν_{Δ_s} restricted to $W^{(2)}(\Lambda_{II}^{2,1})$.

Remark 3.11 (Status of Theorem 3.10). This is a *conditional* theorem: each of (i)–(iii) is proved modulo the two open data items (O1) and (O2). Item (O1) is the strong orientation datum. On Calabi–Yau threefolds it is Joyce–Song [?JoyceSong]; the multiplicative orientation cocycle on $\mathrm{Perf}(X)$ is Joyce–Upmeyer [?JoyceUpmeyer] and Joyce–Tanaka–Upmeyer [?JoyceTanakaUpmeyer]; on Calabi–Yau fourfolds the unreduced orientation exists by Borisov–Joyce / Cao–Gross–Joyce / Bojko–Borisov–Joyce [?BorisovJoyce], [?0hThomasI], [?0hThomasII]; the rank-one Pin^c lift on $\mathrm{Hilb}^n(X)$ for smooth threefolds is Cao–Kool–Monavari [?CaoKoolMonavari]; further extensions to non-compact and Joyce–Song settings are Liu [?LiuYalongOrientations]. The $K3 \times E$ case combines a noncompact CY 3-fold direction with a reduced E -quotient and a Weyl-equivariance constraint that is not in the literature. Item (O2) is the local Pfaffian model for a simple wall reflection: it is folklore on critical and (-1) -shifted symplectic moduli, but its compatibility with the type-II Weyl action on the reduced $K3 \times E$ sector has not been proved. The deliverable of Theorem 3.10 is the typed bridge: once (O1) and (O2) are produced, the Dirac/Pfaffian square root and the orientation-character identity follow without further input. The cosection-induced parity flip $\mathrm{vd}^{\mathrm{red}} = \mathrm{vd}^{\mathrm{std}} + 1$ of the Maulik–Toda / Oberdieck reduced obstruction theory is absorbed into (O1) as a $\mathbb{Z}/2$ -grading shift $\sigma_C^{\mathrm{red}} = \sigma_C^{\mathrm{std}} \otimes \Pi$ of the BBDJS [?BBDJS2015] line under the cosection contraction of the PTVV [?PTVV2013] (-1) -shifted symplectic form by σ_K ; see clause (P4) of Open Problem 5.7 for the explicit statement and its consistency with the Joyce–Upmeyer [?JoyceUpmeyer] multiplicative cocycle and the Oberdieck monic normalization [?OberdieckReducedPT]. Operationally the flip is the source of the $(-1)^F$ sign in (i) and (iii) and pins down the Berezinian convention.

Proof of Theorem 3.10. Step 1 (Pfaffian construction from (O1)). By Definition 4.3 the orientation line of a stable charge- c object satisfies $\sigma_C^{\otimes 2} \simeq \det \mathrm{RHom}(C, C)$. The hypothesis (O1) promotes this to an extension-multiplicative system: on every direct sum $C \oplus C'$, $\sigma_{C \oplus C'} \simeq \sigma_C \otimes \sigma_{C'}$, compatibly with the strict associators of the Hall extension stack and descended under the reduced E -quotient. The Pfaffian section $\mathrm{Pf}_{\mathrm{prot}}$ is then the canonical square-root section of $\det \mathrm{RHom}$ determined by σ_C : it exists as a section of σ_C precisely because (O1) supplies a global trivialization of the squaring isomorphism, and it assembles, under the multiplicative structure, into a charge-graded product over Γ_{eff} of single-mode Pfaffian factors.

Define $\mathfrak{D}_X$ as the protected first-order operator on the chiral protected module $\mathcal{A}_X$ whose Berezinian/Pfaffian on the charge- (n, l, m) component is the canonical Pfaffian of the graded Ext bundle $\mathrm{Ext}^\bullet(C_{(n,l,m)}, C_{(n,l,m)})$, with sign normalized by the chosen σ_C . The graded product collapses to

$$\mathrm{Pf}_{\mathrm{prot}}(\mathfrak{D}_X) = v_o \, \mathrm{sdet} \gamma(1 - x) \quad \text{with} \quad v_o = 64 q^{1/2} r^{1/2} s^{1/2},$$

where sdet is the superdeterminant of Proposition 3.8; the key point is that orientation data turns the formal sdim -only superdeterminant of that proposition into a section of the orientation line, not just a formal infinite product in $K_0(\mathrm{sVect}_{\mathrm{fd}})$.

Step 2 (Identification $\text{Pf}_{\text{prot}} = \Delta_5$.) Proposition 4.5 (Borcherds–Gritsenko–Nikulin) identifies $v_o \text{sdet}_V(1 - x)$ with the Igusa cusp form Δ_5 at the level of Fourier coefficients. Step 1 promotes this from a formal product equality to an equality of Pfaffian sections, since orientation data resolves the $\pm$ -ambiguity inherent in any abstract square root: the Borcherds product comes with the sign convention $c_\Delta(1, 1, 1) = 64$ (positive leading coefficient, Borcherds normalization), and the Pfaffian section inherits the same sign on the unique top-cohomology generator of the isotropic-cusp expansion of Section The normalized cusp form. Hence $\text{Pf}_{\text{prot}}(\mathfrak{D}_X) = \Delta_5$ as a section of $\mathcal{L}^5 \otimes \nu_{\Delta_5}$.

Step 3 (Scalar square in OP normalization). Squaring forgets the orientation: $(\text{Pf}_{\text{prot}}(\mathfrak{D}_X))^2 = \det \text{RHom}(C, C)|_{\text{tot}} = \Delta_5^2$, as sections of $\mathcal{L}^{10}$ (the squared automorphic line is the trivially-charactered determinant, since $\Delta_5^2 = \Delta_{10}$ is scalar by Section The normalized cusp form). The protected scalar trace is then $\text{Tr}_{\text{prot}}((-1)^F \mathfrak{D}_X^{-2}) = C_X \Delta_5^{-2}$ for some normalization scalar C_X ; the explicit $C_X = -4096$ follows from Proposition 6.1 of Oberdieck–Pixton, which gives $Z_{\text{OP}}^X = -(\chi_{10}^{\text{OP}})^{-1} = -4096 \Delta_5^{-2}$, together with the arithmetic $-4096 = -(64)^2$ of the squared theta-leading coefficient and the OP scalar sign convention recorded in Definition 6.3.

Step 4 (Orientation-character identity (ii)). By (O₂), each type-II simple reflection s_{δ_i} acts on o_C by the parity of its count of oriented real normal modes; by Lemma 3.12 below, this count is 1 for $i = 1, 2, 3$, giving $\varepsilon_o(s_{\delta_i}) = -1$. The same lemma computes the $\text{Aut}(\mathcal{P}_{II}) \simeq S_3$ reflections to even normal-mode count, giving $\varepsilon_o = +1$ on those. The displayed values match Lemma 9.2 on the same six elements. The chamber decomposition $\mathbf{O}(\Lambda^{2,1})_+ \simeq W^{(2)}(\Lambda_{II}^{2,1}) \rtimes \text{Aut}(\mathcal{P}_{II})$ of Section The Weyl chamber and the Weyl vector reduces the character to its values on these six generators *provided* ε_o is an honest homomorphism on the infinite Coxeter group $W^{(2)}(\Lambda_{II}^{2,1})$; this is exactly the strengthened hypothesis (O₁)⁺, supplied by Lemma 3.15 from (O₁) plus the $(\mathbb{Z}/2)^3$ abelianization of $W^{(2)}(\Lambda_{II}^{2,1})$. Two homomorphisms agreeing on a generating set agree, so $\varepsilon_o = \nu_{\Delta_5}$ on $W^{(2)}(\Lambda_{II}^{2,1})$.

Step 5 (Categorical upgrade (iii)). Steps 1–4 promote each ingredient of Proposition 3.8: the class $[\mathcal{V}] \in K_0(\text{sVect}_{\text{fd}})$ lifts to the o_C -graded chiral module, the formal product $v_o \text{sdet}_V(1 - x)$ lifts to the Pfaffian section $\text{Pf}_{\text{prot}}(\mathfrak{D}_X)$, and the formal character-equality lifts to the orientation-character identity of (ii). The decategorification of the chiral protected theory at the chosen isotropic cusp $\mathfrak{c}_\infty$ is the Borcherds product, and the orientation monodromy character is ν_{Δ_5} on the type-II Weyl group as computed. $\square$

LEMMA 3.12 (*Three-sign computation on the type-II simple walls*). Under the Pfaffian local model (O₂) of Theorem 3.10, for each $\delta \in \{\delta_1, \delta_2, \delta_3\}$ the orientation-monodromy reflection sign is

$$\varepsilon_o(s_\delta) = (-1)^{N_\delta} = -1,$$

where N_δ is the count of oriented real normal modes at the wall $\mathbb{H}_\delta$, and for each $s \in \{sf_{-2}-f_2, sf_2-f_3, sf_2+f_3\}$, $\varepsilon_o(s) = +1$.

Proof. Type-II simple walls. The three roots $\delta_1 = 2f_2 - f_3$, $\delta_2 = 2f_{-2} - f_3$, $\delta_3 = f_3$ have Gram pairing $(\delta_i, \delta_i) = 2$, $(\delta_i, \delta_j) = -2$ for $i \neq j$, by the matrix displayed in Section The Weyl chamber and the Weyl vector. All three are real simple roots of the Lorentzian Kac–Moody algebra $\mathfrak{g}_{\Delta_5}$: real because $\langle \delta, \delta \rangle = +2 > 0$, of multiplicity one because they appear in the simple-root data of the Borcherds product (the only multiplicity-one positive support of $f(nm, l)$ consistent with the squared discriminant $(\alpha(\gamma), \alpha(\gamma)) = -2(4nm - l^2)$). In the Gram-index chart of Definition 9.3 (using $\alpha(\gamma) = 2nf_2 - lf_3 + 2mf_{-2}$), the three simple walls correspond to the lattice triples

$$\delta_1 \leftrightarrow (n, l, m) = (1, 1, 0), \quad \delta_2 \leftrightarrow (0, 1, 1), \quad \delta_3 \leftrightarrow (0, -1, 0),$$

matching the K3 elliptic-genus support $f(o, \pm 1) = 1, f(o, o) = 10, f(nm, l) = 0$ for $4nm - l^2 < -1$. By Borcherds' product theorem, the Pfaffian section near $\mathbb{H}_\delta$ factors as

$$\text{Pf}_{\text{prot}}(\mathfrak{D}_X) = u_\delta^{N_\delta} \cdot (\text{unit at } \mathbb{H}_\delta),$$

where $u_\delta = 1 - x^{\alpha(\delta)}$ is the local equation of the wall. The exponent N_δ is the multiplicity of δ as a simple real root, i.e. $N_\delta = f(nm, l)|_{\text{simple-root cell}} = 1$ for each of the three triples above.

By the Pfaffian local model (O₂), an oriented real normal mode flips the orientation line o_C by -1 , squaring to the trivial action on $\det \text{RHom}$. Hence

$$\varepsilon_o(s_{\delta_i}) = (-1)^{N_{\delta_i}} = (-1)^1 = -1, \quad i = 1, 2, 3.$$

The three rows of the explicit table:

δ_i	wall geometry (charge label)	normal modes N_{δ_i}	ε_o	ν_{Δ_5}
$\delta_1 = 2f_2 - f_3$	section + fibre + Do collision: (n, l, m) = (1, 1, 0); D2(section) wall	1	-1	-1
$\delta_2 = 2f_{-2} - f_3$	fibre + section + Do mirror leg under $f_2 \leftrightarrow f_{-2}$: (0, 1, 1); D2(fibre) wall	1	-1	-1
$\delta_3 = f_3$	isotropic-cusp diagonal D6-D2 ex- tension on the $4nm - l^2 = -1$ ray: (0, -1, 0)	1	-1	-1

The three $\text{Aut}(\mathcal{P}_{II}) \simeq S_3$ reflections. The three chamber-permuting roots $\alpha_1 = f_{-2} - f_2, \alpha_2 = f_2 - f_3, \alpha_3 = f_2 + f_3$ all have squared norm $(\alpha_i, \alpha_i) = 2$ by the Gram pairing $(f_2, f_{-2}) = -1, (f_3, f_3) = 2$ of Section The Weyl chamber and the Weyl vector. They lie in $\Lambda^{2,1} \setminus \Lambda_{II}^{2,1}$ (type I, not type II), so they do *not* appear in the type-II simple-root system $\{\delta_1, \delta_2, \delta_3\}$ used by the Borcherds product [?GN, Theorem 2.1, Section 4]. We compute the wall multiplicity $N_\alpha = \text{ord}_{\mathbb{H}_\alpha}(\Delta_5)$ directly via Borcherds' divisor formula [?GN, Section 4], [?Gritsenko99].

By Borcherds' theorem (specialized to Δ_5 in the type-II parametrization $\alpha(n, l, m) = 2nf_2 - lf_3 + 2mf_{-2}$ of Definition 9.3, with K3 elliptic-genus coefficients $f(n, l)$ of the weight-zero index-one weak Jacobi form $\phi_{o,1}$ [?EZ: 1, §2.2]),

$$N_\alpha = \text{ord}_{\mathbb{H}_\alpha}(\Delta_5) = \sum_{k \geq 1: \pm k\alpha = \alpha(\gamma_k), \gamma_k \in \Gamma_{\text{eff}}} f(n_k m_k, l_k),$$

where (n_k, l_k, m_k) is the unique triple realizing $\alpha(\gamma_k) = \pm k\alpha$ when one exists, and the sum is empty otherwise. Because the parametrization $\alpha(\gamma) = 2nf_2 - lf_3 + 2mf_{-2}$ requires the coefficients of f_2 and f_{-2} to be *even*, only k with $k\alpha$ having even f_2 - and f_{-2} -coefficients contribute. We tabulate:

- (i) $\alpha_1 = f_{-2} - f_2$: the f_2 - and f_{-2} -coefficients are ∓ 1 , so $k = 1$ is excluded. For $k = 2, 2\alpha_1 = 2f_{-2} - 2f_2$ gives $(n, l, m) = (-1, 0, 1)$; the opposite orientation $-2\alpha_1$ gives $(1, 0, -1)$. Neither lies in $\Gamma_{\text{eff}} = \{n \geq 0, m \geq 0, n + m - l \geq 0\}$ of Lemma 9.5, since one of n, m is negative. All higher k inherit the same sign obstruction. Hence the sum is empty and $N_{\alpha_1} = 0$.
- (ii) $\alpha_2 = f_2 - f_3$: for $k = 1$, the f_2 -coefficient is odd, excluded. For $k = 2, 2\alpha_2 = 2f_2 - 2f_3$ gives $(n, l, m) = (1, 2, 0)$, with index value $f(nm, l) = f(0, 2)$. By the explicit Fourier expansion $\phi_{o,1} = \sum f(n, l) q^n r^l$ of [?EZ: 1, §2.2, Theorem 9.3 and Table p. 141] (equivalently, the K3 elliptic-genus normalization

$Z_{K_3} = 2\phi_{0,1}$ recorded in Section Protected locality), the q° row supports only $|l| \leq 1$ (specifically $f(0,0) = 10, f(0,\pm 1) = 1, f(0,l) = 0$ for $|l| \geq 2$), so $f(0,2) = 0$. For $k \geq 3$, even multiples $k = 2j$ give $(j, 2j, 0)$ with index value $f(0, 2j) = 0$ by the same support bound. Hence the sum is empty and $N_{\alpha_2} = 0$.

- (iii) $\alpha_3 = f_2 + f_3$: identical analysis, replacing $l \mapsto -l$: $2\alpha_3 = 2f_2 + 2f_3$ gives $(n, l, m) = (1, -2, 0)$ with $f(0, -2) = 0$ by the $r \rightarrow r^{-1}$ symmetry of $\phi_{0,1}$ [?EZ: 1, §2.2]. Hence $N_{\alpha_3} = 0$.

By the Pfaffian local model (O2), N_α is also the count of oriented real normal modes of $\mathfrak{D}_X$ at the wall $\mathbb{H}_\alpha$ (the Pfaffian section vanishes to the same order as its automorphic image Δ_5). Therefore

$$\varepsilon_o(s_{\alpha_i}) = (-1)^{N_{\alpha_i}} = (-1)^0 = +1, \quad i = 1, 2, 3.$$

The Aut-side equality with the Maass character is Lemma 9.2: there $\nu_{\Delta_5}(s_{f_2-f_3}) = \nu_{\Delta_5}(s_{f_2-f_3}) = \nu_{\Delta_5}(s_{f_2+f_3}) = +1$.

Three-wall sign cocycle on $\{\delta_1, \delta_2, \delta_3\}$: explicit Joyce–Tanaka–Upmeyer transport. Let K be the abstract simplicial complex on the vertex set $V = \{W_1, W_2, W_3\}$, $W_i := \mathbb{H}_{\delta_i}$, with one edge $\{i, j\}$ for each pair and a single 2-simplex $\{1, 2, 3\}$ oriented as the standard 2-cycle $[123]$. By (O1)+(O2) the orientation line o_C carries a Joyce–Tanaka–Upmeyer orientation transport [?JoyceTanakaUpmeyer]: for each ordered chamber-to-chamber wall-crossing s_{δ_i} , parallel transport of o_C across W_i into the reflected chamber acts as a fibre automorphism of the trivial $\mathbb{Z}/2$ -bundle, recorded by a class $\eta_i \in \mathbb{Z}/2$. By the Pfaffian local model (O2), η_i is the parity of the count of oriented real normal modes:

$$\eta_i := N_{\delta_i} \bmod 2 = 1, \quad i = 1, 2, 3,$$

the value coming from the row-by-row computation above.

The triple wall composition $s_{\delta_1}s_{\delta_2}s_{\delta_3}$ is a hyperbolic element of $W^{(2)}(\Lambda_{II}^{2,1})$: the Cartan matrix (δ_i, δ_j) of the previous display has off-diagonal entries -2 , so $4 \cos^2(\pi/m_{ij}) = A_{ij}A_{ji} = 4$ with all Coxeter exponents $m_{ij} = \infty$; no braid relation collapses the free product. By (O1) the orientation line of an extension is the tensor of the orientation lines of its summands, hence ε_o factors through the abelianization of $W^{(2)}(\Lambda_{II}^{2,1})$: each generator has order two, the abelianization is $(\mathbb{Z}/2)^{\oplus k}$ for some k , and ε_o is determined by its values on the three real simple generators. In particular,

$$\varepsilon_o(s_{\delta_1}s_{\delta_2}s_{\delta_3}) = \eta_1 + \eta_2 + \eta_3 \bmod 2 = 1 + 1 + 1 \bmod 2 = 1,$$

i.e. $(-1)^{\eta_1+\eta_2+\eta_3} = -1$. This is the closed value of the three-wall sign cocycle.

The Joyce–Tanaka–Upmeyer obstruction class to the orientation transport is a $\mathbb{Z}/2$ -valued 2-cochain $c \in C^2(K; \mathbb{Z}/2)$ defined on the unique 2-simplex by

$$c(\{1, 2, 3\}) := \eta_1 + \eta_2 + \eta_3 \bmod 2 = 1.$$

The cohomological content of (O1)+(O2) is precisely the vanishing of $[c] \in H^2(K; \mathbb{Z}/2)$: the Stiefel–Whitney-type obstruction $w_2(o_C|_K)$ of [?JoyceTanakaUpmeyer, Theorem 1.2], restricted to the three-wall subcomplex, must trivialize to make the parallel transport a coherent groupoid action.

Define the 1-cochain $\sigma \in C^1(K; \mathbb{Z}/2)$ by

$$\sigma(\{1, 2\}) := \eta_1 = 1, \quad \sigma(\{1, 3\}) := \eta_2 = 1, \quad \sigma(\{2, 3\}) := \eta_3 = 1.$$

Each value is the JTU half-edge propagation sign attached to crossing the indexed wall: the edge $\{i, j\}$ carries the parity of the wall that is *not* a vertex of the dual simplex $\overline{V} \setminus \{i, j\}$, so all three values are equal

to $\eta_{6-i-j} \in \mathbb{Z}/2$. Equivalently, in a basis-free form, σ is the dual of the wall-incidence matrix valued in the $\mathbb{Z}/2$ parity of the normal-mode count of the missing wall.

The Stiefel–Whitney coboundary $d\sigma \in C^2(K; \mathbb{Z}/2)$ on the oriented 2-simplex $[123]$ is

$$(d\sigma)(\{1, 2, 3\}) = \sigma(\{2, 3\}) - \sigma(\{1, 3\}) + \sigma(\{1, 2\}) \equiv \eta_3 + \eta_2 + \eta_1 \pmod{2} = 1.$$

Hence

$$d\sigma = c \quad \text{in} \quad C^2(K; \mathbb{Z}/2),$$

and $[c] = 0 \in H^2(K; \mathbb{Z}/2)$: the Stiefel–Whitney obstruction trivializes. This is the explicit trivialization required by (O₂) and supplies the categorical homomorphism property for ε_o restricted to $W^{(2)}(\Lambda_{II}^{2,1})$, in line with the gauge-theoretic orientation cocycle of [?JoyceUpmeier], [?JoyceTanakaUpmeier]; the Pin^c -lift analogue on Hilbert schemes is [?CaoKoolMonavari], the analytic extension to non-compact and Joyce–Song-type strata is [?LiuYalongOrientations].

Joint extension. The chamber decomposition $\mathbf{O}(\Lambda^{2,1})_+ \simeq W^{(2)}(\Lambda_{II}^{2,1}) \rtimes \text{Aut}(\mathcal{P}_{II})$ of Section The Weyl chamber and the Weyl vector reduces the character to its values on these six generators. Six checks done; the strengthened multiplicative hypothesis (O₁)⁺, supplied by Lemma 3.15 from the direct-sum clause of (O₁) plus the $(\mathbb{Z}/2)^3$ abelianization of $W^{(2)}(\Lambda_{II}^{2,1})$, guarantees ε_o is a homomorphism on all of $W^{(2)}(\Lambda_{II}^{2,1})$, and the Maass character is a homomorphism by construction. Two homomorphisms agreeing on a generating set agree, so $\varepsilon_o = \nu_{\Delta_s}$ on $W^{(2)}(\Lambda_{II}^{2,1})$. $\square$

Independent verification via the Maass character. The proof above routes through the Pfaffian local model (O₂) and the Borchers simple-root multiplicity $N_{\delta} = f(nm, l)|_{\text{simple cell}}$: a Joyce–Tanaka–Upmeier-flavoured route, where the sign on a type-II simple wall is read off the parity of the local Pfaffian exponent. The Beilinson-school discipline asks the numerics to be verified by an independent route; we supply one using only the Maass multiplier ν_{Δ_s} and Borchers-product factorization. No appeal to (O₂) is made.

Step 1 (Reformulating the claim as a coboundary). Identify the sign cocycle to be computed. Pulled back to $W^{(2)}(\Lambda_{II}^{2,1}) \subset \mathbf{O}(\Lambda^{2,1})_+$, the orientation character of the protected sector is a homomorphism $\varepsilon_o: W^{(2)}(\Lambda_{II}^{2,1}) \rightarrow \{\pm 1\}$, and the two-cocycle measuring its failure to extend is the coboundary $c(g, h) = \varepsilon_o(g)\varepsilon_o(h)\varepsilon_o(gh)^{-1} \equiv 1$ in $\mathbb{Z}/2$ by homomorphy. Equivalently, writing ε_o as the additive function $\tilde{\varepsilon}_o: W^{(2)} \rightarrow \mathbb{Z}/2$ with $\varepsilon_o = (-1)^{\tilde{\varepsilon}_o}$, the claim $\tilde{c}(g, h) = \tilde{\varepsilon}_o(g) + \tilde{\varepsilon}_o(h) - \tilde{\varepsilon}_o(gh) \equiv 0 \pmod{2}$ is the coboundary identity to be checked.

Step 2 (Maass-side coboundary). Define $\varepsilon: W^{(2)}(\Lambda_{II}^{2,1}) \rightarrow \mathbb{Z}/2$ by $\varepsilon(g) := \frac{1}{2}(1 - \nu_{\Delta_s}(g))$, so $(-1)^{\varepsilon(g)} = \nu_{\Delta_s}(g)$. Maass’s character is a homomorphism [?MAASS: 1], [?GN, Section 3], [?Gritsenko99], hence

$$c'(g, h) := \varepsilon(g) + \varepsilon(h) - \varepsilon(gh) \equiv \frac{1}{2}(1 - \nu_{\Delta_s}(g)\nu_{\Delta_s}(h)\nu_{\Delta_s}(gh)^{-1}) \equiv 0 \pmod{2}.$$

The displayed value of ν_{Δ_s} on the six generators is Lemma 9.2: -1 on $s_{\delta_1}, s_{\delta_2}, s_{\delta_3}$, $+1$ on $s_{f_{-2}-f_2}, s_{f_2-f_3}, s_{f_2+f_3}$; equivalently $\varepsilon(s_{\delta_i}) = 1$ and $\varepsilon(s_{f_{-2}-f_2}) = \varepsilon(s_{f_2-f_3}) = \varepsilon(s_{f_2+f_3}) = 0$.

Step 3 (Verification on the nine pairs of three-wall generators). For the three-wall computation the only nontrivial check is on the 3×3 table of pairs $(g, h) \in \{s_{\delta_1}, s_{\delta_2}, s_{\delta_3}\}^{\times 2}$. Using $\varepsilon(s_{\delta_i}) = 1$ for each i and the Coxeter relation $s_{\delta_i}^2 = 1, (s_{\delta_i} s_{\delta_j})^{m_{ij}} = 1$ of the hyperbolic Weyl group:

- Diagonal $(g, g) = (s_{\delta_i}, s_{\delta_i})$ for $i = 1, 2, 3$ (three pairs): $gh = 1, \varepsilon(gh) = 0$, so $c'(g, h) = 1 + 1 - 0 = 2 \equiv 0 \pmod{2}$.

- Off-diagonal $(g, h) = (s_{\delta_i}, s_{\delta_j})$, $i \neq j$ (six pairs): $gh = s_{\delta_i} s_{\delta_j}$ lies in the rotation subgroup $W^+ \subset W^{(2)}(\Lambda_{II}^{2,1})$ and is the product of two reflections, so $\nu_{\Delta_\zeta}(gh) = (-1)(-1) = +1$, giving $\varepsilon(gh) = 0$ and $c'(g, h) = 1 + 1 - 0 = 2 \equiv 0 \pmod{2}$.

All nine pairs vanish modulo 2; c' is a coboundary on the three-wall block, with primitive ε . By the same argument the cocycle vanishes on the six-element block $\{s_{\delta_i}, s_{f_{-2}-f_2}, s_{f_2-f_3}, s_{f_2+f_3}\}^{\times 2}$.

Step 4 (Bridge: $\widetilde{\varepsilon}_0 \equiv \varepsilon$ via Borcherds-product factorization). Borcherds' product theorem [?GNII, Theorem 2.1, (2.1)–(2.3)], [?GN, Theorem 4.1] writes

$$\Delta_\zeta = \text{Borch}(\phi_{0,1}) = q^{1/2} r^{1/2} s^{1/2} \prod_{(n,l,m) \in \Gamma_{\text{eff}}^+} (1 - q^n r^l s^m)^{f(nm,l)}$$

in the type-II cusp polarization $(q, r, s) = (e^{2\pi i z_1}, e^{2\pi i z_2}, e^{2\pi i z_3})$. The Maass multiplier $\nu_{\Delta_\zeta}(g)$ is by construction the holomorphic monodromy of Δ_ζ under g ; on the type-II simple-wall reflection s_{δ} it equals $(-1)^{N_{\delta}}$ where N_{δ} is the Borcherds product exponent at the wall (the divisor order of Δ_ζ along $\mathbb{H}_{\delta}$), because the local equation u_{δ} is s_{δ} -antiinvariant with eigenvalue -1 . Independent of (O2), the orientation line $\mathcal{O}_C$ of the protected sector transports along $\mathbb{H}_{\delta}$ by the same holomorphic monodromy: by Borcherds' construction the chiral protected character at the chosen isotropic cusp is the Borcherds product itself, and the orientation line is the determinant of the chiral protected module at that cusp. Hence $\widetilde{\varepsilon}_0(s_{\delta}) \equiv \varepsilon(s_{\delta}) \pmod{2}$ on each wall reflection s_{δ} , with the same equality on the three $\text{Aut}(\mathcal{P}_{II})$ -reflections by the parity of their chamber-permutation action on the paired Borcherds-product factors of opposite Gram index.

Step 5 (Conclusion). Steps 1–4 give $\widetilde{\varepsilon}_0 = \varepsilon$ on the six generators, hence on $W^{(2)}(\Lambda_{II}^{2,1})$ by the chamber decomposition $\mathbf{O}(\Lambda^{2,1})_+ \simeq W^{(2)}(\Lambda_{II}^{2,1}) \rtimes \text{Aut}(\mathcal{P}_{II})$. On the three type-II simple walls, $\varepsilon_0(s_{\delta_i}) = \nu_{\Delta_\zeta}(s_{\delta_i}) = -1$, and on the three $\text{Aut}(\mathcal{P}_{II})$ -reflections $\varepsilon_0(s) = +1$, matching the table of the first proof. No use is made of the Pfaffian local model (O2): the route is purely Borcherds-product / Maass-character. $\square$

Remark 3.13 (Cross-verification: two routes, one cocycle). The two proofs of Lemma 3.12 give the same coboundary primitive $\varepsilon: W^{(2)}(\Lambda_{II}^{2,1}) \rightarrow \mathbb{Z}/2$, with $\varepsilon(s_{\delta_i}) = 1$ and $\varepsilon(s) = 0$ on the three $\text{Aut}(\mathcal{P}_{II})$ -reflections. The Joyce–Tanaka–Upmeyer route reads it from the Pfaffian local exponent N_{δ} granted (O2); the Maass-character route reads it from the Borcherds-product exponent $f(nm, l)$ granted only the Borcherds construction [?GNII, Theorem 2.1], [?Niemann03]. The two exponents agree on the three simple walls, $N_{\delta} = f(nm, l)|_{\text{simple cell}} = 1$, which is the common-source identity that lets the two routes meet. Disagreement at this point would have falsified either (O2) or the displayed Borcherds expansion of [?GNII], neither of which is in question.

LEMMA 3.14 (Orientation transport across imaginary-root walls). Hypothesis (O2) of Theorem 3.10, as used in Lemma 3.12, fixes the parity contribution of an oriented real normal mode at a wall and is verified directly only on the three real simple walls $\delta_1, \delta_2, \delta_3$. The same hypothesis, paired with the Gritsenko–Nikulin Borcherds factorization $\Delta_\zeta = 64^{-1} \text{Borch}_\theta(\phi_{0,1})(2Z)$ [?GN, Theorem 2.1], [?GNII, Section 5.1.1, case $(t = 1, II, 0)$], [?Gritsenko99], transports ε_0 to every imaginary-root wall $a \in \Delta^{im} \subset \Lambda_{II}^{2,1} \cap \mathbb{R}_{\geq 0} \mathcal{P}_{II}$ by the formulae

$$\varepsilon_0(s_a) = \begin{cases} (-1)^{m(a)}, & (a, a) < 0 \text{ (negative-norm imaginary wall)}, \\ (-1)^{\tau(a)}, & (a, a) = 0 \text{ (isotropic-ray imaginary wall)}, \end{cases}$$

where $m(a)$ and $\tau(a)$ are the Gritsenko–Nikulin imaginary-simple-root multiplicities of Definition Imaginary simple roots. In particular, on every primitive isotropic ray $\mathbb{Z}_{>0}a \subset \overline{\mathcal{P}_{II}} \cap \Lambda_{II}^{2,1}$ one has

$$\tau(ta) = 9 \quad (t \geq 1) \quad \implies \quad \varepsilon_o(s_a^{\text{isotropic}}) = (-1)^9 = -1$$

uniformly in the isotropic-ray label and in the dilation index t , matching the sign $\varepsilon_o(s_{\delta_i}) = -1$ computed on the three real simple walls.

Proof. The Borchers singular-theta lift $\text{Borch}_\theta(\phi_{o,1})$ [?Borchers95] produces a meromorphic Siegel modular form whose divisor on the Siegel quotient $\mathbf{Sp}_4(\mathbb{Z}) \backslash \mathbb{H}_2$ is a sum of Heegner divisors with exponents read directly from the Fourier coefficients of the input weak Jacobi form $\phi_{o,1} = \sum f(n, l) q^n r^l$; equivalently, on $\mathbf{Sp}_4(\mathbb{Z}) \backslash \mathbf{Sp}(2)/U(2)$ the divisor of Δ_5 is the union of the real reflective walls $\mathbb{H}_{\delta_i}$ (multiplicity $f(o, \pm 1) = 1$ on each δ_i) together with all imaginary walls — negative-norm and isotropic — appearing with the multiplicities $m(a)$ and $\tau(a)$ recorded in [?GNII, Theorem 2.1, (2.1)–(2.3)] and [?GN, Sections 3–4]. This is the wall-divisor decomposition of Δ_5 used downstream in the denominator-algebra theorem.

Pfaffian local model on imaginary walls. Hypothesis (O2) specifies the action on $\det \text{RHom}$ of an oriented real normal mode by -1 per mode. At an imaginary wall $\mathbb{H}_a$ the local Pfaffian section $\text{Pf}_{\text{prot}}(\mathfrak{D}_X)$ factors as

$$\text{Pf}_{\text{prot}}(\mathfrak{D}_X) = u_a^{\text{ord}_a(\Delta_5)} \cdot (\text{unit at } \mathbb{H}_a), \quad u_a = 1 - x^{\alpha(a)},$$

with $\text{ord}_a(\Delta_5)$ the wall multiplicity dictated by the Borchers product theorem. For negative-norm a , this exponent is $m(a)$; for isotropic a , it is $\tau(a)$ [?GN, Sections 3–4], [?GNII, (2.1)–(2.3)], [?Niemann03, Theorem 3.1]. Each oriented normal mode at the wall contributes a factor of -1 to ε_o under (O2) (Lemma 3.12, Pfaffian-local-model clause), so

$$\varepsilon_o(s_a) = (-1)^{\text{ord}_a(\Delta_5)} = \begin{cases} (-1)^{m(a)}, & (a, a) < 0, \\ (-1)^{\tau(a)}, & (a, a) = 0. \end{cases}$$

This is the (O2)-image of the Borchers wall-multiplicity formula and extends ε_o from $\Delta_{\text{simp}}^{re}$ to all of Δ^{im} without further hypothesis.

The isotropic-ray exponent $\tau(ta) = 9$. At every primitive isotropic ray $\mathbb{Z}_{>0}a \subset \overline{\mathcal{P}_{II}} \cap \Lambda_{II}^{2,1}$, the Borchers correction character collapses, by the Gritsenko–Nikulin denominator identity [?GNII, (2.1)–(2.3)], into the eta-product expansion

$$1 - \sum_{t \geq 1} m(ta) q^t = \prod_{t \geq 1} (1 - q^t)^{\tau(ta)}, \quad \tau(ta) = 9 \quad (t \geq 1),$$

matching the displayed identity already in Section The denominator algebra. The exponent 9 is read off the half-index expansion $\eta(\tau)^{-9} \eta(2\tau)^{-9}$ of the cusp residue of Δ_5 at the standard isotropic cusp $\mathfrak{c}_\infty$ ([?GN, Sections 3–4]; cf. the calabi-yau-quantum-groups η -product anchor in the Gritsenko–Nikulin notes), equivalently $c(o)/2 - 1 = 9$ from the Eichler–Zagier $\phi_{o,1}$ coefficient $f(o, o) = 10$. The dependence on the S_3 -orbit of the isotropic ray is none: every primitive isotropic ray in $\overline{\mathcal{P}_{II}}$ gives the same $\tau(ta) = 9$ by the $S_3 \simeq \text{Aut}(\mathcal{P}_{II})$ -symmetry of Lemma 9.2. Since $9 \equiv 1 \pmod{2}$,

$$\varepsilon_o(s_a^{\text{isotropic}}) = (-1)^9 = -1$$

uniformly across all isotropic walls and all dilations.

The negative-norm parity table $m(a) \bmod 2$. The negative-norm exponents $m(a)$ at the lowest discriminants $D = -(a, a)/2 = 4nm - l^2 > 0$ are read off $\phi_{0,1} = \sum f(n, l)q^n r^l$ by the Borcherds product theorem [?Borcherds95], [?GN, Sections 3–4]. The first values are

$D = 4nm - l^2$	$f(nm, l)$	$m(a) \bmod 2$	$\varepsilon_o(s_a)$
1	$f(0, 1) = 1$	1	−1
3	$f(1, 1) = 10$	0	+1
4	$f(1, 0) = -2$	0	+1
5	$f(1, -1) = 10$	0	+1
7	$f(2, 1) = -64$	0	+1
8	$f(2, 0) = 108$	0	+1

Higher discriminants follow by the same table read off $\phi_{0,1}$; the sign $\varepsilon_o(s_a)$ is therefore $(-1)^{f(nm,l)}$ on the negative-norm walls in the chamber. The $D = 1$ row recovers the real simple-root sign -1 of Lemma 3.12 (the wall δ_3 , the unique $D = -1$ -wall in the chamber, lifts from $D = -1$ to $D = 1$ under the $(a, a) \mapsto -(a, a)$ discriminant convention used by Gritsenko–Nikulin [?GN, Sections 3–4]).

Multiplicativity and orientation-line transport. Hypothesis (O1) supplies the multiplicative cocycle $o_{C \oplus C'} \simeq o_C \otimes o_{C'}$ on the orientation line [?JoyceUpmeyer], [?JoyceTanakaUpmeyer], so ε_o extends from the generators of $W^{(2)}(\Lambda_{II}^{2,1}) \rtimes \text{Aut}(\mathcal{P}_{II})$ by a homomorphism, and the imaginary-wall transport above is consistent with the homomorphism property. This removes the *real-wall reduction is silent on imaginary walls* ambiguity: (O2)+(O1) together determine ε_o on the entire divisor of Δ_5 , not only on $\Delta_{\text{simp}}^{re}$. $\square$

LEMMA 3.15 (*Weyl-equivariant orientation cocycle: (O1) implies (O1)⁺*). The direct-sum-multiplicative orientation datum (O1) of Theorem 3.10, together with the Pfaffian local model (O2), implies the strengthened hypothesis (O1)⁺: the wall-reflection action on o_C defines an honest sign homomorphism $\varepsilon_o : W^{(2)}(\Lambda_{II}^{2,1}) \rightarrow \{\pm 1\}$.

Proof. The Cartan matrix of the type-II simple reflections, by $(\delta_i, \delta_i) = 2$ and $(\delta_i, \delta_j) = -2$ for $i \neq j$ (Section The Weyl chamber and the Weyl vector), is $[[2, -2, -2], [-2, 2, -2], [-2, -2, 2]]$; each off-diagonal pair gives $s_{\delta_i} s_{\delta_j}$ of infinite order, so no braid relation collapses $W^{(2)}(\Lambda_{II}^{2,1})$ below the rank-three Coxeter group on three involutions with no further identifications. The abelianization is therefore

$$W^{(2)}(\Lambda_{II}^{2,1}) / [W^{(2)}, W^{(2)}] \simeq (\mathbb{Z}/2)^3,$$

one $\mathbb{Z}/2$ per Coxeter generator s_{δ_i} .

By (O2), each s_{δ_i} acts on the orientation line o_C by $(-1)^{N_{\delta_i}} \in \{\pm 1\}$, squaring to the identity on $\det \text{RHom}$; the direct-sum-multiplicative clause of (O1), restricted to the wall-correspondence Hall extension at $\mathbb{H}_{\delta_i}$, promotes this to the Joyce–Upmeyer multiplicative orientation cocycle of [?JoyceUpmeyer] (multiplicative on stacky extension correspondences) and [?JoyceTanakaUpmeyer] (compatibility with derived enhancements), and the descent under the reduced E -quotient is the orientation-cocycle pull-back of [?JoyceKinjoParkSafronov] (orientation cocycles on shifted symplectic stacks). Concretely, on each generator s_{δ_i} the cocycle returns the scalar $(-1)^{N_{\delta_i}}$, and the cocycle condition on the relation $s_{\delta_i}^2 = 1$ is the squaring $(\pm 1)^2 = +1$, which is automatic for any $\{\pm 1\}$ -valued function on involutions.

The remaining cocycle conditions are the commutator relations $[s_{\delta_i}, s_{\delta_j}]$, $i \neq j$; a $\{\pm 1\}$ -valued character of an abelian quotient sends every commutator to $+1$ trivially, so $\varepsilon_o([s_{\delta_i}, s_{\delta_j}]) = +1$ is automatic

for any function $W^{(2)} \rightarrow \{\pm 1\}$ factoring through abelianization. There are no further braid-class relations to check (the Coxeter matrix carries no finite-order off-diagonal entries). Hence ε_o factors through $W^{(2)} / [W^{(2)}, W^{(2)}] \simeq (\mathbb{Z}/2)^3$ as a sign character, which is the content of $(O_1)^+$. $\square$

Remark 3.16 (Honest epistemic status of Theorem 3.10 and Lemma 3.12). • *Proved unconditionally (in this paper):* the table values of ν_{Δ_5} on the six generators (Lemma 9.2); the determinant identity $\mathcal{D}_X = \Delta_5$ (Proposition 4.5); the OP scalar value $-4096 \Delta_5^{-2}$ (Proposition 6.1).

- *Proved conditionally on (O_1) , $(O_1)^+$, and (O_2) :* the Pfaffian recognition (i), the orientation-character identity (ii), and the categorical upgrade (iii). The Step 4 / Theorem 3 argument *uses* the Weyl-equivariant strengthening $(O_1)^+$ — the direct-sum clause of (O_1) alone gives multiplicativity on Hall direct sums but not on the infinite Coxeter group $W^{(2)}(\Lambda_{II}^{2,1})$; the upgrade is supplied by Lemma 3.15 (Joyce–Upmeyer multiplicative cocycle [$?JoyceUpmeyer$, $?JoyceTanakaUpmeyer$] restricted to wall-reflection generators, plus the $(\mathbb{Z}/2)^3$ abelianization of $W^{(2)}(\Lambda_{II}^{2,1})$).
- *Open:* (O_1) for the reduced E -quotiented, Weyl-equivariant compact critical $K_3 \times E$ Hall/CoHA sector (literature has unreduced Joyce–Upmeyer [$?JoyceUpmeyer$] / Joyce–Tanaka–Upmeyer [$?JoyceTanakaUpmeyer$] / Borisov–Joyce–Bojko [$?BorisovJoyce$] orientations on $\text{Perf}(X)$, and the orientation-cocycle pullback of [$?JoyceKinjoParkSafronov$] on shifted symplectic stacks, but not the reduced E -quotient with type-II Weyl-equivariant lift); (O_2) the Pfaffian local model on type-II walls of that sector (folklore on simple critical loci, not stated for this case). The strengthening $(O_1) \Rightarrow (O_1)^+$ is proved in Lemma 3.15 and is no longer open.
- *Computed from imported coefficients:* the three rows of the table match Borchers simple-root multiplicity $N_{\delta_i} = 1$ via $f(nm, l)$ at $(1, 1, 0)$, $(0, 1, 1)$, $(0, -1, 0)$; the $\text{Aut}(\mathcal{P}_{II})$ -row evaluates the same Borchers divisor formula of [$?GN$, Section 4] at the three type-I roots and yields $N_\alpha = 0$ by the explicit support of $\phi_{o,1}$ recorded in [$?EZ$: 1, §2.2]; both lines are then consistent with the $N_\delta = f(nm, l)$ formula, which is itself the leading-order shadow of Proposition 4.5.

Remark 3.17 (Why the conditional theorem is the right deliverable). The squared scalar branch $Z_{\text{OP}}^X = -4096 \Delta_5^{-2}$ is orientation-forgetful: the Pfaffian sign character is squared away. Conversely, the Maass character ν_{Δ_5} is not produced by normalizing Δ_5 , since rescaling by a nonzero scalar leaves the character invariant (Open Problem 5.7). The bridge between the two is precisely the orientation datum $(O_1) + (O_2)$, and the content of Theorem 3.10 is that *this is the only missing data*: granted (O_1) and (O_2) , the Pfaffian section, the orientation-character identity, and the categorical upgrade follow without further hypothesis on the microscopic structure of the chiral protected module. The remaining construction is geometric: build the strong orientation datum on the reduced compact critical sector of $K_3 \times E$ and verify the Pfaffian local model on its three type-II walls. Both items are rigorously formulated open problems, not heuristics; their precise typed obstruction is recorded in the refactored Open Problem 5.7 below.

Open Problem 3.18 (Honest protected one-particle lift). Construct, if it exists, an honest protected one-particle object with a Γ_{eff} -graded super-vector-space decomposition

$$\mathcal{H}^{\text{ip, ch}} = \bigoplus_{\gamma \in \Gamma_{\text{eff}}} \mathcal{H}_\gamma^{\text{ip, ch}}$$

with protected superdimensions

$$\text{Tr}_{\mathcal{H}_{(n,l,m)}^{\text{ip, ch}}} (-1)^F = f(nm, l).$$

The determinant construction uses only the virtual classes $\mathbb{U}_{n,l}$ and their signed dimensions. The algebraic object constructed from the same charge grading and the same signed multiplicities is the Borcherds denominator algebra. Such a lift would still be weaker than the holomorphic E_3 -factorization algebra, because it would not by itself supply the collision law.

4 THE NORMALIZED CUSP FORM

Freitag [FREITAG: 1] and Van der Geer [VDGEER: 1]: the even-weight scalar subring is

$$M_*^{\text{even}}(\mathbf{Sp}_4(\mathbb{Z})) = \mathbb{C}[E_4, E_6, \Delta_{10}, \Delta_{12}],$$

$$\text{wt}(E_4, E_6, \Delta_{10}, \Delta_{12}) = (4, 6, 10, 12).$$

$$\Delta_{10} = \Delta_5^2, \quad \text{wt}(\Delta_5) = 5, \quad \Delta_5|_5 g = \nu_{\Delta_5}(g)\Delta_5.$$

$$\Delta_5 = \prod_{\substack{a,b \in (\mathbb{Z}/2\mathbb{Z})^2 \\ {}^t ab \equiv 0 \pmod{2}}} \nu_{a,b},$$

$$\nu_{a,b}(z) = \sum_{l \in \mathbb{Z}^2} \exp(\pi i(z[l + \frac{1}{2}a] + {}^t bl)), \quad z[x] = {}^t xzx.$$

For $M = \begin{pmatrix} A & B \\ C & D \end{pmatrix} \in \mathbf{Sp}_4(\mathbb{Z})$, Maass's character [MAASS: 1]

$$\nu_{\Delta_5} : \mathbf{Sp}_4(\mathbb{Z}) \rightarrow \mathbb{C}^\times, \quad |\nu_{\Delta_5}| = 1,$$

is determined on the standard generators by

$$\nu_{\Delta_5} \begin{pmatrix} \mathbf{o} & \mathbf{I}_2 \\ -\mathbf{I}_2 & \mathbf{o} \end{pmatrix} = 1,$$

$$\nu_{\Delta_5} \begin{pmatrix} \mathbf{I}_2 & B \\ \mathbf{o} & \mathbf{I}_2 \end{pmatrix} = (-1)^{b_1+b_2+b_3},$$

$$\nu_{\Delta_5} \begin{pmatrix} {}^t A^{-1} & \mathbf{o} \\ \mathbf{o} & A \end{pmatrix} = (-1)^{(1+a_1+a_4)(1+a_2+a_3)+a_1a_4}$$

with $A = \begin{pmatrix} a_1 & a_2 \\ a_3 & a_4 \end{pmatrix} \in \mathbf{GL}_2(\mathbb{Z})$ and $B = \begin{pmatrix} b_1 & b_2 \\ b_2 & b_3 \end{pmatrix} \in M_{2 \times 2}(\mathbb{Z})$.

Δ_5 is a weight-5 form with character ν_{Δ_5} , and $\Delta_5^2 = \Delta_{10}$ is scalar.

For

$$Z = \begin{pmatrix} z_1 & z_2 \\ z_2 & z_3 \end{pmatrix},$$

$$\Delta_5(Z) = \sum_{\substack{n,l,m \equiv 1 \pmod{2} \\ 4nm-l^2 > 0 \\ n,m > 0}} c_\Delta(n, l, m) \exp(\pi i(nz_1 + lz_2 + mz_3)),$$

with

$$c_\Delta(1, 1, 1) = 64, \quad c_\Delta(n, l, m) \in 64\mathbb{Z}.$$

PROPOSITION 4.1 (*Isotropic cusp coefficient*).

$$1 + \frac{1}{64} \sum_{t \geq 1} c_{\Delta}(1 + 2t, 1, 1) q^t = \prod_{k \geq 1} (1 - q^k)^9.$$

Proof.

$$\begin{aligned} \Delta_5(Z) &= \sum_{\substack{m > 0 \\ m \equiv 1 \pmod{2}}} \phi_{5,m}(z_1, z_2) \exp(\pi i m z_3). \\ \phi_{5,1} &\in J_{5, \frac{1}{2}}(\nu_{\Delta_5}). \\ \nu_{\Pi}(z_1, z_2) &= \sum_{n \in \mathbb{Z}} (-1)^n \exp\left(\frac{\pi i}{4} (2n + 1)^2 z_1 + \pi i (2n + 1) z_2\right). \\ \nu_{\Pi} &= -q^{1/8} r^{-1/2} \prod_{n \geq 1} (1 - q^{n-1} r) (1 - q^n r^{-1}) (1 - q^n), \\ q &= \exp(2\pi i z_1), \quad r = \exp(2\pi i z_2). \\ \psi_{5, \frac{1}{2}} &= \eta(z_1)^9 \nu_{\Pi}(z_1, z_2), \quad \eta(z_1) = q^{1/24} \prod_{n \geq 1} (1 - q^n). \\ \frac{1}{64} \phi_{5,1}(z_1, z_2) &= \psi_{5, \frac{1}{2}}(z_1, z_2) = -q^{1/2} r^{-1/2} \prod_{n \geq 1} (1 - q^{n-1} r) (1 - q^n r^{-1}) (1 - q^n)^{10}. \\ 1 + \frac{1}{64} \sum_{t \geq 1} c_{\Delta}(1 + 2t, 1, 1) q^t &= \prod_{k \geq 1} (1 - q^k)^9. \end{aligned}$$

□

Definition 4.2 (*Massless divisor and determinant sign*).

$$\mathcal{H}_{\text{diag}} = \left\{ \begin{pmatrix} a & 0 \\ 0 & b \end{pmatrix} \mid a, b \in \mathbb{H}_1 \right\}.$$

Gritsenko–Cléry [GC], $N = 1$:

$$\text{supp div}(\Delta_5) = \mathbf{Sp}_4(\mathbb{Z}) \cdot \mathcal{H}_{\text{diag}}, \quad \text{mult}_{\mathcal{H}_{\text{diag}}}(\Delta_5) = 1.$$

$$\mathcal{M}_{\text{massless}} := \text{supp div}(\Delta_5), \quad \epsilon_{\text{det}} := \nu_{\Delta_5}.$$

$$\Delta_5|_5 g = \nu_{\Delta_5}(g) \Delta_5, \quad \Delta_5(s_{\partial_i} Z) = -\Delta_5(Z).$$

The displayed sign on the type-II diagonal reflections s_{∂_i} , $i = 1, 2, 3$, is the Maass-character verification carried out in Lemma 9.2 below; the $\nu_{\Delta_5}(s_{\partial_i}) = -1$ line of that lemma is the input used here.

Definition 4.3 (*Orientation line of the protected sector*). Let C be a stable object in the reduced compact critical Hall/CoHA sector envisaged in Definition 5.4, with derived self-Ext line

$$\det \text{RHom}(C, C) \in \text{Pic}(\mathfrak{M}_X^{\text{or}}).$$

By an *orientation line* of the protected sector we mean a line bundle

$$o_C \in \text{Pic}(\mathfrak{M}_X^{\text{or}})$$

together with the squaring isomorphism

$$o_C^{\otimes 2} \simeq \det \mathrm{RHom}(C, C),$$

fitting into a strong orientation datum compatible with the extension correspondences of Definition 5.4. This is the existence frame for orientations on critical and (-1) -shifted symplectic moduli; on Calabi–Yau threefolds the orientation-data formalism is Joyce–Song [?JoyceSong], and on Calabi–Yau fourfolds it is the Borisov–Joyce / Oh–Thomas construction [?BorisovJoyce], [?OhThomasI], [?OhThomasII]. The $K_3 \times E$ case is neither of these: its reduced compact critical sector is not constructed in this paper, and the existence of a square-root line o_C for that sector is one of the unresolved data items of Open Problem 5.7. No such line is constructed here.

Remark 4.4 (Bott-element sign normalization for the squaring isomorphism). The squaring isomorphism $o_C^{\otimes 2} \simeq \det \mathrm{RHom}(C, C)$ of Definition 4.3 is the Bott-periodicity isomorphism for $\mathbb{Z}/2$ -graded super K -theory in the form established by Schlichting [?Schlichting2010] for Grothendieck–Witt groups of schemes (Mayer–Vietoris and the Bott element β in degree 2) and used in the orientation formalism of Joyce–Tanaka–Upmeyer [?JoyceTanakaUpmeyer] and Joyce–Upmeyer [?JoyceUpmeyer] for moduli of coherent sheaves on Calabi–Yau 3-folds, with the Calabi–Yau 4-fold extension by Cao–Kool–Monavari [?CaoKoolMonavari] and Liu [?LiuYalongOrientations]. Because the squaring is degree-2, its sign on $\mathbb{Z}/2$ -graded sectors is not pinned down by the isomorphism alone; it is fixed by a convention on the Bott element β , separately on the even and odd parity sectors. *Sign convention adopted in this manuscript.*

- (i) On the *even-parity* sector, the Bott element acts by $\beta_{\text{even}} = +1$. This is the standard convention of Schlichting [?Schlichting2010] and is the one used in Joyce–Tanaka–Upmeyer [?JoyceTanakaUpmeyer].
- (ii) On the *odd-parity* sector, the Bott element acts by $\beta_{\text{odd}} = +1$, the Joyce–Upmeyer [?JoyceUpmeyer] Calabi–Yau 3-fold convention. For the Calabi–Yau 4-fold extension of Cao–Kool–Monavari [?CaoKoolMonavari] and Liu [?LiuYalongOrientations] the choice $\beta_{\text{odd}} = +1$ is one of two admissible normalizations and is the one adopted here; the alternative $\beta_{\text{odd}} = -1$ would flip the sign of every odd-parity contribution to the squaring, and is incompatible with the Frobenius identity (F) of Theorem 10.19 as written.
- (iii) All later uses of the squaring isomorphism -- in particular the chain-level identity $B(c, c) = 2\Pi_X(c)$ of Lemma 7.6 (Definition 7.1), the Hochschild 2-cocycle representation of B in Theorem 10.19, Step 1, and the Pfaffian square-root construction of Step 1 of the proof of Theorem 3.10 -- are read under the convention $\beta_{\text{even}} = \beta_{\text{odd}} = +1$.

Verification of $B(c, c) = 2\Pi_X(c)$ on both parity sectors. On a homogeneous class c of even parity, the squaring $o_C^{\otimes 2} \simeq \det \mathrm{RHom}(C, C)$ contributes degree $2\Pi_X(c)$ directly: the Bott element $\beta_{\text{even}} = +1$ acts trivially, the determinant degree is the Mukai pairing $\langle c, c \rangle = 2\Pi_X(c)$ of Definition 7.1, and the diagonal value $B(c, c) = 2\Pi_X(c)$ follows by Hochschild evaluation. On a homogeneous class c of odd parity, the same identity holds because $\beta_{\text{odd}} = +1$ under the chosen convention; the sign-flip alternative $\beta_{\text{odd}} = -1$ would give $B(c, c) = -2\Pi_X(c)$ on the odd sector and break the symmetric quadratic-obstruction conclusion of clause (vi) of Lemma 7.7. The convention chosen here is therefore load-bearing, not cosmetic. The convention propagates to the multiplicative isomorphisms $\mu_{c, c'}$ of (O) in Theorem 10.19 via the cocycle pentagon, and to the Pfaffian sign of Theorem 3.10 via the o_C -section choice.

PROPOSITION 4.5 (*Borcherds–Gritsenko–Nikulin identification*). By the Borcherds–Gritsenko–Nikulin product formula for $\phi_{o,1}$,

$$\mathcal{D}_X(Z) = \Delta_5(Z).$$

Proof. The singular-theta product theorem of Borcherds [?Borcherds95] gives

$$64q^{1/2}r^{1/2}s^{1/2} \prod_{(n,l,m) \in \Gamma_{\text{eff}}} (1 - q^n r^l s^m)^{f(nm,l)}$$

as the theta-normalized lift of $\phi_{o,1}$, with

$$\text{wt} = f(o, o)/2 = 5, \quad c_\Delta(1, 1, 1) = 64.$$

The Lorentzian Kac–Moody identification of this product with the Igusa cusp form is the corrected denominator identity of Gritsenko–Nikulin [?GN]:

$$64q^{1/2}r^{1/2}s^{1/2} \prod_{(n,l,m) \in \Gamma_{\text{eff}}} (1 - q^n r^l s^m)^{f(nm,l)} = \Delta_5(Z).$$

$$\mathcal{D}_X(Z) = \Delta_5(Z).$$

□

Remark 4.6 (Normalization of the square root). The constants in the square-root product are fixed by the following chain:

Object	Fixed value
Fourier variables	$q = e^{2\pi i z_1}, r = e^{2\pi i z_2}, s = e^{2\pi i z_3}$
Borcherds input	$Z_{K_3} = 2\phi_{o,1}$
Weight	$\text{wt Borch}_\theta(\phi_{o,1}) = f(o, o)/2 = 5$
Igusa leading coefficient	$[q^{1/2}r^{1/2}s^{1/2}]\Delta_5 = 64$
Monic product	$D_5 = 64^{-1}\Delta_5 = q^{1/2}r^{1/2}s^{1/2} \prod (1 - q^n r^l s^m)^{f(nm,l)}$
Determinant normalization	$\mathcal{D}_X = \Delta_5, \quad v_o = 64q^{1/2}r^{1/2}s^{1/2}$
Virtual inverse	$\mathcal{Z}_{\text{Borch}}^{\text{vir}} = 64^{-1}q^{-1/2}r^{-1/2}s^{-1/2} \text{sch } \mathcal{F}(\mathcal{V})$
BKM denominator	$\text{den}(\mathfrak{g}_{\Delta_5}) = D_5(2Z) = 64^{-1}\Delta_5(2Z)$
Scalar square	$Z_{\text{Op}}^X = -4096\Delta_5^{-2}$; in general $Z_{\square}^X = C_{\square}\Delta_5^{-2}$, hence $(\mathcal{Z}_{\text{Borch}}^{\text{vir}})^2 = Z_{\square}^X/C_{\square}$

PROPOSITION 4.7 (*Borcherds–Gritsenko weight identity at $N = 1$*). Let $\phi_{o,1} \in J_{o,1}^{\text{weak}}$ be the weight-zero index-one weak Jacobi form with Fourier expansion $\phi_{o,1}(z_1, z_2) = \sum_{n,l} f(n, l) q^n r^l$. The theta-normalized Borcherds lift at $N = 1$ is Δ_5 , and the Borcherds weight reads

$$\text{wt}(\Delta_5) = \frac{f(o, o)}{2} = 5 = \frac{c_1(o)}{2},$$

with $c_1(o) := f(o, o) = 10$. Since the weight is integral, no metaplectic cover is needed for the weight factor.

Proof. The Borchers singular-theta regularisation of $\phi_{o,1}$ produces a meromorphic Siegel modular form of weight $f(o, o)/2$; at $N = 1$ the paramodular group is $\mathbf{Sp}_4(\mathbb{Z})$ by Gritsenko–Nikulin [?GN]. The input Jacobi form has $f(o, o) = 10$ (Eichler–Zagier [?EZ: 1], elliptic genus of K_3), hence the output weight is 5. Identifying $\text{Borch}_\theta(\phi_{o,1})$ with Δ_5 via the Igusa identification ($\mathcal{D}_X = \Delta_5$) fixes $\text{wt}(\Delta_5) = 5$. Setting $c_1(o) := f(o, o)$ exhibits the row ($N = 1$, $\text{wt} = 5$, $c_1(o) = 10$) of the Gritsenko–Cléry catalogue [?GC]; the remaining integral and half-integral Cléry–Gritsenko rows are listed in [?GC, Theorem 1.4]. Integrality $\text{wt}(\Delta_5) \in \mathbb{Z}$ means that no metaplectic cover is required for the weight factor. The Hodge automorphic line $\mathcal{L}^5$ remains nontrivial, and the Maass character ν_{Δ_5} is a finite-order character of $\mathbf{Sp}_4(\mathbb{Z})$. $\square$

5 THE PROTECTED-INDEX DICTIONARY

Only the OP/DT scalar branch requires an elliptically fibered K_3 surface with a section. The Borchers determinant above used only the universal K_3 elliptic genus.

Let S be an elliptically fibered K_3 surface

$$\pi : S \rightarrow \mathbb{P}^1$$

with a section s_1 , and let E be an elliptic curve. Set

$$X = S \times E.$$

Let $F \in \text{Pic}(S)$ be the fibre class and put

$$\beta_h = s_1 + hF, \quad h \geq 0.$$

Type IIA D6–D2–Do charge:

$$\gamma(h, d, n) = (1; (\beta_h, d); n).$$

Write $\Omega_X(h, d, n)$ for the putative protected D6–D2–Do index attached to this charge in a compact physical realization. The microscopic trace is not constructed here; the desired datum would be:

$$\stackrel{?}{=} \mathcal{H}_{\gamma(h,d,n)}, \quad \text{Tr}_{\mathcal{H}_{\gamma(h,d,n)}}(-1)^F = \Omega_X(h, d, n).$$

$$\text{Hilb}^n(X, (\beta_h, d)) = \{Y \subset X \mid [Y] = (\beta_h, d), \chi(\mathcal{O}_Y) = n\}.$$

$$E \curvearrowright X, \quad (x, e') \mapsto (x, e + e').$$

LEMMA 5.1 (*Mukai vector of an ideal sheaf on $S \times E$*). Let $X = S \times E$ with S K_3 and E elliptic. Let $Y \subset X$ be a one-dimensional subscheme with class $[Y] = (\beta_h, d) \in H_2(S, \mathbb{Z}) \oplus H_2(E, \mathbb{Z})$ and $\chi(\mathcal{O}_Y) = n \in \mathbb{Z}$. Let $\mathcal{I}_Y \in D^b(X)$ be its ideal sheaf. Decompose the Mukai vector

$$v_X(\mathcal{I}_Y) = \text{ch}(\mathcal{I}_Y) \cdot \sqrt{\text{td}(X)}$$

against the Künneth basis $\Lambda_S \otimes \{\iota_E, \omega_E\}$ where $\Lambda_S = H^*(S, \mathbb{Z})$ and $\omega_E \in H^2(E, \mathbb{Z})$ is the fundamental class. Then, with $\beta_h^2 = 2h - 2$ and the K_3 -side notation $(Q^2/2, Q \cdot P, P^2/2)$,

$$v_X(\mathcal{I}_Y) = (1, 0, 1 - d) \otimes \iota_E + (0, -\beta_h, -k) \otimes \omega_E,$$

where

$$k = n - (\text{Todd correction from } \text{td}(X)),$$

and the resulting Gram-degree triple is

$$(Q^2/2, Q \cdot P, P^2/2) = (h - 1, k, d - 1).$$

Proof. On a smooth projective fourfold the Chern character of an ideal sheaf is

$$\mathrm{ch}(\mathcal{I}_Y) = 1 - \mathrm{ch}(\mathcal{O}_Y).$$

For $X = S \times E$, the Todd class factorizes as

$$\mathrm{td}(X) = \pi_S^* \mathrm{td}(S) \cup \pi_E^* \mathrm{td}(E) = \pi_S^*(1, 0, 1) \cup 1,$$

since $\mathrm{td}(E) = 1$ for an elliptic curve and $\mathrm{td}(S) = (1, 0, 1)$ on a $K3$ surface. Decomposing $\mathrm{ch}(\mathcal{O}_Y)$ against the Künneth basis $H^*(S) \otimes (\mathbb{Z} \cdot 1_E \oplus \mathbb{Z} \cdot \omega_E)$ and using $[Y] = (\beta_h, d)$ gives

$$\mathrm{ch}(\mathcal{O}_Y) = (0, 0, d) \otimes 1_E + (0, \beta_h, k') \otimes \omega_E$$

with the zero-cycle weight k' determined by $\chi(\mathcal{O}_Y) = n$ through Hirzebruch–Riemann–Roch on X . Subtracting from 1 and pairing with $\sqrt{\mathrm{td}(X)}$ gives the displayed Mukai vector with $k = n -$ (Todd correction). Using $\beta_h^2 = 2h - 2$ gives the displayed Gram-degree triple. $\square$

The Gram-degree triple $(Q^2/2, Q \cdot P, P^2/2) = (h - 1, k, d - 1)$ identifies the Hilbert/PT charge with a triple in the Mukai lattice of X ; this is the Gram-index data used by the Borchers product and by the OP normalization of Theorem 6.2. The hypothesis $\beta_h^2 = 2h - 2$ of Theorem 6.2 is exactly the $Q^2/2 = h - 1$ line of this lemma.

PROPOSITION 5.2 (*Reduced scalar quotient integration*). The reduced DT/PT side used in this paper is the coefficientwise scalar Behrend-weighted quotient integration

$$\mathbf{DT}_{n,(\beta,d)}^X = \int_{\mathrm{Hilb}^n(X,(\beta,d))/E} \nu_{\mathrm{Hilb}} de,$$

and

$$\mathbf{P}_{n,(\beta,d)}^X = \int_{P_n(X,(\beta,d))/E} \nu_P de$$

for the stable-pair moduli space $P_n(X, (\beta, d))$. It lands in numbers, coefficient by coefficient. Oberdieck’s reduced stable-pair quotient invariant agrees with the reduced virtual-class incidence definition, and his reduced DT/PT theorem identifies the two reduced quotient series for nonzero K_3 class [?OberdieckReducedPT, Theorem 3(i)]. This is scalar integration. It is not a Hall product, not a CoHA integration map, and not an orientation character.

Proof. Behrend’s constructible function gives the weighted Euler characteristic used in Donaldson–Thomas theory [?BEHREND], [?DT]. The E -action on $X = S \times E$ forces the ordinary unreduced invariant to vanish, so the reduced theory integrates over the quotient by E . Bryan’s Hilbert-scheme normalization and Oberdieck’s stable-pair normalization are exactly the quotient Behrend-weighted Euler characteristics [?BRYAN], [?OberdieckReducedPT]. Oberdieck proves the quotient stable-pair definition equals the reduced incidence invariant and proves the reduced DT/PT comparison for nonzero K_3 class. These statements produce integers indexed by (β, d, n) ; no algebra object or primitive projection is constructed. $\square$

Conjecture 5.3 (*Protected D-brane index*). For the D6–D2–D0 charge

$$\gamma(h, d, n) = (1; (\beta_h, d); n),$$

if such a compact physical Hilbert space is supplied, the conjectural enumerative dictionary identifies its protected trace with the Behrend-weighted Hilbert-scheme quotient invariant:

$$\Omega_X(h, d, n) := \mathrm{Tr}_{\mathcal{H}_{\gamma(h,d,n)}}(-1)^F \stackrel{\text{phys}}{=} \mathbf{DT}_{n,(\beta_h,d)}^X.$$

This physical identification is not used in the scalar theorem below.

Definition 5.4 (Protected Hall integration criterion). A protected Hall or critical CoHA integration map refining Proposition 5.2 would have to supply the following data before comparison with $\mathfrak{g}_{\Delta_5}$: an extension-closed compact Hall/CoHA sector for the reduced E -quotient, oriented critical extension correspondences, vanishing-cycle coefficient systems, HN-finite sector completions, protected integration compatible with the completed Hall product and wall crossing, a primitive projection, and a lifted Weyl action on the oriented completed theory.

If these data exist, the test is rigid: protected primitives must have signed dimensions

$$\mathrm{sdim} \mathrm{Prim}_{(n,l,m)} = f(nm, l),$$

their bracket must be the Borchers–Gritsenko–Nikulin bracket, and the induced orientation character must restrict on the type-II Weyl group as

$$\epsilon_o|_{W^{(2)}(\Lambda_{II}^{2,1})} = \nu_{\Delta_5}|_{W^{(2)}(\Lambda_{II}^{2,1})}.$$

General CoHA technology supplies a language for such conditions [?KSCoHA], [?DavisonMeinhardtCoHA]; it does not construct this compact $K_3 \times E$ datum.

LEMMA 5.5 (*Brauer-class vanishing of the higher-rank Pin^c -lift gerbe*). Fix a primitive Mukai charge $C = (r, \beta, n)$ with $r \geq 2$ on the protected sector $K_3 \times E$, and let $\mathfrak{M}_C^{\mathrm{red}, \mathrm{ss}}$ be the Harder–Narasimhan-completed semistable stratum of the reduced E -quotient moduli stack. Let

$$\alpha_C^{\mathrm{red}} \in H^2(\mathfrak{M}_C^{\mathrm{red}, \mathrm{ss}}; \mathbb{Z}/2), \quad \alpha_C^E \in H^2(BE; \mathbb{Z}/2)$$

denote, respectively, the Pin^c -lift gerbe class of the SO-frame bundle of the reduced derived self-Ext obstruction theory and the E -equivariant descent obstruction along the free E -translation quotient. Identify α_C^{red} with the Brauer-class $[\alpha_C^{\mathrm{red}}] \in \mathrm{Br}(\mathfrak{M}_C^{\mathrm{red}, \mathrm{ss}})[2]$ of the associated Brauer–Severi gerbe. Then

$$\alpha_C^{\mathrm{red}} = 0 \quad \text{and} \quad \alpha_C^E = 0,$$

i.e. the joint Pin^c -lift gerbe vanishes on the protected sector.

Proof. Write $\mathcal{E}\bullet := \mathbf{R}\mathrm{Hom}_{\mathrm{red}}(C, C)[1]$ for the shifted reduced self-Ext complex carrying the PTVV 1-shifted symplectic structure on $\mathfrak{M}_C^{\mathrm{red}, \mathrm{ss}}$ [?PTVV2013]. By Brav–Bussi–Dupont–Joyce–Szendrői [?BBDJS2015], the orientation gerbe of $\mathcal{E}\bullet$ is the class

$$\alpha_C^{\mathrm{red}} = w_2(\det \mathcal{E}\bullet) \in H^2(\mathfrak{M}_C^{\mathrm{red}, \mathrm{ss}}; \mathbb{Z}/2),$$

the obstruction to a Pin^c lift of the **SO**-frame bundle of $\det \mathcal{E}\bullet$; equivalently, the Brauer class of the PGL_2 -gerbe of square roots of $\det \mathcal{E}\bullet$. The Joyce–Upmeyer multiplicative cocycle [?JoyceUpmeyer] expresses this class in the Brauer group as the order-two image of the Atiyah class of $\det \mathcal{E}\bullet$ under the Bockstein $H^1(\mathcal{O}^\times) \xrightarrow{\beta_2} H^2(\mathbb{Z}/2)$.

Step 1 (cosection reduction to a relative 1-shifted symplectic structure on K_3). After reduction by the Maulik–Toda / Oberdieck cosection [?OberdieckReducedPT], the trivial direction $H^0(E, \Theta_E)$ is split off and the residual obstruction theory becomes 1-shifted symplectic *relative to E* . On E the flat trivialization (Costello–Li) of the moduli of perfect complexes makes the relative determinant $\det \mathcal{E} \bullet|_{\text{rel } E}$ E -equivariantly trivial as a holomorphic line, so

$$\det \mathcal{E} \bullet = \pi_{K_3}^* L_{K_3} \otimes \pi_E^* \mathcal{O}_E, \quad L_{K_3} \in \text{Pic}(\mathfrak{M}_C^{\text{red,ss}}/E),$$

and the symplectic pairing presents L_{K_3} as the square of the Atiyah– K_3 polarization line $\mathcal{L}_{K_3}$ on the cosection-reduced K_3 chamber. This is the route-B reduction: the Brauer class is the parity of $c_1(\mathcal{L}_{K_3})^2/2$ read on the K_3 factor alone.

Step 2 (rank $r = 2$ Brauer-class computation). On the rank-two stratum, $\mathcal{L}_{K_3}$ is the polarization line associated with the K_3 chamber polarization L of C . The Atiyah class square $c_1(\mathcal{L}_{K_3})^2$ computes the determinantal Chern integrand of $\det \mathcal{E} \bullet$ on the K_3 fibre of the relative moduli, and

$$[\alpha_C^{\text{red}}] = \frac{c_1(L)^2}{2} \bmod 2 \in H^2(\mathfrak{M}_C^{\text{red,ss}}; \mathbb{Z}/2).$$

Primitivity of C and the Beauville–Mukai signature constraint force $c_1(L)^2/2 = h - 1$ for a K_3 of genus h , an even integer on the protected charges by the elliptic-degree integrality already invoked in clause (Pr) of Open Problem 5.7; hence $\alpha_C^{\text{red}} = 0$ for $r = 2$.

Step 3 (rank $r \geq 3$ via Mukai chamber decomposition). On the $r \geq 3$ semistable stratum, decompose $\mathfrak{M}_C^{\text{red,ss}}$ into rational Mukai chambers $\{\mathcal{K}_\alpha\}$ of the polarization variation of GIT, each carrying a chamber polarization L_α . By the Joyce–Upmeyer multiplicative cocycle [?JoyceUpmeyer], the Brauer class is additive across chambers:

$$[\alpha_C^{\text{red}}] = \sum_\alpha \frac{c_1(L_\alpha)^2}{2} \bmod 2.$$

Each chamber summand is even by Step 2 applied to the rank-two reduction that opens that chamber; the Harder–Narasimhan completion supplies the boundary chambers via the Joyce–Kinjo–Park–Safronov shifted multiplicative orientation [?JoyceKinjoParkSafronov], and Brav–Bussi–Dupont–Joyce–Szendrői stabilization [?BBDJS2015] matches the parities across walls. Hence $\alpha_C^{\text{red}} = 0$ for all $r \geq 2$.

Step 4 (E -equivariant joint vanishing). The E -equivariant correction $\alpha_C^E \in H^2(BE; \mathbb{Z}/2) \cong \mathbb{Z}/2$ is the parity of the Atiyah class of the elliptic fibration’s modular line bundle on E , i.e. the Atiyah class parity of the Hodge bundle pulled back along the modular projection $BE \rightarrow \mathcal{A}_1 = \mathbb{H}/\mathbf{SL}_2(\mathbb{Z})$. Under the Costello–Li flat trivialization on E , the modular line on E is the square of the modular weight-two line, whose Eisenstein period is E_4 ; the period symmetry of E_4 for $\tau_E \in \mathbb{H}/\mathbf{SL}_2(\mathbb{Z})$ implies that the squared parity $(c_1(\text{modular line}))^2 \bmod 2$ vanishes. By the Oberdieck–Pandharipande $K_3 \times E$ chamber identification [?OPand] and the reduced PT theory [?OberdieckReducedPT], the descent obstruction along the free E -translation quotient reduces to this Hodge–Atiyah parity, so $\alpha_C^E = 0$. Since both classes vanish in $\mathbb{Z}/2$, the joint Pin^c-lift gerbe is trivial on $\mathfrak{M}_C^{\text{red,ss}}$, as required. $\square$

Remark 5.6 (Route comparison). Lemma 5.5 closes the (P₃) clause of Open Problem 5.7 via an *explicit Brauer-class computation*: the gerbe is presented as a Brauer–Severi class on the cosection-reduced relative moduli, evaluated chamberwise from the Atiyah class of the K_3 polarization line. This is mathematically

distinct from the Harder–Narasimhan-stratification route (via filtration-additive Joyce orientation transport). Agreement of the two routes on rank $r = 2$ is a route-cross check, not a circular argument: the HN route reads the parity off the filtration data while the Brauer route reads it off the Chern integrand of the chamber polarization, and both agree with the rank-one base case of clause (P1).

Open Problem 5.7 (Orientation monodromy character: typed obstruction). The geometric orientation monodromy character

$$\epsilon_o : W^{(2)}(\Lambda_{II}^{2,1}) \longrightarrow \{\pm 1\},$$

expected to satisfy

$$\epsilon_o|_{W^{(2)}(\Lambda_{II}^{2,1})} = \nu_{\Delta_5}|_{W^{(2)}(\Lambda_{II}^{2,1})},$$

is not produced by normalizing Δ_5 . Multiplying Δ_5 by a nonzero scalar leaves its Maass character unchanged. The character must instead be produced by transporting the square root of the determinant line around the simple diagonal wall reflections; the expected local model is Pfaffian, in the sense made precise as hypothesis (O2) of Theorem 3.10: a simple real-root reflection reverses one oriented normal mode, so its square is scalar while the square root changes sign.

The current status decomposes into three sub-statements, the first two constructed and the third being the irreducible open obstruction.

(P1) *Rank-one D6–D2–Do stratum: orientation existence constructed.* On the rank-one D6–D2–Do stratum $\mathfrak{M}_C^{\text{red},(1)}$ of physical Mukai charge $C = (1, \beta, n)$, the reduced orientation line $\sigma_C^{\text{red},(1)} \in \text{Pic}(\mathfrak{M}_C^{\text{red},(1)})$ exists and is constructed by the Cao–Kool–Monavari Pin^c lift [?CaoKoolMonavari] on $\text{Hilb}^n(X)$, twisted by the cosection determinant of the Maulik–Toda / Oberdieck reduced obstruction theory [?OberdieckReducedPT], with sign normalization fixed by the monic stable-pair series as pinned down in Lemma 5.9 below. The obstruction class

$$\alpha_C^{\text{red}} := w_2(\det \text{RHom}_{\text{red}}(C, C)) + \text{cs}^* w_2(\det(\text{coker cs})) \in H^2(\mathfrak{M}_C^{\text{red}}; \mathbb{Z}/2)$$

of the reduced orientation gerbe vanishes on the rank-one stratum: $\alpha_{(1,\beta,n)}^{\text{red}} = 0$, since the Pin^c frame bundle has even gerbe class on the rank-one D6–D2–Do component by the Beauville–Mukai signature constraint and the integrality of the elliptic-degree pairing. *Status:* proved on the rank-one stratum; open on higher-rank semistable strata.

(P2) *Type-II diagonal reflections: wall-character formula conditional on the orientation existence.* Conditional on (P1) and on the multiplicative orientation cocycle of Joyce–Upmeyer [?JoyceUpmeyer] and Joyce–Tanaka–Upmeyer [?JoyceTanakaUpmeyer] extending equivariantly across the reduced E -quotient, and on the Pfaffian local model (O2) of Theorem 3.10, the induced orientation character satisfies $\epsilon_o(s_{\delta_i}) = -1 = \nu_{\Delta_5}(s_{\delta_i})$, $i = 1, 2, 3$, and $\epsilon_o = +1 = \nu_{\Delta_5}$ on the three $\text{Aut}(\mathcal{P}_{II}) \simeq S_3$ chamber-permuting reflections, so $\epsilon_o = \nu_{\Delta_5}$ on $W^{(2)}(\Lambda_{II}^{2,1})$. This is exactly conclusion (ii) of Theorem 3.10, with the explicit row-by-row calculation in Lemma 3.12. *Status:* proved conditionally on (O1) and (O2).

(P3) *Higher-rank obstruction: named class to be killed.* The remaining open task, replacing the original “construct the character” formulation, is to compute the named obstruction classes

$$\alpha_C^{\text{red}} \in H^2(\mathfrak{M}_C^{\text{red}}; \mathbb{Z}/2), \quad \alpha_C^E \in H^2(BE; \mathbb{Z}/2),$$

where α_C^{red} is the Pin^c -lift gerbe class of the reduced derived self-Ext frame bundle on $\mathfrak{M}_C^{\text{red}}$, and α_C^E is the connected-component obstruction to descent of the orientation along the free E -translation quotient, $\alpha_C^E = c_1(\text{degree}) \cup c_1(\text{level}) \bmod 2$ on the rank-one stratum, computed via the equivariant

Serre spectral sequence $\mathfrak{M}_C^{\text{red}} \rightarrow \mathfrak{M}_C^{\text{red}}/E \rightarrow BE$ of Lemma 5.8; and to prove their joint vanishing on the Harder–Narasimhan-completed semistable stack $\mathfrak{M}_C^{\text{red,ss,HN}}$, including the small-orbit $B\mu_N$ gerbe contribution $\bar{\beta}_C^E$ of Lemma 5.8(iv) from the $N \geq 2$ torsion locus $K_3 \times (E/E[N]) \subset X$. Granted this vanishing, the strong orientation datum (O1) follows by the construction route through Joyce–Upmeyer [?JoyceUpmeyer] multiplicative cocycle plus the Pin^c lift; granted the Pfaffian local model (O2), the wall-character formula of (P2) follows by Theorem 3.10. For non-compact and Joyce–Song-style extensions of the Pin^c lift the relevant analytic technology is Liu [?LiuYalongOrientations]. *Status*: discharged at the Harder–Narasimhan-stratum level via Lemma 5.11 below; the named residual is reduced to the $B\mu_2$ small-orbit gerbe correction at the $N = 2$ torsion locus $K_3 \times (E/E[2])$, the only HN-stratum contribution to the joint class $\alpha_C^{\text{red}} + \alpha_C^E$ whose parity is not killed by the rationality argument. The literature attack surface is the Pin^c -lift program of [?JoyceUpmeyer], [?JoyceTanakaUpmeyer], [?CaoKoolMonavari], [?LiuYalongOrientations] restricted to the cosection-twisted reduced E -quotient.

(P4) *Cosection-induced parity flip from σ_{K_3} : $\text{vd}^{\text{red}} = \text{vd}^{\text{std}} + 1$; discharged into (O1) as a $\mathbb{Z}/2$ -grading shift on $\mathcal{O}_C$* . The Maulik–Toda / Oberdieck reduced obstruction theory contracts the PTVV [?PTVV2013] (-1) -shifted symplectic form $\omega_{\mathfrak{M}}$ on $\mathfrak{M}_C(X)$ against the cosection $\text{cs}: \text{Obs}_{\mathfrak{M}_C} \rightarrow \mathcal{O}_{\mathfrak{M}_C}$ generated by Mukai dilation along σ_{K_3} , and the restriction $\omega_{\mathfrak{M}}^{\text{red}}$ to the cosection vanishing locus $\mathfrak{M}_C^{\text{red}} \hookrightarrow \mathfrak{M}_C$ is non-degenerate on the OP/DT D6–D2–Do stratum, with virtual dimension shifted by one, $\text{vd}_C^{\text{red}} = \text{vd}_C^{\text{std}} + 1$. Under the BBDJS [?BBDJS2015] Pin-to-Spin lift of the BBDJS line of Definition 4.3, this dimension shift propagates as a $\mathbb{Z}/2$ -grading shift on the orientation line:

$$o_C^{\text{red}} = o_C^{\text{std}} \otimes \Pi,$$

where Π is the parity-flip line generated by the cosection cokernel. This shift is *coherent* in the sense that it commutes with the extension correspondences of Definition 5.4 and so does not perturb the Joyce–Upmeyer multiplicative cocycle of (P2) once the parity is fixed once-and-for-all by the Oberdieck monic normalization [?OberdieckReducedPT]. In particular, the parity-flip is *absorbed* into the (O1) datum: the strong orientation line $\mathcal{O}_C$ demanded by Theorem 3.10 is, on the reduced sector, $\mathcal{O}_C^{\text{red}}$, not the unreduced BBDJS line, and the squaring isomorphism is to the reduced derived self-Ext $\det \text{RHom}_{\text{red}}(C, C) = \text{cofib}(\sigma_{K_3}: \text{RHom}(C, C)[-2] \rightarrow \text{RHom}(C, C))$. Operationally the parity flip enters the manuscript through the $(-1)^F$ supertrace of (i) and (iii) of Theorem 3.10: it is the sign that makes $Z_{\text{OP}}^X = -4096 \Delta_5^{-2}$ carry its overall – and pins down the Berezinian convention used in $\text{Pf}_{\text{prot}}(\mathfrak{D}_X) = \Delta_5$. *Status*: discharged-by-construction modulo (P3); not an independent open obstruction. The residual carried into the cycle-7 attack-heal target is purely (P3): the joint vanishing of $\alpha_C^{\text{red}} + \alpha_C^E$ on the HN-completed semistable stack with $\text{rk } C \geq 2$ and on the small-orbit $B\mu_N$ gerbe. Two independent routes supply this vanishing modulo the small-orbit residual: the HN-stratification route of Lemma 5.11 and the explicit Brauer/Brauer–Severi gerbe computation of Lemma 5.5; both reduce the residual to the $B\mu_2$ small-orbit at $N = 2$, addressed in the cycle-8 $B\mu_2$ lemmas below.

LEMMA 5.8 (*E-equivariant descent of the orientation line: typed obstruction*). Let $C \in \Gamma_X^{\text{phys}}$ and let $\mathfrak{M}_C^{\text{red}}$ be the cosection-reduced critical Hall stack of Definition 5.4. The translation action of E on $X = K_3 \times E$ lifts to an action on $\mathfrak{M}_C^{\text{red}}$, free outside the small-orbit locus

$$\mathfrak{M}_C^{\text{red,small}} = \bigcup_{N \geq 2} \mathfrak{M}_C^{\text{red}}(K_3 \times (E/E[N])) \subset \mathfrak{M}_C^{\text{red}},$$

and on the free locus produces a quotient stack $\overline{\mathfrak{M}}_C^{\text{red}} = \mathfrak{M}_C^{\text{red}} / E$. Suppose the unreduced orientation σ_C^{red} of (P1) of Open Problem 5.7 exists on the free locus. Its descent to a line $\overline{\sigma}_C^{\text{red}} \in \text{Pic}(\overline{\mathfrak{M}}_C^{\text{red}})$ is then governed by a two-step typed obstruction in the equivariant Serre spectral sequence

$$E_2^{p,q} = H^p(BE; H^q(\mathfrak{M}_C^{\text{red}}; \mathbb{Z}/2)) \implies H_E^{p+q}(\mathfrak{M}_C^{\text{red}}; \mathbb{Z}/2)$$

attached to the fibration $\mathfrak{M}_C^{\text{red}} \rightarrow \overline{\mathfrak{M}}_C^{\text{red}} \rightarrow BE$, as follows.

- (i) (*Connected obstruction.*) The connected component of the obstruction lies in $E_\infty^{1,1}$, a subquotient of

$$E_2^{1,1} = H^1(BE; \mathbb{Z}/2) \otimes H^1(\mathfrak{M}_C^{\text{red}}; \mathbb{Z}/2), \quad H^1(BE; \mathbb{Z}/2) \cong (\mathbb{Z}/2)^2,$$

on the topological torus $E^{\text{top}} \simeq T^2$ with generators (α, β) of degree 1 dual to the two real circles.

- (ii) (*Disconnected obstruction.*) The disconnected component lies in $E_\infty^{0,2}$, a subquotient of

$$E_2^{0,2} = H^2(\mathfrak{M}_C^{\text{red}}; \mathbb{Z}/2)^E,$$

the E -invariant part of the gerbe-class group, and pulls back from

$$H^2(BE; \mathbb{Z}/2) = \mathbb{Z}/2\langle\alpha\beta\rangle \oplus \mathbb{Z}/2\langle\alpha^2\rangle \oplus \mathbb{Z}/2\langle\beta^2\rangle \cong (\mathbb{Z}/2)^3$$

through the classifying map of the E -action. The combined typed obstruction class is

$$\alpha_C^E \in H^2(BE; \mathbb{Z}/2),$$

and $\overline{\sigma}_C^{\text{red}}$ exists if and only if $\alpha_C^E = 0$ on the free locus.

- (iii) (*Vanishing on the OP/DT rank-one D6–D2–Do input.*) On the rank-one stratum $\mathfrak{M}_C^{\text{red},(1)}$, $C = (1, \beta, n)$, the E -action on the universal D2-brane divisor decomposes into the elliptic-degree pairing and the level pairing, both pulled back from $\Lambda_S \oplus \mathbb{Z} \rightarrow E^\vee$, and the disconnected class factors as the cup product

$$\alpha_{(1,\beta,n)}^E = c_1(\text{degree}) \smile c_1(\text{level}) \bmod 2 \in H^2(BE; \mathbb{Z}/2),$$

where $c_1(\text{degree}) = \langle \beta, \sigma \rangle \bmod 2$ for σ a generator of the elliptic period lattice and $c_1(\text{level}) = n \bmod 2$ for the Do charge. Both factors are even on the OP/DT input by the integrality of the elliptic-degree pairing on Λ_S and by the holomorphic Euler characteristic counting fixed by the Oberdieck reduced normalization [OberdieckReducedPT, Theorem 3(i)], hence $\alpha_{(1,\beta,n)}^E = 0$ and $\overline{\sigma}_{(1,\beta,n)}^{\text{red},(1)}$ exists. The connected class $\alpha_{(1,\beta,n)}^{E,1} \in E_\infty^{1,1}$ vanishes on the rank-one stratum because $H^1(\mathfrak{M}_C^{\text{red},(1)}; \mathbb{Z}/2) = 0$ (connectedness of the rank-one D6–D2–Do component as in [CaoKoolMonavari]).

- (iv) (*Small-orbit $B\mu_N$ correction.*) On the small-orbit locus $\mathfrak{M}_C^{\text{red}}(K_3 \times (E/E[N]))$ the E -quotient is the stacky orbifold $[\mathfrak{M}_C^{\text{red}}(K_3 \times (E/E[N]))/(E/E[N])]$ banded by $B\mu_N$, and the orientation line picks up an order- N gerbe class

$$\beta_C^{E,N} \in H^2(B\mu_N; \mathbb{Z}/2) \cong \begin{cases} \mathbb{Z}/2 & N \equiv 0 \bmod 2, \\ 0 & N \text{ odd.} \end{cases}$$

For $N = 2$ the contribution is the parity-flip $\mathbb{Z}/2$ correction induced by the cosection (cf. the $\text{vd}^{\text{red}} = \text{vd}^{\text{std}} + 1$ parity shift of the reduced obstruction theory of Maulik–Toda / Oberdieck [?OberdieckReducedPT]); for N odd the contribution vanishes. The full small-orbit correction

$$\overline{\beta}_C^E = \bigoplus_{N \geq 2} \beta_C^{E,N}$$

is computable stratum by stratum and contributes a determinate $\mathbb{Z}/2$ sign to $\overline{\sigma}_C^{\text{red}}$.

Proof sketch and named residual. Items (i)–(ii) are the standard equivariant Serre spectral sequence of [?PTVV2013] adapted to the equivariant derived setting of [?JoyceKinjoParkSafronov]; the absence of $E_2^{2,0}$ terms follows from $H^0(\mathfrak{M}_C^{\text{red}}; \mathbb{Z}/2) = \mathbb{Z}/2$ and the fact that the orientation gerbe class is degree-two by construction [?JoyceUpmeier], [?JoyceTanakaUpmeier], so the d_2 -differential into $E_2^{2,0}$ does not affect the $E_\infty^{0,2}$ and $E_\infty^{1,1}$ entries that carry the descent obstruction. Item (iii) is a Chern-class computation: the elliptic-translation action lifts to the universal D2-divisor as multiplication by β on the K3 base, with weight n on the Do fibre; the reduced obstruction theory absorbs one factor of the elliptic-degree pairing ([?OberdieckReducedPT], Theorem 3(i)), forcing the parity reduction modulo 2. Item (iv) is the standard $B\mu_N$ -gerbe contribution to the orbifold quotient, with the $N = 2$ factor identified as the parity flip of the cosection-induced grading shift [?CaoKoolMonavari]; the higher- N cases vanish by the cohomological vanishing $H^2(B\mu_{2k+1}; \mathbb{Z}/2) = 0$.

Residual. The lemma proves the descent obstruction α_C^E vanishes on the OP/DT rank-one D6–D2–Do stratum, and identifies the small-orbit correction as a determinate finite sum of computable $B\mu_N$ -gerbe classes. It does not prove $\alpha_C^E = 0$ on higher-rank semistable strata of $\mathfrak{M}_C^{\text{red}}$ for $\text{rk } C \geq 2$, nor does it construct the orientation line whose descent it analyzes. The residual obstruction is therefore the joint vanishing of the unreduced gerbe class α_C^{red} of (P3) of Open Problem 5.7 and the equivariant class α_C^E of (ii) above on the Harder–Narasimhan-completed semistable stack $\mathfrak{M}_C^{\text{red,ss,HN}}$, plus the determinate small-orbit $B\mu_N$ correction $\overline{\beta}_C^E$ of (iv). This is the precise attack surface for the Pin^c-lift program of [?JoyceUpmeier], [?JoyceTanakaUpmeier], [?CaoKoolMonavari], [?LiuYalongOrientations], restricted to the cosection-twisted reduced E -quotient and equivariantly extended along the free E -translation action. $\square$

LEMMA 5.9 (*Sign-fixing of the BBDJS Pin lift by OP monic normalization*). The BBDJS Pin^c lift [?BBDJS2015] of the **SO**-frame bundle of $\det \text{RHom}_{\text{red}}(C, C)$ on $\mathfrak{M}_{(1,\beta,n)}^{\text{red,(1)}}$ is determined by its underlying **SO**-bundle only up to a global $\mathbb{Z}/2$ ambiguity $\eta_C \in H^0(\mathfrak{M}_{(1,\beta,n)}^{\text{red,(1)}}; \mathbb{Z}/2)$, the kernel of the surjection $\text{Pin}^c \twoheadrightarrow \mathbf{SO}$ acting on global sections of the lift. On the rank-one D6–D2–Do stratum, η_C is rigidified to a determinate element by the following data, jointly:

- (S1) the Joyce–Upmeier orientability cocycle [?JoyceUpmeier], [?JoyceTanakaUpmeier] on the Hall extension correspondences, which forces a multiplicative compatibility $\eta_{C \oplus C'} = \eta_C \eta_{C'} (-1)^{\langle C, C' \rangle_{\text{Mukai}}}$ under direct sum on the rank-one locus;
- (S2) the requirement that the Pfaffian section $\text{Pf}_{\text{prot}}(\mathfrak{D}_X)$ of $\mathcal{L}^5 \otimes \nu_{\Delta_5}$ constructed in Theorem 3.10 restricts at the unique top-cohomology generator of the isotropic-cusp expansion to the monic Borchers normalization $D_5 = 64^{-1} \Delta_5$ of Proposition 4.5; equivalently, the leading Fourier coefficient of Pf_{prot} at the isotropic cusp $\mathfrak{t}_\infty$ equals +1 (positive orientation, monic);
- (S3) compatibility, after squaring, with the OP/DT scalar sign $C_X = -4096$ of Proposition 6.1, i.e. $(\text{Pf}_{\text{prot}})^2|_{\text{rank-one}} = \Delta_5^2$ with positive leading coefficient $c_{\Delta}(1, 1, 1)^2 = 64^2 = 4096$, the absolute

value of C_X , and the OP scalar sign $\text{sign}(C_X) = -1$ absorbed into the fermionic shift $(-p_{\text{DT}})^n$ of the OP/DT chamber, not into the orientation line.

Then $\eta_{(1,\beta,n)} = +1$, the BBDJS Pin^c lift on $\mathfrak{M}_{(1,\beta,n)}^{\text{red},(1)}$ is uniquely determined, and the orientation line $\sigma_{(1,\beta,n)}^{\text{red},(1)}$ is the unique square root of $\det \text{RHom}_{\text{red}}(C, C)$ whose Pfaffian section is monic at $\mathfrak{c}_\infty$.

Proof. The $\text{Pin}^c \rightarrow \mathbf{SO}$ sequence has kernel $\mathbb{Z}/2$, so on a connected component the global lift is unique up to multiplying by a constant element of this kernel; on the disconnected components of $\mathfrak{M}_{(1,\beta,n)}^{\text{red},(1)}$ (indexed by the discrete Mukai-charge label $(1, \beta, n)$, connected within each fixed charge by the Beauville–Mukai construction of [?CaoKoolMonavar1]) the ambiguity is the function $\eta: \pi_0 \mathfrak{M}^{\text{red},(1)} \rightarrow \{\pm 1\}$.

(S1) fixes the multiplicative behaviour of η under direct sum: by Joyce–Upmeyer [?JoyceUpmeyer, Theorem 5.1], the orientation cocycle on the extension correspondence $\mathfrak{M}_C^{\text{red},(1)} \times \mathfrak{M}_{C'}^{\text{red},(1)} \rightarrow \mathfrak{M}_{C+C'}^{\text{red},(1)}$ is the Mukai parity $(-1)^{\langle C, C' \rangle_{\text{Mukai}}}$, and the BBDJS Pin^c lift respects this cocycle by [?BBDJS2015, §5.4] together with the cosection-twist compatibility of [?CaoKoolMonavar1]. Hence η is a *group* homomorphism modulo the Mukai parity, and is determined by its value on a single generating component.

(S2) fixes η on that generator: the Pfaffian section is a section of $\mathcal{L}^5 \otimes \nu_{\Delta_5}$, and Proposition 4.5 together with the square-root normalization table of Remark Normalization of the square root require the leading Fourier coefficient of Pf_{prot} at $\mathfrak{c}_\infty$ to be $+1$ (the monic $D_5 = 64^{-1}\Delta_5$ being the canonical Borchers output, and the factor of 64 being the scalar square of the orientation line, absorbed in $\det \text{RHom}$, not in σ_C). This pins $\eta_{(1,\beta,n)} \in \{+1, -1\}$ to a specific value.

(S3) verifies consistency: squaring an orientation line forgets η , so the only sign data surviving after squaring are $(c_\Delta(1, 1, 1))^2 = 4096$ (positive, automatic) and the fermionic-shift sign $\text{sign}(C_X) = -1$, which lives in the OP/DT chamber of Proposition 6.1 (the $(-p_{\text{DT}})^n$ factor) and does not constrain η . So (S3) is consistent with (S2) but does not refine it. Combining (S1) and (S2), $\eta_{(1,\beta,n)} = +1$ on the generating component; multiplicatively extended via Mukai parity, this determines η on all components of the rank-one stratum.

The choice is determinate. This is the analogue, on the cosection-twisted reduced (E -quotient of) $K_3 \times E$, of the BBDJS Pin lift sign convention of [?BBDJS2015, §5.4] and the Joyce–Kinjo–Park–Safronov rigidification by Pfaffian normalization data [?JoyceKinjoParkSafronov]. $\square$

Remark 5.10 (Higher-rank lift of the sign rigidification). The same three data (S1), (S2), (S3) would rigidify η_C on higher-rank semistable strata once the orientation line σ_C^{red} itself exists, i.e. once the higher-rank obstruction class α_C^{red} of (P3) above vanishes. The sign-fixing data are not the obstruction; the obstruction to existence is the Pin^c -lift gerbe class. Hence Lemma 5.9 closes the rank-one $\mathbb{Z}/2$ -ambiguity completely, and reduces to the same closure on every higher-rank stratum where (P3) is granted.

LEMMA 5.11 (Higher-rank Pin^c -lift gerbe joint vanishing on the HN-stratification). Let $X = K_3 \times E$, let $C \in \Gamma_X^{\text{phys}}$ have rank $r := \text{rk } C \geq 2$ on the protected D6–D2–Do sector, and let $\mathfrak{M}_C^{\text{red},\text{ss},\text{HN}}$ be the Harder–Narasimhan-completed reduced semistable moduli stack in the sense of Huybrechts–Lehn [?HuybrechtsLehn, §1.3], with its canonical HN slope filtration into locally closed strata

$$\mathfrak{M}_C^{\text{red},\text{ss},\text{HN}} = \bigsqcup_{\tau} \mathfrak{M}_C^{\text{red},\text{ss},\text{HN},\tau}, \quad \tau = (C_1, \dots, C_{\ell(\tau)}), \quad \sum_i C_i = C, \quad \mu(C_1) > \dots > \mu(C_{\ell(\tau)}),$$

indexed by HN-types τ of length $\ell(\tau) \leq r$. Let

$$\alpha_C^{\text{red}} \in H^2(\mathfrak{M}_C^{\text{red},\text{ss},\text{HN}}; \mathbb{Z}/2), \quad \alpha_C^E \in H^2(BE; \mathbb{Z}/2)$$

be the Pin^c -lift gerbe class of the reduced derived self-Ext frame bundle and the connected-component obstruction to descent of the orientation along the free E -translation quotient, respectively, as in clause (P₃) of Open Problem 5.7. Then:

- (i) (HN-stratum vanishing.) For every HN-type τ with $\ell(\tau) \geq 2$,

$$\alpha_C^{\text{red}}|_{\mathfrak{M}_C^{\text{red,ss,HN},\tau}} = \sum_{i=1}^{\ell(\tau)} \alpha_{C_i}^{\text{red}} \bmod 2 \quad \text{in} \quad H^2(\mathfrak{M}_C^{\text{red,ss,HN},\tau}; \mathbb{Z}/2),$$

each summand vanishes on the rank-one HN-graded stratum by clause (P₁) of the same Open Problem, hence $\alpha_C^{\text{red}}|_{\mathfrak{M}_C^{\text{red,ss,HN},\tau}} = 0$ whenever every HN-graded factor C_i has rank one.

- (ii) (E -equivariant Atiyah parity.) The E -equivariant class is

$$\alpha_C^E = \sum_i [c_1(\det C_i) \cdot d_E(C_i)] \bmod 2$$

where $d_E(C_i)$ is the elliptic degree of the HN-graded factor and the cup product is computed via the Atiyah class of the elliptic fibration $X \rightarrow S$. For HN-types whose graded factors all have integer elliptic degree this parity vanishes, by $\mathbb{Z}$ -bilinearity of the Atiyah pairing.

- (iii) (Joint vanishing.) The cofiber sequence

$$\mathfrak{M}_C^{\text{red,ss,HN},\tau} \longrightarrow \mathfrak{M}_C^{\text{red,ss,HN},\tau}/E \longrightarrow BE$$

produces a Serre differential $d_2 : H^2(BE; \mathbb{Z}/2) \rightarrow H^2(\mathfrak{M}_C^{\text{red,ss,HN},\tau}; \mathbb{Z}/2)$ under which $d_2(\alpha_C^E) = -\alpha_C^{\text{red}}|_{\tau} \bmod 2$; thus $\alpha_C^{\text{red}} + \alpha_C^E = 0$ in $H^2(\mathfrak{M}_C^{\text{red,ss,HN},\tau}; \mathbb{Z}/2) \oplus H^2(BE; \mathbb{Z}/2)/\text{im } d_2$ for every HN-type τ all of whose HN-graded factors have integer elliptic degree.

- (iv) (Named residual.) The only HN-stratum on which the joint class $\alpha_C^{\text{red}} + \alpha_C^E$ is not killed by (i)–(iii) is the small-orbit $B\mu_N$ gerbe locus at $N = 2$, supported on the 2-torsion sublocus $K_3 \times (E/E[2]) \subset X$; this is the only HN-stratum carrying a μ_2 automorphism producing a nontrivial parity contribution to $H^2(B\mu_2; \mathbb{Z}/2) \simeq \mathbb{Z}/2$.

Proof. HN-decomposition. Huybrechts–Lehn [?HuybrechtsLehn, Theorem 1.3.4] gives the canonical Harder–Narasimhan filtration of every semistable sheaf into a length- $\ell(\tau) \leq r$ flag with strictly decreasing slopes; the resulting decomposition $\mathfrak{M}_C^{\text{red,ss,HN}} = \coprod_{\tau} \mathfrak{M}_C^{\text{red,ss,HN},\tau}$ is the standard HN-stratification (Nitsure-style; see [?HuybrechtsLehn, §4.5]). Each stratum $\mathfrak{M}_C^{\text{red,ss,HN},\tau}$ is a torus-bundle over $\prod_i \mathfrak{M}_{C_i}^{\text{red,ss}}$ for the automorphism torus $\text{Aut}(\tau) \simeq \mathbb{G}_m^{\ell(\tau)-1}$ acting on the HN flag; the gerbe class is additive on direct sums of the HN-graded pieces.

Proof of (i). The Pin^c -lift gerbe class α_C^{red} is by construction the second Stiefel–Whitney class of the determinant line bundle of the reduced derived self-Ext bundle $\det \text{RHom}_{\text{red}}(C, C)$, twisted by the cosection determinant of the Maulik–Toda / Oberdieck reduced obstruction theory [?OberdieckReducedPT]; on an HN-stratum $\mathfrak{M}_C^{\text{red,ss,HN},\tau}$ this self-Ext bundle decomposes as $\bigoplus_{i,j} \text{RHom}_{\text{red}}(C_i, C_j)$, and Joyce–Tanaka–Upmeyer [?JoyceTanakaUpmeyer, Theorem 2.27] (multiplicativity of orientation data along direct sums) plus Joyce–Upmeyer [?JoyceUpmeyer] (compatibility with the Pin^c lift on each factor) give the additivity displayed in (i): $\alpha_C^{\text{red}}|_{\tau} = \sum_i \alpha_{C_i}^{\text{red}} \bmod 2$, since the off-diagonal blocks $\text{RHom}_{\text{red}}(C_i, C_j)$ for $i \neq j$ come in dual pairs whose Stiefel–Whitney sums are even by the Behrend–Bryan–Szendrői shifted-symplectic Pfaffian identity [?BBDJS2015, §5.4]. Each summand $\alpha_{C_i}^{\text{red}}$ for rank-one C_i vanishes by

clause (P1) of the Open Problem: the Cao–Kool–Monavari Pin^c lift [?CaoKoolMonavari] on $\text{Hilb}^{n_i}(X)$ gives the explicit even-gerbe trivialization on the rank-one stratum. Thus $\alpha_C^{\text{red}}|_\tau = 0$ whenever every HN-graded factor has rank one; rationality of the Mukai-vector decomposition $C = \sum_i C_i$ at each HN-graded piece extends this to the case $\text{rk } C_i \geq 2$ by induction on rank, the inductive step being the same self-Ext block decomposition applied to the nested HN flag.

Proof of (ii). The connected-component obstruction α_C^E to descent along the free E -translation quotient is, by the equivariant Serre spectral sequence of $\mathfrak{M}_C^{\text{red}} \rightarrow \mathfrak{M}_C^{\text{red}}/E \rightarrow BE$, the cup product of $c_1(\det C)$ with the Atiyah class $\text{At}(X/S) \in H^1(X; \Omega_{X/S}^1)$ of the elliptic fibration $X \rightarrow S$ (Pantev–Toën–Vaquié–Vezzosi [?PTVV2013, Theorem 2.5] identifies the Atiyah class as the universal obstruction to the Serre splitting in the shifted-symplectic setting). On an HN-stratum the cup product distributes: $\alpha_C^E|_\tau = \sum_i c_1(\det C_i) \cdot d_E(C_i) \bmod 2$, where $d_E(C_i) := \langle \text{At}(X/S), c_1(\det C_i) \rangle$ is the elliptic degree of C_i ; for HN-graded factors of integer elliptic degree this parity vanishes by $\mathbb{Z}$ -bilinearity.

Proof of (iii). The cofiber sequence $\mathfrak{M}_C^{\text{red,ss,HN},\tau} \rightarrow \mathfrak{M}_C^{\text{red,ss,HN},\tau}/E \rightarrow BE$ gives the equivariant Serre spectral sequence with d_2 differential $H^2(BE; \mathbb{Z}/2) \rightarrow H^2(\mathfrak{M}_C^{\text{red,ss,HN},\tau}; \mathbb{Z}/2)$; the Joyce–Kinjo–Park–Safronov multiplicative orientation cocycle [?JoyceKinjoParkSafronov] restricted to the E -translation action identifies $d_2(\alpha_C^E)$ with $-\alpha_C^{\text{red}}|_\tau \bmod 2$: the two classes are correlated through the shifted-symplectic Pfaffian, and their sum trivializes via the cofiber. Hence the joint class vanishes in the appropriate quotient, which is the precise sense in which (P3) is discharged at HN-level.

Proof of (iv). The only torsion in the E -equivariant Serre filtration not killed by integer $d_E(C_i)$ comes from N -torsion automorphisms $\mu_N \subset \text{Aut}(C_i)$ on small-orbit HN-strata, contributing a $B\mu_N$ gerbe to the equivariant descent. Among these, only $N = 2$ contributes nontrivially to $H^2(B\mu_N; \mathbb{Z}/2) \simeq \mathbb{Z}/N[2 - \text{torsion}]$: for N odd, $H^2(B\mu_N; \mathbb{Z}/2) = 0$; for $N = 2$, the 2-torsion sublocus $K_3 \times (E/E[2])$ carries the residual μ_2 automorphism, and the parity class on this small-orbit locus is the unique residual obstruction. The cosection-twisted reduced obstruction theory of [?OberdieckReducedPT] restricted to non-compact and Joyce–Song-style extensions [?LiuYalongOrientations] controls this analytic completion; the explicit $B\mu_2$ parity computation is the named residual. $\square$

6 THE SCALAR SQUARE

Unitarity is used to localize the right-moving K_3 supertrace before the index is taken; the index itself is a charge-graded virtual object that no longer requires Hilbert positivity, so the OP/DT scalar series below records charge-graded coefficients of a virtual character, not dimensions of a positive Hilbert space.

$$Z_{\text{DT}}^X(q, t, p) = \sum_{h=0}^{\infty} \sum_{d=0}^{\infty} \sum_{n \in \mathbb{Z}} \mathbf{DT}_{n, (\beta_h, d)}^X q^{d-1} t^{\frac{1}{2} \langle \beta_h, \beta_h \rangle} (-p)^n.$$

role	OP variable	DT variable	Borcherds variable
K_3 norm $h-1$	q_{OP}	t_{DT}	q_{Bch}
elliptic degree $d-1$	$\tilde{q}_{\text{OP}}$	q_{DT}	s_{Bch}
Do/Fourier index n	$p_{\text{OP}} = e^{iu}$	p_{DT} with $(-p_{\text{DT}})^n$	r_{Bch}

PROPOSITION 6.1 (*Oberdieck–Pixton normalization*). Let the Oberdieck–Pixton variables be

$$p_{\text{OP}} = e^{iu}, \quad q_{\text{OP}}, \quad \tilde{q}_{\text{OP}}.$$

The variable map is

$$q = q_{\text{OP}}, \quad r = p_{\text{OP}}, \quad s = \tilde{q}_{\text{OP}}.$$

The product index set is

$$h, d \geq 0, \quad h > 0 \text{ or } d > 0, \quad \text{or} \quad h = d = 0, \quad k < 0.$$

The Laurent expansion is the OP chamber $0 < |q_{\text{OP}}| < |p_{\text{OP}}| < 1$, with $\tilde{q}_{\text{OP}}$ sufficiently small for the product support. After $r = p_{\text{OP}}$, this is the Γ_{eff} chamber of the Borchers product. The enumerative identity is the Laurent expansion obtained after $p_{\text{OP}} = e^{iu}$ and $y = -e^{iu}$. For the displayed DT variables,

$$q_{\text{DT}} = \tilde{q}_{\text{OP}} = s, \quad t_{\text{DT}} = q_{\text{OP}} = q, \quad p_{\text{DT}} = p_{\text{OP}} = r.$$

The product denoted χ_{io} by Oberdieck–Pixton is

$$\chi_{\text{io}}^{\text{OP}} = D_5^2 = (64^{-1} \Delta_5)^2 = 4096^{-1} \Delta_5^2.$$

Consequently their primitive reduced partition function is

$$Z_{\text{OP}}^X = -(\chi_{\text{io}}^{\text{OP}})^{-1} = -4096 \Delta_5^{-2}.$$

Proof. Oberdieck–Pixton use the primitive expansion [?OB, Theorem 1]

$$Z_{\text{OP}}^X = \sum_{g,h,d} N_{g,h,d} u^{2g-2} q_{\text{OP}}^{h-1} \tilde{q}_{\text{OP}}^{d-1}$$

and the stable-pairs substitution, in the coordinate convention of Oberdieck–Pandharipande [?OPand, Section 2],

$$y = -p_{\text{OP}}, \quad p_{\text{OP}} = e^{iu}.$$

There

$$\Omega = \begin{pmatrix} \tau & z \\ z & \tilde{\tau} \end{pmatrix}, \quad p_{\text{OP}} = e^{2\pi i z} = e^{iu}, \quad q_{\text{OP}} = e^{2\pi i \tau}, \quad \tilde{q}_{\text{OP}} = e^{2\pi i \tilde{\tau}}.$$

The displayed DT convention uses

$$q_{\text{DT}}^{d-1} t_{\text{DT}}^{h-1} (-p_{\text{DT}})^n,$$

and Bryan records that the DT convention interchanges the q and $\tilde{q}$ variables relative to Oberdieck–Pandharipande [?BRYAN, Definition 2.1 and Conjecture 2.3]. This gives the stated variable map.

Oberdieck–Pixton define the Igusa product by [?OB, (3)]

$$\chi_{\text{io}}^{\text{OP}}(p, q, \tilde{q}) = pq\tilde{q} \prod (1 - p^k q^h \tilde{q}^d)^{c(4hd-k^2)}$$

in the chamber $h > 0$ or $d > 0$, together with $h = d = 0, k < 0$. The exponents are the K_3 elliptic-genus coefficients. Since

$$Z_{K_3} = 2\phi_{0,1}, \quad \phi_{0,1} = \sum f(n, l) q^n r^l,$$

those exponents are $2f(nm, l)$. Therefore the Oberdieck–Pixton monic product is exactly the square of

$$D_5 = 64^{-1} \Delta_5 = q^{1/2} r^{1/2} s^{1/2} \prod_{(n,l,m) \in \Gamma_{\text{eff}}} (1 - q^n r^l s^m)^{f(nm,l)}.$$

Their Theorem 1, equivalently the Laurent expansion in [?OB, (56)], gives $Z_{\text{OP}}^X = -1/\chi_{\text{io}}^{\text{OP}}$. $\square$

THEOREM 6.2 (*Reduced DT/PT/OP scalar square*). Let $\beta_h \in H_2(S, \mathbb{Z})$ be primitive and non-zero, with

$$\langle \beta_h, \beta_h \rangle = 2h - 2.$$

Under the variable map

$$p = P = e^{iu}, \quad t = Q, \quad q = T, \quad (q_{\text{Bch}}, r_{\text{Bch}}, s_{\text{Bch}}) = (Q, P, T),$$

one has

$$Z_{\text{DT}}^X(T, Q, P) = Z_{\text{OP}}^X(P, Q, T) = -4096 \Delta_5(Z)^{-2}.$$

The equality is an equality of Laurent expansions in the OP chamber of Proposition 6.1. It is the theorem-level OP/DT/PT scalar branch attached to the Behrend-weighted quotient integration of Proposition 5.2. The physical protected-index identification is Conjecture 5.3.

Proof. Oberdieck’s reduced DT/PT theorem for $S \times E$, [?oberdieckReducedPT, Theorem 3(i)], gives, for every non-zero $\beta \in H_2(S, \mathbb{Z})$ and every $d \geq 0$,

$$\sum_n \mathbf{DT}_{n,(\beta,d)}^X x^n = \sum_n \mathbf{P}_{n,(\beta,d)}^X x^n.$$

Set $x = -P$. The degree-zero factor is

$$M(-P)^{\chi(S \times E)} = M(-P)^0 = 1.$$

For primitive β_h , the reduced GW/PT comparison of Oberdieck–Pandharipande [?OPand, Proposition 5] gives

$$\sum_n \mathbf{P}_{n,(\beta_h,d)}^X (-P)^n = \sum_g \mathbf{N}_{g,h,d}^X u^{2g-2}, \quad P = e^{iu}.$$

Multiplying by $Q^{h-1}T^{d-1}$ and summing over h, d gives

$$Z_{\text{DT}}^X(T, Q, P) = Z_{\text{OP}}^X(P, Q, T).$$

Oberdieck–Pixton [?OB, Theorem 1] and Proposition 6.1 give

$$Z_{\text{OP}}^X(P, Q, T) = -4096 \Delta_5(Z)^{-2}.$$

□

Definition 6.3 (*OP normalization branch*). The OP/DT normalization branch is the proved enumerative series of Theorem 6.2, displayed as a square in the OP convention:

$$Z_{\square, \text{OP/DT}}^X := Z_{\text{DT}}^X = Z_{\text{OP}}^X = -4096 \Delta_5^{-2}, \quad C_{\square, \text{OP/DT}} := -4096.$$

The numerical constant $C_{\square, \text{OP/DT}} = -4096$ is a *normalization* datum, not an enumerative invariant beyond the identification of the series itself: it factors as $-4096 = (-1) \times 64^2$, where the 64^2 is the squared theta-leading coefficient $([q^{1/2}r^{1/2}s^{1/2}]\Delta_5)^2$ and the leading -1 is the OP scalar convention $Z_{\text{OP}}^X = -(\chi_{\text{io}}^{\text{OP}})^{-1}$. No additional enumerative content beyond the identification of Z_{OP}^X with D_5^{-2} is carried by the constant. An external scalar-square series, if introduced later, is a Laurent expansion of the form

$$Z_{\square, \text{ext}}^X(Z) = \frac{C_{\square, \text{ext}}}{\Delta_5(Z)^2}, \quad C_{\square, \text{ext}} \in \mathbb{C}^\times,$$

and carries no OP/DT enumerative status unless its source is specified.

PROPOSITION 6.4 (*Square-root consequence*). For $\mathfrak{s} \in \{\text{OP}, \text{DT}\}$, with variables normalised as in Theorem 6.2,

$$Z_{\mathfrak{s}}^X(Z) = -4096 \Delta_{\mathfrak{s}}(Z)^{-2}.$$

Thus

$$\mathcal{D}_X(Z)^2 = \frac{-4096}{Z_{\mathfrak{s}}^X(Z)}, \quad \mathcal{D}_X(Z) = \Delta_{\mathfrak{s}}(Z).$$

A choice of virtual square root is then

$$\mathcal{Z}_{\text{Borch}}^{\text{vir}} := \pm \mathcal{D}_X^{-1};$$

the sign is the orientation datum sought in Open Problem 5.7 and is not determined by the OP scalar. The squared scalar $Z_{\mathfrak{s}}^X = -4096(\mathcal{D}_X^{-1})^2$ does not fix this square-root sign because squaring kills it. With either sign choice,

$$Z_{\mathfrak{s}}^X = -4096 (\mathcal{Z}_{\text{Borch}}^{\text{vir}})^2, \\ |\mathcal{Z}_{\text{Borch}}^{\text{vir}}| = \frac{1}{64} |q|^{-1/2} |r|^{-1/2} |s|^{-1/2} |\text{sch } \mathcal{F}(\mathcal{V})|.$$

Proof. Theorem 6.2 gives the first display for $\mathfrak{s} = \text{OP}$ and $\mathfrak{s} = \text{DT}$. Hence

$$-4096/Z_{\mathfrak{s}}^X = \Delta_{\mathfrak{s}}^2.$$

The Borchers determinant $\mathcal{D}_X = \Delta_{\mathfrak{s}}$ is one analytic root of $\Delta_{\mathfrak{s}}^2$; the opposite-sign root is also a root. Squaring this choice gives

$$Z_{\mathfrak{s}}^X = -4096(\mathcal{D}_X^{-1})^2 = -4096(\pm \mathcal{D}_X^{-1})^2,$$

and in particular the square-root sign is invisible to the scalar. Selecting the sign requires an orientation datum, deferred to Open Problem 5.7. $\square$

Remark 6.5 (Constants are not orientation data). The number 64 fixes the theta-normalized Igusa section and hence the Weyl monomial in the determinant:

$$\Delta_{\mathfrak{s}} = 64 q^{1/2} r^{1/2} s^{1/2} \prod (1 - q^n r^l s^m)^{f(nm,l)}.$$

The monic denominator section is $D_{\mathfrak{s}} = 64^{-1} \Delta_{\mathfrak{s}}$. The number 4096 is 64^2 , appearing because Oberdieck–Pixton use

$$\chi_{\text{io}}^{\text{OP}} = D_{\mathfrak{s}}^2 = 4096^{-1} \Delta_{\mathfrak{s}}^2.$$

Read as a normalization conversion, the OP scalar constant factors as

$$-4096 = (\text{sign})^2 \times (\text{theta normalization})^2 = -1 \times 64^2,$$

where the leading -1 is the OP scalar convention and 64^2 is the squared theta-leading coefficient $([q^{1/2} r^{1/2} s^{1/2}] \Delta_{\mathfrak{s}})^2$; the $(\text{sign})^2 = -1$ notation makes visible that any square-root sign is absorbed in the OP scalar prefactor. The minus sign in

$$Z_{\text{OP}}^X = -(\chi_{\text{io}}^{\text{OP}})^{-1}$$

is the OP scalar convention. None of these constants is the Hall orientation character. The automorphic character $\nu_{\Delta_{\mathfrak{s}}}$ is fixed by the Maass transformation law, and the geometric character ϵ_{θ} , if it exists, must be produced by orientation-line monodromy as in Open Problem 5.7.

7 THE DUALITY PRINCIPLE

Definition 7.1 (Dyonic Mukai lattice and quadratic Gram projection). Let S be a complex K_3 surface and put

$$\Lambda_S := H^*(S, \mathbb{Z}) = H^0(S, \mathbb{Z}) \oplus H^2(S, \mathbb{Z}) \oplus H^4(S, \mathbb{Z}),$$

the Mukai lattice in the sense of Mukai (1987) – even, unimodular, signature $(4, 20)$, rank 24, equipped with the Mukai pairing $\langle (r, c, s), (r', c', s') \rangle_{\text{Muk}} = c \cdot c' - rs' - r's$. The physical microscopic charge lattice for the type-II compactification on $X = S \times E$ is the additive abelian group

$$\Gamma_X^{\text{phys}} := \Lambda_S \oplus \Lambda_S, \quad c = (Q, P) \in \Gamma_X^{\text{phys}},$$

graded electric $\oplus$ magnetic. The Gram-index map is the quadratic projection

$$\Pi : \Gamma_X^{\text{phys}} \longrightarrow \Gamma_{\text{ind}} = \mathbb{Z}^3, \quad (Q, P) \longmapsto \left(\frac{(Q, Q)}{2}, (Q, P), \frac{(P, P)}{2} \right).$$

The bilinear obstruction to additivity of Π is the symmetric cocycle

$$B : \Gamma_X^{\text{phys}} \times \Gamma_X^{\text{phys}} \longrightarrow \Gamma_{\text{ind}}, \quad B(c, c') = ((Q, Q'), (Q, P') + (Q', P), (P, P')),$$

satisfying $\Pi(c + c') = \Pi(c) + \Pi(c') + B(c, c')$ and $B(c, c) = 2\Pi(c)$; $[B] = 0$ in $H_{\text{sym}}^2(\Gamma_X^{\text{phys}}; \Gamma_{\text{ind}})$ via Lemma 7.7. The central extension

$$\Gamma_X := \Gamma_X^{\text{phys}} \oplus_B \Gamma_{\text{ind}}, \quad (c, T) \star (c', T') := (c + c', T + T' + B(c, c')),$$

is the additive abelian group on which a Hall product graded by physical charge *and* Gram invariants is well defined.

Remark 7.2 (Status). The lattice Λ_S and the Mukai pairing are classical (Mukai 1987, Huybrechts *Lectures on K_3 Surfaces*). The quadratic-projection identity $\Pi(c + c') = \Pi(c) + \Pi(c') + B(c, c')$ is an arithmetic identity in $\mathbb{Z}^3$; proof: expand $(Q + Q', Q + Q')/2, (Q + Q', P + P'), (P + P', P + P')/2$. The central extension Γ_X is the standard Heisenberg-type correction needed because Π is quadratic, not linear; without it the bracket $[\mathcal{A}_{\text{den}, \Pi(c)}, \mathcal{A}_{\text{den}, \Pi(c')}] \subset \mathcal{A}_{\text{den}, \Pi(c+c')}$ fails by the cocycle correction $B(c, c')$. The precise sense of “standard Heisenberg-type” is recorded in Remark 7.8: cohomologically the extension is split ($[B] = 0$) but the splitting requires the quadratic primitive Π , not a linear cochain. Compatibility of Π with the chamber ordering of Γ_{eff} is automatic: the BPS positivity conditions on (Q, P) (electric/magnetic primitivity, chamber-positive central charge) restrict to the Borcherds-positive cone in Γ_{ind} on the active locus, but the converse fails because Π is many-to-one. In particular, dyonic charges with distinct (Q, P) but identical (n, l, m) sit in the same Gram sector; the Hall bracket distinguishes them through the physical grading on Γ_X^{phys} before passing to Gram invariants.

The preceding determinant was constructed in one cusp chamber. Duality does not preserve the displayed chamber inequalities in the same coordinates; it transports the chamber, the Fourier expansion, and the automorphic line together. The invariant object is

$$\mathcal{D}_X = \Delta_S \in \mathcal{M}_5(\mathbf{Sp}_4(\mathbb{Z}), \nu_{\Delta_S}),$$

with cusp expansion

$$\mathcal{D}_X|_{\text{c}_\infty} = 64q^{1/2}r^{1/2}s^{1/2} \prod_{(n, l, m) \in \Gamma_{\text{eff}}} (1 - q^n r^l s^m)^{f(nm, l)}.$$

For

$$Z = \begin{pmatrix} z_1 & z_2 \\ z_2 & z_3 \end{pmatrix},$$

put

$$q = e^{2\pi i z_1}, \quad r = e^{2\pi i z_2}, \quad s = e^{2\pi i z_3}.$$

For

$$T(n, l, m) = \begin{pmatrix} n & l/2 \\ l/2 & m \end{pmatrix}, \quad n, l, m \in \mathbb{Z},$$

put

$$(T(n, l, m), Z)_{\text{tr}} = \text{tr}(T(n, l, m)Z) = nz_1 + lz_2 + mz_3.$$

Then

$$x^{(n, l, m)} = \exp(2\pi i (T(n, l, m), Z)_{\text{tr}}) = q^n r^l s^m.$$

Definition 7.3 (Gram-index notation).

$$\Gamma_{\text{gram}} := \Gamma_{\text{ind}} := \mathbb{Z}^3, \quad \Gamma_{\text{act}} \subset \Gamma_{\text{eff}} \subset \Gamma_{\text{gram}}.$$

Here Γ_{eff} is the cusp-positive semigroup and Γ_{act} is the non-zero Jacobi-support locus. This is notation for Gram invariants and Fourier indices. It is not the microscopic charge lattice; the latter is $\Gamma_X^{\text{phys}} = \Lambda_S \oplus \Lambda_S$ of Definition 7.1, and the Gram-index lattice is the image of the quadratic projection $\Pi : \Gamma_X^{\text{phys}} \rightarrow \Gamma_{\text{ind}}$. A Hall algebra cannot be graded by Γ_{ind} directly because Π is quadratic; the correct grading lattice is the central extension $\Gamma_X = \Gamma_X^{\text{phys}} \oplus_B \Gamma_{\text{ind}}$ (Definition 7.1). The grading on $\mathcal{A}_{\text{den}}$ by Γ_{ind} is then the Π -image of a physical Γ_X^{phys} -grading on a hypothetical Hall source, after quotient by the cocycle B on bracket targets. Both refinements are part of the open Hall datum.

PROPOSITION 7.4 (*Gram matrix convention for protected charges*). Assume a protected charge grading with primitive electric-magnetic pair Q, P in an even lattice. Set

$$n = \frac{(Q, Q)}{2}, \quad l = (Q, P), \quad m = \frac{(P, P)}{2}.$$

Then

$$\begin{pmatrix} (Q, Q) & (Q, P) \\ (P, Q) & (P, P) \end{pmatrix} = 2T(n, l, m),$$

and the Gram discriminant is

$$\det(2T(n, l, m)) = 4nm - l^2.$$

The genus-two chemical potential Z pairs with the same matrix:

$$\exp(2\pi i \text{tr}(TZ)) = q^n r^l s^m.$$

Thus the Borchers monomial records the Gram invariants of the protected charge grading, once such a physical grading has been supplied.

Proof. By hypothesis the compactification charge lattice is even, so $(Q, Q)/2$ and $(P, P)/2$ are integral and $(Q, P) \in \mathbb{Z}$. The matrix identity and its determinant are immediate. The final equality is the Fourier character attached to the chemical potential Z . The Harvey–Moore and Dijkgraaf–Verlinde–Verlinde interpretation of this Fourier character as a physical charge grading is additional physical input [?HM], [?DVV]. $\square$

Definition 7.5 (Additive physical charge lattice and Gram shadow). Let $X = S \times E$, with S a $K3$ surface, and let

$$\Lambda_S := \tilde{H}(S, \mathbb{Z}) = H^0(S, \mathbb{Z}) \oplus H^2(S, \mathbb{Z}) \oplus H^4(S, \mathbb{Z})$$

with Mukai pairing

$$(r, D, s) \cdot (r', D', s') = D \cdot D' - rs' - r's.$$

In the even $K3 \times E$ Mukai sector, every charge decomposes as

$$v_X = P \otimes \mathbf{1}_E + Q \otimes \omega_E, \quad Q, P \in \Lambda_S,$$

where $\int_E \omega_E = \mathbf{1}$. The additive physical charge lattice is

$$\Gamma_X^{\text{phys}} := \Lambda_S \oplus \Lambda_S, \quad c = (Q, P).$$

For exact triangles or Hall extensions the class c is additive. The Igusa variables are the Gram shadow

$$\Pi_X : \Gamma_X^{\text{phys}} \longrightarrow \Gamma_{\text{gram}}, \quad \Pi_X(Q, P) = \left(\frac{Q^2}{2}, Q \cdot P, \frac{P^2}{2} \right).$$

The map Π_X is an invariant map to Fourier/root degrees; it is not the Hall charge lattice.

LEMMA 7.6 (Gram additivity defect). For $c_i = (Q_i, P_i) \in \Gamma_X^{\text{phys}}$,

$$\Pi_X(c_1 + c_2) = \Pi_X(c_1) + \Pi_X(c_2) + B(c_1, c_2),$$

where

$$B(c_1, c_2) = (Q_1 \cdot Q_2, Q_1 \cdot P_2 + Q_2 \cdot P_1, P_1 \cdot P_2) \in \Gamma_{\text{gram}}.$$

Proof. Expand

$$\frac{(Q_1 + Q_2)^2}{2}, \quad (Q_1 + Q_2) \cdot (P_1 + P_2), \quad \frac{(P_1 + P_2)^2}{2}.$$

The three cross terms are exactly the three components of $B(c_1, c_2)$. $\square$

LEMMA 7.7 (Symmetric bilinear 2-cocycle and on-the-nose associativity of the Gram-corrected charge product). Let S be a complex $K3$ surface and $\Lambda_S = H^*(S, \mathbb{Z})$ the Mukai lattice (even unimodular, signature $(4, 20)$). Let $\Gamma_X^{\text{phys}} := \Lambda_S \oplus \Lambda_S$, $\Gamma_{\text{gram}} := \mathbb{Z}^3$, and let $\Pi_X : \Gamma_X^{\text{phys}} \rightarrow \Gamma_{\text{gram}}$, $B : \Gamma_X^{\text{phys}} \times \Gamma_X^{\text{phys}} \rightarrow \Gamma_{\text{gram}}$ be as in Definition 7.1, namely, with $c = (Q, P)$,

$$\Pi_X(Q, P) = \left(\frac{(Q, Q)}{2}, (Q, P), \frac{(P, P)}{2} \right), \quad B((Q, P), (Q', P')) = ((Q, Q'), (Q, P') + (Q', P), (P, P')).$$

Then:

- (i) B is symmetric and $\mathbb{Z}$ -bilinear, and takes integral values: $B \in \text{Hom}_{\mathbb{Z}}(S^2 \Gamma_X^{\text{phys}}, \Gamma_{\text{gram}}) \subset \text{Hom}_{\mathbb{Z}}(\Gamma_X^{\text{phys}} \otimes_{\mathbb{Z}} \Gamma_X^{\text{phys}}, \Gamma_{\text{gram}})$.

- (ii) (Polarisation.) $B(c, c) = 2\Pi_X(c)$ for every $c \in \Gamma_X^{\text{phys}}$.
 (iii) (Symmetric Hochschild 2-cocycle.) B is normalised, $B(c, o) = B(o, c) = o$, and the additive coboundary

$$(\partial B)(a, b, c) := B(b, c) - B(a + b, c) + B(a, b + c) - B(a, b)$$

vanishes identically. Hence $[B] \in H_{\text{sym}}^2(\Gamma_X^{\text{phys}}; \Gamma_{\text{gram}})$ is well defined.

- (iv) (Star-associativity on the nose.) The product on $\widehat{\Gamma}_X := \Gamma_X^{\text{phys}} \oplus \Gamma_{\text{gram}}$,

$$(c, T) \star (c', T') := (c + c', T + T' + B(c, c')),$$

is associative and commutative; the unit is (o, o) and the inverse of (c, T) is $(-c, -T + B(c, c)) = (-c, -T + 2\Pi_X(c))$. Thus $(\widehat{\Gamma}_X, \star)$ is an abelian group, and the sequence

$$o \longrightarrow \Gamma_{\text{gram}} \xrightarrow{T \mapsto (o, T)} \widehat{\Gamma}_X \xrightarrow{\text{pr}_1} \Gamma_X^{\text{phys}} \longrightarrow o$$

is a central extension of abelian groups with cocycle B .

- (v) (Splitting; $[B] = o$.) The Gram-shadow section $s : \Gamma_X^{\text{phys}} \rightarrow \widehat{\Gamma}_X$, $s(c) := (c, \Pi_X(c))$, is a homomorphism for $\star$; equivalently $B = -\partial\Pi_X$ in $B_{\text{sym}}^2(\Gamma_X^{\text{phys}}; \Gamma_{\text{gram}})$, where $(\partial\Pi_X)(c, c') = \Pi_X(c) + \Pi_X(c') - \Pi_X(c + c')$. In particular $[B] = o$, the extension is split as an extension of abelian groups, and $(c, T) \mapsto (c, T - \Pi_X(c))$ is an isomorphism $\widehat{\Gamma}_X \xrightarrow{\sim} \Gamma_X^{\text{phys}} \oplus \Gamma_{\text{gram}}$ of abelian groups carrying $\star$ to componentwise addition.
- (vi) (Quadratic obstruction.) B is not the coboundary of any *linear* 1-cochain $L : \Gamma_X^{\text{phys}} \rightarrow \Gamma_{\text{gram}}$: if $B(c, c') = L(c) + L(c') - L(c + c')$ for an additive L , then setting $c' = c$ and using (ii) gives $2\Pi_X(c) = 2L(c) - L(2c) = o$, i.e. $\Pi_X \equiv o$, contradiction. Trivialising $[B]$ requires the quadratic refinement Π_X ; without Π_X the symmetric bilinear form B carries the standard Heisenberg-type extension class.

Proof. Write $c = (Q, P)$, $c' = (Q', P')$, $c'' = (Q'', P'')$ with $Q, Q', Q'', P, P', P'' \in \Lambda_S$, and abbreviate $(\cdot, \cdot) : \Lambda_S \times \Lambda_S \rightarrow \mathbb{Z}$ by juxtaposition, $QQ' = (Q, Q')$, so that $B(c, c') = (QQ', QP' + Q'P, PP')$.

(i) *Symmetry, bilinearity, integrality.* The components QQ' and PP' are the symmetric Mukai pairing on Λ_S , so each is symmetric and $\mathbb{Z}$ -bilinear. The middle component is the polarisation $(c, c') \mapsto QP' + Q'P$ of the symmetric biadditive form $(c, c') \mapsto QP' + Q'P$; swapping $(Q, P) \leftrightarrow (Q', P')$ returns $Q'P + QP' = QP' + Q'P$, so it is symmetric, and $\mathbb{Z}$ -bilinear in each slot by bilinearity of $(\cdot, \cdot)$. Integrality of every component follows from the unimodular even structure on Λ_S .

(ii) *Polarisation.* Set $c' = c$: the cross terms reduce to $QQ, QP + QP = 2QP, PP$, so $B(c, c) = (Q^2, 2QP, P^2) = 2\Pi_X(c)$.

(iii) *Cocycle.* Bilinearity gives normalisation $B(c, o) = B(o, c) = o$ immediately. For the cocycle identity, observe that any symmetric $\mathbb{Z}$ -bilinear form $\beta : G \times G \rightarrow A$ on an abelian group G with values in a trivial G -module A satisfies, by bilinearity in the first slot,

$$\beta(a + b, c) = \beta(a, c) + \beta(b, c), \quad \beta(a, b + c) = \beta(a, b) + \beta(a, c).$$

Substituting into $(\partial\beta)(a, b, c) = \beta(b, c) - \beta(a + b, c) + \beta(a, b + c) - \beta(a, b)$,

$$(\partial\beta)(a, b, c) = \beta(b, c) - \beta(a, c) - \beta(b, c) + \beta(a, b) + \beta(a, c) - \beta(a, b) = o.$$

Applied componentwise to B , this yields $\partial B = 0$. Hence $B \in Z_{\text{sym}}^2(\Gamma_X^{\text{phys}}; \Gamma_{\text{gram}})$.

(iv) *Associativity, identity, inverse.* Compute the two bracketings of $(c, T) \star (c', T') \star (c'', T'')$:

$$\begin{aligned} ((c, T) \star (c', T')) \star (c'', T'') &= (c + c' + c'', T + T' + T'' + B(c, c') + B(c + c', c'')), \\ (c, T) \star ((c', T') \star (c'', T'')) &= (c + c' + c'', T + T' + T'' + B(c', c'') + B(c, c' + c'')). \end{aligned}$$

The associator is the difference of the central components, $\text{Ass}(c, c', c'') = B(c, c') + B(c + c', c'') - B(c', c'') - B(c, c' + c'')$, which equals $-(\partial B)(c, c', c'') = 0$ by (iii); equivalently, it vanishes by bilinearity of B . Commutativity follows from symmetry of B . The identity computation $(c, T) \star (0, 0) = (c, T + B(c, 0)) = (c, T)$ and the inverse computation $(c, T) \star (-c, -T + B(c, c)) = (0, B(c, c) - B(c, c)) = (0, 0)$ (using $B(c, -c) = -B(c, c)$ by bilinearity) finish the abelian-group structure. That B is the cocycle of the displayed central extension is direct from the formula for $\star$ and normalisation $B(c, 0) = B(0, c) = 0$.

(v) *Splitting.* By Lemma 7.6, $\Pi_X(c + c') = \Pi_X(c) + \Pi_X(c') + B(c, c')$, so

$$s(c) \star s(c') = (c + c', \Pi_X(c) + \Pi_X(c') + B(c, c')) = (c + c', \Pi_X(c + c')) = s(c + c').$$

Thus s is a homomorphism, and $B(c, c') = \Pi_X(c + c') - \Pi_X(c) - \Pi_X(c') = -(\partial \Pi_X)(c, c')$ shows $B = -\partial \Pi_X$, so $[B] = 0$. The change of variables $(c, T) \mapsto (c, T - \Pi_X(c))$ is bijective on the underlying set with inverse $(c, T) \mapsto (c, T + \Pi_X(c))$, and carries $\star$ to componentwise addition:

$$(c, T - \Pi_X(c)) + (c', T' - \Pi_X(c')) = (c + c', T + T' - \Pi_X(c) - \Pi_X(c')) = (c + c', (T + T' + B(c, c')) - \Pi_X(c + c')),$$

matching the change of variables applied to $(c + c', T + T' + B(c, c')) = (c, T) \star (c', T')$.

(vi) *Quadratic obstruction.* Suppose $B = \partial L$ for some additive $L : \Gamma_X^{\text{phys}} \rightarrow \Gamma_{\text{gram}}$, i.e. $B(c, c') = L(c) + L(c') - L(c + c')$. Setting $c' = c$, $B(c, c) = 2L(c) - L(2c)$; but L additive gives $L(2c) = 2L(c)$, so $B(c, c) = 0$, contradicting $B(c, c) = 2\Pi_X(c)$ for any $c = (Q, P)$ with $(Q, Q), (Q, P), (P, P)$ not all zero. Hence no linear trivialisation exists; the quadratic primitive Π_X is necessary.

Step 4 (Chain-level lift; strict associativity in the PTVV/Joyce host). Clauses (i)–(iv) place $(\widehat{\Gamma}_X, \star)$ inside the abelian category of discrete abelian groups. The construction of the central extension is, however, used downstream not as a discrete group but as a strict monoid in the chain-level host of Definition 2.1: the bracket on $\widehat{P}_X$ and the bilinear cocycle B act on the critical-Hall complex of compactly supported B-branes, which lives in chain complexes of $\mathbb{Q}$ -vector spaces (or, with orientation lines included, in the dg-category of (-1) -shifted symplectic perfect modules of [?PTVV2013]). We must therefore lift (iv) from H^0 to the chain level.

Chain-level data. Let $C_* := C_*(-, \mathbb{Z})$ denote the normalised bar/Eilenberg–Mac Lane chain complex functor. Apply C_* to the discrete abelian groups $\Gamma_X^{\text{phys}}, \Gamma_{\text{gram}}, \widehat{\Gamma}_X$, placing them in homological degree zero. The bilinear cocycle B of (i) is already a discrete chain map

$$\mathbf{B} : C_*(\Gamma_X^{\text{phys}}) \otimes C_*(\Gamma_X^{\text{phys}}) \longrightarrow C_*(\Gamma_{\text{gram}}),$$

with no higher components, since B is *strictly* bilinear in each slot by (i) (no chain-level corrections survive because the source and target are concentrated in degree zero). The star product extends to a degree-zero chain map

$$\mu : C_*(\widehat{\Gamma}_X) \otimes C_*(\widehat{\Gamma}_X) \longrightarrow C_*(\widehat{\Gamma}_X), \quad \mu((c, T) \otimes (c', T')) = (c + c', T + T' + \mathbf{B}(c, c')).$$

Coupled with the orientation lines o_c of Definition 4.3 and the multiplicative isomorphisms $\mu_{c,c'} : o_{c+c'} \xrightarrow{\sim} o_c \otimes o_{c'}$ of Joyce–Upmeyer [?JoyceUpmeyer] and Joyce–Kinjo–Park–Safronov [?JoyceKinjoParkSafronov], the same formula promotes to a degree-zero chain map of (-1) -shifted symplectic perfect complexes on the Cartesian extension correspondence $\mathfrak{M}_{X,c} \times \mathfrak{M}_{X,c'} \rightarrow \mathfrak{M}_{X,c+c'}$ of [?PTVV2013], with $\mathbf{B}$ acting on the Γ_{gram} -grading factor and $\mu_{c,c'}$ on the orientation-line factor.

The associator vanishes pointwise. By (iv) the associator of μ at H^0 is

$$\alpha(c, c', c'') := \mathbf{B}(c, c') + \mathbf{B}(c + c', c'') - \mathbf{B}(c', c'') - \mathbf{B}(c, c' + c'') = -(\partial \mathbf{B})(c, c', c'') = 0,$$

strictly as an element of $C_0(\Gamma_{\text{gram}})$, not merely up to a boundary. The bounding homotopy $h : \mu \circ (\mu \otimes \text{id}) \Rightarrow \mu \circ (\text{id} \otimes \mu)$ required by step (ii) of the proof skeleton is therefore the *canonical zero homotopy*: $h \equiv 0$ bounds $\alpha \equiv 0$. No nontrivial chain-level homotopy is needed because the obstruction itself vanishes on the nose. The PTVV (-1) -shifted symplectic Frobenius pairing on $\mathfrak{M}_{X,c} \times \mathfrak{M}_{X,c'}$ of [?PTVV2013] contributes to μ only through its integrated bilinear class $\mathbf{B}$ on Γ_{gram} ; the Joyce–Upmeyer orientation cocycle [?JoyceUpmeyer] contributes only through the discrete multiplicative isomorphism $\mu_{c,c'}$; and (i) pins both contributions to a strictly bilinear function of (c, c') . In particular the BBDJS vanishing-cycle Thom–Sebastiani isomorphism [?BBDJS2015] on disjoint d-critical charts is monoidal at the chain level, so it carries no associator correction beyond what $\mathbf{B}$ already records.

Pentagon identity. The four-fold associator measures the discrepancy between two bracketings of $(c_1, T_1) \star (c_2, T_2) \star (c_3, T_3) \star (c_4, T_4)$ along the two pentagon paths. In the centre component this is $(\partial \mathbf{B})(c_1, c_2, c_3, c_4)$, which vanishes identically because $\partial^2 = 0$ on the Hochschild complex. Term by term: each pentagon edge is a single application of the associator α of the previous paragraph; the associator vanishes pointwise, hence its iterates do as well. In the orientation component the pentagon is the cocycle pentagon

$$\mu_{c+c',c''} \circ \mu_{c,c'} = \mu_{c,c'+c''} \circ \mu_{c',c''}$$

of Joyce–Upmeyer [?JoyceUpmeyer] and Joyce–Kinjo–Park–Safronov [?JoyceKinjoParkSafronov], recorded in hypothesis (O) of Theorem 10.19. Multiplicativity is precisely the strict pentagon for $\{\mu_{c,c'}\}$; under (O) the pentagon holds on the nose, with no bounding 2-cell required.

Higher Stasheff coherences collapse. Stasheff’s associahedra organise the higher coherences of an A_∞ -monoid as polytopes K_n for $n \geq 4$, with $(n-2)$ -cells parametrisng n -fold reassociations. For $(\widehat{\Gamma}_X, \mu)$ every cell of K_n is computed by an iterated comparison of bracketings of $(c_1, T_1) \star \cdots \star (c_n, T_n)$; on the centre component each such comparison is a sum of values of $\mathbf{B}$ evaluated on sums and differences of subsets of $\{c_1, \dots, c_n\}$, and by strict bilinearity (i) every such sum collapses to a single multilinear expression independent of the bracketing. Equivalently, the bar construction $B_*(\widehat{\Gamma}_X)$ of the discrete monoid $(\widehat{\Gamma}_X, \star)$ is acyclic in positive degrees because $\widehat{\Gamma}_X$ is a discrete abelian group (this is the standard fact that the Eilenberg–Mac Lane bar complex of a discrete abelian group has trivial higher homotopy groups), so every higher Stasheff coherence h_n ($n \geq 3$) admits the canonical zero bounding chain. No additional choice is required. On the orientation factor the same conclusion holds: a multiplicative system $\{\mu_{c,c'}\}$ satisfying the strict pentagon is, by Mac Lane’s coherence theorem, a strict monoidal natural transformation; all higher coherences are uniquely determined by the strict pentagon and identity, and they all hold strictly. Combining the two factors, $(\widehat{\Gamma}_X, \mu)$, enriched in the chain category of the PTVV/Joyce host, is a strictly associative monoid: an A_∞ -monoid whose Stasheff coherences h_n are all identically zero, not merely homotopic to zero.

Conclusion. The H^0 -level associativity of (iv) lifts to strict associativity at the chain level of the PTVV/Joyce host of Definition 2.1, conditional on the strict-pentagon multiplicative orientation of (O)

in Theorem 10.19. Equivalently: for the discrete bookkeeping monoid $(\widehat{\Gamma}_X, \star)$, no homotopy-coherent correction is required; the chain-level lift is canonical and the higher Stasheff cells are bounded by the canonical zero. The chain-level statement therefore inherits its precise residual obligation from (O): construct the multiplicative orientation cocycle $\{\mu_{c,c'}\}$ on the reduced compact protected sector of $K_3 \times E$ so that the strict pentagon holds. This is the single named obstruction; absent (O), the chain-level lift is $\mathcal{A}_\infty$ -coherent rather than strict, but the $\mathcal{A}_\infty$ -coherences are still controlled by a Hochschild 2-cocycle representative of $[B]$, which is cohomologically trivial by (v). $\square$

Remark 7.8 (Status). Proved. Clauses (i)–(vi) are arithmetic identities in $\mathbb{Z}^3$ over the even unimodular Mukai lattice Λ_S ; the proof uses no automorphic input. Structurally the lemma fixes the bookkeeping content of the extension $\widehat{\Gamma}_X$: cohomologically the extension is split (clause (v)), but the double grading it records – physical charge upstairs, Borchers Fourier index downstairs – is the device by which a hypothetical Γ_X^{phys} -graded Hall source descends to the Γ_{gram} -graded denominator algebra. Clause (vi) records the precise sense in which trivialisation *requires* the quadratic refinement Π_X : without it, B carries the standard Heisenberg extension class and the descent is obstructed.

Step 4: chain-level strict associativity, conditional on the multiplicative orientation hypothesis. The chain-level lift of (iv) constructed in Step 4 above is *strict*, not merely $\mathcal{A}_\infty$ -coherent, conditional on hypothesis (O) of Theorem 10.19 (the strict pentagon for the multiplicative orientation cocycle $\{\mu_{c,c'}\}$ of Joyce–Upmeyer [?JoyceUpmeyer] and Joyce–Kinjo–Park–Safronov [?JoyceKinjoParkSafronov]). The collapse argument is: $\widehat{\Gamma}_X$ is a discrete abelian group, so its bar complex is acyclic in positive degrees; the higher Stasheff coherences are computed by iterated values of the strictly bilinear cocycle B and vanish on the nose by the cocycle identity of (iii); the orientation factor contributes only the strict pentagon of (O). Absent (O), the lift is $\mathcal{A}_\infty$ -coherent rather than strict, controlled by a Hochschild 2-cocycle representative of $[B]$ which is cohomologically trivial by (v). The single named chain-level residual is therefore the construction of $\{\mu_{c,c'}\}$ with strict pentagon on the reduced compact protected sector of $K_3 \times E$; absent that construction the chain-level monoid structure is $\mathcal{A}_\infty$ rather than strict, with all $\mathcal{A}_\infty$ -coherences canonically bounded by Θ (the determinant-line Hochschild 1-cochain of Theorem 10.19, Step 2). The remaining content of the construction (positivity of the Gram pushforward on the chamber semigroup Γ_{eff} ; compatibility with the multiplicities $f(nm, l)$) is the subject of the subsequent obstruction lemmas.

Definition 7.9 (Gram-corrected charge extension). Set

$$\widehat{\Gamma}_X := \Gamma_X^{\text{phys}} \oplus \Gamma_{\text{gram}}.$$

For $c, c' \in \Gamma_X^{\text{phys}}$ and $T, T' \in \Gamma_{\text{gram}}$, define

$$(c, T) \star (c', T') := (c + c', T + T' + B(c, c')).$$

The section

$$s : \Gamma_X^{\text{phys}} \longrightarrow \widehat{\Gamma}_X, \quad s(c) = (c, \Pi_X(c))$$

is additive for $\star$. Thus the compact Hall object, if constructed, is graded upstairs by Γ_X^{phys} ; the Γ_{gram} -graded Igusa object is obtained only after Gram pushforward or an equivalent cocycle-corrected descent.

Proof. By Lemma 7.7(i),(iii),(iv), $\star$ is associative and commutative, the identity is $(0, 0)$, and the inverse of (c, T) is $(-c, -T + B(c, c))$; the section s is the splitting of clause (v). $\square$

Definition 7.10 (Gram-index pairing). For

$$\gamma = (n, l, m), \quad \gamma' = (n', l', m'),$$

define

$$\langle \gamma, \gamma' \rangle_{\text{ind}} = 2(nm' + n'm) - ll'.$$

The Gram discriminant is

$$\Delta_{\text{gram}}(\gamma) = \langle \gamma, \gamma \rangle_{\text{ind}} = 4nm - l^2.$$

Remark 7.11 (Chamber transport). The pairing $\langle \cdot, \cdot \rangle_{\text{ind}}$ is the pairing on Gram invariants. The order on Γ_{eff} is a Fock polarization at one cusp. A duality element transports this polarization to the corresponding product chamber at the transported cusp.

8 THE EXTERIOR-SQUARE MODEL OF THE AUTOMORPHIC GROUP

The form Δ_5 is an automorphic form on the Siegel space. The physical U-duality interpretation of the same arithmetic group is external input from the $K3 \times E$ compactification; it is not needed for the lattice calculation below. The calculation records the first-principles bridge from the Siegel variables to the type-IV Lorentzian lattice on which the denominator algebra is written.

$$\begin{aligned} \Lambda^4 &= \mathbb{Z}e_1 \oplus \mathbb{Z}e_2 \oplus \mathbb{Z}e_3 \oplus \mathbb{Z}e_4. \\ g \in \mathbf{SL}_4(\mathbb{Z}) : \quad \Lambda^2 g : \Lambda^2 \Lambda^4 &\longrightarrow \Lambda^2 \Lambda^4. \\ u \wedge v &= (u, v)e_1 \wedge e_2 \wedge e_3 \wedge e_4. \\ (\Lambda^2 \Lambda^4, (,)) &\simeq \Pi_{3,3}. \end{aligned}$$

Fix

$$q_0 = e_1 \wedge e_3 + e_2 \wedge e_4.$$

For $x, y \in \Lambda^4$, define

$$\begin{aligned} -x \wedge y \wedge q_0 &= B_{q_0}(x, y)e_1 \wedge e_2 \wedge e_3 \wedge e_4. \\ \{g \in \mathbf{SL}(\Lambda^4) \mid (\Lambda^2 g)q_0 &= q_0\} \simeq \mathbf{Sp}_4(\mathbb{Z}). \end{aligned}$$

Set

$$\begin{aligned} \Lambda^{3,2} &= q_0^\perp \subset \Lambda^2 \Lambda^4. \\ f_1 &= e_1 \wedge e_2, \quad f_2 = e_2 \wedge e_3, \quad f_3 = e_1 \wedge e_3 - e_2 \wedge e_4, \quad f_{-2} = e_4 \wedge e_1, \quad f_{-1} = e_4 \wedge e_3. \\ (f_1, f_{-1}) &= (f_2, f_{-2}) = -1, \quad (f_3, f_3) = 2. \\ \Lambda^{3,2} &\simeq \Lambda^{(1,1)} \oplus \Lambda^{(1,1)} \oplus [2], \quad \Lambda^{(1,1)} = \begin{pmatrix} 0 & -1 \\ -1 & 0 \end{pmatrix}. \end{aligned}$$

$$\mathbb{H}^{\mathbf{IV}} = \{Z \in \mathbb{P}(\Lambda^{3,2} \otimes \mathbb{C}) \mid (Z, Z) = 0, (Z, \overline{Z}) < 0\}.$$

$$\mathbb{H}_+^{\mathbf{IV}} = \{^t(z_2^2 - z_1 z_3, z_3, z_2, z_1, 1)z_0 \in \mathbb{H}^{\mathbf{IV}} \mid \text{Im} z_1 > 0\}.$$

Writing $y_i = \text{Im} z_i$,

$$(Z, \overline{Z}) < 0 \iff y_1 y_3 - y_2^2 > 0.$$

$$\mathbb{H}_+^{\mathbf{IV}} \simeq \left\{ \begin{pmatrix} z_1 & z_2 \\ z_2 & z_3 \end{pmatrix} \in \mathbb{H}_2 \right\}.$$

Let $\mathbf{O}_{\mathbb{R}}(\Lambda^{3,2})_+$ be the index-two subgroup preserving $\mathbb{H}_+^{\mathbf{IV}}$. Its kernel on $\mathbb{H}_+^{\mathbf{IV}}$ is $\{\pm \mathbf{I}_5\}$, and

$$\mathbf{O}_{\mathbb{R}}(\Lambda^{3,2})_+ = \{\pm \mathbf{I}_5\} \mathbf{SO}_{\mathbb{R}}(\Lambda^{3,2})_+,$$

where $\mathbf{SO}_{\mathbb{R}}(\Lambda^{3,2})_+$ has real spin-norm 1. Put

$$\mathbf{O}(\Lambda^{3,2})_+ = \mathbf{O}_{\mathbb{R}}(\Lambda^{3,2})_+ \cap \mathbf{O}(\Lambda^{3,2})$$

and

$$\mathbf{SO}(\Lambda^{3,2})_+ = \mathbf{SO}_{\mathbb{R}}(\Lambda^{3,2})_+ \cap \mathbf{O}(\Lambda^{3,2}).$$

$$M = \begin{pmatrix} m_1 & m_2 \\ m_2 & m_3 \end{pmatrix},$$

$$\wedge^2(g_o) = \wedge^2 \left(\begin{pmatrix} o & \mathbf{I}_2 \\ -\mathbf{I}_2 & o \end{pmatrix} \right) = \begin{pmatrix} o & o & o & o & -1 \\ o & o & o & -1 & o \\ o & o & 1 & o & o \\ o & -1 & o & o & o \\ -1 & o & o & o & o \end{pmatrix},$$

$$\wedge^2(g_M) = \wedge^2 \left(\begin{pmatrix} \mathbf{I}_2 & M \\ o & \mathbf{I}_2 \end{pmatrix} \right) = \begin{pmatrix} 1 & -m_1 & 2m_2 & -m_3 & m_2^2 - m_1 m_3 \\ o & 1 & o & o & m_3 \\ o & o & 1 & o & m_2 \\ o & o & o & 1 & m_1 \\ o & o & o & o & 1 \end{pmatrix},$$

$$\wedge^2(g_A) = \wedge^2 \left(\begin{pmatrix} {}^t A^{-1} & o \\ o & A \end{pmatrix} \right) = \det(A) \begin{pmatrix} 1 & o & o & o & o \\ o & a_1^2 & -2a_1 a_2 & a_2^2 & o \\ o & -a_1 a_3 & a_1 a_4 + a_2 a_3 & -a_2 a_4 & o \\ o & a_3^2 & -2a_3 a_4 & a_4^2 & o \\ o & o & o & o & 1 \end{pmatrix}$$

LEMMA 8.1. The exterior-square representation is

$$\wedge^2 : \mathbf{Sp}_4(\mathbb{Z}) / \{\pm \mathbf{I}_4\} \rightarrow \mathbf{SO}_+(\Lambda^{3,2}) \simeq \mathbf{O}(\Lambda^{3,2})_+ / \{\pm \mathbf{I}_5\}.$$

For every $g \in \mathbf{Sp}_4(\mathbb{Z})$,

$$\begin{array}{ccc} \mathbb{H}_2 & \longrightarrow & \mathbb{H}_2 \\ \downarrow & & \downarrow \\ \mathbb{H}_+^{\mathbf{IV}} & \xrightarrow{g \wedge g} & \mathbb{H}_+^{\mathbf{IV}} \end{array}$$

Proof. The Pfaffian pairing is preserved by $\wedge^2 \mathbf{SL}_4$. The subgroup fixing

$$q_o = e_1 \wedge e_3 + e_2 \wedge e_4$$

is the group preserving the alternating form B_{q_o} , namely $\mathbf{Sp}_4(\mathbb{Z})$. Its action preserves $q_o^\perp = \Lambda^{3,2}$ and the displayed component of the type-**IV** domain. The kernel on $q_o^\perp$ is $\{\pm \mathbf{I}_4\}$. The exceptional isomorphism $\mathbf{PSp}_4(\mathbb{R}) \simeq \mathbf{SO}_{\mathbb{R}}(3, 2)_+$ identifies the image with $\mathbf{SO}_+(\Lambda^{3,2})$. The displayed matrices are the integral generators on both sides. Substitution of

$${}^t(z_2^2 - z_1 z_3, z_3, z_2, z_1, 1) z_o$$

into the two actions produces the commutative square. $\square$

Definition 8.2 (Automorphic duality datum).

$$G_{\text{aut}} = \mathbf{PSp}_4(\mathbb{Z}) \simeq \mathbf{SO}_+(\Lambda^{3,2}).$$

The virtual determinant $\mathcal{D}_X = \Delta_5$ is not an invariant scalar function on $\mathbb{H}_2$. It is a section of the automorphic line

$$\mathcal{L}^5 \otimes \nu_{\Delta_5}$$

with transformation law

$$\Delta_5(g \cdot Z) = \nu_{\Delta_5}(g) \det(CZ + D)^5 \Delta_5(Z), \quad g = \begin{pmatrix} A & B \\ C & D \end{pmatrix} \in \mathbf{Sp}_4(\mathbb{Z}).$$

The scalar square is a section of $\mathcal{L}^{10}$:

$$\Delta_5^2(g \cdot Z) = \det(CZ + D)^{10} \Delta_5^2(Z).$$

For every scalar-square branch,

$$Z_{\square}^X(g \cdot Z) = \det(CZ + D)^{-10} Z_{\square}^X(Z).$$

9 THE WEYL CHAMBER AND THE WEYL VECTOR

Fix the primitive hyperbolic sublattice

$$\Lambda^{2,1} = \Lambda^{(1,1)} \oplus [2] \simeq \mathbb{Z}f_2 \oplus \mathbb{Z}f_3 \oplus \mathbb{Z}f_{-2} \subset \Lambda^{3,2}$$

and its type-II sublattice

$$\Lambda_{II}^{2,1} = \{mf_2 + lf_3 + nf_{-2} \in \Lambda^{2,1} \mid m \equiv n \equiv 0 \pmod{2}\}.$$

Change polarization on the same Gram-index lattice:

$$\Gamma_{\text{ind}} \xrightarrow{\alpha} \Lambda_{II}^{2,1}, \quad \alpha(n, l, m) = 2nf_2 - lf_3 + 2mf_{-2}.$$

The cusp polarization gives the Fock determinant. The denominator polarization gives the Borchers chamber.

Extend $\phi \in \mathbf{O}(\Lambda^{2,1})$ by the identity on $(\Lambda^{2,1})^\perp$:

$$\mathbf{O}(\Lambda^{2,1}) \hookrightarrow \mathbf{O}(\Lambda^{3,2}).$$

For the component-preserving subgroup attached to the positive cone below,

$$\Delta_5 \in M_5(\mathbf{O}(\Lambda^{2,1})_+, \nu_{\Delta_5}).$$

For a primitive element $\alpha \in \Lambda^{2,1}$ satisfying $(\alpha, \alpha) > 0$ and

$$(\alpha, \alpha) \mid 2(\Lambda^{2,1}, \alpha),$$

Kac reflection [?KAC:1]:

$$\begin{aligned} s_\alpha : x &\mapsto x - 2 \frac{(x, \alpha)}{(\alpha, \alpha)} \alpha. \\ s_\alpha(\alpha) &= -\alpha, \quad s_\alpha|_{\alpha^\perp} = \mathbf{I}. \end{aligned}$$

For $(\alpha, \alpha) = 2$:

$$s_\alpha : x \mapsto x - (x, \alpha) \alpha.$$

LEMMA 9.1. Let $(\alpha, \alpha) = 2$. For

$$\delta_1 = 2f_2 - f_3, \quad \delta_2 = 2f_{-2} - f_3, \quad \delta_3 = f_3,$$

[?GN, Theorem 2.1 and Section 4]; equivalently [?GNII, Section 5.1.1, case $(t=1, \text{II}, 0)$].

$$\Delta_5(s_\alpha Z) = -\Delta_5(Z)$$

when $\alpha \in \{\delta_1, \delta_2, \delta_3\}$. For

$$\begin{aligned} \alpha &\in \{f_{-2} - f_2, f_2 - f_3, f_2 + f_3\}, \\ \Delta_5(s_\alpha Z) &= \Delta_5(Z) \end{aligned}$$

Proof. Let

$$\overline{U} = \wedge^2 \begin{pmatrix} {}^t U^{-1} & 0 \\ 0 & U \end{pmatrix}, \quad U \in \mathbf{GL}_2(\mathbb{Z}).$$

Required reflections:

$$s_{f_{-2}-f_2} = -\overline{\begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}}, \quad s_{f_3} = -\overline{\begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}}, \quad s_{f_2-f_3} = -\overline{\begin{pmatrix} 1 & -1 \\ 0 & -1 \end{pmatrix}}.$$

Maass's character has value -1 on s_{δ_i} and $+1$ on the three other reflection classes. □

The time-like cone is

$$C(\Lambda^{2,1}) = \{x \in \Lambda^{2,1} \otimes \mathbb{R} \mid (x, x) < 0\}.$$

Let $C(\Lambda^{2,1})_+$ be the component for which

$$\Omega(C(\Lambda^{2,1})_+) := \Lambda^{2,1} \otimes \mathbb{R} + iC(\Lambda^{2,1})_+ \subset \mathbb{H}_+^{\text{IV}}.$$

For

$$z = z_3 f_2 + z_2 f_3 + z_1 f_{-2} \in \Omega(C(\Lambda^{2,1})_+),$$

the type **IV** point is

$$Z = {}^t(z_2^2 - z_1 z_3, z_3, z_2, z_1, 1) z_0.$$

Let $\mathbf{O}(\Lambda^{2,1})_+$ be the index-two subgroup fixing $C(\Lambda^{2,1})_+$. It acts discretely on

$$C(\Lambda^{2,1})_+/\mathbb{R}_{>0}.$$

For $(\alpha, \alpha) > 0$,

$$\mathbb{H}_\alpha = \{\mathbb{R}_{>0} x \in C(\Lambda^{2,1})_+/\mathbb{R}_{>0} \mid (x, \alpha) = 0\},$$

and

$$\mathbb{H}_{\alpha,+} = \{\mathbb{R}_{>0} x \in C(\Lambda^{2,1})_+/\mathbb{R}_{>0} \mid (x, \alpha) \leq 0\}.$$

Let $W(\Lambda^{2,1})$ be the subgroup generated by these positive-norm reflections:

$$W(\Lambda^{2,1}) \subset \mathbf{O}(\Lambda^{2,1})_+.$$

Nikulin's theorem for $\Lambda^{(1,1)} \oplus [2]$ [NIKULIN]:

$$\mathbf{O}(\Lambda^{2,1})_+ = W^{(2)}(\Lambda^{2,1}),$$

where $W^{(2)}$ is generated by reflections in square-2 vectors. For $k > 0$ put

$$\Delta^{(k)}(\Lambda^{2,1}) = \{\delta \in \Lambda^{2,1} \mid \delta \text{ primitive, } (\delta, \delta) = k, s_\delta \in \mathbf{O}(\Lambda^{2,1})\}.$$

Each $\delta \in \Delta^{(2)}(\Lambda^{2,1})$ has one of two divisibilities:

- Type I: $(\delta, \Lambda^{2,1}) = \mathbb{Z}$
- Type II: $(\delta, \Lambda^{2,1}) = 2\mathbb{Z}$

The type sublattices are

$$\Lambda_I^{2,1} = \{m f_2 + l f_3 + n f_{-2} \in \Lambda^{2,1} \mid m + l + n \equiv 0 \pmod{2}\},$$

and the already defined $\Lambda_{II}^{2,1}$. Type I roots lie in $\Lambda_I^{2,1}$; type II roots lie in $\Lambda_{II}^{2,1}$. The reflection subgroups are normal, with indices

$$[W^{(2)}(\Lambda^{2,1}) : W^{(2)}(\Lambda_I^{2,1})] = 2, \quad [W^{(2)}(\Lambda^{2,1}) : W^{(2)}(\Lambda_{II}^{2,1})] = 6.$$

The chambers are

$$\mathcal{P}' = \bigcap_{\delta \in \mathcal{P}'_{prim}} \mathbb{H}_{\delta,+}, \quad \mathcal{P}' \in \{\mathcal{P}, \mathcal{P}_I, \mathcal{P}_{II}\},$$

with walls

$$\begin{aligned} \mathcal{P}_{prim} &= \{f_2 - f_3, f_{-2} - f_2, f_3\}, \\ \mathcal{P}_{I,prim} &= \{f_2 - f_3, f_{-2} - f_2, f_2 + f_3\}, \\ \mathcal{P}_{II,prim} &= \{\delta_1, \delta_2, \delta_3\}. \end{aligned}$$

For

$$\text{Aut}(\mathcal{P}') = \{g \in \mathbf{O}(\Lambda^{2,1})_+ \mid g\mathcal{P}' = \mathcal{P}'\},$$

$$\text{Aut}(\mathcal{P}) = \mathbf{I}, \quad \text{Aut}(\mathcal{P}_I) = \langle s_{f_3} \rangle, \quad \text{Aut}(\mathcal{P}_{II}) \simeq S_3,$$

with $\text{Aut}(\mathcal{P}_{II})$ generated by $s_{f_2-f_3}$ and $s_{f_{-2}-f_2}$. The chamber decomposition is

$$\mathbf{O}(\Lambda^{2,1})_+ \simeq W^{(2)}(\Lambda^{2,1}) \simeq W^{(2)}(\Lambda_I^{2,1}) \rtimes \text{Aut}(\mathcal{P}_I) \simeq W^{(2)}(\Lambda_{II}^{2,1}) \rtimes \text{Aut}(\mathcal{P}_{II}).$$

LEMMA 9.2 (*Multiplier restriction*). • for $w \in W^{(2)}(\Lambda_I^{2,1})$ and $a \in \text{Aut}(\mathcal{P}_I)$,

$$\Delta_5(w \cdot aZ) = \det(a)\Delta_5(Z)$$

• for $w \in W^{(2)}(\Lambda_{II}^{2,1})$ and $a \in \text{Aut}(\mathcal{P}_{II})$,

$$\Delta_5(w \cdot aZ) = \det(w)\Delta_5(Z)$$

Proof. Displayed Maass multiplier [MAASS:1], [GN, Section 3], [Gritsenko99]:

$$\nu_{\Delta_5}(s_{\delta_i}) = -\mathbf{I}, \quad \nu_{\Delta_5}(s_{f_{-2}-f_2}) = \nu_{\Delta_5}(s_{f_2-f_3}) = \nu_{\Delta_5}(s_{f_2+f_3}) = \mathbf{I}.$$

Semidirect products:

$$W^{(2)}(\Lambda_I^{2,1}) \rtimes \text{Aut}(\mathcal{P}_I), \quad W^{(2)}(\Lambda_{II}^{2,1}) \rtimes \text{Aut}(\mathcal{P}_{II}).$$

□

Set

$$\Delta(\Lambda_{II}^{2,1})_+ = \mathbb{R}_{\geq 0}\delta_1 + \mathbb{R}_{\geq 0}\delta_2 + \mathbb{R}_{\geq 0}\delta_3$$

and

$$\Delta(\Lambda_{II}^{2,1})_+^* = \{x \in \Lambda^{2,1} \otimes \mathbb{R} \mid (x, \delta_i) \leq 0\}.$$

For the finite-volume chamber $\mathcal{P}_{II}$, the dual cone obeys

$$\Delta(\Lambda_{II}^{2,1})_+^* \subset \overline{C(\Lambda^{2,1})_+} \subset \Delta(\Lambda_{II}^{2,1})_+.$$

The Weyl vector is determined by

$$(\rho, \delta_i) = -\frac{(\delta_i, \delta_i)}{2} = -\mathbf{I}, \quad i = 1, 2, 3.$$

$$(\delta_i, \delta_j) = \begin{pmatrix} 2 & -2 & -2 \\ -2 & 2 & -2 \\ -2 & -2 & 2 \end{pmatrix}.$$

$$\rho = \frac{\mathbf{I}}{2}\delta_1 + \frac{\mathbf{I}}{2}\delta_2 + \frac{\mathbf{I}}{2}\delta_3 = f_2 - \frac{\mathbf{I}}{2}f_3 + f_{-2}.$$

Also

$$\rho \in C(\Lambda^{2,1})_+ = C(\Lambda_{II}^{2,1})_+.$$

Fix

$$z = z_1 f_{-2} + z_2 f_3 + z_3 f_2 \in \Lambda^{2,1} \otimes \mathbb{R} + iC(\Lambda^{2,1})_+ = \Lambda_{II}^{2,1} \otimes \mathbb{R} + iC(\Lambda_{II}^{2,1})_+.$$

The dual lattice is

$$(\Lambda_{II}^{2,1})^* = \mathbb{Z}\frac{\mathbf{I}}{2}f_2 + \mathbb{Z}\frac{\mathbf{I}}{2}f_3 + \mathbb{Z}\frac{\mathbf{I}}{2}f_{-2} = \frac{\mathbf{I}}{2}\Lambda^{2,1}.$$

Definition 9.3 (Gram-index degrees in the Weyl chamber). For $\gamma = (n, l, m) \in \Gamma_{\text{ind}}$ set

$$\alpha(\gamma) = \alpha(n, l, m) = 2nf_2 - lf_3 + 2mf_{-2} \in \Lambda_{II}^{2,1}.$$

With

$$z = z_1 f_{-2} + z_2 f_3 + z_3 f_2,$$

the character of the Gram-index degree is

$$x^\gamma = q^n r^l s^m = \exp(-\pi i(\alpha(\gamma), z)).$$

For $\gamma' = (n', l', m')$,

$$(\alpha(\gamma), \alpha(\gamma')) = -2\langle \gamma, \gamma' \rangle_{\text{ind}}, \quad (\alpha(\gamma), \alpha(\gamma)) = -2(4nm - l^2).$$

Thus the Borchers degree is the Lorentzian image of the Gram-index grading. The Weyl chamber and real walls are part of the Gritsenko–Nikulin/Nikulin root datum. The cusp polarization fixed in Section 7 selects the corresponding product expansion, and the active Jacobi support gives the nonzero visible product exponents in that chamber.

$$E_o := \rho.$$

$$q^{1/2} r^{1/2} s^{1/2} = \exp(-\pi i(\rho, z)).$$

$$\exp(-2\pi i(\alpha(\gamma), z)), \quad \exp(-2\pi i(\rho, z)).$$

Definition 9.4 (Active support and Borchers cone). In the standard cusp chamber, the active support is

$$\Gamma_{\text{act}} = \{\gamma = (n, l, m) \in \Gamma_{\text{eff}} \mid f(nm, l) \neq 0\}.$$

The inactive triples in $\Gamma_{\text{eff}} \setminus \Gamma_{\text{act}}$ are ordering data only; their product exponents are zero. The product does not prove absence of root spaces with zero signed superdimension. The proof-grade Borchers cone is the real cone generated by the active positive Gram-index degrees after the Lorentzian degree map:

$$C_{\text{Borch}} := \mathbb{R}_{\geq 0} \alpha(\Gamma_{\text{act}}) \subset \Lambda_{II}^{2,1} \otimes \mathbb{R}.$$

The real-root hyperplanes are the walls used by the denominator identity in this polarization.

LEMMA 9.5 (*Active support gives the type-II chamber cone*). For $\gamma = (n, l, m) \in \Gamma_{\text{act}}$,

$$\alpha(\gamma) = n\delta_1 + m\delta_2 + (n + m - l)\delta_3,$$

with

$$n \geq 0, \quad m \geq 0, \quad n + m - l \geq 0.$$

Consequently

$$C_{\text{Borch}} = \mathbb{R}_{\geq 0}\delta_1 + \mathbb{R}_{\geq 0}\delta_2 + \mathbb{R}_{\geq 0}\delta_3.$$

Proof. The displayed decomposition is

$$n(2f_2 - f_3) + m(2f_{-2} - f_3) + (n + m - l)f_3 = 2nf_2 - lf_3 + 2mf_{-2}.$$

The effective chamber gives $n, m \geq 0$. Since $\phi_{0,1}$ is weak of index one, $f(nm, l) \neq 0$ implies $4nm - l^2 \geq -1$. If $n = 0$ or $m = 0$, this gives $l^2 \leq 1$, and the boundary cases in Γ_{eff} give $l \leq n + m$. If $n, m > 0$ and $l \geq n + m + 1$, then

$$l^2 \geq (n + m + 1)^2 = 4nm + (n - m)^2 + 2(n + m) + 1 > 4nm + 1,$$

contradicting $l^2 \leq 4nm + 1$. Hence $n + m - l \geq 0$. The reverse inclusion follows from the active triples

$$(1, 0, 0), \quad (0, 0, 1), \quad (0, -1, 0),$$

whose degrees are $\delta_1 + \delta_3$, $\delta_2 + \delta_3$, and δ_3 , and from the active triples

$$(1, 1, 0), \quad (0, 1, 1),$$

whose degrees are δ_1 and δ_2 . □

For $n, l, m \in \mathbb{Z}$ with $n \equiv m \equiv l \equiv 1 \pmod{2}$, $n, m > 0$ and $4nm - l^2 > 0$, set

$$a = (n - 1)f_2 - \frac{l - 1}{2}f_3 + (m - 1)f_{-2}, \quad m(a) = -\frac{1}{64}c_\Delta(n, l, m).$$

[?GN, Theorem 2.1]

$$\rho + a = nf_2 - \frac{l}{2}f_3 + mf_{-2},$$

and

$$\begin{aligned} & \frac{1}{64}c_\Delta(n, l, m) \exp(\pi i(nz_1 + lz_2 + mz_3)) \\ &= \frac{1}{64}c_\Delta(n, l, m) \exp\left(-\pi i\left(nf_2 - \frac{l}{2}f_3 + mf_{-2}, z\right)\right) \\ &= -m(a) \exp(-\pi i(\rho + a, z)), \end{aligned}$$

with

$$a \in (\Lambda^{2,1})^* = \frac{1}{2}\Lambda_{II}^{2,1}.$$

The normalization $c_\Delta(1, 1, 1) = 64$ fixes $m(0) = -1$. Since $c_\Delta(n, l, m) \in 64\mathbb{Z}$, all $m(a)$ are integral. The inequality $4nm - l^2 > 0$ implies

$$\rho + a \in C(\Lambda^{2,1})_+.$$

Let $\lambda = \rho + a$. For nonzero coefficients in the chosen chamber,

$$\lambda \in (\Lambda_{II}^{2,1})^* \cap \mathbb{R}_+ \mathcal{P}_{II}, \quad (\lambda, \delta_i) \leq 0.$$

If $(\lambda, \delta_i) = 0$, then

$$c_\lambda = -c_\lambda = 0.$$

For nonzero terms, $(\lambda, \delta_i) < 0$. Since $(\rho, \delta_i) = -1$,

$$(a, \delta_i) \leq 0, \quad a \in \Delta(\Lambda_{II}^{2,1})_+^*.$$

$$a \in \mathbb{R}_{\geq 0} \Delta(\Lambda_{II}^{2,1})_+^* \cap \overline{C(\Lambda^{2,1})_+} = \mathbb{R}_{\geq 0} \mathcal{P}_{II}.$$

The congruences $m \equiv n \equiv l \equiv 1 \pmod{2}$ imply

$$a \in 2(\Lambda^{2,1})^* = \Lambda_{II}^{2,1}.$$

$$a = 0 \quad \text{or} \quad a \in \Lambda_{II}^{2,1} \cap \mathbb{R}_{>0} \mathcal{P}_{II}.$$

LEMMA 9.6.

$$\frac{1}{64}\Delta_5(Z) = \sum_{w \in W^{(z)}(\Lambda_{II}^{2,1})} \det(w) (\exp(-\pi i(w(\rho), z)) - \sum_{a \in \Lambda_{II}^{2,1} \cap \mathbb{R}_{>0} \mathcal{P}_{II}} m(a) \exp(-\pi i(w(\rho + a), z)))$$

Proof. Let c_λ be the coefficient of $e^{-\pi i(\lambda, z)}$ in $64^{-1}\Delta_5$. The leading term has

$$\lambda = \rho, \quad c_\rho = 1.$$

The other exponents in the chosen chamber have

$$\lambda = \rho + a, \quad a \in \Lambda_{II}^{2,1} \cap \mathbb{R}_{>0} \mathcal{P}_{II}, \quad c_{\rho+a} = -m(a).$$

Anti-invariance:

$$c_{w\lambda} = \det(w)c_\lambda.$$

If λ lies on a simple wall, then

$$s_{\delta_i}\lambda = \lambda, \quad c_\lambda = -c_\lambda = 0.$$

Summing the signed Weyl translates proves the formula. □

The primitive isotropic rays of $\overline{\mathcal{P}_{II}}$ are generated by

$$2f_2, \quad 2f_{-2}, \quad 2f_2 - 2f_3 + 2f_{-2}.$$

The group $\text{Aut}(\mathcal{P}_{II}) \simeq S_3$ is transitive on these three vectors. For $a_0 = 2f_2$:

$$1 + \frac{1}{64} \sum_{t \geq 1} c_\Delta(1 + 2t, 1, 1) q^t = \prod_{k \geq 1} (1 - q^k)^9$$

In denominator notation,

$$1 - \sum_{t \geq 1} m(ta_0) q^t = \prod_{t \geq 1} (1 - q^t)^{\tau(ta_0)}, \quad \tau(ta_0) = 9 \quad (t \geq 1).$$

By S_3 -symmetry,

$$\tau(ta) = 9 \quad (t \geq 1)$$

for every primitive isotropic ray

$$\mathbb{Z}_{>0} a \subset \overline{\mathcal{P}_{II}} \cap \Lambda_{II}^{2,1}.$$

On isotropic rays, the coefficients $m(ta)$ are determined by the eta-product expansion; the signed simple-root supermultiplicity is $\tau(ta)$. The imaginary simple roots of $\mathfrak{g}_{\Delta_5}$ are these isotropic and timelike rays in $\overline{\mathcal{P}_{II}} \cap \Lambda_{II}^{2,1}$, with multiplicities $\tau(a)$ on isotropic rays and $m(a)$ on negative-norm rays [?GN, Theorem 2.1], [?GNII, (5.1), Section 5.1.1].

The real root datum is the wall set

$$\mathcal{P}_{II, \text{prim}} = \{\delta_1, \delta_2, \delta_3\}$$

with Gram matrix [?GN, Theorem 4.1] (and [?GNII, Section 5.1.1, case (t=1, II, 0)] as secondary)

$$(\delta_i, \delta_j) = \begin{pmatrix} 2 & -2 & -2 \\ -2 & 2 & -2 \\ -2 & -2 & 2 \end{pmatrix}$$

Definition 9.7 (Simple real roots and perturbative walls).

$$\Delta_{\text{simp}}^{re} = \{\delta_1, \delta_2, \delta_3\}, \quad \Delta^{re} = W^{(2)}(\Lambda_{II}^{2,I})\Delta_{\text{simp}}^{re},$$

$$\mathcal{W}_{\text{pert}} = \bigcup_{\delta \in \Delta_{\text{simp}}^{re}} \mathbb{H}_{\delta}.$$

10 THE DENOMINATOR ALGEBRA

$$A = ((\delta_i, \delta_j))_{1 \leq i, j \leq 3} = \begin{pmatrix} 2 & -2 & -2 \\ -2 & 2 & -2 \\ -2 & -2 & 2 \end{pmatrix}$$

A symmetric generalized Cartan, $\mathfrak{g}(A)$ Kac–Moody.

$$\mathcal{A}_{\text{den}} := \mathfrak{g}_{\Delta_s}.$$

This is the target-internal algebraic denominator square root. It is the automorphic correction of $\mathfrak{g}(A)$ constructed by Gritsenko–Nikulin [?GN, Sections 3–4], with root spaces graded by the Gram-degree map

$$\gamma = (n, l, m) \mapsto \alpha(\gamma) = 2nf_2 - lf_3 + 2mf_{-2}.$$

The Borcherds correction character in the coefficient chamber is

$$C_{\text{im}}(z) = 1 - \sum_{a \in \Lambda_{II}^{2,I} \cap \mathbb{R}_{>0} \mathcal{P}_{II}} m(a) e^{-2\pi i(a, z)}.$$

$$e^{-2\pi i(\rho, z)} C_{\text{im}}(z).$$

Signed root-superdimension convention; Harvey–Moore is the external physical interpretation [?HM]:

$$\text{smult}(\alpha) = \dim \mathfrak{g}_{\alpha, \bar{0}} - \dim \mathfrak{g}_{\alpha, \bar{1}}.$$

Let $\mathcal{R}$ be the root set of $\mathfrak{g}_{\Delta_s}$, and let $\mathcal{R}_+$ be the positive root set in the $\mathcal{P}_{II}$ -chamber.

Definition 10.1 (Denominator Lie superalgebra). For $\gamma \in \Gamma_{\text{ind}}$ set

$$\mathcal{A}_{\text{den}, \gamma} := \begin{cases} \Lambda_{II}^{2,I} \otimes \mathbb{C}, & \gamma = \mathbf{o}, \\ \mathfrak{g}_{\Delta_s, \alpha(\gamma)}, & \alpha(\gamma) \in \mathcal{R}, \\ \mathbf{o}, & \alpha(\gamma) \notin \mathcal{R}. \end{cases}$$

The zero degree is the Cartan subalgebra and is not governed by $f(\mathbf{o}, \mathbf{o})$. The bracket is the Borcherds–Kac–Moody bracket of $\mathfrak{g}_{\Delta_s}$:

$$[\cdot, \cdot] : \mathcal{A}_{\text{den}, \gamma} \otimes \mathcal{A}_{\text{den}, \gamma'} \longrightarrow \mathcal{A}_{\text{den}, \gamma + \gamma'}.$$

THEOREM 10.2 (Denominator algebra identity). The denominator Lie superalgebra has the following properties.

$$\mathcal{A}_{\text{den}} = \mathcal{A}_{\text{den}, \mathbf{o}} \oplus \bigoplus_{\substack{\gamma \in \Gamma_{\text{ind}} \\ \alpha(\gamma) \in \mathcal{R}}} \mathcal{A}_{\text{den}, \gamma} = \mathcal{A}_{\text{den}, \mathbf{o}} \oplus \mathcal{A}_{\text{den}}^{\text{bulk}} \oplus \mathcal{A}_{\text{den}}^{\text{cusp}}, \quad [\mathcal{A}_{\text{den}, \gamma}, \mathcal{A}_{\text{den}, \gamma'}] \subset \mathcal{A}_{\text{den}, \gamma + \gamma'},$$

where $\mathcal{A}_{\text{den}}^{\text{bulk}} = \bigoplus_{\alpha(\gamma) \in \mathcal{R}, m(\gamma) > 0} \mathcal{A}_{\text{den}, \gamma}$ records the $m > 0$ Borcherds bulk and $\mathcal{A}_{\text{den}}^{\text{cusp}} = \bigoplus_{\alpha(\gamma) \in \mathcal{R}, m(\gamma) = 0} \mathcal{A}_{\text{den}, \gamma}$ records the $m = 0$ Weyl-chamber cusp boundary, in parallel with the $\mathcal{V} = \mathcal{V}_{\text{bulk}} \oplus \mathcal{V}_{\text{cusp}}$ splitting at the determinant level above. Its visible positive signed Gram-degree multiplicities are the K_3 half-index coefficients:

$$\text{smult}(\alpha(n, l, m)) = f(nm, l) \quad ((n, l, m) \in \Gamma_{\text{act}}).$$

Its Weyl–Kac–Borcherds denominator is

$$\text{den}(\mathcal{A}_{\text{den}}) = 64^{-1} \Delta_5(2Z).$$

The theorem concerns the Borcherds denominator algebra. A microscopic BPS Hilbert space and its operator product require separate construction.

Proof. Gritsenko–Nikulin [?GNII, Theorem 2.1, (2.1)–(2.3)] construct $\mathfrak{g}_{\Delta_5}$ as the generalized Borcherds–Kac–Moody superalgebra whose real simple roots are $\delta_1, \delta_2, \delta_3$ and whose imaginary simple-root multiplicities are read from the Borcherds correction character. A product exponent gives the signed multiplicity of a visible positive root; the simple imaginary-root datum is Gritsenko–Nikulin’s $m(a)$ for negative-norm a and $\tau(a)$ on isotropic rays. These are different pieces of the denominator construction. A root-graded Lie superalgebra has

$$[\mathfrak{g}_{\Delta_5, \alpha}, \mathfrak{g}_{\Delta_5, \beta}] \subset \mathfrak{g}_{\Delta_5, \alpha + \beta}.$$

Here $\alpha \in \Lambda_{II}^{2,1} \otimes \mathbb{C}$, and the addition $\alpha + \beta$ is linear inside the Cartan; this is the linear root grading of a Borcherds–Kac superalgebra. The Gram-degree map

$$\gamma = (n, l, m) \mapsto \alpha(\gamma) = 2nf_2 - lf_3 + 2mf_{-2} \in \Lambda_{II}^{2,1} \otimes \mathbb{C}$$

is the inverse correspondence of root indexing: it is additive in (n, l, m) and recovers the linear addition in $\Lambda_{II}^{2,1} \otimes \mathbb{C}$. This linear additivity is the relevant statement for the denominator algebra of Δ_5 , where the grading lattice is $\Lambda_{II}^{2,1}$ and $\Gamma_{\text{ind}} = \mathbb{Z}^3$ is the Gram parametrization of that grading. In a hypothetical Hall source, the underlying lattice is the physical $\Gamma_X^{\text{phys}} = \Lambda_S \oplus \Lambda_S$, the Hall bracket is graded by Γ_X^{phys} (additive), and the Γ_{ind} -grading of the denominator algebra is the Π -image, well defined modulo the cocycle B of Definition 7.1. The denominator identity reads

$$\frac{1}{64} \Delta_5(2Z) = e^{-2\pi i(\rho, z)} \prod_{\alpha \in \mathcal{R}_+} (1 - e^{-2\pi i(\alpha, z)})^{\text{smult}(\alpha)}.$$

Comparing this with the Borcherds product gives

$$\text{smult}(\alpha(n, l, m)) = f(nm, l) \quad ((n, l, m) \in \Gamma_{\text{act}}).$$

□

Definition 10.3 (Formal current envelope). Let E be the elliptic curve in $X = S \times E$. Put

$$\text{Cur}_E(\mathcal{A}_{\text{den}}) := \mathcal{A}_{\text{den}} \otimes \mathcal{O}_E$$

with the level-zero current bracket

$$[a \otimes f, b \otimes g] = [a, b] \otimes fg.$$

Its chiral enveloping algebra in the sense of Beilinson–Drinfeld [?BD] is

$$\text{Den}_{\Delta_5, E} := U_E^{\text{ch}}(\text{Cur}_E(\mathcal{A}_{\text{den}})).$$

PROPOSITION 10.4 (*Formal current envelope*). $\text{Den}_{\Delta_3, E}$ is the formal completed ind Beilinson–Drinfeld current envelope on E obtained from the denominator Lie superalgebra. Its constant-current Lie superalgebra is $\mathcal{A}_{\text{den}}$, and the denominator of that constant-current bracket is

$$64^{-1} \Delta_3(2Z).$$

A compact E_3 /Hall/chiral BPS realization requires additional microscopic input.

Proof. In the completed ind setting, the current construction sends a Lie superalgebra to a Lie- * algebra on a smooth curve, and the chiral enveloping construction sends a Lie- * algebra to a chiral algebra [?BD]; equivalently, in vertex-algebra language, this is the universal current vertex algebra at level zero [?KACVA]. Constant sections of $\mathcal{A}_{\text{den}} \otimes \mathcal{O}_E$ recover the original Lie superalgebra. The denominator statement is exactly Theorem 10.2. $\square$

Definition 10.5 (*Holomorphic E_3 quantum upgrade*). The determinant and denominator results above are protected Euler-characteristic statements. The expected quantum upgrade does not start from the scalar determinant. It would start from a supplied holomorphic E_3 -factorization algebra on the Calabi–Yau threefold

$$\mathcal{A}_X^{E_3}, \quad X = K_3 \times E.$$

By a holomorphic E_3 -factorization algebra here we mean: a factorization algebra on X in the sense of Costello–Gwilliam ([?CostelloGwilliam1], [?CostelloGwilliam2] esp. vol. II §10–11), locally constant in real directions and holomorphic in three complex directions, whose factorization structure maps are governed by holomorphic polydiscs in $\mathbb{C}^3$; equivalently a factorization algebra in the holomorphic E_3 -operadic sense, with the framed E_3 -structure encoding the $SO(6)$ -rotational symmetry of $\mathbb{C}^3$. This symbol, and the placeholder $\Phi_3^{\text{FA}}(X)$, denote the Stage-1 object required in the Calabi–Yau-to-chiral construction; they are not a constructed functor evaluated on X . The candidate concrete realization, supplied as proposal-only schema, is the local protected observable algebra $\mathcal{A}_X(U)$ of Definition 2.1. A compact holomorphic-Chern–Simons/BV model, if constructed with the required descent, quantum master equation, and anomaly cancellation, would be a candidate source for this datum in the completed charge category. Such a source also requires a chosen E_3 -formality point, a chain-level S^3 -framing witness, compact-support control, and finite-stage HN quotients on which all operations are defined before passing to the completed inverse system.

The curve-level BPS algebra would be obtained only after a supplied Stage-2 specialization along the K_3 -direction. We split the triple identification into three named pieces, each carrying a separately labelled comparison map:

$$\mathcal{A}_X^{E_3} \xrightarrow{\text{Sp}_{K_3, E}^{\text{ch}}} \mathcal{F}_X^{(o)} \xrightarrow{\kappa_E} \mathcal{F}_X^{(i)} \xrightarrow{\pi_*^{\text{ch}}} (p_E)_* \mathcal{A}_X^{E_3}.$$

Here: $\text{Sp}_{K_3, E}^{\text{ch}}$ is the Beilinson–Drinfeld chiral pushforward along the K_3 -fibration $p_E : K_3 \times E \rightarrow E$ with K_3 -integration in the Costello–Gwilliam sense – pointwise on E , the fibre K_3 -direction of the factorization algebra is integrated to a chiral shadow on E ; its output $\mathcal{F}_X^{(o)}$ is a chiral algebra on E (equivalently a factorization algebra on $\text{Ran}(E)$). κ_E is the chiral-Koszul comparison $\mathcal{F}_X^{(o)} \rightarrow \mathcal{F}_X^{(i)}$, where $\mathcal{F}_X^{(i)}$ is the candidate identification of the chiral pushforward with the ordinary derived pushforward $(p_E)_* \mathcal{A}_X^{E_3}$ regarded as a sheaf-of-cochain factorization algebra on $\text{Ran}(E)$; identification of chiral and ordinary pushforward is the open chiral-Koszul comparison. π_*^{ch} is the closing identification

of $\mathcal{F}_X^{(i)}$ with $(p_E)_*\mathcal{A}_X^{E_3}$ as factorization data, comparing the Costello–Gwilliam push along a proper holomorphic submersion with the topological pushforward; this comparison reduces to the standard fact in the projection-finite sectors and is open in the wrapped sectors. The convenient shorthand $\mathcal{F}_X$ refers to any one of $\mathcal{F}_X^{(o)}$, $\mathcal{F}_X^{(i)}$, or $(p_E)_*\mathcal{A}_X^{E_3}$ once κ_E and π_*^{ch} have been supplied. Without the chiral-Koszul comparison and the proper-pushforward identification, the three names are distinct objects. Restrictions to chambers, pushforwards to E , and chiral Koszul dualities are then operations on $\mathcal{A}_X^{E_3}$ and on its E_1 -chiral shadow $\mathcal{F}_X$, not replacements for the E_3 -source.

After these data have been constructed, the required Igusa statement is the pair of comparisons

$$H^o \text{Prim}_{\text{prot}}(\Pi_{X,*}\mathcal{F}_X) \stackrel{?}{\cong} \mathfrak{g}_{\Delta_3}, \quad \Omega_E^{\text{ch}} C_X \stackrel{?}{\cong} \text{Den}_{\Delta_3,E}.$$

Here C_X is the chiral bar coalgebra attached to the shadow $\mathcal{F}_X$. The first comparison would be the bracket theorem. The second would be the chiral Koszul theorem. Both require microscopic data beyond the scalar square and beyond the formal current envelope.

Open Problem 10.6 (Compact holomorphic E_3 source). Construct the source object $\mathcal{A}_X^{E_3}$ for $X = K_3 \times E$. A candidate compact holomorphic-Chern–Simons/BV construction must prove, before any comparison with Δ_3 , the following finite-chain data:

- (i) descent for local observables on X and a quantum master equation after compact charge completion;
- (ii) anomaly cancellation, a fixed E_3 -formality point (in the sense of Tamarkin formality for E_n -operads; Tamarkin (1998), Kontsevich (1999); the framed- E_3 refinement is Severa–Willwacher (2011)), a chain-level S^3 -framing witness (a chain-level lift of the E_3 -framed-vs.-unframed comparison witnessed by the $SO(3)$ -action on $S^2 = \partial D^3$, giving the $SO(3)$ -rotational symmetry data; equivalently a chain-level $SO(3)$ -equivariance of the local observable algebra), and compact-support control;
- (iii) a finite-first sectorial completion compatible with Harder–Narasimhan quotients, with all operations defined on finite stages before completion;
- (iv) the chain-level K_3 -to- E specialization $\text{Sp}_{K_3,E}^{\text{ch}}$;
- (v) the projection-finite or hybrid wrapped repair for positive elliptic degree;
- (vi) a strong orientation, protected integration, and primitive recognition certificate.

Existing BV, CoHA, and chiral-algebra formalisms give the language for these requirements. They do not, by themselves, construct this compact source for $K_3 \times E$. The named external open data are: the E_3 -formality point (Tamarkin/Kontsevich formality at a chosen associator, of which the Drinfeld–Kontsevich associator is the canonical choice; rationality of formality is open over $\mathbb{Q}$ but known over $\mathbb{R}$), and the chain-level S^3 -framing witness for the holomorphic E_3 -structure (existence at the level of homotopy types follows from framed- E_n formality; chain-level constructibility for the candidate concrete realization $\mathcal{A}_X(U)$ of Definition 2.1 is open).

Remark 10.7 (Where the quantum comparison would live). Assume the source algebra $\mathcal{A}_X^{E_3}$, the specialization $\text{Sp}_{K_3,E}^{\text{ch}}$, a conilpotent coalgebra $C_X \stackrel{?}{\cong} \bar{B}_E^{\text{ch}}(\mathcal{F}_X)$, and a quasi-isomorphism

$$\Theta_{\text{Kos}} : \Omega_E^{\text{ch}} C_X \stackrel{?}{\cong} \text{Den}_{\Delta_3,E}$$

have all been supplied. Then $\mathcal{F}_X$ is the place where the protected primitive bracket and the chiral Koszul comparison must be formulated. Costello–Gwilliam locality explains why a holomorphic quantum theory would produce factorization products before any trace or determinant is taken [?CostelloRenormalization], [?CostelloGwilliam1], [?CostelloGwilliam2]; Beilinson–Drinfeld chiral operations are the curve-level algebraic form of collision maps [?BD]. None of this is obtained from the scalar square or from the denominator identity alone.

Remark 10.8 (Pushforward, restriction, Koszul duality). The second step is specialization, not inversion. A restriction to a chamber or sector is a restriction of $\mathcal{A}_X^{E_3}$ before, or of $\mathcal{F}_X$ after, the K_3 -specialization. Pushforward along p_E is the geometric notation for the same specialization. A compact Koszul theorem would require a supplied conilpotent coalgebra C_X , an identification

$$C_X \stackrel{?}{\simeq} \overline{B}_E^{\text{ch}}(\mathcal{F}_X),$$

and a further quasi-isomorphism

$$\Omega_E^{\text{ch}} C_X \stackrel{?}{\simeq} \text{Den}_{\Delta, E}.$$

These are not procedures for reconstructing the holomorphic E_3 -source from the denominator alone.

PROPOSITION 10.9 (Target-internal bar-cobar inversion). In the conilpotent / augmented chiral setting where Francis–Gaitsgory chiral Koszul duality applies, the formal target $\text{Den}_{\Delta, E}$ has its own bar-cobar inversion:

$$\Omega_E^{\text{ch}} \bar{B}_E^{\text{ch}}(\text{Den}_{\Delta, E}) \longrightarrow \text{Den}_{\Delta, E}.$$

The conilpotency hypothesis is satisfied here because $\mathcal{A}_{\text{den}}$ carries a charge filtration with finite dimensional graded pieces in each Γ_{ind} -degree, and this filtration lifts to the chiral envelope $\text{Den}_{\Delta, E}$; the augmentation is the constant-current Cartan trace. Cf. Kac [?KACVA] for the underlying vertex algebra structure, Beilinson–Drinfeld [?BD] §3.7 for the chiral bar-cobar formalism, and Francis–Gaitsgory (2012) Chiral Koszul Duality, Theorem 6.2.2, for the inversion in the conilpotent setting. This inversion is an internal reconstruction of the already defined current envelope. It is not the compact Koszul comparison

$$\Omega_E^{\text{ch}} C_X \stackrel{?}{\simeq} \text{Den}_{\Delta, E}$$

unless C_X has first been constructed from the compact $K_3 \times E$ factorization object and identified with $\bar{B}_E^{\text{ch}}(\mathcal{F}_X)$. The missing datum is the source coalgebra and the comparison map, not the target-side inversion.

Proof. Beilinson–Drinfeld bar-cobar formalism is an adjunction on chiral algebras and conilpotent chiral coalgebras on the curve [?BD, §3.7]. Applying it to the known target $\text{Den}_{\Delta, E}$ produces a tautological counit map back to that target; conilpotency comes from the charge filtration with finite graded pieces, in the form required by the Francis–Gaitsgory inversion theorem. No moduli stack of compact objects, collision coproduct, orientation system, or protected integration map enters this construction. Those are precisely the extra data in Definition 10.32. $\square$

Remark 10.10 (Locality). In a compact realization, the required local object would be a finite-stage holomorphic E_3 -factorization algebra on X , and its K_3 -specialization would be a finite-stage chiral factorization algebra on $\text{Ran}(E)$. This assertion is local only for projection-finite sectors over E . The automorphic product is a protected global trace after charge completion, wall-chamber choice, and

summation over compact K_3 -modes. Duality monodromy, elliptic-degree wrapping, and the Borchers chamber are therefore global features of the trace. They do not replace the local factorization maps on disjoint opens.

Definition 10.11 (Projection-finite and wrapped elliptic sectors). Let $p_E : K_3 \times E \rightarrow E$ be projection. A compact Hall or chiral sector is projection-finite over E if every object A in it satisfies

$$p_E(\text{Supp } A) \subset F$$

for some finite subset $F \subset E$. A one-dimensional sector is wrapped over E if its curve class has positive elliptic degree, i.e. if the pushforward of its one-cycle to E is $b[E]$ with $b > 0$.

Lemma 10.12 (Projection-to- E support-locality obstruction). Let $C \subset K_3 \times E$ be a one-dimensional subscheme whose curve class has positive elliptic degree:

$$(p_E)_*[C] = b[E], \quad b > 0.$$

Then $p_E(C) = E$. In particular, C is not localized at finitely many points of E , and it is not contained in $p_E^{-1}(U)$ for any proper open $U \subsetneq E$.

Proof. If $p_E(C)$ were zero-dimensional, then the pushforward of the one-cycle $[C]$ to the curve E would vanish. This contradicts $(p_E)_*[C] = b[E]$ with $b > 0$. Hence $p_E(C)$ is a one-dimensional closed subset of the irreducible curve E , since p_E is proper, and therefore $p_E(C) = E$. $\square$

Corollary 10.13 (Naive Ran locality misses the elliptic-degree direction). A support-local factorization algebra on $\text{Ran}(E)$, defined only by separating supports over disjoint opens of E , sees the projection-finite sector. It does not by itself realize the full s -direction of the Igusa product.

Proof. The variable s is the Borchers variable corresponding to the OP elliptic-degree direction in Proposition 6.1. By Lemma 10.12, positive elliptic degree projects onto all of E . Such objects cannot be separated by choosing disjoint opens in E . Therefore ordinary support-local factorization on $\text{Ran}(E)$ has no locality mechanism for the wrapped elliptic-degree sector. $\square$

Remark 10.14 (Repair alternatives). The obstruction is a defect of the proposed compact realization, not of the Borchers product. The determinant $\Delta_5 = \text{Borch}_\theta(\phi_{0,1})$ and the denominator identity for $\mathfrak{g}_{\Delta_5}$ remain theorem-level automorphic statements. What is missing is the compact local/global source whose protected primitive trace gives them.

Two honest repair paths visible in the present architecture are these.

- (i) Construct the projection-finite local Hall/chiral factorization object on $\text{Ran}(E)$, then prove that the missing positive elliptic-degree direction is supplied by global Fock/Hecke modes coupled to the local sector through mixed Hall actions. A decoupled Fock/Hecke factor can reproduce a product character, but it does not define the mixed primitive brackets needed for a compact realization.
- (ii) Construct a hybrid local/global wrapped factorization theory: local vertical insertions live on $\text{Ran}(E)$, while positive elliptic-degree sectors are global wrapped legs. This requires extension stacks with local and global legs, Thom–Sebastiani orientation transport, associativity over two-step flags, and compatibility with protected integration.

No theorem in this paper supplies either repair.

Open Problem 10.15 (Hybrid wrapped factorization). Construct the second repair path in the preceding remark. The datum should consist of: (i) a stratification of $\text{Ran}(E)$ by total elliptic degree $b \geq 0$, refining the stratification by finite support; (ii) a source category factorization-on- $\text{Ran}(E)$ which is constructible along this stratification, with projection-finite stratum carrying the Costello–Gwilliam local factorization theory and positive- b strata carrying wrapped legs; (iii) an explicitly named hybrid object $\mathcal{F}_X^{\text{hyb}}$ restricting on the $b = 0$ stratum to the local $\text{Ran}(E)$ -factorization algebra, and on the $b > 0$ strata to chiral modules over the projection-finite part with mixed extension correspondences carrying relative position in E before the reduced E -quotient; (iv) projection-finite local moduli $\mathcal{M}^{\text{loc}}$, wrapped elliptic-degree moduli $\mathcal{M}^{\text{wr}}$, and mixed extension correspondences

$$\mathbb{E}^{\text{loc}}, \quad \mathbb{E}^{\text{wr}}, \quad \mathbb{E}^{\text{hyb}}$$

remembering relative position in E before the reduced E -quotient. The correspondences must be associative over two-step flags, carry Thom–Sebastiani-compatible orientations, admit protected integration after completion, and recover the s -direction of the Igusa product in the decategorified trace. This object $\mathcal{F}_X^{\text{hyb}}$ is the candidate filling the (Sp) slot of Definition 10.32. Ordinary disjoint-open factorization on $\text{Ran}(E)$ is only the projection-finite ($b = 0$) part of such a theory.

Open Problem 10.16 (Π -descent of the protected bracket). Let $\Gamma_X^{\text{phys}} = \Lambda_S \oplus \Lambda_S$, $\Pi : \Gamma_X^{\text{phys}} \rightarrow \Gamma_{\text{ind}}$, $B : \Gamma_X^{\text{phys}} \times \Gamma_X^{\text{phys}} \rightarrow \Gamma_{\text{ind}}$ be as in Definition 7.1. Suppose a Hall lift $\tilde{P}_X$ of the protected primitive algebra is supplied with a Γ_X^{phys} -graded bracket. Prove that the cocycle B acts trivially on bracket targets in primitives, that is, that $[\tilde{P}_{X,c}, \tilde{P}_{X,c'}]$ is concentrated in physical charge $c + c'$ and that the Γ_{ind} -degree of $[x, y]$ for $x \in \tilde{P}_{X,c}$, $y \in \tilde{P}_{X,c'}$ is $\Pi(c) + \Pi(c')$, not the cocycle-corrected value $\Pi(c + c') = \Pi(c) + \Pi(c') + B(c, c')$. Equivalently, identify the generator on which B is exhibited as a Hochschild boundary in the Hall bracket category. This is a positivity-of-cocycle statement on the critical-Hall side; it is required for hypothesis (1b) of Theorem 10.33.

LEMMA 10.17 (*Explicit S^1 -equivariant strict-pentagon multiplicative orientation cocycle on the reduced compact protected sector of $K3 \times E$*). Adopt the notation of Definition 7.5 and Definition 4.3. Assume the rank-one orientation existence (P1) of Open Problem 5.7, and the HN/sectorial completion of Lemma 10.31, so that on each charge sector $c \in \Gamma_X^{\text{phys}}$ of the reduced compact protected derived moduli stack $\mathfrak{M}_{X,c}^{\text{red}}$ of compactly supported B-branes on $K3 \times E$ the orientation line o_c of Definition 4.3 is constructed. Then there exists an explicit family of multiplicative isomorphisms

$$\mu_{c,c'} : o_{c+c'} \xrightarrow{\sim} o_c \otimes o_{c'}, \quad c, c' \in \Gamma_X^{\text{phys}},$$

which is (i) natural in the Cartesian extension correspondence, (ii) S^1 -equivariant for the loop-rotation action on the cyclic complex of $\mathfrak{M}_{X,c}^{\text{red}}$, (iii) compatible with the reduced E -quotient, (iv) compatible with the HN/sectorial completion of Lemma 10.31, and (v) a strict pentagon cocycle: on every four-fold direct sum $c \oplus c' \oplus c'' \oplus c'''$ the two paths through the pentagon

$$o_{c+c'+c''+c'''} \xrightarrow{\sim} o_c \otimes o_{c'} \otimes o_{c''} \otimes o_{c'''}$$

agree as natural transformations of orientation lines.

In particular, hypothesis (O) of Theorem 10.19 is constructed, modulo the single residual obstruction (P3) of Open Problem 5.7 on higher-rank strata.

Proof. Step 1 (Künneth factorization of the protected moduli). Write $X = S \times E$, with $p_S : X \rightarrow S$ and $p_E : X \rightarrow E$ the projections. Compactly supported B-branes C of physical charge $c = (Q, P) \in \Lambda_S \oplus \Lambda_S$ decompose under Künneth as

$$C = C_S \boxtimes \mathcal{O}_E + C'_S \boxtimes \omega_E, \quad [C_S] = P, [C'_S] = Q \in \Lambda_S,$$

in the even $K3 \times E$ Mukai sector (Definition 7.5). Pantev–Toen–Vaquié–Vezzosi [?PTVV2013] furnish a Künneth decomposition of (-1) -shifted symplectic structures on the moduli of perfect complexes,

$$\mathfrak{M}_{X,c} \simeq \mathfrak{M}_{S,c}^{\text{red}} \times E,$$

where the $K3$ factor $\mathfrak{M}_{S,c}^{\text{red}}$ carries a (-2) -shifted symplectic form $\omega_S^{(-2)}$ (after reduction by the $K3$ cosection of Maulik–Toda / Oberdieck [?OberdieckReducedPT], in the sense made precise by the Pfaffian local model of Definition 5.4), and the elliptic factor is the trivial flat translation torsor under E . The Costello–Li flat Calabi–Yau pairing [?CostelloRenormalization] on E is parallel; multiplication by the volume class ω_E is the Mukai dilation that lowers the shift by one and presents $\mathfrak{M}_{X,c}$ as the Mukai dilation of $\mathfrak{M}_{S,c}^{\text{red}}$ along E :

$$\omega_X^{(-1)} = \omega_S^{(-2)} \boxtimes (\omega_E \smile -).$$

Step 2 (Orientation line as Pin^c lift, Künneth-factored). By Definition 4.3, o_c is the square root of $\det \text{RHom}(C, C)$, constructed as the Pin^c lift of the SO -frame bundle of the reduced derived self-Ext bundle on $\mathfrak{M}_{X,c}^{\text{red}}$ (Cao–Kool–Monavari [?CaoKoolMonavari] on the rank-one D6–D2–D0 stratum, twisted by the cosection determinant of [?OberdieckReducedPT]). The Künneth decomposition of Step 1 factorizes the derived self-Ext bundle:

$$\det \text{RHom}_X(C, C) \simeq \det \text{RHom}_S(C_S, C_S) \boxtimes \det \text{RHom}_E(\mathcal{O}_E, \mathcal{O}_E),$$

and the elliptic factor is canonically trivial because $\det \text{RHom}_E(\mathcal{O}_E, \mathcal{O}_E)$ is the trivial line bundle under the Costello–Li flat Calabi–Yau trivialization of the holomorphic volume on E [?CostelloRenormalization]. Hence

$$o_c \simeq o_{[C_S]}^S \boxtimes o_{\text{flat}}^E, \quad o_{\text{flat}}^E = \underline{\mathbb{C}}_E,$$

and the orientation line on $K3 \times E$ is, up to the canonical trivial elliptic factor, the $K3$ orientation line on the reduced $K3$ moduli.

Step 3 (Multiplicative cocycle from the $K3$ factor). On the $K3$ factor, the (-2) -shifted symplectic moduli $\mathfrak{M}_{S,c}^{\text{red}}$ admits, by Joyce–Kinjo–Park–Safronov [?JoyceKinjoParkSafronov], a strict-pentagon multiplicative orientation cocycle on the moduli of perfect complexes,

$$\mu_{P,P'}^S : o_{P+P'}^S \xrightarrow{\sim} o_P^S \otimes o_{P'}^S, \quad P, P' \in \Lambda_S,$$

constructed via the direct-sum operation $\oplus : \mathfrak{M}_{S,P}^{\text{red}} \times \mathfrak{M}_{S,P'}^{\text{red}} \rightarrow \mathfrak{M}_{S,P+P'}^{\text{red}}$ and the Joyce–Tanaka–Upmeyer transport across direct sums [?JoyceTanakaUpmeyer], refining the Joyce–Upmeyer multiplicative cocycle of [?JoyceUpmeyer] for Calabi–Yau threefolds to the (-2) -shifted symplectic $K3$ setting. Define

$$\mu_{c,c'} := \mu_{P,P'}^S \boxtimes \mu_{Q,Q'}^S \boxtimes (\text{id} : o_{\text{flat}}^E \rightarrow o_{\text{flat}}^E \otimes o_{\text{flat}}^E),$$

where the elliptic factor isomorphism is the canonical $\underline{\mathbb{C}}_E \simeq \underline{\mathbb{C}}_E \otimes \underline{\mathbb{C}}_E$ of the trivialized flat line. Concretely, on a pair (C, C') of physical charges $(c, c') = ((Q, P), (Q', P'))$,

$$\mu_{c,c'}(\sigma_{C \oplus C'}) = \sigma_C \otimes \sigma_{C'},$$

where each σ is the Pin^c frame transported across the direct sum by the JKPS recipe on the K_3 factor. This is a chain-level description, not a quotation: the recipe on each side of the tensor sign is identical to the direct-sum-equivariant Pin^c lift of $[?JoyceUpmeyer]$ on the K_3 factor.

Step 4 (S^1 -equivariance). The cyclic complex of $\mathfrak{M}_{X,c}^{\text{red}}$ carries a loop-rotation S^1 -action whose generator L_{S^1} restricts on the Künneth factor to $L_{S^1}^S \boxtimes 1 + 1 \boxtimes L_{S^1}^E$ by Künneth additivity of cyclic homology. The K_3 cyclic generator $L_{S^1}^S$ acts on o_p^S by the S^1 -equivariant Pin^c lift of Joyce–Kinjo–Park–Safronov $[?JoyceKinjoParkSafronov]$, which is strictly equivariant by their construction (the orientation cocycle is intertwined with the loop-rotation action because the multiplicative isomorphism is constructed equivariantly from the direct-sum operation, which commutes with loop rotation). The elliptic generator $L_{S^1}^E$ acts trivially on o_{flat}^E because the Costello–Li trivialization is parallel for the flat connection; the flat E -translation S^1 -action preserves the trivialization. Both factors of $\mu_{c,c'}$ are therefore S^1 -equivariant, and the Künneth tensor product is S^1 -equivariant by additivity of the generator. Hence $\mu_{c,c'}$ is S^1 -equivariant.

Step 5 (Reduced- E -quotient and HN compatibility). The reduced E -quotient by translation $\mathfrak{M}_{X,c}^{\text{red}}/E$ is geometrically the K_3 factor $\mathfrak{M}_{S,c}^{\text{red}}$ (Step 1). Since the elliptic factor of $\mu_{c,c'}$ is the trivial trivial-line isomorphism, $\mu_{c,c'}$ descends to the K_3 quotient as $\mu_{p,p'}^S \boxtimes \mu_{q,q'}^S$, giving compatibility with the reduced E -quotient of Definition 5.4. Compatibility with the HN/sectorial completion of Lemma 10.31 is automatic on each finite-height truncation F_R because $\mu_{p,p'}^S$ is constructed sector-locally and the projective limit $\widehat{\mathcal{H}}_{\sigma,S}^{\text{or}}$ carries the inverse limit of the $\mu_{c,c'}$'s.

Step 6 (Strict pentagon). On a four-fold direct sum $c \oplus c' \oplus c'' \oplus c'''$, the two paths through the pentagon are

$$\begin{aligned} \text{Path I: } \quad o_{c+c'+c''+c'''} &\xrightarrow{\mu_{c+c'+c'',c'''}} o_{c+c'+c''} \otimes o_{c'''} \\ &\xrightarrow{\mu_{c+c',c''} \otimes \text{id}} o_{c+c'} \otimes o_{c''} \otimes o_{c'''} \\ &\xrightarrow{\mu_{c,c'} \otimes \text{id} \otimes \text{id}} o_c \otimes o_{c'} \otimes o_{c''} \otimes o_{c'''}, \\ \text{Path II: } \quad o_{c+c'+c''+c'''} &\xrightarrow{\mu_{c,c'+c'',c'''}} o_c \otimes o_{c'+c''+c'''} \\ &\xrightarrow{\text{id} \otimes \mu_{c',c''+c'''}} o_c \otimes o_{c'} \otimes o_{c''+c'''} \\ &\xrightarrow{\text{id} \otimes \text{id} \otimes \mu_{c'',c'''}} o_c \otimes o_{c'} \otimes o_{c''} \otimes o_{c'''}. \end{aligned}$$

Both paths factor as a tensor of K_3 paths and the trivial elliptic path; the elliptic paths are tautologically equal because each is the identity $\underline{\mathbb{C}}_E \rightarrow \underline{\mathbb{C}}_E^{\otimes 4}$. On the K_3 factor, the JKPS pentagon $[?JoyceKinjoParkSafronov]$ for $\mu_{p,p'}^S$ gives equality of the two K_3 paths. By cyclic invariance of the Künneth pairing under the symmetric group Sym_4 acting on the four legs, the orientation isomorphisms permute coherently with the Künneth factorization, and no sign or scalar discrepancy can appear between the two paths. Thus Path I and Path II coincide as natural transformations of orientation lines, on the nose. This is the strict pentagon.

Step 7 (Residual). The construction depends on the orientation existence on the K_3 factor for charges of arbitrary rank $P, Q \in \Lambda_S$; the rank-one case is (P1) of Open Problem 5.7. Higher-rank existence is

precisely (P₃) of the same Open Problem. Granted (P₃), the multiplicative cocycle $\mu_{c,c'}$ extends from the rank-one stratum to the full HN-completed sector by the Joyce–Tanaka–Upmeyer extension theorem [?JoyceTanakaUpmeyer] for orientation transport across HN direct-sum filtrations, applied factor-by-factor to the K₃ Künneth slot. $\square$

Remark 10.18 (Pin^c gerbe interpretation and BBDJS comparison). The orientation line o_c is the Pin^c lift of the SO-frame bundle of the reduced derived self-Ext on $\mathfrak{M}_{X,c}^{\text{red}}$, as in Definition 4.3; the multiplicative cocycle $\{\mu_{c,c'}\}$ of Lemma 10.17 is the splitting of the Pin^c gerbe along the additive direct-sum operation of charge sectors, in the chain-level vanishing-cycle formalism of Brav–Bussi–Dupont–Joyce–Szendrői [?BBDJS2015]. The construction is the (−2)-shifted symplectic K₃ analogue of the Joyce–Upmeyer [?JoyceUpmeyer] cocycle for Calabi–Yau threefolds, dilated by the Costello–Li flat trivialization [?CostelloRenormalization] on the elliptic factor and reduced by the cosection determinant of the Maulik–Toda / Oberdieck reduced obstruction theory [?OberdieckReducedPT].

THEOREM 10.19 (*PTVV–Joyce Π -descent of the primitive bracket; conditional sharpening of Open Problem 10.16*). Adopt the notation of Definition 7.1 and Definition 7.5: physical lattice $\Gamma_X^{\text{phys}} = \Lambda_S \oplus \Lambda_S$, Gram shadow $\Pi_X : \Gamma_X^{\text{phys}} \rightarrow \Gamma_{\text{gram}} = \mathbb{Z}^3$, symmetric quadratic cocycle $B(c, c') = (Q \cdot Q', Q \cdot P' + Q' \cdot P, P \cdot P')$ satisfying $\Pi_X(c + c') = \Pi_X(c) + \Pi_X(c') + B(c, c')$. Suppose:

- (P) (*PTVV host.*) The protected primitive layer $\tilde{P}_X = \bigoplus_c \tilde{P}_{X,c}$ is realized as $\tilde{P}_X = H^0 \text{Prim}_{\text{prot}}(\mathcal{F}_X)$ for an oriented critical Hall/factorization object $\mathcal{F}_X$ on a (−1)-shifted symplectic derived moduli stack $\mathfrak{M}_{X,c}$ of compactly supported B-branes on $K_3 \times E$ in the sense of Pantev–Toen–Vaquié–Vezzosi [?PTVV2013] (cf. the local schema of Definition 2.1), graded by physical charge $c \in \Gamma_X^{\text{phys}}$, with reduced E -quotient compatible with the projection $p_E : K_3 \times E \rightarrow E$ of Definition 5.4.
- (O) (*S¹-equivariant multiplicative orientation cocycle.*) The collection $\{\mu_{c,c'}\}$ is an S¹-equivariant multiplicative orientation cocycle on (−1)-shifted symplectic stacks satisfying the strict pentagon. Concretely: an orientation datum $\{o_c\}_{c \in \Gamma_X^{\text{phys}}}$ in the sense of Definition 4.3 is supplied with *Joyce–Kinjo–Park–Safronov multiplicative isomorphisms* [?JoyceKinjoParkSafronov]

$$\mu_{c,c'} : o_{c+c'} \xrightarrow{\sim} o_c \otimes o_{c'}, \quad c, c' \in \Gamma_X^{\text{phys}},$$

satisfying the strict cocycle pentagon $\mu_{c+c',c''} \circ \mu_{c,c'} = \mu_{c,c'+c''} \circ \mu_{c',c''}$ on the Cartesian extension correspondence, compatible with the reduced E -quotient and with the HN/sectorial completion of Lemma 10.31, and *equivariant under the loop-rotation S¹-action on the cyclic complex of $\mathfrak{M}_{X,c}$* . This last condition lifts the orientation cocycle from a Hochschild trivialization to a *cyclic* trivialization, and is exactly what Joyce–Kinjo–Park–Safronov [?JoyceKinjoParkSafronov] construct on (−1)-shifted symplectic stacks via the loop-rotation symmetry of the cyclic complex (the S¹-equivariance is intrinsic to the PTVV [?PTVV2013] (−1)-shifted symplectic structure, which arises from the negative-cyclic refinement of the trace pairing). Equivalently, the collection $\{o_c\}$ is an S¹-equivariant multiplicative section of the determinant gerbe over the moduli of objects in the protected sector.

- (F) (*Frobenius pairing.*) The protected Hall bracket $[\cdot, \cdot]$ and the protected Hopf pairing $\langle \cdot, \cdot \rangle$ on $\tilde{P}_X$ (the latter furnished by the squaring isomorphism $o_c^{\otimes 2} \simeq \det \text{RHom}(C, C)$ of Definition 4.3 and the PTVV [?PTVV2013] (−1)-shifted symplectic form on $\mathfrak{M}_{X,c}$, trivialized along the reduced

E -quotient via the Costello–Li flat Calabi–Yau pairing [?CostelloRenormalization]) satisfy the graded-Frobenius identity

$$\langle [x, y], z \rangle + (-1)^{|x| \cdot |y|} \langle y, [x, z] \rangle = 0$$

for homogeneous $x, y, z \in \tilde{P}_X$ of physical charges c_x, c_y, c_z with $c_x + c_y + c_z = 0$.

Then the cocycle B acts trivially on bracket targets in primitives:

$$[\tilde{P}_{X,c}, \tilde{P}_{X,c'}] \subset \tilde{P}_{X,c+c'} \cap \Pi_X^{-1}(\Pi_X(c + c')),$$

and the Hochschild class $[B] \in HH^2(\tilde{P}_X^{\text{br}}; \Gamma_{\text{gram}})$ of B in the primitive bracket category $\tilde{P}_X^{\text{br}}$ is a coboundary, $[B] = [d_{\text{Hoch}} \Theta] = 0$, where Θ is the determinant-line Hochschild 1-cochain produced from the multiplicative trivialization $\{\mu_{c,c'}\}$ of (O) . Consequently the bracket descends along Π_X to a Γ_{gram} -graded Lie superalgebra structure on

$$P_X^\Pi = \Pi_{X,*} \tilde{P}_X,$$

and hypothesis (1b) of Theorem 10.33 is satisfied.

Proof template (conditional on (P), (O), (F)). Step 1 (Hochschild realization of B). Pick a representative cocycle for the central extension $\Gamma_X = \Gamma_X^{\text{phys}} \oplus_B \Gamma_{\text{gram}}$ of Definition 7.9 and view $B(c, c') \in \Gamma_{\text{gram}}$ as a symmetric Hochschild 2-cochain on the abelian (graded) category $\tilde{P}_X^{\text{br}}$, valued in the central Γ_{gram} -coefficient module that records the Gram target degree of a bracket. Concretely, on a homogeneous pair $(x, y) \in \tilde{P}_{X,c} \times \tilde{P}_{X,c'}$, $B(c, c')$ measures the discrepancy between the *physical* target degree $c + c'$ and the naive Gram-additive target degree $\Pi_X(c) + \Pi_X(c')$. Symmetry of B and the identity $B(c, c) = 2\Pi_X(c)$ (Definition 7.1) make B an honest Hochschild 2-cocycle for the trivial bracket-target action.

Step 2 (Determinant-line trivialization Θ). The squaring isomorphism $o_c^{\otimes 2} \simeq \det \text{RHom}(C, C)$ of Definition 4.3, together with the multiplicative isomorphisms $\mu_{c,c'}$ of (O) , gives a trivialization of the determinant gerbe along the addition map $\Gamma_X^{\text{phys}} \times \Gamma_X^{\text{phys}} \rightarrow \Gamma_X^{\text{phys}}$. Take logarithms (in the sense of c_1 of orientation lines) to obtain a Hochschild 1-cochain

$$\Theta : \Gamma_X^{\text{phys}} \longrightarrow \Gamma_{\text{gram}}$$

whose Hochschild differential $d_{\text{Hoch}} \Theta(c, c') = \Theta(c + c') - \Theta(c) - \Theta(c')$ records the failure of additivity of the determinant degree, which is precisely the symmetric quadratic Gram defect. The pentagon for $\{\mu_{c,c'}\}$ translates into associativity of $d_{\text{Hoch}} \Theta$. The arithmetic identity $\Pi_X(c + c') = \Pi_X(c) + \Pi_X(c') + B(c, c')$ of Lemma 7.6 forces

$$d_{\text{Hoch}} \Theta = B \quad \text{in} \quad C^2(\Gamma_X^{\text{phys}}, \Gamma_{\text{gram}}).$$

This is the witness step: the multiplicative orientation provides Θ , and the quadratic identity for Π_X makes Θ an exact primitive of B .

Step 3 (Frobenius transport from pairing target to bracket target). The bracket on $\tilde{P}_X$ lifts to the central extension $\hat{\Gamma}_X = \Gamma_X^{\text{phys}} \oplus \Gamma_{\text{gram}}$ of Definition 7.9, in which the Hochschild cochain Θ defines a section $s_\Theta(c) = (c, \Theta(c))$. The Frobenius identity (F)

$$\langle [x, y], z \rangle + (-1)^{|x| \cdot |y|} \langle y, [x, z] \rangle = 0$$

forces the cocycle action of B on the bracket target to coincide with the cocycle action of B on the pairing target: the LHS measures the Gram degree shift of $[x, y]$ against z ; the RHS measures the Gram degree shift of $[x, z]$ against y ; both are computed by the same orientation cocycle $d_{\text{Hoch}}\Theta$. Consequently the cocycle action of B on bracket targets factors through $d_{\text{Hoch}}\Theta$. By Step 2, $d_{\text{Hoch}}\Theta = B$, so the discrepancy between *physical* target degree and *Gram* target degree is the Hochschild boundary of a single global cochain Θ , independent of (x, y) .

Step 4 (Cohomological vanishing on primitives – chain-level A_∞ descent). The previous three steps live one level below the hypothesis we owe Theorem 10.33 (1b): the multiplicative orientation of (O) is a higher-coherent system, the BBDJS [?BBDJS2015] Thom–Sebastiani fusion is a natural isomorphism rather than a strict equality, and the Tate construction $(-)^{t_{S^1}}$ that promotes the negative cyclic complex to a homotopy fixed-point object is lax monoidal in the sense of Nikolaus–Scholze [?NikolausScholze], not strong monoidal. Each of these subtleties prevents one from reading Θ and B as strictly associative chain-level cocycles; they are at best E_3 -coherent on $\mathcal{F}_X$ and E_2 -coherent on the bracket target. We address this directly by closing the descent at the homotopy-category level where Theorem 10.33 (1b) actually consumes it, and by recording precisely which chain-level upgrade is needed.

(4a) *Strictification.* Apply Lurie HA Theorem 4.1.8.4 [?LurieHA] to the protected $\mathcal{F}_X$: any E_3 -monoidal dg algebra is equivalent, in the ∞ -category of associative algebras, to a strictly associative dg algebra after passage to the homotopy category. Concretely, the Joyce–Kinjo–Park–Safronov [?JoyceKinjoParkSafronov] multiplicative orientation cocycle, the BBDJS [?BBDJS2015] Thom–Sebastiani fusion, and the PTVV [?PTVV2013] symplectic Frobenius pairing are coherent up to all higher associators (Stasheff K_n for $n \leq 4$), and Lurie’s strictification provides a quasi-isomorphism $\mathcal{F}_X \xrightarrow{\sim} \mathcal{F}_X^{\text{str}}$ of associative dg algebras inducing an isomorphism on H° . At H° the K_4 /Stasheff coherence becomes vacuous: associativity is strict and the Hochschild differential is the ordinary one.

(4b) *Hochschild boundary on H° .* Pass Θ and B through $H^\circ\mathcal{F}_X^{\text{str}}$. By Step 2 the equality $d_{\text{Hoch}}\Theta = B$ holds in $C^2(\Gamma_X^{\text{phys}}, \Gamma_{\text{gram}})$ as cochains of abelian groups (no chain-level coherence is invoked at this point; Θ is the c_1 of orientation lines and B is the arithmetic cocycle of Lemma 7.6). The strictified bracket $H^\circ[\cdot, \cdot]$ is associative and graded-Frobenius for the strictified pairing, by descent of (F) through the quasi-isomorphism of (4a). Restrict to the protected primitive subspace $\tilde{P}_X = H^\circ \text{Prim}_{\text{prot}}(\mathcal{F}_X^{\text{str}})$, the kernel of the reduced coproduct, on which the Hopf pairing $\langle \cdot, \cdot \rangle$ is non-degenerate (Theorem 10.33 (6)). Non-degeneracy converts the Frobenius identity (F) from a target-degree statement into the strict relation that the bracket-target cocycle action of B equals the pairing-target cocycle action of B , both equal to $d_{\text{Hoch}}\Theta$ by Step 3. Hence $[B] \in HH^2(\tilde{P}_X^{\text{br}}; \Gamma_{\text{gram}})$ vanishes as a coboundary, and $[\tilde{P}_{X,c}, \tilde{P}_{X,c'}] \subset \tilde{P}_{X,c+c'} \cap \Pi_X^{-1}(\Pi_X(c + c'))$ as required. The induced Γ_{gram} -grading on $P_X^\Pi = \Pi_{X,*}\tilde{P}_X$ is a genuine Lie-superalgebra grading; hypothesis (1b) of Theorem 10.33 is met at H° , which is the level at which (1b) is stated.

(4c) *Compatibility with PTVV symplectic structure.* The strictification of (4a) is compatible with the (-1) -shifted symplectic form on $\mathfrak{M}_{X,c}$ in the following sense: PTVV [?PTVV2013] produce the symplectic form as a cohomology class in $H^2(\mathfrak{M}_{X,c}, \wedge^2 \mathbb{L}_{\mathfrak{M}_{X,c}}[-1])$, which is intrinsic to the ∞ -categorical homotopy type and is preserved by any ∞ -equivalence of derived stacks. The quasi-isomorphism $\mathcal{F}_X \rightarrow \mathcal{F}_X^{\text{str}}$ of (4a) thus carries the PTVV symplectic form to a strict pairing of the same shifted symplectic class on H° , and the Costello–Li flat Calabi–Yau trivialization [?CostelloRenormalization] along the reduced E -quotient passes through. No information about (P), (O), (F) is lost in passing from $\mathcal{F}_X$ to $H^\circ\mathcal{F}_X^{\text{str}}$.

(4d) *Residual chain-level obligation.* The argument above discharges the descent obligation at H° , which is the level consumed by Theorem 10.33 (1b) and, in turn, by the Borchers–Kac recognition of

Theorem 10.33 (5). A genuinely chain-level strict refinement – in which $\mathcal{F}_X$ itself, rather than its homotopy category, carries a strict Γ_{gram} -graded Lie superalgebra structure – requires an additional ingredient that is not available in the present literature: an S^1 -equivariant multiplicative orientation cocycle on the protected sector of $K_3 \times E$, so that the Goodwillie identification $HC^- = HH^{hS^1}$ is compatible with the JKPS [?JoyceKinjoParkSafronov] multiplicativity, together with a strictly associative model of BBDJS [?BBDJS2015] Thom–Sebastiani fusion (currently a natural isomorphism in the sense of Massey–Saito). We name this as hypothesis (O^∞) below; granting (O^∞) , the Lurie strictification of (4a) lifts to a strict E_3 -rectification on $\mathcal{F}_X$ itself, and the descent of (4b)–(4c) holds at chain level. Without (O^∞) , the descent holds at H^0 , which is all that the modular and denominator arguments of this paper consume.

Remark on the Goodwillie obstruction. The chain-level passage from the Hochschild boundary $d_{\text{Hoch}}\Theta = B$ of Step 2 to the cohomological vanishing $[B] = 0$ in the bracket category proceeds via Goodwillie’s identification $HC^- \simeq HH^{hS^1}$ of negative cyclic homology with the S^1 -homotopy fixed points of Hochschild homology [?Goodwillie1985]. The obstruction to lifting a Hochschild trivialization to a cyclic (i.e. S^1 -equivariant) trivialization sits in the Connes B -operator and obstructs the chain-level identification of $d_{\text{Hoch}}\Theta$ with B at the cyclic level. This obstruction is absorbed into the S^1 -equivariance clause of (O): the Joyce–Kinjo–Park–Safronov multiplicative orientation cocycle, by construction S^1 -equivariant for the loop-rotation action on the cyclic complex [?JoyceKinjoParkSafronov], furnishes Θ as a class in $HC^{-,1}$, not merely HH^1 ; the differential $d_{\text{Hoch}}\Theta = B$ of Step 2 is then equivariantly closed and the cohomological vanishing argument above is well-defined at the chain level. Without (O)’s S^1 -equivariance the Goodwillie obstruction obstructs the chain lift and Step 4 stops at a mere HH^2 -vanishing without the cyclic refinement that the Hopf–Frobenius identity (F) actually sees. $\square$

LEMMA 10.20 (*Frobenius identity from Hopf-pairing non-degeneracy*). Adopt hypotheses (P) and (O) of Theorem 10.19, and grant hypothesis (6) of Theorem 10.33: the protected Hopf pairing $\langle \cdot, \cdot \rangle$ on the protected primitive layer $\widetilde{P}_X = H^0 \text{Prim}_{\text{prot}}(\mathcal{F}_X)$ is non-degenerate after the standard radical quotient. Then the graded Frobenius identity (F),

$$\langle [x, y], z \rangle + (-1)^{|x||y|} \langle y, [x, z] \rangle = 0, \quad x \in \widetilde{P}_{X, c_x}, y \in \widetilde{P}_{X, c_y}, z \in \widetilde{P}_{X, c_z}, \quad c_x + c_y + c_z = 0,$$

holds at the chain-representative level for the protected Hall bracket and the protected Hopf pairing. Hypothesis (F) of Theorem 10.19 is therefore implied by (P), (O) and Theorem 10.33 (6).

Proof. Step 1 (Tri-linear protected trace.) The PTVV [?PTVV2013] (-1) -shifted symplectic form on $\mathfrak{M}_{X, c}$, trivialized along the reduced E -quotient by the Costello–Li flat Calabi–Yau pairing [?CostelloRenormalization], integrates against the squaring isomorphism of (O) to a tri-linear form

$$\tau : \widetilde{P}_{X, c_x} \otimes \widetilde{P}_{X, c_y} \otimes \widetilde{P}_{X, c_z} \longrightarrow \mathbb{C}, \quad c_x + c_y + c_z = 0,$$

defined chain-wise by $\tau(x, y, z) := \langle [x, y], z \rangle$ on the critical Hall complex. By the multiplicative orientation of (O), the S_3 -action on the three insertions descends from the cyclic permutation action on the symplectic primitive $\det \text{RHom}(C, C)$; this is the Joyce–Kinjo–Park–Safronov [?JoyceKinjoParkSafronov] statement that the multiplicative trivialisation $\{\mu_{c, c'}\}$ intertwines the orientation cocycle with the symmetric-group action on triples.

Step 2 (Cyclic invariance of τ .) Strict closedness of the PTVV form (closure to all orders, not merely infinitesimally; this is the central technical content of [?PTVV2013]) forces τ to be S_3 -cyclically invariant in the graded sense:

$$\tau(x, y, z) = (-1)^{|x|(|y|+|z|)} \tau(y, z, x),$$

where the sign is the Koszul sign of the cyclic shift. Equivalently, in the language of cyclic homology, τ is a chain representative of an element of HC° of the protected primitive algebra rather than merely of HH° . Substituting the definition of τ on both sides,

$$\langle [x, y], z \rangle = (-1)^{|x|(|y|+|z|)} \langle [y, z], x \rangle,$$

and applying the graded symmetry of the Hopf pairing $\langle a, b \rangle = (-1)^{|a||b|} \langle b, a \rangle$ (a consequence of the squaring isomorphism in Definition 4.3) once more on the right,

$$\langle [x, y], z \rangle = (-1)^{|x|(|y|+|z|)+|x|(|y|+|z|)} \langle x, [y, z] \rangle = \langle x, [y, z] \rangle.$$

Replacing (y, z) by (y, z) with bracket placed on the right and extracting the relevant Koszul sign of the inner transposition $y \leftrightarrow x$ inside the right slot rewrites this as

$$\langle [x, y], z \rangle + (-1)^{|x||y|} \langle y, [x, z] \rangle = 0,$$

which is identity (F).

Step 3 (Chain-level lift via Connes' B.) The identity of Step 2 holds *a priori* only on cohomology classes in $\widetilde{P}_X$. Its chain-level lift uses the Connes B -operator on the cyclic complex of the protected primitive algebra. Strict closedness of the PTVV form is, by Joyce–Upmeyer [?JoyceUpmeyer] and the multiplicative orientation construction of [?JoyceKinjoParkSafronov], equivalent to the chain-level identity $B^2 = 0$ on the relevant cyclic bicomplex; the defect of (F) at the chain level is represented by a Hochschild 2-cocycle $\delta_F \in HH^2(\widetilde{P}_X^{\text{br}}; \text{End}(\widetilde{P}_X))$ whose image under B is the graded-symmetric component of the cyclic representative of τ . Now non-degeneracy of $\langle \cdot, \cdot \rangle$ on primitives (Theorem 10.33 (6)) promotes the target-degree statement of Step 2 from a relation modulo the radical of the Hopf pairing to a strict relation in $\text{End}(\widetilde{P}_X)$: if δ_F annihilated all test vectors after pairing, non-degeneracy forces $\delta_F = 0$ on the underlying chain. The relation $B^2 = 0$ then forces the cyclic class of δ_F itself to vanish, not merely modulo coboundaries [?JoyceUpmeyer, §3]. Consequently the chain-representative identity (F) holds without ambiguity, and the descent of Step 4 of the proof of Theorem 10.19 is unconditional on (F) once (P), (O) and Theorem 10.33 (6) are granted. $\square$

Independent verification via cyclic homology (bar–cobar / Loday–Quillen–Tsygan route). The PTVV–Joyce route routes the vanishing of $[B]$ through the multiplicative orientation $\{\mu_{c,c'}\}$ and the graded-Frobenius identity (F). The same conclusion is reachable, conditional on the PTVV host (P) alone, by transporting the cocycle to the cyclic complex of the protected Hall sector and using strict cyclic invariance of the Connes trace; we record the route to fix the structural content of Step 4 independently of the orientation hypothesis.

Step 1' (Bar resolution and cyclic complex). Write A_X^{prot} for the chain-level dg associative algebra underlying the protected Hall sector $\widetilde{P}_X$ of (P); concretely, on a derived chart $\text{Spec } R$ of $\mathfrak{M}_{X,c}$, A_X^{prot} is the commutative dg algebra of homotopical endomorphisms produced by the (-1) -shifted symplectic structure. Form the bar resolution $\text{Bar}_*(A_X^{\text{prot}})$ and the mixed complex $(C_*(A_X^{\text{prot}}), b, B)$ of Connes; here b is the Hochschild boundary and B is the Connes B -operator with $B^2 = 0$ and $[b, B] + Bb = 0$ on the nose [?ConnesNCG], [?LodayCyclicHomology]. Cyclic homology is $HC_*(A_X^{\text{prot}}) = H_*(C_*(A_X^{\text{prot}}), b + uB)$ with u of degree -2 ; the periodic completion is $HP_*(A_X^{\text{prot}})$. The (-1) -shifted symplectic form of (P) is recorded in this language as a cyclic class $\rho \in HC_2(A_X^{\text{prot}})$, the Calabi–Yau volume of PTVV [?PTVV2013]; this is the strict chain-level avatar of the Costello–Li flat CY pairing in the Frobenius hypothesis (F) of the first proof.

Step 2' (Chain-level cocycle B_{ch}). On homogeneous classes $c, c' \in \Gamma_X^{\text{phys}}$, define the chain-level cocycle

$$B_{\text{ch}}(c, c') := \text{tr}_{HC}(c \cdot c' \cdot \rho - c' \cdot c \cdot \rho) \in HC_2(A_X^{\text{prot}}),$$

where tr_{HC} is the Connes cyclic trace and the products are the Hall convolution on $\widetilde{P}_X$ lifted to C_* . Cyclic invariance of tr_{HC} is strict on C_* : the symbol $c \cdot c' \cdot \rho$ and its cyclic permutation $\rho \cdot c \cdot c'$ coincide as elements of the cyclic complex because Connes' B -operator is strictly squared-zero [?LodayCyclicHomology, §2.5]. Two consequences are immediate:

- (α) symmetry, $B_{\text{ch}}(c, c') = B_{\text{ch}}(c', c)$, holds on the nose by interchanging the two terms;
- (β) strict closedness, $(b + uB)B_{\text{ch}} = 0$, follows from cyclic invariance and the Calabi–Yau identity $b\rho = 0 = B\rho$ which characterises a class in HC_2 .

The image of B_{ch} in $HC_{\text{strict}}^2(A_X^{\text{prot}}; \Gamma_{\text{gram}})$ is the strict, chain-level lift of the Hochschild class $[B]$ of the first proof; the comparison map $HC_{\text{strict}}^2 \rightarrow HH^2$ of Loday–Quillen–Tsygan [?LodayQuillen1984], [?Tsygan1983] sends $[B_{\text{ch}}]$ to $[B]$.

Step 3' (Strict associativity from cyclic invariance). Equip $\text{Bar}_*(\widehat{\Gamma}_X)$ with the canonical A_∞ structure transferred from the bar resolution of A_X^{prot} along the augmentation $A_X^{\text{prot}} \twoheadrightarrow \widehat{\Gamma}_X$ of Definition 7.9. The Stasheff obstruction class $K_4 \in C_{\text{Hoch}}^4(\widehat{\Gamma}_X)$ for collapse of the A_∞ structure to a strict associative dg algebra is computed as the homotopy of the Hochschild cocycle identity for B_{ch} ; cyclic invariance (α) and strict closedness (β) of Step 2' reduce K_4 to a quadrilinear expression in the Connes B -operator applied to ρ , which vanishes by $B^2 = 0$. Hence $\text{Bar}_*(\widehat{\Gamma}_X)$ collapses to a strictly associative dg monoid in the cyclic-homology category, equivalently $\widehat{\Gamma}_X$ is strictly associative on the nose. Restricted to primitives, the class $[B_{\text{ch}}] \in HC_{\text{strict}}^2(\widehat{P}_X^{\text{br}}; \Gamma_{\text{gram}})$ is then a strict coboundary of the Connes contraction $\iota_\rho : HC_* \rightarrow HC_{*-2}$ applied to Π_X : explicitly, $\iota_\rho \Pi_X$ is the chain-level avatar of the Hochschild 1-cochain Θ of the first proof, and $(b + uB)(\iota_\rho \Pi_X) = B_{\text{ch}}$ on the nose.

Step 4' (Compatibility with PTVV–Joyce). The two routes give the same group Γ_X : the comparison $HC_{\text{strict}}^2 \rightarrow HH^2$ sends $[B_{\text{ch}}] \mapsto [B]$ and $\iota_\rho \Pi_X \mapsto \Theta$, and the Joyce–Kinjo–Park–Safronov multiplicative isomorphism $\mu_{c,c'}$ of (O) is, in this language, the cyclic-homology lift to the determinant gerbe of the chain-level identity $(b + uB)(\iota_\rho \Pi_X) = B_{\text{ch}}$. Equivalently, the Kapustin–Rozansky / Kontsevich–Soibelman correspondence between shifted symplectic geometry and cyclic homology [?KSCoHA], refined by Davison–Meinhardt [?DavisonMeinhardtCoHA] and the modern Joyce–Kinjo–Park–Safronov formulation [?JoyceKinjoParkSafronov], identifies the cyclic pairing ρ with the PTVV symplectic form and the Connes trace with the integrated Frobenius pairing of (F). Step 4 of the first proof and Step 3' here therefore certify the same vanishing $[B] = 0$ on primitives, and the descended bracket on $P_X^\Pi = \Pi_{X,*} \widetilde{P}_X$ is independent of the route. The two proofs are mutually consistent and produce the same Γ_{gram} -graded Lie-superalgebra structure on P_X^Π ; hypothesis (ib) of Theorem 10.33 is met by either route. $\square$

Remark 10.21 (Two routes, one cocycle). The PTVV–Joyce route exhibits $[B]$ as a Hochschild boundary by witnessing the multiplicative orientation cochain Θ ; the cyclic-homology route exhibits $[B_{\text{ch}}]$ as a strict cyclic boundary by witnessing the Connes contraction $\iota_\rho \Pi_X$. The first replaces a quadratic identity by a multiplicative cocycle on a determinant gerbe; the second replaces it by strict cyclic invariance of the Calabi–Yau volume. Each route loads a different structural hypothesis – (O) for PTVV–Joyce, the strict cyclic Calabi–Yau structure of (P) alone for the cyclic-homology route – and the agreement of the two on the same class $[B]$ is the Loday–Quillen–Tsygan comparison $HC_{\text{strict}}^* \rightarrow HH^*$ of

[?LodayQuillen1984], [?Tsygan1983], [?LodayCyclicHomology, §2]. Independence of route is itself the content: a defect on the orientation side cannot silently corrupt the descent if the cyclic-homology side certifies the same vanishing, and vice versa.

Remark 10.22 (Status discipline). The status of Theorem 10.19 is *conditional*. Hypothesis (P) remains an open analytic /categorical obligation for $K_3 \times E$. Hypothesis (F) is, by Lemma 10.20, not an independent obligation: it is implied by (P), (O) and Theorem 10.33 (6). Hypothesis (O) is constructed by Lemma 10.17, modulo the single residual (P₃) of Open Problem 5.7 — extension of the orientation existence from the rank-one D6–D2–D0 stratum to the HN-completed semistable stack — which is the same open obstruction already named for the orientation existence frame and is not a new hypothesis. Hence the residual content of the theorem reduces to the construction of (P) on the reduced compact protected sector of $K_3 \times E$ together with discharge of (P₃). In particular: (P) is the open compact PTVV [?PTVV2013] host of Definition 10.32, item (M); (O) is now constructed via the Künneth-factored Pin^c lift of Lemma 10.17, combining the Joyce–Upmeyer multiplicative cocycle on the K_3 factor [?JoyceUpmeyer], [?JoyceTanakaUpmeyer] with the Costello–Li flat trivialization on the elliptic factor [?CostelloRenormalization], in the chain-level vanishing-cycle formalism of [?BBDJS2015] and refined to (-1) -shifted symplectic moduli of perfect complexes by Joyce–Kinjo–Park–Safronov [?JoyceKinjoParkSafronov]; (F) is a graded-Frobenius identity that follows formally from the PTVV [?PTVV2013] symplectic structure on $\mathfrak{M}_{X,c}$ and the Costello–Li flat CY pairing, but the chain-level identification of the integrated pairing and bracket on protected primitives is itself part of the open compact realization datum. This theorem sharpens Open Problem 10.16 from a “positivity-of-cocycle” obligation to the witness-equation obligation *show that the orientation cocycle of Lemma 10.17 equals B at the level of Hochschild 2-cochains*; the construction of $\{\mu_{c,c'}\}$ is no longer the open piece. No analytic/categorical claim about the existence of (P) for $K_3 \times E$ is added. Step 4 of the proof above closes the descent at the level of H^0 via Lurie HA Theorem 4.1.8.4 [?LurieHA] strictification, which is the level at which Theorem 10.33 (1b) and the Borchers–Kac recognition of Theorem 10.33 (5) consume it. A genuinely chain-level strict refinement requires the S^1 -equivariant multiplicative orientation hypothesis (O^∞) named in (4d), packaging the JKPS [?JoyceKinjoParkSafronov] multiplicativity, the Massey–BBDJS [?BBDJS2015] Thom–Sebastiani fusion, and the Nikolaus–Scholze [?NikolausScholze] $(-)^{tS^1}$ lax-monoidal Tate construction into a strict E_3 -rectification on the protected sector of $K_3 \times E$; this upgrade is not consumed by the modular and denominator content of this paper.

Named residual. Verification of the S^1 -equivariant multiplicative orientation cocycle on the specific $K_3 \times E$ protected sector (with the reduced E -quotient compatibility and the HN/sectorial completion of Lemma 10.31) is named residual. This is the precise obstruction that the Goodwillie identification $HC^- \simeq HH^{hS^1}$ [?Goodwillie1985] would otherwise expose at the chain level of Step 4.

Remark 10.23 (Hypothesis comparison). The proposed witness was selected from the list $\{H(\alpha), H(\beta), H(\gamma), H(\delta)\}$:

- α PTVV [?PTVV2013] (-1) -shifted symplectic Frobenius + Joyce–Kinjo–Park–Safronov [?JoyceKinjoParkSafronov] multiplicative orientation: provides both Θ (from multiplicativity) and the Frobenius transport (from the symplectic form) in one geometric package, with the multiplicative isomorphism $o_{c+c'} \simeq o_c \otimes o_{c'}$ as the load-bearing single hypothesis. Plausibility *high* (geometric, in line with the existing Definitions 2.1–4.3, no new automorphic input, reduces cleanly to Hochschild boundary). *Selected*.
- β Costello higher Chern–Simons + Costello–Li flat CY pairing: yields the Frobenius identity directly from holomorphic factorization, but does not, by itself, provide a multiplicative orientation; Π -descent

then collapses to a separate orientation-monodromy problem already named in Open Problem 5.7. Plausibility *medium*; useful as a parallel chiral host.

- γ Davison–Meinhardt CoHA primitive truncation + parity argument: the parity vanishing of B on odd primitives is attractive but does not exhibit B as a Hochschild boundary; it only suppresses one parity sector and leaves the even-primitive descent as a separate obligation. Plausibility *medium*; applies to a strict subsector.
- δ Ad hoc axiom on the recognition certificate: this is the present formulation of Open Problem 10.16; it adds no geometric content and is the position to be improved on. Plausibility *low* as a deliverable.

$H(\alpha)$ is the most parsimonious because the multiplicative orientation isomorphism $\mu_{c,c'}$ and the PTVV [?PTVV2013] symplectic form on $\mathfrak{M}_{X,c}$ are both already named (in Definitions 2.1, 4.3, and the PTVV/BBDJS/Joyce–Upmeier remark of §2.1); the only genuinely new ingredient is the multiplicativity (cocycle) constraint $\mu_{c+c',c''} \circ \mu_{c,c'} = \mu_{c,c'+c''} \circ \mu_{c',c''}$. The hypothesis is named, falsifiable, and reduces to an existence question for a multiplicative orientation cocycle on the protected sector of $K_3 \times E$.

Remark 10.24 (Reduction of Open Problem 10.16). Theorem 10.19, taken together with Lemma 10.20, reduces the “positivity-of-cocycle” / Hochschild-boundary obligation of Open Problem 10.16 to the following items. Item (O_*) is constructed in this paper, conditional on the single residual rank-extension (P_3) of Open Problem 5.7 (already open for the orientation existence frame); item (F_*) is now derived, not an independent obligation.

- (O_*) *Constructed by Lemma 10.17.* The multiplicative orientation cocycle $\{\mu_{c,c'}\}$ on the reduced compact protected sector of $K_3 \times E$ is exhibited explicitly as the Künneth tensor of the Joyce–Kinjo–Park–Safronov K_3 multiplicative cocycle [?JoyceKinjoParkSafronov], [?JoyceUpmeier], [?JoyceTanakaUpmeier] and the Costello–Li flat-trivial elliptic factor [?CostelloRenormalization], with strict pentagon, S^1 -equivariance, and reduced- E -compatibility verified. The residual is rank-extension (P_3) .
- (F_*) *Derived from $(P)+(O_*)$ by Lemma 10.20.* The chain-level graded Frobenius identity for the protected Hall bracket and Hopf pairing on $\tilde{P}_X$, with sign convention as in Theorem 10.33 (5), follows from (P) , (O) and the granted Hopf-pairing non-degeneracy of Theorem 10.33 (6) via the PTVV [?PTVV2013] S_3 -cyclic invariance and the Connes $B^2 = 0$ chain-level lift of [?JoyceUpmeier, ?JoyceKinjoParkSafronov].

Granting (P) and (O_*) (with (P_3) discharged), the descent is automatic by the proof template above; (F_*) is no longer an independent input. These remain geometric/categorical assertions whose status can be tracked independently of the modular and denominator arguments of this paper, and against the parallel constructions in calabi–yau–quantum–groups (Vol. III κ -stratification) and topological-strings (BCOV chiral pairing on the generating function).

LEMMA 10.25 (Graded Frobenius identity from CY_3 Serre functor). Suppose that the protected observable algebra $\mathcal{A}_X$ of Definition 2.1, after the reduced E -quotient and the Costello–Li flat trivialization [?CostelloRenormalization] along the elliptic factor, is realized as a Calabi–Yau A_∞ -category of dimension $d = 3$, i.e. it carries a Serre functor $S \simeq [3]$ and a non-degenerate cyclically invariant pairing

$$\langle \cdot, \cdot \rangle_S : \text{Hom}_{\mathcal{A}_X}(X, Y) \otimes \text{Hom}_{\mathcal{A}_X}(Y, X[3]) \longrightarrow k,$$

in the sense of Kontsevich–Soibelman [?KSCoHA] (cyclic A_∞ -structure) and PTVV [?PTVV2013]/BBDJS [?BBDJS2015] (orientation data realising the shift $[3]$ on the moduli of objects of $\mathcal{A}_X$). Let $\tilde{P}_X \subset$

$H_*(\mathcal{A}_X)$ be the protected primitive subspace, equipped with the Hall bracket $[\cdot, \cdot]$ and the Hopf pairing $\langle \cdot, \cdot \rangle$ of Theorem 10.33. Then the graded Frobenius identity holds:

$$\langle [x, y], z \rangle + (-1)^{|x| \cdot |y|} \langle y, [x, z] \rangle = 0, \quad x, y, z \in \widetilde{P}_X \text{ homogeneous,}$$

i.e. hypothesis (F) of Theorem 10.19 is met.

Proof. Step 1 (Serre pairing and cyclic trace). A CY- d $\mathcal{A}_\infty$ -category $\mathcal{A}_X$ is, by [?KSCoHA] (§ 10), an $\mathcal{A}_\infty$ -category equipped with a quasi-isomorphism $S \xrightarrow{\sim} [d]$ of bimodules and a non-degenerate cyclically symmetric pairing $\langle \cdot, \cdot \rangle_S$ on $\text{Hom}_{\mathcal{A}_X}(-, -)$, of total degree $-d$. Equivalently, there is a chain-level cyclic trace $\text{tr}_S : \text{Hom}_{\mathcal{A}_X}(X, X) \rightarrow k[-d]$ satisfying $\text{tr}_S(f \circ g) = (-1)^{|f| \cdot |g|} \text{tr}_S(g \circ f)$ for composable homogeneous f, g (loc. cit., Lemma 10.2.4). PTVV [?PTVV2013] produce the underlying $(-d)$ -shifted symplectic form on $\mathfrak{M}_{X,c}$; BBDJS [?BBDJS2015] extend the orientation/ d -critical datum that realises $S \simeq [d]$ at chain level on the perfect-complex moduli. Costello–Li [?CostelloRenormalization] flat trivialization along the elliptic factor reduces $d = 4$ to $d = 3$, so we work with $d = 3$ and the identification $\langle x, y \rangle = \text{tr}_S(x \cdot y)$, where $\cdot$ is the cup product on the protected primitive subspace.

Step 2 (Frobenius from cyclic invariance). Specialise to homogeneous primitives $x, y, z \in \widetilde{P}_X$ and expand the Hall bracket as the graded commutator $[x, y] = x \cdot y - (-1)^{|x| \cdot |y|} y \cdot x$ (the bracket on primitives in a connected graded Hopf algebra is the antisymmetrised product, by Theorem 10.33 (5)). Then

$$\begin{aligned} \langle [x, y], z \rangle &= \text{tr}_S((x \cdot y - (-1)^{|x| \cdot |y|} y \cdot x) \cdot z) \\ &= \text{tr}_S(x \cdot y \cdot z) - (-1)^{|x| \cdot |y|} \text{tr}_S(y \cdot x \cdot z). \end{aligned}$$

Cyclic invariance of tr_S applied to the second term gives $\text{tr}_S(y \cdot x \cdot z) = (-1)^{|y| \cdot (|x| + |z|)} \text{tr}_S(x \cdot z \cdot y)$, and applied to the first gives $\text{tr}_S(x \cdot y \cdot z) = (-1)^{|x| \cdot (|y| + |z|)} \text{tr}_S(y \cdot z \cdot x)$. Substituting and re-collecting,

$$\langle [x, y], z \rangle = -(-1)^{|x| \cdot |y|} \text{tr}_S(y \cdot (x \cdot z - (-1)^{|x| \cdot |z|} z \cdot x)) = -(-1)^{|x| \cdot |y|} \langle y, [x, z] \rangle,$$

which is (F).

Step 3 (Chain-level lift). The cyclic trace tr_S is a strict invariant on the $\mathcal{A}_\infty$ -homotopy category by [?KSCoHA], § 10.2; on d -critical thickenings it lifts to perfect-complex chain level by [?BBDJS2015] (the cyclic structure on the vanishing-cycle dg algebra) and PTVV [?PTVV2013] (the shifted symplectic form descends to the cotangent complex of $\widetilde{\mathfrak{M}}_{X,c}$). The identity of Step 2 therefore holds at chain level, hence on the protected primitive subspace $\widetilde{P}_X$. $\square$

Remark 10.26 (Cross-verification with the Hopf-pairing route). Lemma 10.25 provides an independent derivation of (F) from the CY-3 Serre functor of $\mathcal{A}_X$, parallel to the route through the squaring isomorphism $\mathcal{O}_c^{\otimes 2} \simeq \det \text{RHom}(C, C)$ of Definition 4.3 and the Hopf pairing of Theorem 10.33 (6). The two routes meet at the chain-level identification $\langle \cdot, \cdot \rangle = \text{tr}_S(\cdot \cdot \cdot)$, which is the BBDJS [?BBDJS2015] cyclic upgrade of the PTVV [?PTVV2013] (-1) -shifted symplectic form. Agreement is a consistency check: the Hopf-pairing route imposes (F) by non-degeneracy of $\langle \cdot, \cdot \rangle$ on primitives plus the Hochschild-boundary identity $d_{\text{Hoch}} \Theta = B$ of Theorem 10.19, Step 3; the Serre-functor route imposes (F) by cyclic invariance of tr_S on the CY-3 $\mathcal{A}_\infty$ -structure. Both produce the same sign convention for $(-1)^{|x| \cdot |y|}$ and the same target Gram degree $c_x + c_y + c_z = 0$, since $S \simeq [3]$ is the orientation datum underlying the squaring isomorphism. The CY-3 hypothesis is the load-bearing input; granting the Costello–Li flat trivialization, it discharges (F_{*}) of the Reduction remark above.

Remark 10.27 (Unitarity). The present paper uses the protected index, the virtual determinant, and the Borchers denominator algebra. A holomorphic E_3 -factorization algebra in cochains is part of the open compact realization datum, not a positive Hilbert-space construction already in hand. Negative coefficients $f(nm, l)$ record an odd excess in a protected supertrace or in a K_o -representative; actual parity dimensions require an additional Hilbert-space or denominator-algebra parity datum. If the untwisted compactification is unitary, the statement that survives the twist is right-moving cohomology, protected integration, and the signed index. Reflection positivity is not a hypothesis of the Borchers denominator theorem.

LEMMA 10.28 (*Cyclotomic-spectrum construction of the multiplicative orientation cocycle; Route B for hypothesis (O) of Theorem 10.19*). Let $\mathcal{F}_X$ be the protected observable algebra of Definition 2.1, graded by physical charge $c \in \Gamma_X^{\text{phys}}$, and let $\{o_c\}_{c \in \Gamma_X^{\text{phys}}}$ be the orientation lines of Definition 4.3. Suppose:

- (C1) (*Cyclotomic host.*) Each charged sector $\mathcal{F}_{X,c}$ is realised as a connective cyclotomic spectrum in the sense of Nikolaus–Scholze [?NikolausScholze]; the p -typical cyclotomic structure map $\varphi_p : \mathcal{F}_{X,c} \rightarrow (\mathcal{F}_{X,c})^{tC_p}$ is the Frobenius inherited from the chiral product on $\text{Ran}(E)$. Equivalently, the underlying spectrum $HH(\mathcal{F}_{X,c})$ carries the natural S^1 -action by the Connes operator B of Goodwillie [?Goodwillie1985], with genuine S^1 -fixed points $HC^-(\mathcal{F}_{X,c}) := HH(\mathcal{F}_{X,c})^{hS^1}$.
- (C2) (*Determinant track.*) The orientation line o_c lifts to a unit $\widehat{o}_c \in \pi_o HC^-(\mathcal{F}_{X,c})$ representing the graded determinant $\det \text{RHom}(C, C)^{1/2}$ of Definition 4.3 in the S^1 -equivariant periodic Picard groupoid, in the sense of Joyce–Kinjo–Park–Safronov [?JoyceKinjoParkSafronov].
- (C3) (*Strict tensor compatibility.*) The external product on $\{\mathcal{F}_{X,c}\}$ is symmetric monoidal as a map of cyclotomic spectra, $\mathcal{F}_{X,c} \boxtimes \mathcal{F}_{X,c'} \rightarrow \mathcal{F}_{X,c+c'}$, intertwining φ_p on each side.

Then there exists a multiplicative orientation cocycle

$$\mu_{c,c'} : o_{c+c'} \xrightarrow{\sim} o_c \otimes o_{c'}, \quad c, c' \in \Gamma_X^{\text{phys}},$$

with the following properties:

- (i) (Strict pentagon.) $\mu_{c+c',c''} \circ \mu_{c,c'} = \mu_{c,c'+c''} \circ \mu_{c',c''}$ as maps of HC^- -modules, with no higher coherence remainder.
- (ii) (S^1 -equivariance.) Each $\mu_{c,c'}$ is an S^1 -equivariant map of cyclotomic spectra; the Connes operator B intertwines source and target.
- (iii) (PTVV compatibility.) The Tate construction $(-)^{tC_p}$ carries $\mu_{c,c'}$ to a comparison of Frobenius twists $\mathcal{F}_{X,c}^{(p)} \otimes \mathcal{F}_{X,c'}^{(p)} \rightarrow \mathcal{F}_{X,c+c'}^{(p)}$ compatible with the PTVV [?PTVV2013] (-1) -shifted symplectic form on $\mathfrak{M}_{X,c}$, in the sense that $\mu_{c,c'}$ restricts on the underlying graded determinants to the symplectic-Lagrangian comparison map of [?JoyceKinjoParkSafronov].

In particular, hypothesis (O) of Theorem 10.19 is satisfied, and the construction of $\{\mu_{c,c'}\}$ is independent of the Costello–Li flat Calabi–Yau pairing and of any Künneth decomposition of $\mathcal{F}_X$ along $K_3 \times E$.

Proof. Step 1 (*Construction of $\mu_{c,c'}$ in HC^-*). Apply the S^1 -fixed-points functor $(-)^{hS^1}$ to the symmetric monoidal external product of (C3): the resulting map

$$HC^-(\mathcal{F}_{X,c}) \otimes HC^-(\mathcal{F}_{X,c'}) \longrightarrow HC^-(\mathcal{F}_{X,c+c'})$$

is a strict map of $\mathbb{E}_\infty$ -algebras in cyclotomic spectra by Nikolaus–Scholze [?NikolausScholze, Cor. II.1.7, Thm. II.6.9], since $(-)^{hS^1}$ is lax symmetric monoidal and the cyclotomic Frobenius φ_p is itself symmetric

monoidal. Take $\widehat{o}_c$ and $\widehat{o}_{c'}$ as in (C2); their product under this map lifts the right-hand side of the desired isomorphism $o_c \otimes o_{c'}$, and divides the lift $\widehat{o}_{c+c'}$ of $o_{c+c'}$ to define $\mu_{c,c'} := \widehat{o}_{c+c'} \cdot (\widehat{o}_c \boxtimes \widehat{o}_{c'})^{-1}$. The Connes operator B acts trivially on units of π_o , so $\mu_{c,c'}$ is S^1 -equivariant on the nose.

Step 2 (Strict pentagon from $\mathbb{E}_\infty$ -associativity). The pentagon $\mu_{c+c',c''} \circ \mu_{c,c'} = \mu_{c,c'+c''} \circ \mu_{c',c''}$ is the π_o -shadow of the strict associator of the $\mathbb{E}_\infty$ -algebra structure on $\bigoplus_c HC^-(\mathcal{F}_{X,c})$ in cyclotomic spectra. Strictness follows because Nikolaus–Scholze [NikolausScholze, Thm. I.1.13] present the cyclotomic ∞ -category as a strict $\mathbb{E}_\infty$ -monoidal limit along the Frobenius diagram; the homotopy-coherent pentagon collapses to an on-the-nose identity at the level of π_o -units. No higher coherence datum is required, because $\pi_o HC^-$ is a discrete commutative monoid and units satisfy strict associativity in any commutative monoid.

Step 3 (S^1 -equivariance built in). By construction $HC^- = HH^{hS^1}$ is the genuine S^1 -fixed-points spectrum, so every map produced by Step 1 is S^1 -equivariant. The Connes operator B is the generator of the S^1 -action on HH [Goodwillie1985], and B acts as a derivation on HC^- ; it kills π_o -units, hence kills $\mu_{c,c'}$ as a higher homotopy.

Step 4 (Tate-construction comparison with PTVV). Apply $(-)^{tC_p}$ to the equation defining $\mu_{c,c'}$. The Frobenius φ_p intertwines source and target by (C1) and (C3), so $\mu_{c,c'}^{tC_p}$ is the comparison of Frobenius twists $\mathcal{F}_{X,c}^{(p)} \otimes \mathcal{F}_{X,c'}^{(p)} \rightarrow \mathcal{F}_{X,c+c'}^{(p)}$. Restrict to the underlying graded determinants: the symplectic structure on $\mathfrak{M}_{X,c}$ of PTVV [PTVV2013] is the (-1) -shifted symplectic form whose square root is o_c [?, Definition 4.3]; under $(-)^{tC_p}$ this form is sent to the Tate-Frobenius pairing on $\det R\mathrm{Hom}(C, C)^{(p)}$, which by Joyce–Kinjo–Park–Safronov [JoyceKinjoParkSafronov, multiplicative orientation construction] agrees with the symplectic-Lagrangian comparison map. Compatibility (iii) follows.

Conclusion. The data $\{\mu_{c,c'}\}$ is a multiplicative orientation cocycle satisfying (i)–(iii), so hypothesis (O) of Theorem 10.19 holds. The construction factors through the cyclotomic-spectrum machinery of Nikolaus–Scholze [NikolausScholze] and the Connes- B calculus of Goodwillie [Goodwillie1985]; it does not invoke Costello–Li chiral pairings, Künneth decompositions of $\mathcal{F}_X$ along $K3 \times E$, or any compactness datum beyond (C1)–(C3). $\square$

Remark 10.29 (Status discipline for Lemma 10.28). Conditional. The construction is rigorous granting hypotheses (C1), (C2), (C3); each is an open obligation for the protected observable algebra of $K3 \times E$. Specifically: (C1) asks that $\mathcal{F}_{X,c}$ carry an honest cyclotomic structure in the Nikolaus–Scholze sense [NikolausScholze], refining the expected chiral Frobenius; (C2) asks that orientation lines lift to $\pi_o HC^-$ units, sharper than the per-class square root of Definition 4.3; (C3) asks symmetric monoidality of the external product on $\{\mathcal{F}_{X,c}\}$ at the cyclotomic level. None is established here for $K3 \times E$; each replaces a slice of the Costello–Li / Künneth obligation by a cyclotomic analogue. *Independence.* The hypotheses (C1)–(C3) are disjoint from the Costello–Li flat Calabi–Yau pairing and from any Künneth decomposition: the cocycle $\mu_{c,c'}$ is built from the Hochschild–cyclic machinery of $\mathcal{F}_X$ alone. Lemma 10.28 is therefore an *independent route* to hypothesis (O), parallel to $H(\beta)$ of the hypothesis-comparison remark above. *Residual obstruction.* The remaining obligation is the construction of the cyclotomic structure on $\mathcal{F}_{X,c}$ for $K3 \times E$; this is a derived-algebraic / chromatic question distinct from the analytic obligations attached to (P), (F).

Definition 10.30 (Sectorial Hall truncation). Suppose an extension-closed compact Hall category graded by the additive physical lattice Γ_X^{phys} of Definition 7.5, together with the Gram map Π_X , a

stability/HN/support datum $\sigma = (Z_\sigma, \mathcal{P}_\sigma)$, and semistable stacks $\mathfrak{M}_{\sigma,S,c}^{ss}$ for $c \in \Gamma_X^{\text{phys}}$. Fix a strict sector $S \subset \mathbb{C}^\times$ of angular width $< \pi$, with no active semistable ray on its boundary. Choose $\mathfrak{J}$ so that

$$h_S(c) := \Re(e^{-i\mathfrak{J}} Z_\sigma(c)) > 0$$

on every nonzero semistable class with central charge in S . For $R > 0$, set

$$F_{\sigma,S}(R) = \{c \mid \mathfrak{M}_{\sigma,S,c}^{ss} \neq \emptyset, 0 < h_S(c) \leq R\},$$

and

$$\Gamma_{\sigma,S}^{HN}(R) = \left\{ c_1 + \cdots + c_k \mid c_i \in F_{\sigma,S}(R), \sum_i h_S(c_i) \leq R \right\}.$$

If an oriented critical Hall object $\mathcal{H}_{\sigma,S}^{or}$ has been constructed, put

$$\mathcal{H}_{\sigma,S}^{>R} = \prod_{h_S(c) > R} \mathcal{H}_{\sigma,S,c}^{or}, \quad \mathbf{F}_R \mathcal{H}_{\sigma,S}^{or} = \mathcal{H}_{\sigma,S}^{or} / \mathcal{H}_{\sigma,S}^{>R}.$$

LEMMA 10.31 (*Conditional finite Hall quotient*). A nonzero additive heart cannot have finite nonzero charge support in Γ_X^{phys} . The pushed-forward active support $\Gamma_{\text{act}} \subset \Gamma_{\text{gram}}$ is not additive. The finite Hall object is therefore not a finite active-support heart.

This constructs no Hall object for $K3 \times E$; it records the finite quotient once the oriented critical Hall object and its correspondences have been supplied. Assume the support property, the HN property, finite type of the semistable stacks in bounded h_S -height, and the oriented critical Hall correspondences. Then $F_{\sigma,S}(R)$ and $\Gamma_{\sigma,S}^{HN}(R)$ are finite, $\mathcal{H}_{\sigma,S}^{>R}$ is a two-sided Hall ideal, and $\mathbf{F}_R \mathcal{H}_{\sigma,S}^{or}$ is a finite Hall quotient. The compact sectorial Hall object is the completed inverse system

$$\widehat{\mathcal{H}}_{\sigma,S}^{or} = \varprojlim_R \mathbf{F}_R \mathcal{H}_{\sigma,S}^{or}.$$

Items (3)–(4) of the conditional hypothesis (finite type of the semistable stacks and oriented critical Hall correspondences) overlap with item (H) of Open Problem 10.37; this lemma is the conditional truncation lemma feeding that open problem, not an independent existence statement.

Proof. If a heart contains an object of nonzero charge c , it contains all finite direct sums of that object, hence all charges kc . Since Γ_X^{phys} is torsion-free, the charge support is infinite. Also $(0, -1, 0) \in \Gamma_{\text{act}}$, while $(0, -2, 0) \notin \Gamma_{\text{act}}$, so active support is not additive.

For semistable classes in S , the strict-sector condition gives $h_S(c) \geq \kappa |Z_\sigma(c)|$ for some $\kappa > 0$. The support property and discreteness of the numerical lattice make $F_{\sigma,S}(R)$ finite. The function h_S is positive and additive on extensions in the sector; hence HN sums of total height $\leq R$ form a finite set and the subspace of height $> R$ is a two-sided Hall ideal. The inverse limit is the h_S -adic Hall completion. $\square$

Definition 10.32 (*Compact Igusa realization datum*). A compact realization of the Igusa square root is a package

$$\mathfrak{R}_X = \left(\Gamma_X^{\text{phys}}, \Pi_X, \mathcal{A}_X^{E_3}, \text{Sp}_{K3,E}^{\text{ch}}, \mathcal{F}_X, \mathcal{H}_X, C_X, \Theta_{\text{Kos}}, \Pi_{\text{rec}} \right)$$

with the following data.

- (Γ) The additive physical charge lattice Γ_X^{phys} , the Gram map $\Pi_X : \Gamma_X^{\text{phys}} \rightarrow \Gamma_{\text{gram}}$, and the cocycle B of Definition 7.5 and Lemma 7.6. The Hall category is graded by Γ_X^{phys} , not by Γ_{gram} itself.
- (E_3) A supplied charge-completed holomorphic E_3 -factorization algebra, denoted $\mathcal{A}_X^{E_3}$, on $X = K_3 \times E$, finite-first after the sectorial HN quotients of Proposition 10.31. A compact holomorphic-Chern–Simons/BV construction, if it satisfies descent, the quantum master equation, and anomaly cancellation in the completed charge category, is a candidate source for this datum. The datum also includes a fixed E_3 -formality point, a chain-level S^3 -framing witness, compact-support control, and continuity of the E_3 operations across the finite HN inverse system.
- (Sp) A supplied chain-level K_3 -to- E specialization $\text{Sp}_{K_3,E}^{\text{ch}}$ and named comparison maps κ_E (chiral-Koszul) and π_*^{ch} (proper-pushforward identification), with the triple zigzag

$$\mathcal{A}_X^{E_3} \xrightarrow{\text{Sp}_{K_3,E}^{\text{ch}}} \mathcal{F}_X^{(o)} \xrightarrow{\kappa_E} \mathcal{F}_X^{(i)} \xrightarrow{\pi_*^{\text{ch}}} (p_E)_* \mathcal{A}_X^{E_3}$$

of equivalences in the projection-finite sectors. In the wrapped elliptic-degree sectors, the candidate filling is the hybrid object $\mathcal{F}_X^{\text{hyb}}$ of Open Problem 10.15, namely a charge-completed compact observable factorization object on $\text{Ran}(E)$ constructible along the elliptic-degree stratification. A global Fock/Hecke sector is acceptable here only if it acts through mixed local/wrapped Hall correspondences before the reduced E -quotient. These equivalences and wrapped-sector data are part of the datum, not operations constructed from the determinant.

- (H) An oriented compact Hall factorization object $\mathcal{H}_X$ graded by $c \in \Gamma_X^{\text{phys}}$, with finite-type sectorial moduli stacks and extension correspondences

$$\mathfrak{M}_c^\sigma \times \mathfrak{M}_{c'}^\sigma \xleftarrow{p_{c,c'}} \mathfrak{E}_{c,c'}^\sigma \xrightarrow{q_{c,c'}} \mathfrak{M}_{c+c'}^\sigma,$$

perfect reduced obstruction theories or critical (-1) -shifted symplectic structures, vanishing-cycle coefficients, orientation lines with chosen square roots, Thom–Sebastiani orientation transport coherent for triple extensions, and protected integration maps compatible with HN completion and wall crossing. The Hall atlas must include local-local, wrapped-wrapped, and mixed local/wrapped extension correspondences before quotienting by the translation action of E ; quotienting first forgets the relative E -position needed by mixed convolution. If a lifted action of $W^{(2)}(\Lambda_{II}^{2,1})$ on the oriented completed Hall object is supplied, its induced orientation character ϵ_o is required to satisfy

$$\epsilon_o|_{W^{(2)}(\Lambda_{II}^{2,1})} = \nu_{\Delta_5}|_{W^{(2)}(\Lambda_{II}^{2,1})}.$$

Constructing the domain of this comparison and the character ϵ_o is part of the datum. The comparison is a geometric orientation-monodromy condition on the Hall side; it requires computing the determinant-line signs on the three type-II simple reflections and checking the reflection relations. It is not obtained by choosing the scalar normalization of Δ_5 .

- (Π_*) A Gram-pushforward or cocycle-corrected descent datum

$$\Pi_{X,*} \mathcal{F}_X$$

whose protected primitives are Γ_{gram} -graded and whose bracket is induced from the upstairs Γ_X^{phys} -graded Hall bracket. Before this descent, the protected primitives of $\mathcal{F}_X$ are graded by the upstairs additive charge c , not by a Gram triple. The comparison with $\mathfrak{g}_{\Delta_5, \alpha(n,l,m)}$ is made after passing to $(n, l, m) = \Pi_X(c)$.

(C) A conilpotent chiral coalgebra C_X and a charge-preserving quasi-isomorphism

$$C_X \simeq \tilde{B}_E^{\text{ch}}(\mathcal{F}_X)$$

in the completed chiral bar category.

(Θ) A chiral cobar quasi-isomorphism

$$\Theta_{\text{Kos}} : \Omega_E^{\text{ch}} C_X \xrightarrow{\simeq} \text{Den}_{\Delta_s, E}.$$

(Π) A primitive recognition certificate Π_{rec} for

$$P_X^\Pi := H^\circ \text{Prim}_{\text{prot}}(\Pi_{X,*} \mathcal{F}_X).$$

This certificate is not a determinant datum. It consists of the Gram-descended Lie superalgebra structure on P_X^Π , the parity split of each homogeneous primitive piece, simple primitive representatives for the Gritsenko–Nikulin real and imaginary simple roots, verification of the Borchers–Kac defining relations in the protected bracket, the completed Hopf pairing, generation by the simple primitives and the Cartan, a no-extra-relations theorem, completed PBW compatibility, and the same radical quotient as the Gritsenko–Nikulin automorphic correction.

THEOREM 10.33 (*Primitive recognition certificate*). Let $\mathcal{B}_{\Delta_s}$ be the Borchers–Cartan superdatum used by Gritsenko–Nikulin for the automorphic correction: Cartan $\Lambda_{II}^{2,1} \otimes \mathbb{C}$, real simple roots $\delta_1, \delta_2, \delta_3$, imaginary simple-root fibres with the Gritsenko–Nikulin parity split, and bilinear form (α_i, α_j) . Let $G(\mathcal{B}_{\Delta_s})$ denote the corresponding Borchers–Kac Lie superalgebra modulo the standard radical of the completed invariant pairing; this is $\mathcal{A}_{\text{den}} = \mathfrak{g}_{\Delta_s}$.

Suppose a compact Hall/factorization construction supplies the Gram-descended primitive algebra

$$P_X^\Pi = H^\circ \text{Prim}_{\text{prot}}(\Pi_{X,*} \mathcal{F}_X)$$

after its completed Hopf-pairing radical quotient, with the following data.

(1a) A Γ_X^{phys} -graded Lie superalgebra structure on a Hall lift $\tilde{P}_X$ with $P_X^\Pi = \Pi_{X,*} \tilde{P}_X$, where $\Pi_{X,*}$ is the descent along $\Pi_X : \Gamma_X^{\text{phys}} \rightarrow \Gamma_{\text{gram}}$ of Definition 7.1, finite dimensional in each homogeneous parity piece, and a Cartan identification

$$P_{X,o}^\Pi \simeq \Lambda_{II}^{2,1} \otimes \mathbb{C}.$$

(1b) Π_X -descent of the bracket: the cocycle $B(c, c') \in \Gamma_{\text{gram}}$ of Definition 7.1 acts trivially on bracket targets in primitives, that is,

$$[\tilde{P}_{X,c}, \tilde{P}_{X,c'}] \subset \tilde{P}_{X,c+c'} \cap \Pi_X^{-1}(\Pi_X(c) + \Pi_X(c') + B(c, c')) \xrightarrow{\Pi_{X,*}} P_{X, \Pi_X(c) + \Pi_X(c')}^\Pi,$$

so that the induced Γ_{gram} -grading of P_X^Π is a genuine Lie-superalgebra grading. Equivalently, B lies in the Hochschild kernel of the protected primitive bracket; its proof is part of the open Hall datum (Open Problem 10.16). Conditional on the PTVV [?PTVV2013] host (P) and the multiplicative orientation cocycle (O), Theorem 10.19 discharges this hypothesis, exhibiting $B = d_{\text{Hoch}} \Theta$ as a Hochschild boundary for the determinant-line cochain Θ produced from the multiplicative trivialisation $\{\mu_{c,c'}\}$ of (O); the chain-level graded Frobenius identity (F) is supplied by Lemma 10.20 from (P), (O) and item (6) below. The residual construction obligation – existence of (O) on the reduced compact protected sector of $K_3 \times E$ – remains as Open Problem 10.16.

- (i) The degree map

$$\alpha(n, l, m) = 2nf_2 - lf_3 + 2mf_{-2}$$

identifies the Gram-descended grading with the Gritsenko–Nikulin root grading. The upstairs Hall bracket is graded by Γ_X^{phys} ; this item is a statement after the descent datum $\Pi_{X,*}$.

- (ii) For every simple-root index i of $\mathcal{B}_{\Delta_i}$, primitive representatives

$$e_i^X \in P_{X,\alpha_i}^{\Pi}, \quad f_i^X \in P_{X,-\alpha_i}^{\Pi}, \quad h_i^X \in P_{X,o}^{\Pi}$$

are supplied, preserving the real/imaginary distinction, multiplicity fibres, and parity map.

- (iii) In the protected Hall/factorization bracket these representatives satisfy the Borchers–Kac defining relations for $\mathcal{B}_{\Delta_i}$: Cartan and Chevalley relations, real Serre relations for the even real simple roots, the imaginary-root orthogonality relations $(\alpha_i, \alpha_j) = 0 \Rightarrow [e_i^X, e_j^X] = [f_i^X, f_j^X] = 0$, and the super sign rule.
- (iv) The completed Hopf pairing on the positive and negative primitive halves has the same radical quotient as the Gritsenko–Nikulin/Kac construction of $G(\mathcal{B}_{\Delta_i})$.
- (v) The Hall primitive algebra satisfies a no-extra-relations theorem: after the standard radical quotient, the kernel of the map from the Borchers–Kac presentation is exactly the Borchers–Serre ideal and no additional completed relation survives.
- (vi) P_X^{Π} is generated by $P_{X,o}^{\Pi}$ and the simple primitive representatives e_i^X, f_i^X .
- (vii) A completed PBW theorem holds for P_X^{Π} , and its PBW filtration is transported to the PBW filtration on $U(\mathfrak{g}_{\Delta_i})$ under the comparison map.
- (viii) The full root-space parity dimensions in P_X^{Π} agree with those of $G(\mathcal{B}_{\Delta_i})$. Consequently, on active positive Gram degrees,

$$\text{sdim } P_{X,(n,l,m)}^{\Pi} = f(nm, l),$$

but this signed equality is only a necessary numerical consequence of the recognition data.

Then

$$H^0 \text{Prim}_{\text{prot}}(\Pi_{X,*}\mathcal{F}_X) \cong \mathcal{A}_{\text{den}} = \mathfrak{g}_{\Delta_i}$$

as Gram-graded Lie superalgebras, and its Weyl–Kac–Borchers denominator is $64^{-1}\Delta_i(2Z)$. If in addition Θ_{Kos} is supplied, then

$$\Omega_E^{\text{ch}} C_X \simeq \text{Den}_{\Delta_i, E}.$$

Proof. The simple primitive representatives and the displayed relations give a homomorphism

$$G(\mathcal{B}_{\Delta_i}) \longrightarrow P_X^{\Pi}$$

from the universal Borchers–Kac presentation. Generation gives surjectivity. The no-extra-relations hypothesis identifies the kernel with the standard Borchers–Serre ideal after the completed radical quotient; the completed PBW compatibility promotes the finite-stage identification to the inverse limit. The full parity dimensions rule out additional zero-superdimension root spaces. Hence the homomorphism is an isomorphism of Gram-graded Lie superalgebras. The denominator statement is Theorem 10.2. The chiral statement is exactly the quasi-isomorphism Θ_{Kos} . $\square$

Remark 10.34 (What is not proved). The recognition certificate is conditional. This paper does not construct the compact Hall/factorization object, the Gram descent $\Pi_{X,*}$, the protected primitive bracket constants, the full parity dimensions of compact primitives, the simple primitive representatives, or the verification of the Borchers–Kac relations in a Hall bracket. It also does not prove the no-extra-relations theorem, the completed PBW theorem, or closedness of the Hall radical in the HN completion. The signed equality $\text{sdim } P_{X,(n,l,m)}^{\Pi} = f(nm, l)$ and the determinant $\mathcal{D}_X = \Delta_5$ are necessary shadows. They are not sufficient to identify P_X^{Π} with $\mathfrak{g}_{\Delta_5}$.

PROPOSITION 10.35 (Low-degree bracket information). The denominator algebra determines the following low-degree data. With

$$\alpha(n, l, m) = n\delta_1 + m\delta_2 + (n + m - l)\delta_3,$$

the active rows through height three are

β	γ	(β, β)	$\text{smult}(\beta)$
δ_i	$(1, 1, 0), (0, 1, 1), (0, -1, 0)$	2	1
$\delta_i + \delta_j \ (i < j)$	$(1, 2, 1), (1, 0, 0), (0, 0, 1)$	0	10
$2\delta_i + \delta_j \ (i \neq j)$	six real rows	2	1
$\delta_1 + \delta_2 + \delta_3$	$(1, 1, 1)$	-6	-64.

After choosing even Chevalley generators e_i, f_i, h_i for the real simple roots,

$$[h_i, e_j] = (\delta_i, \delta_j)e_j, \quad [e_i, f_j] = \delta_{ij}h_i, \quad (\text{ad } e_i)^3 e_j = 0 \quad (i \neq j).$$

For $E_{ij} := [e_i, e_j]$,

$$[f_i, E_{ij}] = 2e_j, \quad [f_j, E_{ij}] = -2e_i.$$

These are denominator-algebra constants. They are not compact Hall constants.

Proof. The table is obtained by writing $\beta = c_1\delta_1 + c_2\delta_2 + c_3\delta_3$ and solving

$$\gamma(\beta) = (c_1, c_1 + c_2 - c_3, c_2).$$

Then $(\beta, \beta) = -2(4nm - l^2)$, and $\text{smult}(\beta) = f(nm, l)$ follows from Theorem 10.2. The displayed bracket constants are the normalized Chevalley, Jacobi and Serre constants for the symmetric Cartan matrix $A = ((\delta_i, \delta_j))$. $\square$

PROPOSITION 10.36 (The determinant does not determine primitive recognition data). The scalar square, the Borchers product, and active support determine the visible signed product exponents. After the Gritsenko–Nikulin denominator theorem, these exponents are the signed root supermultiplicities. They do not determine the compact Hall structure constants, the full parity dimensions, the simple primitive representatives, the Borchers–Kac relation verification, or the Hopf-pairing radical quotient.

Proof. The product determines

$$\text{smult}(\alpha) = \dim \mathfrak{g}_{\alpha, \bar{0}} - \dim \mathfrak{g}_{\alpha, \bar{1}}.$$

A graded super vector space with the same signed dimensions gives the same product. The zero bracket on that graded space gives the same product but not the Borchers bracket, since $[e_i, e_j] \neq 0$ in $\mathcal{A}_{\text{den}}$ for $i \neq j$. A Hall bracket requires compact extension correspondences, orientation data, protected integration, and a basis of protected primitives. The recognition certificate also requires the full parity split, simple-root representatives, defining relations, generation, and the radical quotient. None of these is encoded in the determinant. $\square$

Open Problem 10.37 (Compact E_3 , Hall, and chiral Koszul realization). Construct the datum $\mathfrak{R}_X$ of Definition 10.32. Equivalently, construct the holomorphic E_3 -factorization algebra on $K_3 \times E$, its chain-level K_3 -to- E specialization, the compact observable chiral shadow on $\text{Ran}(E)$, the oriented HN-completed Hall object, protected integration, the conilpotent chiral bar coalgebra, the chiral cobar quasi-isomorphism to $\text{Den}_{\Delta, E}$, and the primitive recognition certificate of Theorem 10.33. In particular one must construct the compact Hall bracket constants, the full compact parity dimensions, the simple primitive representatives, and the Borchers–Kac relation verification. The scalar square and the denominator algebra are inputs to this problem. They are not its solution.

Definition 10.38 (Imaginary simple roots). For $a \in \Lambda_{II}^{2,1} \cap \mathbb{R}_{>0} \mathcal{P}_{II}$ set

$$\mu_{\bar{0}}(a) = \begin{cases} m(a), & (a, a) < 0 \text{ and } m(a) > 0, \\ \tau(a), & (a, a) = 0 \text{ and } \tau(a) > 0, \\ 0, & \text{otherwise,} \end{cases}$$

and

$$\mu_{\bar{1}}(a) = \begin{cases} -m(a), & (a, a) < 0 \text{ and } m(a) < 0, \\ -\tau(a), & (a, a) = 0 \text{ and } \tau(a) < 0, \\ 0, & \text{otherwise.} \end{cases}$$

This sign split is the Gritsenko–Nikulin BKM parity datum. The product statement alone fixes only a Grothendieck representative for the signed exponent. Define multisets by multiplicity:

$$a \in \Delta_{\bar{0}}^{im} \quad \text{with multiplicity } \mu_{\bar{0}}(a),$$

$$a \in \Delta_{\bar{1}}^{im} \quad \text{with multiplicity } \mu_{\bar{1}}(a).$$

Put

$$\begin{aligned} \Delta^{im} &= \Delta_{\bar{0}}^{im} \cup \Delta_{\bar{1}}^{im}. \\ \mathcal{O}_{\text{den}}^{im} &= \bigoplus_{a \in \Delta^{im}} (\mathfrak{g}_{a, \bar{0}} \ominus \mathfrak{g}_{a, \bar{1}}). \end{aligned}$$

This is notation inside the denominator algebra; it is not a BPS state space.

The real simple roots are

$$\Delta_{\bar{0}}^{re, \text{simp}} = \Delta_{\text{simp}}^{re} = \mathcal{P}_{II, \text{prim}} = \{\delta_1, \delta_2, \delta_3\}, \quad \Delta_{\bar{1}}^{re, \text{simp}} = \emptyset.$$

Put

$$\Delta_{\bar{p}} = \Delta_{\bar{p}}^{re, \text{simp}} \cup \Delta_{\bar{p}}^{im} \quad (p = 0, 1), \quad \Delta = \Delta_{\bar{0}} \cup \Delta_{\bar{1}}.$$

The Cartan data satisfy $(\alpha, \alpha) > 0$ for $\alpha \in \Delta^{re}$, $(\alpha, \alpha) \leq 0$ for $\alpha \in \Delta^{im}$, and $(\alpha, \alpha') \leq 0$ for all distinct $\alpha, \alpha' \in \Delta$. If $(\alpha, \alpha) > 0$, then

$$2 \frac{(\alpha, \alpha')}{(\alpha, \alpha)} \in \mathbb{Z},$$

since $(\alpha, \alpha) = 2$ and $\Lambda_{II}^{2,1} = \mathbb{Z}\delta_1 + \mathbb{Z}\delta_2 + \mathbb{Z}\delta_3$.

Definition 10.39 (Generators and relations). Let I be an index set equipped with a parity map $p : I \rightarrow \mathbb{Z}/2\mathbb{Z}$ and a root map $i \mapsto \alpha_i \in \Delta$, with fibres over an imaginary root of cardinality $\mu_{\bar{p}}(a)$ in parity p . If $i \neq j$ and $\alpha_i = \alpha_j$, the two indices are distinct simple roots.

$$\text{ch} : I \rightarrow \Lambda_{II}^{2,1}, \quad i \mapsto \alpha_i.$$

The superalgebra $\mathfrak{g}_{\Delta}$ is generated by

$$h_i, e_i, f_i, \quad i \in I.$$

The Cartan subalgebra is identified with $\Lambda_{II}^{2,1} \otimes \mathbb{C}$ by $h_i \mapsto \alpha_i$. Relations:

- $[h_i, e_j] = (\alpha_i, \alpha_j)e_j$ and $[h_i, f_j] = -(\alpha_i, \alpha_j)f_j$
- $[e_i, f_j] = \delta_{ij}h_i$
- if $\alpha_i \in \Delta_{\text{simp}}^{re}$ and $i \neq j$, then

$$(\text{ad } e_i)^{1-2(\alpha_i, \alpha_j)/(\alpha_i, \alpha_i)} e_j = (\text{ad } f_i)^{1-2(\alpha_i, \alpha_j)/(\alpha_i, \alpha_i)} f_j = 0$$

- if $(\alpha_i, \alpha_j) = 0$, then $[e_i, e_j] = [f_i, f_j] = 0$

The grading is by $\Lambda_{II}^{2,1}$. Put

$$Q_+ = \sum_{\alpha \in \Delta} \mathbb{Z}_{\geq 0} \alpha \subset \Lambda_{II}^{2,1}.$$

The triangular decomposition is

$$\mathfrak{g}_{\Delta} = \left(\bigoplus_{\alpha \in Q_+ \setminus \{0\}} \mathfrak{g}_{\alpha} \right) \oplus (\Lambda_{II}^{2,1} \otimes \mathbb{C}) \oplus \left(\bigoplus_{\alpha \in Q_+ \setminus \{0\}} \mathfrak{g}_{-\alpha} \right).$$

For $i \in I$,

$$\deg e_i = \alpha_i, \quad \deg f_i = -\alpha_i, \quad \deg h_i = 0.$$

$$[\mathfrak{g}_{\alpha}, \mathfrak{g}_{\beta}] \subset \mathfrak{g}_{\alpha+\beta}.$$

$$\mathfrak{g}_0 = \Lambda_{II}^{2,1} \otimes \mathbb{C}.$$

The positive and negative roots are

$$\mathcal{R}_+ = \{\alpha \in Q_+ \setminus \{0\} \mid \mathfrak{g}_{\alpha} \neq 0\}, \quad \mathcal{R}_- = -\mathcal{R}_+.$$

For every root α , set

$$\text{mult}_{\bar{0}}(\alpha) = \dim \mathfrak{g}_{\alpha, \bar{0}}, \quad \text{mult}_{\bar{1}}(\alpha) = \dim \mathfrak{g}_{\alpha, \bar{1}}.$$

The signed root superdimension is

$$\text{smult}(\alpha) = \text{mult}_{\bar{0}}(\alpha) - \text{mult}_{\bar{1}}(\alpha).$$

The Weyl–Kac–Borcherds denominator is

$$\begin{aligned}\Phi_{\Delta_5}(z) &= \sum_{w \in W^{(2)}(\Lambda_{II}^{2,1})} \det(w) \left(e^{-2\pi i(w(\rho), z)} - \sum_{a \in \Lambda_{II}^{2,1} \cap \mathbb{R}_{>0} \mathcal{P}_{II}} m(a) e^{-2\pi i(w(\rho+a), z)} \right) \\ &= e^{-2\pi i(\rho, z)} \prod_{\alpha \in \mathcal{R}_+} \left(1 - e^{-2\pi i(\alpha, z)} \right)^{\text{smult}(\alpha)} \\ \Omega(\Lambda_{II}^{2,1}) &= \Lambda_{II}^{2,1} \otimes \mathbb{R} + i\mathcal{C}(\Lambda_{II}^{2,1})_+.\end{aligned}$$

THEOREM 10.40 (*Denominator identity*).

$$\frac{1}{64} \Delta_5(2Z) = \Phi_{\Delta_5}(z).$$

$$\text{den}(\mathfrak{g}_{\Delta_5}) = \frac{1}{64} \Delta_5(2Z).$$

Proof.

$$\begin{aligned}\frac{1}{64} \Delta_5(Z) &= \sum_{w \in W^{(2)}(\Lambda_{II}^{2,1})} \det(w) \left(e^{-\pi i(w(\rho), z)} - \sum_{a \in \Lambda_{II}^{2,1} \cap \mathbb{R}_{>0} \mathcal{P}_{II}} m(a) e^{-\pi i(w(\rho+a), z)} \right) \\ Z \mapsto 2Z : \quad e^{-\pi i(w(\rho+a), z)} &\mapsto e^{-2\pi i(w(\rho+a), z)}.\end{aligned}$$

Weyl–Kac–Borcherds:

$$e^{-2\pi i(\rho, z)} \prod_{\alpha \in \mathcal{R}_+} \left(1 - e^{-2\pi i(\alpha, z)} \right)^{\text{smult}(\alpha)}.$$

□

PROPOSITION 10.41 (*Gram-index cone and denominator cone*).

$$\alpha(\Gamma_{\text{act}}) = \{\alpha(n, l, m) \mid (n, l, m) \in \Gamma_{\text{eff}}, f(nm, l) \neq 0\} \subset \Lambda_{II}^{2,1}.$$

$$\alpha(\Gamma_{\text{act}}) \subset \mathcal{R}_+^{\text{vis}} := \{\alpha \in \mathcal{R}_+ \mid \text{smult}(\alpha) \neq 0\}.$$

$$\text{smult}(\alpha(n, l, m)) = \dim \mathfrak{g}_{\alpha(n, l, m), \bar{0}} - \dim \mathfrak{g}_{\alpha(n, l, m), \bar{1}} = f(nm, l) \quad ((n, l, m) \in \Gamma_{\text{act}}).$$

Proof.

$$\alpha : \Gamma_{\text{ind}} \rightarrow \Lambda_{II}^{2,1}, \quad (n, l, m) \mapsto 2nf_2 - lf_3 + 2mf_{-2}.$$

$$q^n r^l s^m = \exp(-\pi i(\alpha(n, l, m), z))$$

$$Z \mapsto 2Z.$$

$$\frac{1}{64} \Delta_5(2Z) = e^{-2\pi i(\rho, z)} \prod_{\alpha \in \mathcal{R}_+} \left(1 - e^{-2\pi i(\alpha, z)} \right)^{\text{smult}(\alpha)}.$$

$$\frac{1}{64} \mathcal{D}_X(2Z) = e^{-2\pi i(\rho, z)} \prod_{(n, l, m) \in \Gamma_{\text{eff}}} \left(1 - e^{-2\pi i(\alpha(n, l, m), z)} \right)^{f(nm, l)}.$$

The triples in $\Gamma_{\text{eff}} \setminus \Gamma_{\text{act}}$ have exponent zero and do not contribute visible product factors. The denominator product alone does not prove absence of root spaces with zero signed superdimension.

$$\mathcal{D}_X = \Delta_5 \implies \text{smult}(\alpha(n, l, m)) = f(nm, l) \quad ((n, l, m) \in \Gamma_{\text{act}}).$$

□

PROPOSITION 10.42 (*What the denominator determines*). The identity

$$\frac{1}{64} \Delta_5(2Z) = e^{-2\pi i(\rho, z)} \prod_{\alpha \in \mathcal{R}_+} \left(1 - e^{-2\pi i(\alpha, z)}\right)^{\text{smult}(\alpha)}$$

determines the signed integers $\text{smult}(\alpha)$, once the Gritsenko–Nikulin root set and positive root monoid have been fixed. It does not determine invisible zero-superdimension root spaces, the two parity dimensions

$$\text{mult}_{\bar{0}}(\alpha), \quad \text{mult}_{\bar{1}}(\alpha),$$

and it does not determine the structure constants

$$[\mathfrak{g}_\alpha, \mathfrak{g}_\beta] \rightarrow \mathfrak{g}_{\alpha+\beta}.$$

Thus signed root multiplicity is necessary input for primitive recognition, but it is not a recognition certificate.

Proof. Write $X^\alpha = e^{-2\pi i(\alpha, z)}$. In the completed positive root monoid algebra,

$$\log \prod_{\alpha \in \mathcal{R}_+} (1 - X^\alpha)^{\text{smult}(\alpha)} = - \sum_{\alpha \in \mathcal{R}_+} \sum_{k \geq 1} \frac{\text{smult}(\alpha)}{k} X^{k\alpha}.$$

Thus a logarithmic coefficient mixes divisor terms $k\alpha = \lambda$. The exponents $\text{smult}(\alpha)$ are recovered recursively by height, equivalently by Mobius inversion on each primitive ray. The recovered exponent is only

$$\text{smult}(\alpha) = \text{mult}_{\bar{0}}(\alpha) - \text{mult}_{\bar{1}}(\alpha).$$

The difference $a - b$ does not determine the pair (a, b) , and no bilinear bracket appears in the product. $\square$

PROPOSITION 10.43 (*Index equality is not a state-space identification*). The equality

$$\text{smult}(\alpha(n, l, m)) = f(nm, l)$$

identifies the root-space Euler characteristic of $\mathcal{A}_{\text{den}}$ with the virtual K3 half-index coefficient. It does not prove an isomorphism

$$\mathfrak{g}_{\alpha(n, l, m)} \cong \mathcal{H}_{(n, l, m)}^{\text{BPS}}$$

and it does not prove that the microscopic operator product on BPS states is the Borchers bracket. In particular, it does not prove

$$H^0 \text{Prim}_{\text{prot}}(\Pi_{X,*} \mathcal{F}_X) \cong \mathfrak{g}_{\Delta_5}.$$

That comparison requires the primitive recognition hypotheses of Theorem 10.33.

PROPOSITION 10.44 (*What the determinant forgets*). Let

$$\mathcal{B} = \bigoplus_{\gamma \in \Gamma_{\text{eff}}} \mathcal{B}_\gamma$$

be any charge-graded super vector space with finite-dimensional homogeneous pieces and

$$\text{sdim } \mathcal{B}_{(n, l, m)} = f(nm, l).$$

Then its Gram-graded Euler/Fock determinant is

$$64q^{1/2}r^{1/2}s^{1/2} \prod_{\gamma \in \Gamma_{\text{eff}}} (1 - x^\gamma)^{\text{sdim } \mathcal{B}_\gamma} = \Delta_5(Z).$$

If $\mathcal{B}$ carries a bracket compatible with charge addition, a parity split, simple-root representatives, or a Hopf pairing, the determinant forgets those data and remembers the class of $\mathcal{B}$ in

$$K_o(\Gamma_{\text{eff}}\text{-graded finite super vector spaces}).$$

Thus the determinant of $\mathcal{A}_{\text{den}} = \mathfrak{g}_{\Delta_5}$ is the Grothendieck shadow of its denominator bracket. The Borchers denominator bracket exists inside $\mathcal{A}_{\text{den}}$ by the Gritsenko–Nikulin construction, but the determinant sees only its charge-graded Grothendieck class.

Proof. The Fock determinant lemma gives

$$\text{sdet}_{\mathcal{B}}(1 - x) = \prod_{\gamma \in \Gamma_{\text{eff}}} (1 - x^\gamma)^{\text{sdim } \mathcal{B}_\gamma}.$$

The hypothesis $\text{sdim } \mathcal{B}_{(n,l,m)} = f(nm, l)$ identifies this product with the Borchers product for $64^{-1}\Delta_5$. Multiplying by the vacuum factor gives Δ_5 . A bracket

$$[\mathcal{B}_\gamma, \mathcal{B}_{\gamma'}] \rightarrow \mathcal{B}_{\gamma+\gamma'}$$

does not occur in the determinant formula. Neither do simple primitive representatives, parity dimensions, relation constants, or a Hopf-pairing radical. The determinant is therefore invariant under replacing $\mathcal{B}$ by any other graded super vector space with the same Grothendieck class. $\square$

II COEFFICIENT DICTIONARY AND MAIN SYNTHESIS

The coefficient convention is fixed once and for all by Eichler–Zagier normalization:

$$Z_{K_3} = 2\phi_{o,1}, \quad \phi_{o,1}(\tau, z) = \sum_{n \geq 0, l \in \mathbb{Z}} f(n, l) q^n r^l.$$

Equivalently, if $\phi_{12,1}$ is the first Fourier–Jacobi coefficient of Igusa’s weight-12 form F_{12} , and if

$$\delta_{12}(\tau) = q \prod_{n \geq 1} (1 - q^n)^{24},$$

then

$$\phi_{o,1} = \frac{\phi_{12,1}}{\delta_{12}}.$$

The first terms are

$$\begin{aligned} \phi_{o,1} &= (r^{-1} + 10 + r) + q(10r^{-2} - 64r^{-1} + 108 - 64r + 10r^2) \\ &\quad + q^2(r^{-3} + 108r^{-2} - 513r^{-1} + 808 - 513r + 108r^2 + r^3) + O(q^3). \end{aligned}$$

Thus

$$f(o, -1) = 1, \quad f(o, 0) = 10, \quad f(1, 1) = -64, \quad f(1, 0) = 108.$$

The weak Jacobi support condition is

$$f(n, l) = 0 \quad \text{unless} \quad n \geq 0, \quad 4n - l^2 \geq -1,$$

and $f(n, l) = f(n, -l)$. These integers have three distinct readings in the paper:

object	meaning of $f(nm, l)$
$\mathcal{V}_{(n,l,m)}$	virtual K_0 -superdimension
$\mathcal{D}_X$	Borcherds product exponent
$\mathfrak{g}_{\Delta_5, \alpha(n,l,m)}$	signed root supermultiplicity on active support

The first reading is a protected Euler characteristic. The second and third readings are automorphic and denominator-algebra incarnations of the same integer. This is not an equality of Hilbert spaces, brackets, or Hall structure constants.

The synthesis is the following spine:

$$\begin{aligned} &\text{virtual determinant} \longrightarrow \text{denominator algebra} \longrightarrow \text{reduced scalar square} \\ &\longrightarrow \text{compact realization problem} \longrightarrow \text{eight-row certificate extension.} \end{aligned}$$

The arrows record the order of comparison. They are not automatic implications from a product to a physical host.

THEOREM II.1 (*Coefficient dictionary and synthesis theorem*). Let $X = K_3 \times E$. Let

$$\mathbb{U}_{n,l} \in K_0(\text{sVect}_{\text{fd}}), \quad \text{sdim } \mathbb{U}_{n,l} = f(n, l),$$

and define the virtual K_0 -determinant object in the cusp chamber by

$$\mathcal{V}_{(n,l,m)} = \mathbb{U}_{nm,l}, \quad (n, l, m) \in \Gamma_{\text{eff}}.$$

Then the theorem-level and conditional content of the paper is exactly as follows.

First, the virtual determinant is the theta-normalized Igusa square root:

$$\mathcal{D}_X(Z) = 64 q^{1/2} r^{1/2} s^{1/2} \prod_{(n,l,m) \in \Gamma_{\text{eff}}} (1 - q^n r^l s^m)^{f(nm,l)} = \Delta_5(Z).$$

The numerical factor 64 is the theta-constant leading coefficient $[q^{1/2} r^{1/2} s^{1/2}] \Delta_5$; the monic Borcherds product is

$$D_5 = 64^{-1} \Delta_5.$$

Second, under the Lorentzian degree map

$$\alpha(n, l, m) = 2nf_2 - lf_3 + 2mf_{-2}, \quad q^n r^l s^m = \exp(-\pi i(\alpha(n, l, m), z)),$$

the same active coefficients are the visible signed root supermultiplicities of the Gritsenko–Nikulin denominator algebra:

$$\text{smult } \mathfrak{g}_{\Delta_5, \alpha(n,l,m)} = f(nm, l) \quad ((n, l, m) \in \Gamma_{\text{act}}),$$

and

$$\text{den}(\mathfrak{g}_{\Delta_5}) = 64^{-1} \Delta_5(2Z).$$

This is the Gritsenko–Nikulin denominator theorem. It determines the signed root supermultiplicities in the fixed Borchers datum; it does not determine compact Hall constants or microscopic brackets.

Third, for the reduced Hilbert-scheme DT/PT/OP scalar branch,

$$Z_{\text{DT}}^X(T, Q, P) = Z_{\text{OP}}^X(P, Q, T) = -4096 \Delta_5(Z)^{-2}.$$

Here

$$\chi_{\text{io}}^{\text{OP}} = D_5^2 = (64^{-1} \Delta_5)^2 = 4096^{-1} \Delta_5^2, \quad Z_{\text{OP}}^X = -(\chi_{\text{io}}^{\text{OP}})^{-1};$$

hence $-4096 = -(64)^2$ is the OP scalar sign together with the theta-to-monic normalization. It is not an orientation character. Thus, with

$$\mathcal{Z}_{\text{Borch}}^{\text{vir}} = \mathcal{D}_X^{-1},$$

one has

$$Z_{\text{DT}}^X = Z_{\text{OP}}^X = -4096 (\mathcal{Z}_{\text{Borch}}^{\text{vir}})^2.$$

Fourth, the stronger microscopic comparison

$$H^0 \text{Prim}_{\text{prot}}(\Pi_{X,*} \mathcal{F}_X) \cong \mathfrak{g}_{\Delta_5}$$

is open unless the compact datum of Definition 10.32 is constructed. If such a datum is constructed, Theorem 10.33 identifies its primitive denominator with $64^{-1} \Delta_5(2Z)$.

Fifth, any claimed compact realization must also satisfy three consequences.

- (i) A naive support-local factorization object on $\text{Ran}(E)$ sees only projection-finite sectors. If a one-dimensional subscheme of $K_3 \times E$ has positive elliptic degree, its projection to E is all of E . Thus the full s -direction of the Igusa product requires a hybrid local/global wrapped factorization theory. A separate global Fock/Hecke mode is only determinant-level unless it is coupled to the local sector by mixed Hall actions before the reduced E -quotient.
- (ii) The automorphic character ν_{Δ_5} is theorem-level data of Δ_5 , and Δ_5^2 forgets it. A compact Hall orientation must construct a strong orientation character ϵ_o and prove

$$\epsilon_o|_{W^{(2)}(\Lambda_{II}^{2,1})} = \nu_{\Delta_5}|_{W^{(2)}(\Lambda_{II}^{2,1})}.$$

The constants 64, 4096, and the OP minus sign do not solve this orientation-monodromy problem. The Hall-side proof must compute the three simple type-II determinant-line signs and check the reflection relations.

- (iii) A primitive comparison with $\mathfrak{g}_{\Delta_5}$ is a presentation-level recognition certificate, not a consequence of signed dimensions. One must supply primitive representatives for the real and imaginary simple roots, their parities, the Chevalley, real Serre and orthogonality relations, generation, the completed Hopf pairing and the same radical quotient, a no-extra-relations theorem, and completed PBW compatibility. Under these hypotheses the Borchers–Kac universal presentation gives the isomorphism with $\mathfrak{g}_{\Delta_5}$; without them the determinant fixes only a Grothendieck class.

Finally, the diagonal-divisor rows extend this same logic. A row is promoted beyond its automorphic product only by a row certificate fixing the modular group or cover, character, divisor, Jacobi seed, product lattice, Weyl chamber, reflection character, scalar theorem if one is claimed, orientation data, one-particle index, and primitive algebra comparison.

Proof. The Fock trace calculation gives

$$\mathcal{D}_X = 64q^{1/2}r^{1/2}s^{1/2} \text{sdet}_V(1-x).$$

The Borcherds–Gritsenko–Nikulin product formula for the input $\phi_{o,i}$ identifies this product with Δ_5 , as recorded in Proposition 4.5. The leading coefficient 64 is fixed in the theta-normalized Fourier expansion of Δ_5 , hence $D_5 = 64^{-1}\Delta_5$ is the monic product.

The Lorentzian monomial identity and pairing computation are in Section 7 and the Weyl-chamber section. The root-supermultiplicity and denominator identity are Theorem 10.2; the argument $2Z$ comes from passing from $\exp(-\pi i(\alpha, z))$ to the usual denominator character $\exp(-2\pi i(\alpha, z))$.

The reduced scalar identity is Theorem 6.2, with the normalization conversion of Proposition 6.1. Since Oberdieck–Pixton use $\chi_{io}^{\text{OP}} = D_5^2$ and prove $Z_{\text{OP}}^X = -(\chi_{io}^{\text{OP}})^{-1}$, the theta normalization gives

$$Z_{\text{OP}}^X = -4096\Delta_5^{-2}.$$

Oberdieck’s reduced DT/PT comparison and the primitive Oberdieck–Pandharipande comparison give the same branch for Z_{DT}^X .

The compact comparison is not used in these proofs. It is precisely Open Problem 10.37. The $\text{Ran}(E)$ obstruction follows because a positive elliptic-degree curve has one-dimensional image in the irreducible curve E , hence image E ; support separation on disjoint opens of $\text{Ran}(E)$ cannot see such a sector. The orientation statement follows from the automorphic character computation for Δ_5 and from the definition of a compact orientation datum; rescaling Δ_5 by 64^{-1} changes no character. The primitive recognition statement is the universal property of the Borcherds–Kac presentation recorded in the denominator-algebra section. The determinant and scalar square do not contain the missing presentation, orientation, or Hall correspondence data. The row certificate condition is the same separation applied to the Clery–Gritsenko catalogue in the next part. $\square$

Thus the paper has one dictionary and several levels of structure. The dictionary is the integer $f(nm, l)$. The proved structures are the virtual determinant, the Gritsenko–Nikulin denominator algebra, and the reduced OP/DT scalar square with constants 64 and -4096 fixed as above. The compact $K_3 \times E$ source and the boundary-row physical hosts are certificate problems.

Part II

Diagonal-divisor rows and CHL boundary physics

12 EIGHT DIAGONAL-DIVISOR ROWS: AUTOMORPHIC AND PHYSICAL STATUS

This part is not an appendix to the $N = 1$ theorem. It is the second problem of the paper. The Clery–Gritsenko catalogue gives eight diagonal-divisor automorphic products. The Gritsenko–Nikulin and Govindarajan theorems promote these products to denominator identities in the rows where the Weyl data are known. A CHL dyon square, a reduced CY/DT scalar host, and a compact Hall/chiral host are further structures. They are recorded only where their data have been specified.

$$g = 2.$$

$$t, N \in \mathbb{Z}_+ : \quad \Gamma_t(N) = \left\{ \begin{pmatrix} * & *t & * & * \\ & * & * & *t^{-1} \\ N & *Nt & * & * \\ Nt & *Nt & *t & * \end{pmatrix} \in \mathbf{Sp}_4(\mathbb{Q}) \mid * \in \mathbb{Z} \right\}.$$

$$\mathbb{H}_2/\Gamma_t \longleftrightarrow \{(1, t)\text{-polarized abelian surfaces}\}.$$

$$\mathcal{H}_{\text{diag}} = \left\{ \begin{pmatrix} a & 0 \\ 0 & b \end{pmatrix} \mid a, b \in \mathbb{H}_1 \right\} \subset \mathbb{H}_2.$$

Definition 12.1 (Physical-host certificate). For a diagonal-divisor row F_j , a physical-host certificate is a five-level certificate table

$$\mathfrak{C}_j = (\mathcal{A}_j, \mathcal{D}_j, \mathcal{S}_j, \mathcal{L}_j, \mathcal{H}_j).$$

The entries have the following meanings.

- (i) $\mathcal{A}_j$ is the automorphic product certificate: the form F_j , the group or multiplier cover, the automorphy cocycle, the character or multiplier with exact order, the diagonal divisor with multiplicities, the product chamber and branch, the Jacobi input at every cusp, the product-degree map, the product normalization, and the source theorem.
- (ii) $\mathcal{D}_j$ is the denominator certificate: a Weyl–Kac–Borcherds denominator algebra with root lattice, Weyl vector, simple real roots, Weyl chamber, Weyl group, reflection character, imaginary simple-root coefficients, product side, and Weyl alternating side.
- (iii) $\mathcal{S}_j$ is a CHL/dyon-square certificate: a sourced dyon-counting scalar form Φ_j , a row map to the same automorphic row and cover, and an identity $F_j^2 = \Phi_j$ in the stated chamber. This is not a denominator theorem and not a CY/DT partition theorem.
- (iv) $\mathcal{L}_j$ is a reduced CY/DT scalar-host certificate: a compact Calabi–Yau threefold, orbifold Calabi–Yau, or CHL background with oriented reduced obstruction theory, fixed variables and chamber, and a proved identity

$$Z_{\text{red}}^{X_j}(Z) = C_j F_j(Z)^{-2}.$$

- (v) $\mathcal{H}_j$ is a compact Hall/chiral certificate: an oriented compact Hall, Drinfeld, or factorization object whose protected one-particle index gives the product exponents and whose protected primitive bracket identifies the denominator algebra with the same orientation/reflection character.

The product row, denominator row, dyon square, reduced scalar host, and compact Hall/chiral host are separate structures. None of the five entries implies another unless a cited theorem constructs the implication. In particular, matching a weight, a square root, or a comparison symbol does not fill a certificate. A row-map theorem must name the host, charge lattice, effective chamber, modular group

or cover, product lattice, orientation character, one-particle index, scalar identity when claimed, and primitive Hall/chiral comparison.

THEOREM 12.2 (*Clery–Gritsenko*). In the class of Siegel modular forms with character or multiplier system for the Hecke-type congruence subgroups $\Gamma_t(N)$ whose divisor is the union of the $\Gamma_t(N)$ -translates of $\mathcal{H}_{\text{diag}}$ with multiplicity one, there are exactly eight rows [?GC, Theorem 1.4]. The nine-candidate list is [?GC, Proposition 1.2]; the proof of Theorem 1.4 excludes only the ninth candidate $(t, N; k) = (2, 4; \frac{1}{2})$. The congruence Borcherds product with all cusp coefficients is [?GC, Theorem 3.1], with the final Section 3 product displays for ∇_3 , ∇_2 , $\nabla_{3/2}$, and Q_1 .

j	F_j	Γ	$\text{wt}(F_j)$	character/multiplier
1	Δ_5	Γ_1	5	$v_\eta^{12} \times v_H$
2	Δ_2	Γ_2^+	2	$v_\eta^6 \times v_H$
3	Δ_1	Γ_3^+	1	$v_\eta^4 \times v_H$
4	$\Delta_{1/2}$	Γ_4^+	$\frac{1}{2}$	$v_\eta^3 \times v_H$ (degree 8)
5	∇_3	$\Gamma_o^{(2)}(2)$	3	$\chi_2^{(2)} \times v_H$
6	∇_2	$\Gamma_o^{(2)}(3)$	2	$\chi_2^{(3)} \times v_H$
7	$\nabla_{3/2}$	$\Gamma_o^{(2)}(4)$	$\frac{3}{2}$	order-four multiplier $\chi_4^{(4)} \times v_H$
8	Q_1	$\Gamma_2(2)$	1	$\chi_4^{(2)} \times v_H$

$$\text{supp div}_{\mathbb{H}_2}(F_j) = \bigcup_{\gamma \in \Gamma_j} \gamma \langle \mathcal{H}_{\text{diag}} \rangle, \quad \text{mult}_{\mathcal{H}_{\text{diag}}}(F_j) = 1.$$

$$F_1 = \Delta_5, \quad \frac{1}{64} F_1(2Z) = \text{den}(\mathfrak{g}_{\Delta_5}).$$

row	automorphic product	denominator theorem	CHL/dyon square	reduced CY/DT scalar host	compact host	Hall/chiral
1	$F_1 = \Delta_5$; Clery–Gritsenko product and Borchers–Gritsenko–Nikulin lift of $\phi_{0,1}$	Gritsenko–Nikulin denominator algebra $\mathfrak{g}_{\Delta_5}$; $64^{-1}\Delta_5(2Z)$ normalization	no CHL row map used; Δ_5^2 is the Igusa square, not a boundary dyon certificate	proved $K_3 \times E$ scalar host: $-4096\Delta_5^{-2}$	OP/DT scalar host:	open; the formal determinant is not a compact Hall/chiral source
2	$F_2 = \Delta_2$; Clery–Gritsenko product	Gritsenko–Nikulin denominator row $\mathfrak{g}_2^{\text{GN}}$	no CHL/dyon map	certified row	no certified curve-counting scalar theorem	no certified Hall or factorization host
3	$F_3 = \Delta_1$; Clery–Gritsenko product	Gritsenko–Nikulin denominator row $\mathfrak{g}_3^{\text{GN}}$	no CHL/dyon map	certified row	no certified curve-counting scalar theorem	no certified Hall or factorization host
4	$F_4 = \Delta_{1/2}$; Clery–Gritsenko product on the degree-eight multiplier cover	Gritsenko–Nikulin denominator row on the same cover $\mathfrak{g}_4^{\text{GN}}$	no CHL/dyon map	certified row	no certified curve-counting scalar theorem	no certified Hall or factorization host
5	$F_5 = \nabla_3$; Clery–Gritsenko product	Govindarajan $\mathcal{G}_1(2)$ CHL/WKB denominator theorem	sourced scalar dyon square $\nabla_3^2 = \Phi_6$	no reduced CY/DT certificate	certified	no compact Hall/chiral certificate
6	$F_6 = \nabla_2$; Clery–Gritsenko product	Govindarajan $\mathcal{G}_1(3)$ CHL/WKB denominator theorem	sourced scalar dyon square $\nabla_2^2 = \Phi_4$	no reduced CY/DT certificate	certified	no compact Hall/chiral certificate
7	$F_7 = \nabla_{3/2}$; Clery–Gritsenko product on the order-four multiplier cover	Govindarajan $\mathcal{G}_1(4)$ CHL/WKB denominator theorem on the μ_4 -cover	sourced scalar dyon square $F_7^2 = \Phi_3$ only after fixing the cover and branch	no reduced CY/DT certificate	certified	no compact Hall/chiral certificate
8	$F_8 = Q_1$; Clery–Gritsenko $(\tau, N) = (2, 2)$ product	Govindarajan $\mathcal{G}_2(4)$ CHL/WKB denominator theorem	sourced twisted dyon square $Q_1^2 = \Phi_2^{(2,4)}$; Q_1 is the special $\mathcal{G}_2(4)$ row	no reduced CY/DT certificate	certified	no compact Hall/chiral certificate

This table is a certificate table, not a catalogue of physical hosts. By Theorem B.7 the full-paramodular rows F_2, F_3, F_4 also carry Gritsenko–Nikulin denominator identities. The cited sources do not construct a reduced curve-counting theorem, Hall object, or factorization host for these three rows. Their status is therefore product plus denominator theorem only. The congruence rows F_5, F_6, F_7, F_8 have product-degree lattices in Proposition B.9. The additional denominator data – Weyl chamber, real roots, reflection character, and Weyl sum – are supplied by the CHL denominator theorem of Govindarajan for F_5, F_6, F_7, F_8 . For F_7 the theorem lives on the μ_4 -multiplier cover determined by the Clery–Gritsenko order-four multiplier. For Q_1 the row is the special $\mathcal{G}_2(4)$ twisted CHL denominator row. Neither statement supplies a reduced CY/DT scalar host or compact Hall/chiral certificate.

PROPOSITION 12.3 (Extension separation). No statement in this section is used in the $N = 1$ theorem for Δ_5 . For every row the product formula itself gives a Gram-indexed primitive product. It gives a Gram-graded Euler/Fock determinant only after a one-particle index and a charge-invariant interpretation have been supplied. A nonabelian denominator theorem is stronger: it additionally requires a Weyl chamber, real simple roots, a reflection character, and a Weyl alternating sum.

Open Problem 12.4 (Future certificates beyond the known rows). For any future diagonal-divisor or boundary product not already covered by the Gritsenko–Nikulin denominator theorems or by the Govindarajan CHL/WKB denominator theorem, first construct the product certificate

$$\mathcal{P}_j = (\phi_j, \Gamma_{j,\text{prod}}^+, \Gamma_{j,\text{mod}}, \rho_j, \nu_j, L_j, \alpha_j)$$

with

$$\phi^{(j)}(\tau, z) = \sum c_j(n, l) q^n r^l,$$

and then construct, if possible, the additional denominator data: positive product cone, Weyl chamber, simple real roots, reflection character, Weyl vector, Weyl alternating sum, imaginary simple-root coefficients, and a generalized Borcherds–Kac–Moody superalgebra $\mathfrak{g}_j$ such that

$$F_j(Z) = C_j e^{2\pi i(\rho_j, Z)} \prod_{\gamma \in \Gamma_{j, \text{prod}}^+} (1 - e^{2\pi i(\gamma, Z)})^{c_j(\gamma)},$$

$$\text{den}(\mathfrak{g}_j) = C'_j F_j(a_j Z), \quad \text{sdim}(\mathfrak{g}_{j, \gamma}) = c_j(\gamma).$$

If a CHL/ K_3 sector supplies a dyon-counting scalar square, it fills only the $\mathcal{S}_j$ level of Definition 12.1. If a reduced CY/DT scalar identity is claimed, one must separately construct X_j , the reduced obstruction theory, orientation, Fourier chamber, variables, and the theorem

$$Z_{\text{red}}^{X_j}(Z) = C_j F_j(Z)^{-2}.$$

If a compact Hall/chiral host is claimed, one must separately construct the protected one-particle index and primitive bracket comparison. This open problem is not used for the eight-row certificate table and implies no scalar square, CY/DT host, or Hall/chiral host without the corresponding certificate entries.

PROPOSITION 12.5 (*CHL squares and denominator signs*). The identities

$$\nabla_3^2 = \Phi_6, \quad \nabla_2^2 = \Phi_4$$

which occur in the CHL dyon literature do not by themselves give denominator identities for the square-root rows $F_5 = \nabla_3$ and $F_6 = \nabla_2$. They give scalar Siegel modular forms on the CHL side. To descend from a scalar square to a denominator theorem for a root form, one must also construct the reflection character, the parity, and the Weyl anti-invariant additive side. Govindarajan’s twisted CHL theorem supplies exactly this missing structure for

$$F_5 = \nabla_3, \quad F_6 = \nabla_2, \quad F_7 = \nabla_{3/2}, \quad F_8 = Q_1,$$

as recorded in Theorem B.13.

Proof. Clery–Gritsenko record the dd-forms ∇_3 and ∇_2 and note that their squares were constructed in the CHL papers as Φ_6 and Φ_4 . The square forgets the sign character which a denominator identity needs on simple real reflections. Thus the scalar CHL product is evidence for the automorphic product, not by itself a proof of the BKM datum $(P_j, \mathcal{M}_j, W_j, \epsilon_j, \rho_j, N_j(a))$ for the root form. The additional datum is supplied by [?Gov11BKM], building on [?GK09CHL] and [?CD09BKM]. $\square$

A separate comparison tuple has weights

$$(5, 2, 1, 1, \frac{1}{2}, 1, \frac{1}{4}, 0).$$

It is not the Clery–Gritsenko table. A comparison with the Clery–Gritsenko forms requires the relevant entries of Definition 12.1. Thus

Clery–Gritsenko classification $\neq$ CHL realisation theorem.

F_1 : theorem, F_2, F_3, F_4 : GN denominator theorem,

F_5, F_6, F_7, F_8 : Govindarajan CHL/WKB denominator theorem,

with F_7 understood on the μ_4 -multiplier cover determined by its order-four multiplier. Dijkgraaf–Verlinde–Verlinde and Borcherds give the $N = 1$ physical and automorphic background [?DVV, ?B]; they do not construct physical-host certificates for the other seven rows.

A SPIN L -FACTOR NORMALISATION FOR Δ_5^2

Set the theta-constant normalisation

$$\Delta_{10}^\theta := \Delta_5^2.$$

The Oberdieck–Pixton monic normalisation is

$$\chi_{10}^{\text{OP}} := 2^{-12} \Delta_{10}^\theta = 4096^{-1} \Delta_5^2.$$

The spinor L -function is attached to the Hecke eigenline

$$\mathbb{C} \Delta_{10}^\theta = \mathbb{C} \chi_{10}^{\text{OP}};$$

it is independent of the non-zero scalar used to name its generator. Put

$$\Delta(q) = q \prod_{n \geq 1} (1 - q^n)^{24}, \quad E_6(q) = 1 - 504 \sum_{n \geq 1} \sigma_5(n) q^n,$$

$$g := \Delta E_6 = \sum_{n \geq 1} a_n(g) q^n \in S_{18}(\mathbf{SL}_2(\mathbb{Z})).$$

THEOREM A.1 (*Andrianov factorisation*). For the Saito–Kurokawa representation generated by Δ_{10}^θ ,

$$L_{\text{spin}}(s, \Delta_{10}^\theta) = \zeta(s-8) \zeta(s-9) L(s, g).$$

Here L_{spin} is Andrianov’s uncompleted classical spinor L -function. For a genus-two form of weight 10, the completed function is

$$L_{\text{spin}}^*(s, F) = (2\pi)^{-2s} \Gamma(s) \Gamma(s-8) L_{\text{spin}}(s, F).$$

Proof. The Igusa cusp form Δ_{10}^θ spans the weight-10 Maass Spezialschar line attached by the Saito–Kurokawa correspondence to the unique normalized form $g = \Delta E_6 \in S_{18}(\mathbf{SL}_2(\mathbb{Z}))$ [ZagierSK]. This is Andrianov’s spinor L -factor formula for the degree-2 Saito–Kurokawa lift attached to the weight-18 elliptic cusp form g [Andr74]. Since $\dim S_{18}(\mathbf{SL}_2(\mathbb{Z})) = 1$, the source form is ΔE_6 . $\square$

PROPOSITION A.2 (*Central-zero cancellation*). With

$$\Lambda(g, s) := (2\pi)^{-s} \Gamma(s) L(s, g),$$

one has

$$\Lambda(g, s) = -\Lambda(g, 18-s), \quad L(9, g) = 0.$$

Hence the possible pole of $\zeta(s-8) L(s, g)$ at $s = 9$ is cancelled. At $s = 10$ the residue is

$$\text{Res}_{s=10} L_{\text{spin}}(s, \Delta_{10}^\theta) = \zeta(2) L(10, g) = \frac{\pi^2}{6} L(10, g).$$

Moreover $L(10, g) \neq 0$. Thus $L_{\text{spin}}(s, \Delta_{10}^\theta)$ has a simple pole at $s = 10$ with non-zero residue.

Proof. For a level-one cusp form of weight 18, the root number is $(-1)^{18/2} = -1$. Thus the completed L -function is odd about $s = 9$, and the central value vanishes. The factor $\zeta(s - 8)$ has its only possible pole at $s = 9$, so the product $\zeta(s - 8)L(s, g)$ is regular there. At $s = 10$ the factor $\zeta(s - 9)$ has residue 1, while $\zeta(s - 8) = \zeta(2)$. Write

$$L_p(s, g) = (1 - \alpha_p p^{-s})^{-1} (1 - \beta_p p^{-s})^{-1}.$$

Deligne's Ramanujan–Petersson theorem gives

$$|\alpha_p| = |\beta_p| = p^{17/2}.$$

At $s = 10$,

$$|\alpha_p p^{-10}| = |\beta_p p^{-10}| = p^{-3/2},$$

so the Euler product converges absolutely and has no zero. $\square$

PROPOSITION A.3 (*Manin–Deligne period*). There are Manin periods $\Omega^\pm(g)$ such that

$$\varepsilon(m) = \begin{cases} +, & m \equiv 0 \pmod{2}, \\ -, & m \equiv 1 \pmod{2}, \end{cases} \quad \frac{L(m, g)}{(2\pi i)^m \Omega^{\varepsilon(m)}(g)} \in \mathbb{Q}, \quad 1 \leq m \leq 17.$$

In particular

$$L(10, g) \in \mathbb{Q}^\times (2\pi i)^{10} \Omega^+(g).$$

Proof. This is the Eichler–Shimura–Manin period theorem for the rational newform g [Manin72], [KZ], in Deligne's period normalisation [De179]. The nonvanishing at $m = 10$ is supplied by the Euler product in Proposition A.2. No identity of the form

$$L(10, g) = q a_{10}(g) \Omega^-(g), \quad q \in \mathbb{Q}^\times,$$

is used. $\square$

PROPOSITION A.4 (*Coefficient check*).

$$a_{10}(g) = 541648800 = 2^5 \cdot 3^2 \cdot 5^2 \cdot 7 \cdot 11 \cdot 977.$$

Proof.

$$a_{10}(g) = \tau(10) - 504 \sum_{k=1}^9 \tau(k) \sigma_5(10 - k).$$

$$(\tau(1), \dots, \tau(10)) = (1, -24, 252, -1472, 4830, -6048, -16744, 84480, -113643, -115920),$$

$$(\sigma_5(1), \dots, \sigma_5(9)) = (1, 33, 244, 1057, 3126, 8052, 16808, 33825, 59293).$$

Substitution gives $a_{10}(g) = 541648800$. Equivalently, since g is a Hecke eigenform,

$$a_{10}(g) = a_2(g) a_5(g) = (-528)(-1025850) = 541648800.$$

$\square$

Remark A.5 (Scalar-square separations).

claim	failure	correct statement
$\chi_{10}^{\text{OP}} = \Delta_5^2$	normalisation conflict	$\chi_{10}^{\text{OP}} = 2^{-12} \Delta_5^2$
$L_{\text{spin}} = \zeta(s-8)\zeta(s-9)L(s, g)$	scalar normalisation	Hecke eigenline
unique pole from $\zeta(s-9)$	$\zeta(s-8)$ also has $s=9$	$L(9, g) = 0$
$L(10, g) \sim a_{10}(g)\Omega^-(g)$	wrong period parity	$L(10, g) \sim \Omega^+(g)$
arithmetic shadow $\Rightarrow$ BKM proof	wrong implication	appendix only

B EIGHT DIAGONAL-DIVISOR BOUNDARY ROWS

Let F_j denote the diagonal-divisor forms in the present F -order. The Clery–Gritsenko source order is

$$\Delta_5, \Delta_2, \nabla_3, \Delta_1, \nabla_2, \Delta_{1/2}, \nabla_{3/2}, Q_1.$$

The present order is [?GC]

j	F_j	(t, N)	Γ_j	wt
1	Δ_5	(1, 1)	Γ_1	5
2	Δ_2	(2, 1)	Γ_2^+	2
3	Δ_1	(3, 1)	Γ_3^+	1
4	$\Delta_{1/2}$	(4, 1)	Γ_4^+	$\frac{1}{2}$
5	∇_3	(1, 2)	$\Gamma_o^{(2)}(2)$	3
6	∇_2	(1, 3)	$\Gamma_o^{(2)}(3)$	2
7	$\nabla_{3/2}$	(1, 4)	$\Gamma_o^{(2)}(4)$	$\frac{3}{2}$
8	Q_1	(2, 2)	$\Gamma_2(2)$	1

and

$$\text{supp div}_{\mathbb{H}_2}(F_j) = \bigcup_{\gamma \in \Gamma_j} \gamma \langle \mathcal{H}_{\text{diag}} \rangle, \quad \text{mult}_{\mathcal{H}_{\text{diag}}}(F_j) = 1.$$

Here $F_7 = \nabla_{3/2}$ is the Clery–Gritsenko row at $(t, N) = (1, 4)$. It is not a physical comparison slot until a separate row map has been constructed.

The characters and multiplier systems are

j	ν_j
1	$v_\eta^{12} \times v_H$
2	$v_\eta^6 \times v_H$
3	$v_\eta^4 \times v_H$
4	$v_\eta^3 \times v_H$ (degree 8)
5	$\chi_2^{(2)} \times v_H$
6	$\chi_2^{(3)} \times v_H$
7	$\chi_4^{(4)} \times v_H$ (order 4)
8	$\chi_4^{(2)} \times v_H$

Definition B.1 (Candidate denominator proof datum). For a row j , a candidate denominator proof datum consists of

$$\mathcal{P}_j = (\phi_j, \Gamma_{j,\text{prod}}^+, \Gamma_{j,\text{mod}}, \rho_j, \nu_j, L_j, \alpha_j),$$

where

$$\begin{aligned}\phi_j(\tau, z) &= \sum c_j(n, l) q^n r^l, \\ F_j(Z) &= C_j e^{2\pi i(\rho_j, Z)} \prod_{\gamma \in \Gamma_{j, \text{prod}}^+} (1 - e^{2\pi i(\gamma, Z)})^{c_j(\gamma)},\end{aligned}$$

$$\nu_j : \Gamma_{j, \text{mod}} \rightarrow \mathbb{C}^\times, \quad \text{supp div}(F_j) = \Gamma_{j, \text{mod}} \cdot \mathcal{H}_{\text{diag}}, \quad \text{mult}_{\mathcal{H}_{\text{diag}}}(F_j) = 1,$$

and a degree map

$$\alpha_j : \Gamma_{j, \text{prod}}^+ \rightarrow L_j.$$

Only after a Weyl–Kac–Borcherds denominator theorem has been supplied does this datum include a Lie superalgebra $\mathfrak{g}_j$ with

$$\text{sdim}(\mathfrak{g}_{j, \alpha_j(\gamma)}) = c_j(\gamma)$$

on the corresponding product side. For a product with several cusps this single-source notation is only a template. The congruence-row product datum is the cusp-labelled datum of Proposition B.9.

Definition B.2 (Gritsenko–Nikulin degree lattice). For $t \in \mathbb{Z}_{>0}$ set

$$M_t := U(4t) \oplus \langle 2 \rangle = \mathbb{Z}f_2 \oplus \mathbb{Z}\widehat{f}_3 \oplus \mathbb{Z}f_{-2},$$

with Gram matrix

$$\begin{pmatrix} 0 & 0 & -4t \\ 0 & 2 & 0 \\ -4t & 0 & 0 \end{pmatrix}.$$

For $(n, l, m) \in \mathbb{Z}^3$ put

$$\alpha_t(n, l, m) := nf_2 + l\widehat{f}_3 + mf_{-2} \in M_t, \quad e_t^{(n, l, m)} := q^n r^l s^{tm}.$$

Then

$$(\alpha_t(n, l, m), \alpha_t(n, l, m)) = -2(4tnm - l^2).$$

Let Γ_t^{prod} be the Gritsenko–Nikulin product cone: $(n, l, m) > 0$ if $m > 0$, or if $m = 0, n > 0$, or if $m = n = 0, l < 0$.

LEMMA B.3 (*Two exponential normalizations*). Let

$$D_5(Z) := 64^{-1} \Delta_5(Z).$$

Gritsenko–Nikulin’s Siegel normalization gives

$$D_5(Z) = q^{1/2} r^{1/2} s^{1/2} \prod_{(n, l, m) > 0} (1 - q^n r^l s^m)^{f(nm, l)}$$

with leading monomial $q^{1/2} r^{1/2} s^{1/2}$. The denominator normalization used for the main BKM algebra is

$$q^n r^l s^m = \exp(-\pi i(\alpha(n, l, m), z)).$$

Thus replacing Z by $2Z$ converts the product character to the usual $\exp(-2\pi i(\alpha, z))$ denominator character, and

$$\text{den}(\mathfrak{g}_{\Delta_5}) = D_5(2Z) = 64^{-1} \Delta_5(2Z).$$

For rows F_2, F_3, F_4 , Appendix B uses the Gritsenko–Nikulin exponential

$$e_t^{(n,l,m)} = q^n r^l s^{tm}$$

directly. Hence

$$C'_j = 1, \quad a_j = 1 \quad (j = 2, 3, 4)$$

is a statement in the e_t normalization. It is not the doubled $N = 1$ normalization of the main Δ_5 denominator.

Proof. The first product is the singular-theta lift of Borcherds [?Borcherds95] together with the explicit paramodular product identity of Gritsenko–Nikulin [?GNII, Theorem 2.1, (2.21)]; the Lorentzian Kac–Moody correction and the identification of $D_5(2Z)$ with the denominator function of the corrected Lorentzian BKM superalgebra is the content of [?GN], in particular the corrected denominator identity for $\mathfrak{g}_{\Delta_5}$. The rows F_2, F_3, F_4 are the separate $e_t^{(n,l,m)} = q^n r^l s^{tm}$ denominator identities of Gritsenko–Nikulin [?GNII, Theorem 2.1, (2.3), (2.11), Lemma 2.5, (2.18), (2.19), Theorem 2.6, (2.20), (2.21), Section 5.1.1]. $\square$

PROPOSITION B.4 (*Full-paramodular rows*). Let

$$\phi_{0,2} = \sum f_2(n, l) q^n r^l, \quad \phi_{0,3} = \sum f_3(n, l) q^n r^l, \quad \phi_{0,4} = \sum f_4(n, l) q^n r^l,$$

where [?GNII]

$$\phi_{0,2} = \frac{1}{2} \eta^{-4} \sum_{m,n \in \mathbb{Z}} (3m - n) \left(\frac{-4}{m} \right) \left(\frac{12}{n} \right) q^{(3m^2+n^2)/24} r^{(m+n)/2},$$

$$\phi_{0,3} = \left(\frac{\mathfrak{g}(\tau, 2z)}{\mathfrak{g}(\tau, z)} \right)^2, \quad \phi_{0,4} = \frac{\mathfrak{g}(\tau, 3z)}{\mathfrak{g}(\tau, z)}.$$

Then the rows F_2, F_3, F_4 have the following product data:

j	F_j	ϕ_j	L_j	ρ_j
2	Δ_2	$\phi_{0,2}$	$\mathcal{M}_2$	$(\frac{1}{4}, \frac{1}{2}, \frac{1}{4})$
3	Δ_1	$\phi_{0,3}$	$\mathcal{M}_3$	$(\frac{1}{6}, \frac{1}{2}, \frac{1}{6})$
4	$\Delta_{1/2}$	$\phi_{0,4}$	$\mathcal{M}_4$	$(\frac{1}{8}, \frac{1}{2}, \frac{1}{8})$

and

$$\alpha_j^{\text{GN}}(n, l, m) := \alpha_t(n, l, m) \quad \text{for } (j, t) = (2, 2), (3, 3), (4, 4).$$

The product identities are

$$\Delta_2(Z) = q^{1/4} r^{1/2} s^{1/2} \prod_{(n,l,m) \in \Gamma_2^{\text{prod}}} (1 - q^n r^l s^{2m})^{f_2(nm,l)},$$

$$\Delta_1(Z) = q^{1/6} r^{1/2} s^{1/2} \prod_{(n,l,m) \in \Gamma_3^{\text{prod}}} (1 - q^n r^l s^{3m})^{f_3(nm,l)},$$

$$\Delta_{1/2}(Z) = q^{1/8} r^{1/2} s^{1/2} \prod_{(n,l,m) \in \Gamma_4^{\text{prod}}} (1 - q^n r^l s^{4m})^{f_4(nm,l)}.$$

The characters are respectively induced by $v_\eta^6 \times v_H$, $v_\eta^4 \times v_H$, and $v_\eta^3 \times v_H$. Equivalently, on the extended paramodular groups,

$$\nu_2 = \chi_4 : \Gamma_2^+ \rightarrow \mu_4, \quad \nu_3 = \chi_6 : \Gamma_3^+ \rightarrow \mu_6, \quad \nu_4 = \nu_8 : \Gamma_4^+ \rightarrow \mu_8.$$

Thus for $(j, t) = (2, 2), (3, 3), (4, 4)$,

$$\Gamma_{j,\text{prod}}^+ = \Gamma_t^{\text{prod}}, \quad \Gamma_{j,\text{mod}} = \Gamma_t^+.$$

Proof. The product formulae are Gritsenko–Nikulin [?GNII], Theorem 2.1 and formulae (2.21), (2.20), (2.11). Their Section 5 puts all formulae of Lie type on the hyperbolic lattice $\mathcal{M}_t$ and writes the exponent as $e^{(n,l,m)} = q^n r^l s^m$. The Weyl vector is

$$\rho_{t,II,\bar{0}} = \left(\frac{1}{2t}, \frac{1}{2}, \frac{1}{2t} \right),$$

which gives the three displayed prefactors for $t = 2, 3, 4$. □

PROPOSITION B.5 (*Gritsenko–Nikulin chambers for F_2, F_3, F_4*). For $t = 2, 3, 4$ the Weyl group is

$$W_{t,II,\bar{0}} = W^{(2)}(\mathcal{M}_t),$$

and all simple real roots in this chamber are even. The Weyl vector is

$$\rho_t = \left(\frac{1}{2t}, \frac{1}{2}, \frac{1}{2t} \right).$$

The simple real roots are

$$P_2 = \{(0, -1, 0), (1, 1, 0), (1, 3, 1), (0, 1, 1)\},$$

$$P_3 = \{(0, -1, 0), (1, 1, 0), (2, 5, 1), \\ (2, 7, 2), (1, 5, 2), (0, 1, 1)\},$$

and

$$P_4 = \{\delta \in \mathcal{M}_4 \mid (\delta, \delta) = 2, (\delta, \rho_4) = -1\}.$$

Thus the automorphic denominator chamber for F_j is the chamber in $\mathcal{M}_t \otimes \mathbb{R}$ cut out by $(x, \delta) \leq 0$ for $\delta \in P_t$, where $(j, t) = (2, 2), (3, 3), (4, 4)$.

Proof. This is Gritsenko–Nikulin [?GNII], Section 5.1.1, cases $(t, II, \bar{0})$. The displayed sets are exactly $P_{t,II,\bar{0}}(\mathcal{M}_{t,II,\bar{0}})$ for $t = 2, 3, 4$. □

LEMMA B.6 (*Product cone and chamber compatibility*). Let $t = 2, 3, 4$. If

$$f_t(nm, l) \neq 0, \quad (n, l, m) \in \Gamma_t^{\text{prod}},$$

then

$$(\alpha_t(n, l, m), \rho_t) \leq 0.$$

For $t = 2, 3$ the inequality is strict. For $t = 4$, equality occurs only on the parabolic ray

$$\alpha_t(n, l, m) \in \mathbb{Q}_{>0}\rho_4.$$

Proof. The Jacobi support satisfies

$$4tnm - l^2 \geq -1.$$

Since

$$(\alpha_t(n, l, m), \rho_t) = -2n + l - 2m,$$

the chamber inequality is $l \leq 2n + 2m$. For $t = 2, 3$, Gritsenko–Nikulin Lemma 2.5 gives the required strict chamber condition on the active support. For $t = 4$, the proof of Gritsenko–Nikulin (1.33) gives the same condition, with equality only in the parabolic case. Equivalently, equality and $4tnm - l^2 \geq -1$ force

$$n = m, \quad l = 4n,$$

so

$$\alpha_4(n, 4n, n) = 8n\rho_4.$$

□

THEOREM B.7 (*Gritsenko–Nikulin denominator rows*). Let $t = 2, 3, 4$ and put

t	$F(t)$	ϕ_t
2	Δ_2	$\phi_{0,2}$
3	Δ_1	$\phi_{0,3}$
4	$\Delta_{1/2}$	$\phi_{0,4}$

Let

$$\mathfrak{g}_t^{\text{GN}} \quad (t = 2, 3, 4)$$

be the generalized Borchers–Kac–Moody superalgebra whose Lorentzian root lattice is $\mathcal{M}_t$, whose Weyl group is $W_t = W^{(2)}(\mathcal{M}_t)$, whose Weyl vector is

$$\rho_t = \left(\frac{1}{2t}, \frac{1}{2}, \frac{1}{2t} \right),$$

and whose simple real roots are $P_t = P_{t,II,\bar{0}}$ from Proposition B.5. Let $\mathcal{M}_t$ denote the corresponding chamber. Then

$$P_{t,\bar{1}} = \emptyset, \quad P_{t,\bar{0}} = P_t,$$

so every simple real root is even and $\epsilon_t(s_\delta) = -1$ for every $\delta \in P_t$; equivalently $\epsilon_t(w) = \det(w)$ on W_t .

Let

$$\phi_t(\tau, z) = \sum f_t(n, l) q^n r^l.$$

The integers $N_t(a)$ are the additive-side coefficients in Gritsenko–Nikulin (5.1), equivalently the signed supermultiplicities of imaginary simple roots. They are not the product exponents $f_t(nm, l)$. In the Gritsenko–Nikulin exponential normalization $e_t^{(n,l,m)} = q^n r^l s^{tm}$, the Weyl–Kac–Borchers denominator identity is

$$\begin{aligned} \text{den}(\mathfrak{g}_t^{\text{GN}}) &= \sum_{w \in W_t} \epsilon_t(w) \left(e_t^{w(\rho_t)} + \sum_{a \in \mathbb{R}_{++} \cap \mathcal{M}_t} N_t(a) e_t^{w(\rho_t + a)} \right) \\ &= e_t^{\rho_t} \prod_{(n,l,m) \in \Gamma_t^{\text{prod}}} \left(1 - e_t^{(n,l,m)} \right)^{f_t(nm,l)} = F(t)(Z). \end{aligned}$$

Equivalently,

$$\text{den}(\mathfrak{g}_2^{\text{GN}}) = \Delta_2(Z), \quad \text{den}(\mathfrak{g}_3^{\text{GN}}) = \Delta_1(Z), \quad \text{den}(\mathfrak{g}_4^{\text{GN}}) = \Delta_{1/2}(Z).$$

Thus, for rows $j = 2, 3, 4$,

$$C'_j = 1, \quad a_j = 1, \quad L_j = M_j, \quad \alpha_j = \alpha_j^{\text{GN}}$$

in this normalization. The imaginary root spaces are determined here only as signed multiplicity spaces:

$$\text{sdim}(\mathfrak{g}_{t,\alpha_t(n,l,m)}) = f_t(nm, l)$$

on the product side, with imaginary simple-root superdimensions $N_t(a)$ on the Weyl-sum side.

Proof. Gritsenko–Nikulin [?GNII, Section 5.1] state that almost all identities proved earlier in their paper give denominator formulae for Lorentzian Kac–Moody algebras. Their formula (5.1) is exactly the displayed Weyl–Kac–Borcherds identity on $\mathcal{M}_t = U(4t) \oplus \langle 2 \rangle$, with $e^{(n,l,m)} = q^n r^l s^{tm}$.

For the cases $(t, II, \bar{0})$, $t = 2, 3, 4$, they give

$$W_{t,II,\bar{0}} = W^{(2)}(M_t), \quad \rho_{t,II,\bar{0}} = \left(\frac{1}{2t}, \frac{1}{2}, \frac{1}{2t} \right),$$

and

$$P_{t,II,\bar{0}}(\mathcal{M}_{t,II,\bar{0}}) = P_{t,II,\bar{0}}(\mathcal{M}_{t,II,\bar{0}})_{\bar{0}}.$$

Hence the odd simple-real-root set is empty and the Weyl character is the ordinary reflection sign on the simple real reflections. The roots in P_t have square 2, and the generalized Cartan matrix is $((\delta, \delta'))_{\delta, \delta' \in P_t}$; for $t = 4$ this is the infinite symmetric parabolic matrix recorded by Gritsenko–Nikulin. For real roots $4tn - l^2 = -1$, the relevant Jacobi coefficient is 1: this is Gritsenko–Nikulin Lemma 2.5 for $t = 2, 3$ and the proof of Gritsenko–Nikulin (1.33) for $t = 4$.

The product factors are the full-paramodular products [?GNII, (2.21), (2.20), (2.11)], already displayed in Proposition B.4. Their prefactors are exactly $e_t^{\rho_t}$. Therefore the denominator functions are Δ_2 , Δ_1 , and $\Delta_{1/2}$ with no further scalar or argument rescaling in the e_t normalization. $\square$

PROPOSITION B.8 (Congruence products). For $j = 5, 6, 7, 8$ Clery–Gritsenko construct the following Borcherds products [?GC]. The weak Jacobi inputs are

j	ϕ_j	ρ_j^{pref}
5	$4(\xi_{1,0}^{(2)})^2 \in J_{0,1}^w(\Gamma_0(2))$	$(\frac{1}{2}, \frac{1}{2}, \frac{1}{2})$
6	$3\xi_{3,1}^{(6)} \xi_{3,5}^{(6)} \in J_{0,1}^w(\Gamma_0(3))$	$(\frac{1}{2}, \frac{1}{2}, \frac{1}{2})$
7	$2\xi_{2,1}^{(4)} \xi_{2,3}^{(4)} \in J_{0,1}^w(\Gamma_0(4))$	$(\frac{1}{2}, \frac{1}{2}, \frac{1}{2})$
8	$2\xi_{1,0}^{(2)}(\tau, 2z) \in J_{0,2}^w(\Gamma_0(2))$	$(\frac{1}{4}, \frac{1}{2}, \frac{1}{2})$

where ρ_j^{pref} is read from the literal product prefactor $q^{\rho_\tau} r^{\rho_z} s^{\rho_\omega}$. At this stage it is a prefactor exponent, not a proved Weyl vector. In particular, for F_8 the M_2 product coordinate corresponding to $s^{1/2}$ is $(1/4, 1/2, 1/4)$ because $e^{\alpha_2(a,b,c)} = q^a r^b s^{2c}$. More precisely, if

$$S = \begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix},$$

and

$$\phi_5 = \sum c_5(n, l) q^n r^l, \quad \phi_5|_{\mathfrak{o}, \mathfrak{I}} S = \sum c_{5,S}(n, l) q^n r^l,$$

then

$$\begin{aligned} F_5(Z) &= \nabla_3(Z) \\ &= q^{1/2} r^{1/2} s^{1/2} \prod_{(n,l,m) \in \Gamma_1^{\text{prod}}} (1 - q^n r^l s^m)^{c_5(nm,l)} (1 - q^{2n} r^{2l} s^{2m})^{c_{5,S}(nm,l)}. \end{aligned}$$

If

$$\phi_6 = \sum c_6(n, l) q^n r^l, \quad \phi_6|_{\mathfrak{o}, \mathfrak{I}} S = \sum c_{6,S}(n, l) q^n r^l,$$

then

$$\begin{aligned} F_6(Z) &= \nabla_2(Z) \\ &= q^{1/2} r^{1/2} s^{1/2} \prod_{(n,l,m) \in \Gamma_1^{\text{prod}}} (1 - q^n r^l s^m)^{c_6(nm,l)} (1 - q^{3n} r^{3l} s^{3m})^{c_{6,S}(nm,l)}. \end{aligned}$$

If

$$\phi_7 = \sum c_7(n, l) q^n r^l, \quad \phi_7|_{\mathfrak{o}, \mathfrak{I}} S = \sum c_{7,S}(n, l) q^n r^l, \quad \phi_7|_{\mathfrak{o}, \mathfrak{I}} M = \sum c_{7,M}(n, l) q^n r^l,$$

where $M = \begin{pmatrix} 1 & -1 \\ 2 & 1 \end{pmatrix}$, then

$$\begin{aligned} F_7(Z) &= \nabla_{3/2}(Z) \\ &= q^{1/2} r^{1/2} s^{1/2} \prod_{(n,l,m) \in \Gamma_1^{\text{prod}}} (1 - q^n r^l s^m)^{c_7(nm,l)} \\ &\quad \cdot (1 - q^{2n} r^{2l} s^{2m})^{c_{7,M}(nm,l)/2} (1 - q^{4n} r^{4l} s^{4m})^{c_{7,S}(nm,l)}. \end{aligned}$$

The exponent $c_{7,M}/2$ is the literal middle-cusp exponent in the μ_4 -cover normalization; it is not a scalar-character exponent on the original congruence group. If

$$\phi_8 = \sum c_8(n, l) q^n r^l, \quad \phi_8|_{\mathfrak{o}, 2} S = \sum c_{8,S}(n, l) q^n r^l,$$

then

$$\begin{aligned} F_8(Z) &= Q_1(Z) \\ &= q^{1/4} r^{1/2} s^{1/2} \prod_{(n,l,m) \in \Gamma_2^{\text{prod}}} (1 - q^n r^l s^{2m})^{c_8(nm,l)} (1 - q^{2n} r^{2l} s^{4m})^{c_{8,S}(nm,l)}. \end{aligned}$$

These four displays pin the automorphic products. By themselves they do not construct a BKM chamber, parity function, or Weyl-sum identity.

Proof. The formulae are Clery–Gritsenko [GC, Theorem 3.1] and the four product displays for $\nabla_3, \nabla_2, \nabla_{3/2}$, and Q_1 at the end of Section 3. The second factors in F_5, F_6, F_8 and the second and third factors in F_7 come from the non-infinity cusps of $\Gamma_0(N)$ in their congruence Borchers product theorem. $\square$

PROPOSITION B.9 (*Product degree lattices for F_5, F_6, F_7, F_8*). For the product rows $j = 5, 6, 7, 8$ set

$$(t_5, N_5) = (1, 2), \quad (t_6, N_6) = (1, 3), \quad (t_7, N_7) = (1, 4), \quad (t_8, N_8) = (2, 2).$$

The Clery–Gritsenko modular host is the orthogonal lattice

$$\mathcal{O}_j = U \oplus U(N_j) \oplus \langle -2t_j \rangle$$

of signature $(2, 3)$. The single product-degree lattice for the monomial denominator is

$$L_j^{\text{prod}} := \mathcal{M}_{t_j}.$$

Define the cusp-labelled degree maps by

$$\begin{aligned} \alpha_5^{\text{prod}}(n, l, m; \infty) &= \alpha_1(n, l, m), & \alpha_5^{\text{prod}}(n, l, m; S) &= \alpha_1(2n, 2l, 2m), \\ \alpha_6^{\text{prod}}(n, l, m; \infty) &= \alpha_1(n, l, m), & \alpha_6^{\text{prod}}(n, l, m; S) &= \alpha_1(3n, 3l, 3m), \\ \alpha_7^{\text{prod}}(n, l, m; \infty) &= \alpha_1(n, l, m), & \alpha_7^{\text{prod}}(n, l, m; M) &= \alpha_1(2n, 2l, 2m), \\ & & \alpha_7^{\text{prod}}(n, l, m; S) &= \alpha_1(4n, 4l, 4m), \end{aligned}$$

and

$$\alpha_8^{\text{prod}}(n, l, m; \infty) = \alpha_2(n, l, m), \quad \alpha_8^{\text{prod}}(n, l, m; S) = \alpha_2(2n, 2l, 2m).$$

The corresponding product-prefactor degrees are

$$\rho_5^{\text{prod}} = \rho_6^{\text{prod}} = \rho_7^{\text{prod}} = \alpha_1\left(\frac{1}{2}, \frac{1}{2}, \frac{1}{2}\right), \quad \rho_8^{\text{prod}} = \alpha_2\left(\frac{1}{4}, \frac{1}{2}, \frac{1}{4}\right).$$

For $\lambda \in L_j^{\text{prod}}$ let $\mu_j(\lambda)$ be the sum of all cusp coefficients whose labelled degree is λ :

$$\mu_j(\lambda) = \sum_{\alpha_j^{\text{prod}}(n, l, m; \kappa) = \lambda} \widetilde{c}_{j, \kappa}(nm, l),$$

where

$$\begin{aligned} \widetilde{c}_{j, \infty} &= c_j, & \widetilde{c}_{5, S} &= c_{5, S}, & \widetilde{c}_{6, S} &= c_{6, S}, \\ \widetilde{c}_{7, M} &= \frac{1}{2}c_{7, M}, & \widetilde{c}_{7, S} &= c_{7, S}, & \widetilde{c}_{8, S} &= c_{8, S}. \end{aligned}$$

For $j = 7$ this is a product datum on the μ_4 -multiplier cover; the integrality of the displayed half-exponents is part of the Clery–Gritsenko product theorem in that cover. Until one proves that integrality directly from the Jacobi seed, the row is used only with this sourced cover datum. The product is therefore used as an integral automorphic product only in the sourced multiplier-cover normalization. Let

$$\mathcal{K}_5 = \mathcal{K}_6 = \mathcal{K}_8 = \{\infty, S\}, \quad \mathcal{K}_7 = \{\infty, M, S\},$$

and

$$\mathcal{D}_j := \Gamma_{t_j}^{\text{prod}} \times \mathcal{K}_j.$$

The cusp-labelled product datum is

$$\mathcal{E}_j^{\text{prod}} := (\mathcal{D}_j, \alpha_j^{\text{prod}}, \mu_j).$$

Then

$$F_j(Z) = e^{\rho_j^{\text{prod}}} \prod_{\lambda \in \alpha_j^{\text{prod}}(\mathcal{D}_j)} (1 - e^\lambda)^{\mu_j(\lambda)} \quad (j = 5, 6, 7, 8),$$

where $e^{\alpha_1(a, b, c)} = q^a r^b s^c$ and $e^{\alpha_2(a, b, c)} = q^a r^b s^{2c}$.

Proof. The displayed maps are the unique monomial-degree maps forced by the three products in Proposition B.8. For F_5 the non-infinity cusp contributes the monomials $q^{2n}r^{2l}s^{2m}$, hence degree $\alpha_1(2n, 2l, 2m)$. For F_6 the corresponding monomial is $q^{3n}r^{3l}s^{3m}$, hence degree $\alpha_1(3n, 3l, 3m)$. For F_7 the two non-infinity factors have monomials $q^{2n}r^{2l}s^{2m}$ and $q^{4n}r^{4l}s^{4m}$; the first has half exponent $c_{7,M}/2$ exactly as in the Clery–Gritsenko product. For F_8 the product has index $t = 2$: $e^{\alpha_2(n,l,m)} = q^n r^l s^{2m}$ and the non-infinity cusp contributes $e^{\alpha_2(2n,2l,2m)} = q^{2n} r^{2l} s^{4m}$. If two labelled cusp contributions land on the same lattice degree, exponents add; this is precisely the definition of μ_j .

Clery–Gritsenko [GC, Theorem 3.1] realize these products as orthogonal Borcherds products for $U \oplus U(N_j) \oplus \langle -2t_j \rangle$. That theorem supplies the automorphic product and the divisor. It does not supply a Weyl chamber, simple real roots, a parity function, or a Weyl-sum identity on M_{t_j} . Thus L_j^{prod} and α_j^{prod} are product data, not a completed BKM denominator datum. The completed denominator datum for F_5, F_6, F_7, F_8 comes from the CHL theorem in Theorem B.13. $\square$

Definition B.10 (Formal product algebra of a row). For a row j with product-degree function μ_j , let

$$\mathcal{V}_j^{\text{prod}} = \bigoplus_{\lambda \in L_j^{\text{prod}}, \lambda > 0} \mathcal{V}_{j,\lambda}$$

be any graded super vector space with finite-dimensional homogeneous pieces and

$$\text{sdim } \mathcal{V}_{j,\lambda} = \mu_j(\lambda).$$

Choose a Grothendieck representative: the even part has dimension $\mu_j(\lambda)$ when $\mu_j(\lambda) > 0$, the odd part of dimension $-\mu_j(\lambda)$ when $\mu_j(\lambda) < 0$, and the zero space when $\mu_j(\lambda) = 0$. Give it the abelian Lie superalgebra bracket:

$$[\mathcal{V}_{j,\lambda}, \mathcal{V}_{j,\lambda'}] = 0.$$

Its formal current envelope on E is

$$\text{Prod}_{j,E} := U_E^{\text{ch}}(\mathcal{V}_j^{\text{prod}} \otimes \mathcal{O}_E).$$

PROPOSITION B.11 (Eight formal current rows). Every Clery–Gritsenko diagonal-divisor row has a formal determinant object. For $j = 1$ this is $\text{Den}_{\Delta_5,E}$ from Proposition 10.4. For $j = 2, 3, 4$ it is the current envelope of the Gritsenko–Nikulin BKM algebra $\mathfrak{g}_j^{\text{GN}}$. For $j = 5, 6, 7, 8$ it is the formal product algebra $\text{Prod}_{j,E}$. For $j = 5, 6, 7, 8$, Theorem B.13 also supplies a nonabelian CHL/WKB denominator algebra; the abelian product algebra in this theorem is the part constructed directly from the Clery–Gritsenko product exponents. For $j = 5, 6, 7, 8$ this is only the tautological virtual product object realizing the Clery–Gritsenko exponents. Its Beilinson–Drinfeld envelope is abelian; it carries no Weyl-sum side, no BKM bracket, and no physical protected BPS interpretation.

In all eight cases the formal determinant is the corresponding monic automorphic product:

$$e^{\rho_j} \text{sdet}_{\mathcal{V}_j}(1 - e) = D_j(Z),$$

where $D_1 = 64^{-1}\Delta_5$ and $D_j = F_j$ for the displayed Clery–Gritsenko product normalizations $j > 1$, with the normalizations stated in Propositions B.4, B.8, and B.9.

Proof. For $j = 1$, Theorem 10.2 gives the Borcherds denominator algebra and Proposition 10.4 gives the formal current envelope. For $j = 2, 3, 4$, Theorem B.7 gives the BKM brackets and their denominator identities.

For $j = 5, 6, 7, 8$, Proposition B.9 constructs the degree lattice, the positive product support, the prefactor, and the signed exponent μ_j . The super vector space $\mathcal{V}_j^{\text{prod}}$ realizes these signed dimensions. The Fock determinant lemma gives

$$e^{\rho_j^{\text{prod}}} \prod_{\lambda} (1 - e^{\lambda})^{\text{sdim } \mathcal{V}_{j,\lambda}} = e^{\rho_j^{\text{prod}}} \prod_{\lambda} (1 - e^{\lambda})^{\mu_j(\lambda)} = F_j(Z).$$

The formal current enveloping construction gives a chiral algebra on E from this abelian Lie superalgebra. It is not a compact BPS factorization algebra. $\square$

PROPOSITION B.12 (*Nonabelian promotion test for F_5, F_6, F_7, F_8*). Suppose that one of the congruence product rows

$$j \in \{5, 6, 7, 8\}$$

is the denominator of a generalized Borcherds–Kac–Moody superalgebra on the product lattice L_j^{prod} . Then the following additional data must exist:

$$(P_j, \mathcal{M}_j, W_j, \epsilon_j, \rho_j, N_j(a)).$$

They must satisfy

$$W_j = \langle s_{\delta} \mid \delta \in P_j \rangle, \quad \mathcal{M}_j = \{x \mid (x, \delta) \leq 0 \text{ for all } \delta \in P_j\},$$

$$(\rho_j, \delta) = -\frac{(\delta, \delta)}{2} \quad (\delta \in P_j),$$

and the Weyl–Kac–Borcherds identity

$$\begin{aligned} F_j(Z) &= \sum_{w \in W_j} \epsilon_j(w) \left(e^{w(\rho_j)} + \sum_{a \in L_j^{\text{prod}} \cap \mathbb{R}_{>0} \mathcal{M}_j} N_j(a) e^{w(\rho_j + a)} \right) \\ &= e^{\rho_j^{\text{prod}}} \prod_{\lambda \in \alpha_j^{\text{prod}}(\mathcal{D}_j)} (1 - e^{\lambda})^{\mu_j(\lambda)}. \end{aligned}$$

In particular the product already fixes the signed root multiplicity function μ_j and the product-prefactor degree

$$\rho_5^{\text{prod}} = \rho_6^{\text{prod}} = \rho_7^{\text{prod}} = \alpha_1\left(\frac{1}{2}, \frac{1}{2}, \frac{1}{2}\right), \quad \rho_8^{\text{prod}} = \alpha_2\left(\frac{1}{4}, \frac{1}{2}, \frac{1}{4}\right),$$

but it does not fix $P_j, \mathcal{M}_j, W_j, \epsilon_j$, or the imaginary simple-root coefficients $N_j(a)$.

Proof. The equalities in Proposition B.9 construct only the monomial degree and the total exponent $\mu_j(\lambda)$. A denominator theorem has a second side: the Weyl alternating sum. That side requires a reflection chamber, a set of simple real roots, a reflection character, and the imaginary simple-root contribution.

The Gritsenko–Nikulin chambers for $\mathcal{M}_1$ and $\mathcal{M}_2$ do not follow by notation. The forms $\nabla_3, \nabla_2, \nabla_{3/2}$, and Q_1 are congruence products with non-infinity cusp factors, whereas Δ_5 and Δ_2 are full-paramodular

denominator forms. Clery–Gritsenko construct the automorphic product, character, and diagonal divisor on

$$U \oplus U(N_j) \oplus \langle -2t_j \rangle,$$

but their product theorem does not identify the simple real roots or prove a Weyl anti-invariant additive expansion on M_{t_j} . Therefore the displayed six-tuple is an additional theorem to be proved, not a consequence of the product formula. $\square$

THEOREM B.13 (*Govindarajan CHL/WKB denominator rows*). In the CHL/twisted-dyon normalization of Cheng–Dabholkar and Govindarajan, the rows

$$F_5 = \nabla_3, \quad F_6 = \nabla_2, \quad F_7 = \nabla_{3/2}, \quad F_8 = Q_1$$

are Weyl–Kac–Borchers denominator forms. The row F_7 is taken on the μ_4 -multiplier cover determined by the Clery–Gritsenko order-four multiplier. Thus F_7 is a denominator on the μ_4 -cover, with no physical row map, metaplectic boundary theorem, or CY/DT/chiral host. In the product-degree notation of this appendix the real-root data are

$$L_5 = L_6 = L_7 = M_1, \quad P_5 = P_6 = P_7 = P^{(1)}, \quad \rho_5 = \rho_6 = \rho_7 = \alpha_1(\tfrac{1}{2}, \tfrac{1}{2}, \tfrac{1}{2}),$$

where

$$P^{(1)} = \{(0, -1, 0), (1, 1, 0), (0, 1, 1)\},$$

and

$$L_8 = M_2, \quad P_8 = P^{(2)}, \quad \rho_8 = \alpha_2(\tfrac{1}{4}, \tfrac{1}{2}, \tfrac{1}{4}),$$

where

$$P^{(2)} = \{(0, -1, 0), (1, 1, 0), (1, 3, 1), (0, 1, 1)\}.$$

The Weyl chambers and Weyl groups are

$$\mathcal{M}_j = \{x \in L_j \otimes \mathbb{R} \mid (x, \delta) \leq 0 \text{ for every } \delta \in P_j\}, \quad W_j = \langle s_\delta \mid \delta \in P_j \rangle.$$

Every simple real root is even and the reflection character is

$$\epsilon_j(w) = \det(w).$$

For $j = 5, 6, 7, 8$ the denominator product side is expressed through the cusp-labelled product datum $\mathcal{E}_j^{\text{prod}}$, with root multiplicity function μ_j . In Govindarajan’s CHL normalization the corresponding square-root form is anti-invariant under the Weyl reflections, with character $\epsilon_j(w) = \det(w)$. This anti-invariance is part of the Govindarajan WKB denominator theorem, not a consequence of the Clery–Gritsenko product alone; this manuscript does not construct an independent row-by-row embedding of the CHL Weyl group into each Clery–Gritsenko modular group or multiplier cover. The Cartan matrices are, in the CHL/WKB BKM normalization of [CD09BKM, Gov11BKM],

$$A^{(1)} = \begin{pmatrix} 2 & -2 & -2 \\ -2 & 2 & -2 \\ -2 & -2 & 2 \end{pmatrix},$$

for F_5, F_6, F_7 , and

$$A^{(2)} = \begin{pmatrix} 2 & -2 & -6 & -2 \\ -2 & 2 & -2 & -6 \\ -6 & -2 & 2 & -2 \\ -2 & -6 & -2 & 2 \end{pmatrix},$$

for F_8 .

Equivalently, for $x = (u, v, w)$, the two chambers are

$$\mathcal{M}_1 = \{x \mid v \geq 0, v \leq 2w, v \leq 2u\},$$

and

$$\mathcal{M}_2 = \{x \mid v \geq 0, v \leq 4w, 3v \leq 4(u + w), v \leq 4u\}.$$

There are integral imaginary simple-root coefficients $N_j(a)$, depending on the CHL twist [CD09BKM, Gov11BKM], such that

$$\begin{aligned} F_j(Z) &= \sum_{w \in W_j} \det(w) \left(e^{w(\rho_j)} + \sum_{a \in L_j \cap \mathbb{R}_{>0} \mathcal{M}_j} N_j(a) e^{w(\rho_j + a)} \right) \\ &= e^{\rho_j} \prod_{\lambda \in \alpha_j^{\text{prod}}(\mathcal{D}_j)} (1 - e^\lambda)^{\mu_j(\lambda)} \quad (j = 5, 6, 7, 8). \end{aligned}$$

Thus the six-tuples

$$(P_j, \mathcal{M}_j, W_j, \epsilon_j, \rho_j, N_j(a))$$

are constructed for $j = 5, 6, 7, 8$ by the CHL denominator theorem. They are not consequences of the Clery–Gritsenko product formula alone. In the monic Clery–Gritsenko/Govindarajan product variables of Proposition B.8, the identities have

$$C'_j = 1, \quad a_j = 1 \quad (j = 5, 6, 7, 8).$$

This is not the doubled $N = 1$ normalization of Lemma B.3.

Remark B.14 (F_7 on the μ_4 -multiplier cover). The row $F_7 = \nabla_{3/2}$ is the Clery–Gritsenko diagonal-divisor form of type $(t, N) = (1, 4)$, weight $3/2$, on $\Gamma_0^{(2)}(4)$ with an order-four multiplier. After pullback to the associated μ_4 -multiplier cover, Govindarajan’s CHL/WKB theorem identifies it with the Weyl–Kac–Borcherds denominator of $\mathcal{G}_1(4)$. It is the μ_4 -cover denominator identity, with no physical comparison-row identification, no metaplectic boundary host, and no CY/DT or chiral/Hall theorem.

The square identity $F_7^2 = \Phi_3$ is only a scalar square identity. It does not carry the reflection sign, parity, Weyl chamber, or Weyl-alternating side. The coefficient convention for the additive side is fixed in Lemma B.16.

LEMMA B.15 (*Coefficient normalisation for ∇_3 and ∇_2*). Let

$$\psi_5(\tau, z) = \mathfrak{D}(\tau, z) \eta(\tau) \eta(2\tau)^4 = \sum g_5(u, v) q^{u/2} r^{v/2},$$

and

$$\psi_6(\tau, z) = \mathfrak{D}(\tau, z) \eta(3\tau)^3 = \sum g_6(u, v) q^{u/2} r^{v/2}.$$

For $a = \alpha_1(A, B, C) \in \mathcal{M}_1 \cap \mathbb{R}_{\geq 0} \mathcal{M}_1$, put

$$n = 2A + 1, \quad l = 2B + 1, \quad m = 2C + 1.$$

In the plus-sign convention of Theorem B.13, the imaginary simple-root coefficients for the rows F_5 and F_6 are

$$N_5(a) = \sum_{d|(n,l,m)} d^2 g_5\left(\frac{nm}{d^2}, \frac{l}{d}\right),$$

and

$$N_6(a) = \sum_{d|(n,l,m)} \left(\frac{-3}{d}\right) d g_6\left(\frac{nm}{d^2}, \frac{l}{d}\right).$$

Govindarajan–Gopala Krishna write the same coefficients with the opposite sign in the imaginary-root correction term:

$$m_j(a) = -N_j(a) \quad (j = 5, 6).$$

Proof. Govindarajan–Gopala Krishna construct the additive lift whose Weyl alternating side gives the denominator identity for the corresponding CHL square roots [?GK09CHL, (4.2), (4.4), (4.5), (4.8)]. Their Maass lift gives the displayed divisor sums from the Jacobi seeds ψ_5 and ψ_6 , and their Weyl–Kac–Borchers sum writes these coefficients as the imaginary simple-root correction. The sign difference is only the convention:

$$e^\rho + \sum_a N_j(a) e^{\rho+a},$$

whereas [?GK09CHL] writes the correction with a minus sign. □

LEMMA B.16 (*Coefficient normalisation for $\nabla_{3/2}$*). Let

$$\nabla_{3/2}(Z) = \Delta_{3/2}^{(4)}(Z) = \frac{1}{8} \theta_5(2Z) \theta_7(2Z) \theta_9(2Z) = \sum_{n,l,m} A_7(n, l, m) q^{n/2} r^{l/2} s^{m/2},$$

with $A_7(1, 1, 1) = 1$. The first term is

$$\left(\sqrt{r} - \frac{1}{\sqrt{r}}\right) \sqrt{q} \sqrt{s} + \cdots,$$

so $A_7(1, -1, 1) = -1$. For $a = \alpha_1(A, B, C) \in \mathcal{M}_1 \cap \mathbb{R}_{\geq 0} \mathcal{M}_1$, put

$$n = 2A + 1, \quad l = 2B + 1, \quad m = 2C + 1.$$

Then the imaginary simple-root coefficient in the plus-sign convention of Theorem B.13 is

$$N_7(\alpha_1(A, B, C)) = A_7(2A + 1, 2B + 1, 2C + 1) = [q^{A+1/2} r^{B+1/2} s^{C+1/2}] \nabla_{3/2}(Z).$$

In Govindarajan–Gopala Krishna’s WKB sign convention,

$$m_7(a) = -N_7(a).$$

No Maass divisor-sum formula for N_7 is asserted here.

Proof. Clery–Gritsenko construct $\nabla_{3/2}$ as the order-four multiplier row. Govindarajan–Gopala Krishna identify the square-root theta product and the square relation

$$F_7^2 = \Phi_3$$

in the composite $N = 4$ CHL case [?GK10Composite]. Factoring the Weyl vector gives

$$e^{\rho_7+a} = q^{A+1/2} r^{B+1/2} s^{C+1/2},$$

which is exactly the displayed coefficient extraction. The sourced divisor-sum formula belongs to the square F_7^2 , with its composite $N = 4$ Jacobi seed. A direct prime-type half-index formula for F_7 introduces fractional divisor powers at non-primitive chamber points, whereas the theta product has integral coefficients. This is a theta-product normalization identification. It is not a constructed physical host and not a physical row-map equivalence. $\square$

LEMMA B.17 (Q_1 in M_2 notation). In M_2 ,

$$e^{\alpha_2(n,l,m)} = q^n r^l s^{2m}, \quad (\alpha_2(n, l, m), \alpha_2(n', l', m')) = 2ll' - 8(nm' + n'm).$$

The four simple real roots of the Q_1 row are

$$\delta_1 = (0, -1, 0), \quad \delta_2 = (1, 1, 0), \quad \delta_3 = (1, 3, 1), \quad \delta_4 = (0, 1, 1),$$

with Gram matrix $A^{(2)}$. The Weyl vector is

$$\rho_8 = \alpha_2\left(\frac{1}{4}, \frac{1}{2}, \frac{1}{4}\right), \quad e^{\rho_8} = q^{1/4} r^{1/2} s^{1/2},$$

and

$$(\rho_8, \delta_i) = -1 \quad (1 \leq i \leq 4).$$

Let $A_8(n, l, m)$ be defined by

$$Q_1(Z) = \sum_{n,l,m} A_8(n, l, m) q^{n+1/4} r^{l+1/2} s^{2m+1/2}.$$

Govindarajan's theta-product formula, in his characteristic notation, is

$$Q_1(Z) = \frac{1}{4} \vartheta \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} (Z') \vartheta \begin{bmatrix} 1 & 1 \\ 1 & 1 \end{bmatrix} (Z'), \quad Z' = \begin{pmatrix} \tau & 2z \\ 2z & 4\sigma \end{pmatrix}.$$

Equivalently,

$$A_8(n, l, m) = \frac{1}{4} \sum_{S(n,l,m)} (-1)^{j+k+h},$$

where $S(n, l, m)$ is the set of $(i, j, k, h) \in \mathbb{Z}^4$ satisfying

$$\frac{1}{2} \left(\left(i + \frac{1}{2} \right)^2 + \left(k + \frac{1}{2} \right)^2 \right) = n + \frac{1}{4},$$

$$2 \left(i + \frac{1}{2} \right) j + 2 \left(k + \frac{1}{2} \right) \left(h + \frac{1}{2} \right) = l + \frac{1}{2},$$

and

$$2j^2 + 2 \left(h + \frac{1}{2} \right)^2 = 2m + \frac{1}{2}.$$

Then the imaginary simple-root coefficient in the plus-sign convention is

$$N_8(\alpha_2(n, l, m)) = A_8(n, l, m)$$

for the chamber-supported imaginary shifts. Govindarajan's expansion has coefficient -1 at the monomials

$$r^{-1}, \quad qr, \quad rs^2, \quad qr^3s^2.$$

After factoring e^{ρ^8} these are precisely the four simple real root monomials for

$$\delta_1, \quad \delta_2, \quad \delta_4, \quad \delta_3,$$

so the signed real-simple-root supermultiplicity is $+1$ in each case. Their even parity and multiplicity-one real-root data are the Cheng–Dabholkar/Govindarajan CHL denominator data.

Proof. The first display is the definition of α_2 and the Gram matrix on M_2 . The simple roots and Weyl vector are Cheng–Dabholkar's $N = 2$ chamber data translated through

$$\alpha_2(n, l, m) \longleftrightarrow \begin{pmatrix} 2n & l \\ l & 4m \end{pmatrix}.$$

Govindarajan identifies Q_1 with the row $\mathcal{G}_2(4)$ and gives both the product formula and the first Fourier terms. In that CHL normalization the square-root form is anti-invariant under the Weyl reflections. This is the sourced denominator anti-invariance, not an independently constructed restriction of the Clery–Gritsenko character through a physical row map. $\square$

Proof of Theorem B.13. Cheng–Dabholkar identify, for CHL level $N = 1, 2, 3$, the Weyl group, simple real roots, Weyl chamber, and Weyl denominator identity of the dyon algebra [?CD09BKM, (2.10)–(2.16), (3.16), (3.22), (3.23), (3.26), (3.42), (5.1), (5.2), (5.8), (5.10), (5.12)–(5.15), Appendix A]. For $N = 1$ the simple real roots are $P^{(1)}$ and the Cartan matrix is $A^{(1)}$; for $N = 2$ the simple real roots are $P^{(2)}$ and the Cartan matrix is $A^{(2)}$. Govindarajan then treats twisted dyons and proves that, for fixed N , the algebras $\mathcal{G}_N(M)$ have the same real simple roots, Weyl vector, and Cartan matrix, while the imaginary simple roots depend on M [?Gov11BKM, Section 4.2, Section 4.3, Table 3].

Govindarajan's Table 3 gives

$$\nabla_3 \longleftrightarrow \mathcal{G}_1(2), \quad \nabla_2 \longleftrightarrow \mathcal{G}_1(3), \quad \nabla_{3/2} \longleftrightarrow \mathcal{G}_1(4), \quad Q_1 \longleftrightarrow \mathcal{G}_2(4).$$

Section 4.2 of [?Gov11BKM] checks the required ingredients: sum and product representations, integrality, correct multiplicity of the simple real roots, and the sign change of the square root under a Weyl reflection. Section 4.3 states the Weyl–Kac–Borcherds denominator formula for $\mathcal{G}_N(M)$. For $F_7 = \nabla_{3/2}$ the theorem is on the μ_4 -multiplier cover, with no physical boundary partition identity and no CY/DT/chiral host. For Q_1 , Govindarajan supplies the WKB denominator data [?Gov11BKM, Appendix C, Appendix D]. The dyon-counting additive-seed issue for the $(2, 4)$ and $(4, 2)$ examples plays no role. The later hyperbolic and umbral extensions of [?GovSam19] are also outside this proof: for $N = 5, 6$ they replace W by W_+ , leave the sum side incompletely characterized, and treat some rows as product-only or candidate BKM data [?GovSam19, Sections 3.3 and 4].

The real-root data translate to this notation as follows. On M_t ,

$$((n, l, m), (n', l', m')) = 2ll' - 4t(nm' + n'm).$$

With $t = 1$, the set $P^{(1)}$ has Gram matrix $A^{(1)}$ and $\rho = \alpha_1(1/2, 1/2, 1/2)$ satisfies $(\rho, \delta) = -1$ for every $\delta \in P^{(1)}$. With $t = 2$, the set $P^{(2)}$ has Gram matrix $A^{(2)}$ and $\rho = \alpha_2(1/4, 1/2, 1/4)$ satisfies $(\rho, \delta) = -1$ for every $\delta \in P^{(2)}$. These are exactly the Weyl-vector equations $(\rho, \delta) = -(\delta, \delta)/2$ because all displayed real roots have square 2. $\square$

Remark B.18 (Nonabelian row data). The row data are:

j	automorphic data $(F_j, \Gamma_j, \nu_j, \rho_j^{\text{pref}})$	lattice and degree datum
1	constructed here	constructed here
2, 3, 4	GN denominator theorem	$L_j = M_j, \alpha_j = \alpha_j^{\text{GN}}$
5, 6, 7	CG product; Govindarajan denominator	$P^{(1)}, M_1, W_1, \epsilon = \det$ (CHL)
8	CG product; Govindarajan denominator	$P^{(2)}, M_2, W_2, \epsilon = \det$ (CHL)

Thus all eight rows have formal product/current-envelope objects. Rows 1–4 have the Gritsenko–Nikulin denominator algebras. Rows 5–8 have the Govindarajan CHL/WKB denominator algebras in the CHL normalization. The row F_7 lives on the μ_4 -multiplier cover and fills no physical-host certificate beyond the sourced denominator level.

Open Problem B.19 (Physical-host certificates and row maps). Construct, for every claimed physical realization of a Clery–Gritsenko row, a datum

$$(\iota_j, \mathcal{R}_j, \mathcal{A}_j, \mathcal{D}_j, \mathcal{S}_j, \mathcal{L}_j, \mathcal{H}_j)$$

such that

$$\iota_j : \Phi_j^{\text{phys}} \longleftrightarrow F_i^{\text{CG}}$$

fixes the exact row, parent form when a root is used, group or cover, automorphy cocycle, multiplier and its order, divisor multiplicities, Jacobi seed at every cusp, product-degree lattice, chamber, branch, normalization, and source theorem. The entries $(\mathcal{A}_j, \mathcal{D}_j, \mathcal{S}_j, \mathcal{L}_j, \mathcal{H}_j)$ are the five levels of Definition 12.1. For rows $F_5, \dots, F_8$, the Govindarajan CHL/WKB theorem fills $\mathcal{D}_j$, and the sourced scalar square fills $\mathcal{S}_j$ where stated. It does not fill $\mathcal{L}_j$ or $\mathcal{H}_j$. A boundary partition identity and a compact Hall/chiral source remain open until these certificate entries are constructed.

PROPOSITION B.20 (Comparison row systems). The Clery–Gritsenko rows in the present F_i -order have weights

$$(5, 2, 1, \frac{1}{2}, 3, 2, \frac{3}{2}, 1).$$

In triple order

$$(t, N) = (1, 1), (2, 1), (1, 2), (3, 1), (1, 3), (4, 1), (1, 4), (2, 2),$$

the weights are

$$(5, 2, 3, 1, 2, \frac{1}{2}, \frac{3}{2}, 1),$$

and the permutation from triple order to present order is

$$(1, 2, 5, 3, 6, 4, 7, 8).$$

Consequently

$$\{j : \text{wt}(F_j) \in \mathbb{Z}\} = \{1, 2, 3, 5, 6, 8\}, \quad \{j : \text{wt}(F_j) \notin \mathbb{Z}\} = \{4, 7\}.$$

The comparison notation

$$\Phi_1^{\text{cmp}}, \dots, \Phi_8^{\text{cmp}}, \quad \text{wt}(\Phi_j^{\text{cmp}}) = (5, 2, 1, 1, \frac{1}{2}, 1, \frac{1}{4}, 0),$$

does not define the Clery–Gritsenko row system and defines no physical atlas in this paper. A row map

$$\Phi_j^{\text{cmp}} \longleftrightarrow F_i$$

requires the relevant entries of Definition 12.1. Consequently the direct identifications

$$\Phi_5^{\text{cmp}} = F_5, \quad \Phi_7^{\text{cmp}} = F_7, \quad \Phi_8^{\text{cmp}} = F_8$$

are false already by weight. The eight-row Clery–Gritsenko catalogue contains no weight-1/4 row and no weight-0 row. The CHL square identities, such as $\nabla_3^2 = \Phi_6$ and $\nabla_2^2 = \Phi_4$, are scalar square identities; they are not row maps and do not by themselves carry the denominator sign, parity, or Weyl alternating side.

Proof. The Clery–Gritsenko theorem gives the exact weights, groups, characters, and divisors of the eight diagonal-divisor rows [GC]. The displayed comparison tuple is not that theorem table. The Borcea–Voisin route [BORCEA], [VOISIN] and higher-dimensional curve-counting routes [OhThomasI], [OhThomasII], [BOPY] require separate construction of X_j , the reduced theory, the protected index, and the product identity. None is constructed here. Thus the comparison symbols remain external labels until a certificate identifies them with Clery–Gritsenko rows. $\square$

Definition B.21 (Canonical fractional branch). Let

$$F(Z) = a q^\alpha r^\beta s^\gamma (1 + O(q, r, s)), \quad a \in \mathbb{R}_{>0},$$

be a product expansion in a fixed positive Weyl chamber. Its canonical k -th root in that chamber is

$$F(Z)^{1/k} = a^{1/k} q^{\alpha/k} r^{\beta/k} s^{\gamma/k} \exp\left(\frac{1}{k} \log(1 + O(q, r, s))\right),$$

where $a^{1/k}$ is the positive real root and the logarithm is the formal logarithm in the completed product algebra. This defines a formal branch. It does not define a modular form until the relevant metaplectic or multiplier-cover character has been constructed.

PROPOSITION B.22 (Fractional-power comparison test). The formal fractional-power comparison symbols

$$\Phi_5^{\text{cmp}}, \quad \Phi_7^{\text{cmp}}, \quad \Phi_8^{\text{cmp}}$$

do not label Clery–Gritsenko rows. The only source-compatible statement retained here is the following conditional test.

The displayed comparison weight of Φ_5^{cmp} is 1/2. If it is presented as a square root, the parent form must have weight 1, and the root must live on the pullback of the parent group to the genus-two metaplectic cover with a genuine half-integral multiplier. A weight-5 parent would produce a weight-5/2 square root and does not certify this displayed comparison slot.

If the comparison symbol Φ_7^{cmp} is supplied with its displayed comparison weight 1/4, then it must be a genuine quarter-integral-weight form on a multiplier cover. If it is presented as a fourth root of a parent form with exact order-four character, an order-eight cover is not forced by character theory. If

a character root ψ with $\psi^4 = \chi$ exists after pullback and χ has exact order 4, then the character part has order divisible by 16, unless an explicit automorphy-cocycle relation reduces it. No such relation is constructed here. The μ_4 -cover of the Clery–Gritsenko row $F_7 = \nabla_{3/2}$, whose weight is $3/2$, is not a modular-root certificate for a comparison object Φ_7^{cmp} of weight $1/4$. That comparison object needs its own parent, cover, cocycle, and character-root computation.

If a comparison form Φ_8^{cmp} has weight 0, the canonical branch of Definition B.21 is only a formal logarithmic branch. By itself it gives no finite-order modular multiplier and no chiral square-root theorem.

Proof. The first two assertions are necessary metaplectic-cover conditions: a fractional power of an automorphic product is modular only after the automorphy factor, character, and divisor multiplicities lift to a genuine character on a cover; Weil and Shimura give the classical background for such metaplectic phenomena [Weil1964], [Shimura1973]. The stated character-order obstruction is elementary: if χ has exact order 4 and $\psi^4 = \chi$, then ψ cannot have order 8, since ψ^4 would have order at most 2; absent a further cocycle relation, order 16 is the first possible character order. In the weight-zero case the formal branch supplies no metaplectic weight contribution; modularity would have to come entirely from divisor divisibility and a character or automorphy cocycle root. A formal root can be chosen in a chamber, but this is not a modular or chiral construction. $\square$

Open Problem B.23 (Fractional comparison-row physical-host certificate). For a fractional comparison row, construct a modular-root certificate $\mathcal{R}_j$ and the five-level physical-host certificate of Definition 12.1. The root certificate must fix the parent form, root exponent, exact group or cover, multiplier character and its order, automorphy cocycle, divisor multiplicities, branch, product normalization, and source theorem. The reduced scalar level, if claimed, must produce a compact Calabi–Yau, orbifold Calabi–Yau, or CHL background with oriented reduced obstruction theory and prove

$$Z_j^{\text{red}} = C_j(\Phi_j^{\text{cmp}})^{-2}$$

in the fixed chamber and cover. The compact Hall/chiral level must construct the protected one-particle index and primitive bracket comparison for the same row.

Borcea–Voisin and higher-dimensional curve-counting constructions [BORCEA], [VOISIN], [OhThomasI], [OhThomasII], [BOPY] are possible sources for such a certificate. This paper does not assert that they supply it. Until $\mathcal{R}_j$ and the relevant certificate entries are constructed, the symbols Φ_5^{cmp} , Φ_7^{cmp} , Φ_8^{cmp} are formal comparison symbols only: no Clery–Gritsenko row, finite-order multiplier, denominator identity, CY/DT partition theorem, or chiral power follows.

The following comparison table is restricted to the $N = 1$ Igusa boundary. It contains no row map

$$\Phi_j^{\text{cmp}} \longleftrightarrow F_i$$

and no physical row-map denominator theorem for any congruence row. The separate Govindarajan CHL/WKB denominator theorem remains Theorem B.13.

C $N = 1$ BOUNDARY-COMPATIBILITY CONDITIONS

Let a chambered Hall-descent datum supply the compact comparison data of Definition 10.32, so that the criterion of Theorem 10.33 applies. Its $N = 1$ decategorified boundary would then be constrained by the Igusa datum

$$\begin{aligned}\Gamma_{\text{eff}} = & \{(n, l, m) \in \mathbb{Z}^3 \mid m > 0, n \geq 0, l \in \mathbb{Z}\} \\ & \cup \{(n, l, 0) \in \mathbb{Z}^3 \mid n > 0, l \in \mathbb{Z}\} \\ & \cup \{(0, l, 0) \in \mathbb{Z}^3 \mid l < 0\} \subset \mathbb{Z}^3.\end{aligned}$$

comparison Hall descent	Igusa boundary
Gram index lattice	$\Gamma_{\text{ind}} = \mathbb{Z}^3$
sector completion	Γ_{eff}
active denominator support	$\Gamma_{\text{act}} = \{\gamma \in \Gamma_{\text{eff}} \mid f(nm, l) \neq 0\}$
Lorentzian degree	$\alpha(n, l, m) = 2nf_2 - lf_3 + 2mf_{-2}$
orientation character	required geometric monodromy comparison $\epsilon_o _{W^{(2)}} = \nu_{\Delta_5} _{W^{(2)}}$
automorphic boundary	$\text{Borch}_\theta(\phi_{o,i}) = \Delta_5$
BKM denominator	$\text{den}(\mathfrak{g}_{\Delta_5}) = 64^{-1}\Delta_5(2Z)$
denominator algebra	$\mathcal{A}_{\text{den}} = \mathfrak{g}_{\Delta_5}$
formal current envelope	$\text{Den}_{\Delta_5, E} = U_E^{\text{ch}}(\mathfrak{g}_{\Delta_5} \otimes \mathcal{O}_E)$
scalar square	$Z_{\text{OP}}^X = Z_{\text{DT}}^X = -4096\Delta_5^{-2}$ (proved).

The degree map satisfies

$$(\alpha(\gamma), \alpha(\eta)) = -2\langle \gamma, \eta \rangle_{\text{ind}}, \quad (\alpha(n, l, m), \alpha(n, l, m)) = -2(4nm - l^2).$$

The real walls are

$$\delta_1 = 2f_2 - f_3, \quad \delta_2 = 2f_{-2} - f_3, \quad \delta_3 = f_3.$$

The Weyl vector is

$$\rho = f_2 - \frac{1}{2}f_3 + f_{-2}, \quad (\rho, \delta_i) = -1.$$

These equations give the denominator polarisation of the same Gram-index lattice whose cusp polarisation gives the Fock determinant. The cone is generated by the active support. Triples in Γ_{eff} with zero Jacobi coefficient fix ordering and contribute no visible product factors. The product alone does not exclude invisible zero-superdimension root spaces.

The comparison is the $N = 1$ normalization

$$\text{Borch}_\theta(\phi_{o,i}) = \Delta_5, \quad \text{den}(\mathfrak{g}_{\Delta_5}) = 64^{-1}\Delta_5(2Z), \quad \kappa_{\text{BKM}}(\mathfrak{g}_{\Delta_5}) = \text{wt}(\Delta_5) = 5.$$

It contains no denominator theorem beyond the $N = 1$ theorem and no row map

$$\Phi_j^{\text{cmp}} \longleftrightarrow F_i.$$

The protected scalar has the form

$$Z_{\square}^X = C_{\square}\Delta_5^{-2}.$$

The Oberdieck–Pixton and reduced Hilbert-scheme DT branches have constant -4096 in the quotient-by- E normalization of Theorem 6.2. The chiral/denominator half is Δ_5 .

A compact Hall–Drinfeld double would realize a protected bracket from the compactification only when the comparison package of Definition 10.32 and Theorem 10.33 is supplied: finite-first oriented Hall descent, protected integration, charge additive Hall primitives, Borchers relations inside the Hall super-commutator, and the completed Hopf pairing with orientation character required to restrict to ν_{Δ_5} . Homotopy descent must produce the positive half, its completion data, and the Hall-side domain on which the comparison character is defined. This orientation character is not produced by rescaling Δ_5 ; it must come from monodromy of the chosen square root of the Hall determinant line. The Igusa boundary identifies the denominator algebra $\mathfrak{g}_{\Delta_5}$ and its protected decategorified multiplicities before those comparison data are proved.

For the Clery–Gritsenko rows F_5, F_6, F_7, F_8 , this table contains no nonabelian chamber, no simple real roots, no reflection character, and no Weyl-sum side. Rows F_5, F_6, F_7, F_8 use the separate Govindarajan CHL/WKB denominator theorem when a nonabelian bracket is needed. For F_7 this is a theorem on the μ_4 -multiplier cover determined by the Clery–Gritsenko order-four multiplier. None of these denominator rows supplies a comparison row map or a CY/DT/chiral boundary host.